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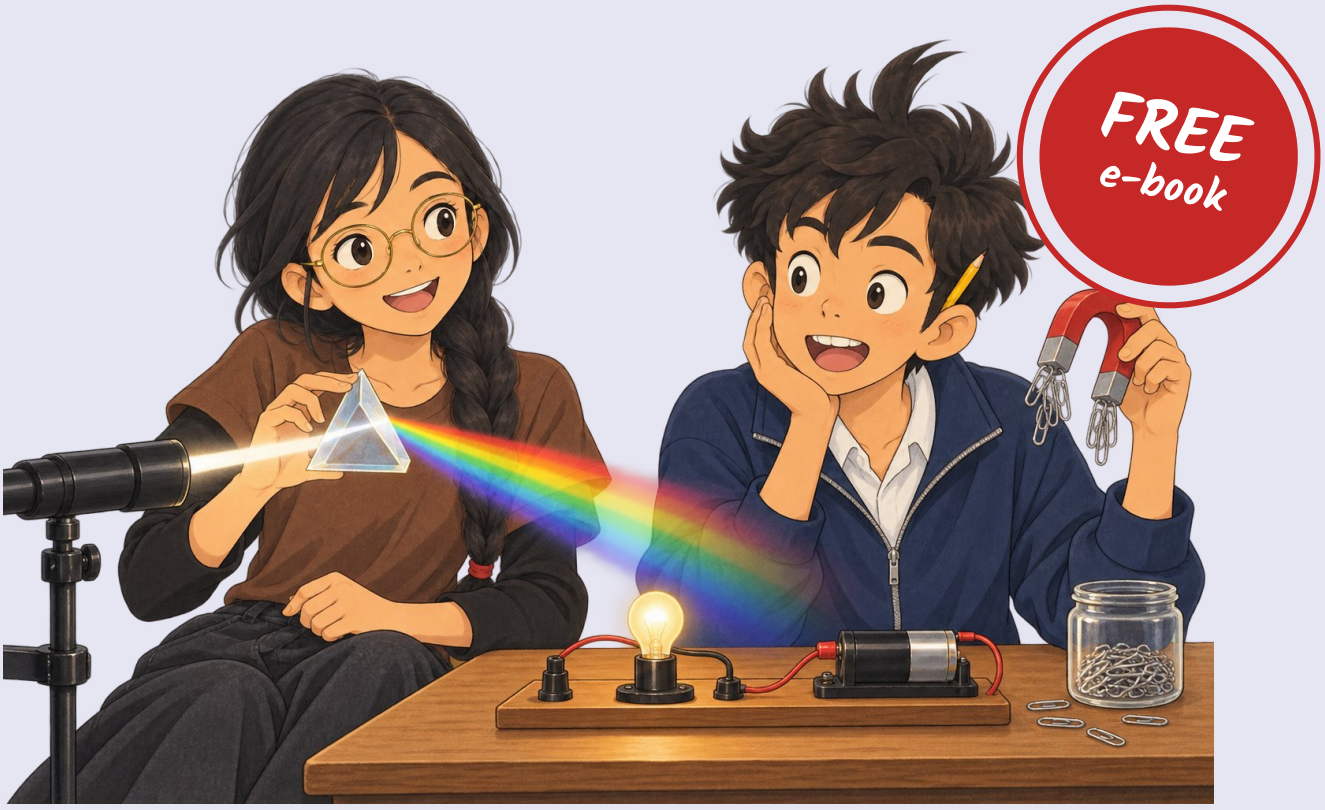
CBSE Class 12

PHYSICS

KHATAM IN 199 PAGES

Session 2026-27



Electrostatics · Current & Magnetism · EMI & AC · Optics · Modern Physics

By Nitesh Bhaiya

One Shot handwritten notes · NCERT diagrams · Most Important Questions in every chapter

Eduspark · Free for students, not for sale

Puri Physics, Ek Notebook Mein

Most asked: 14 chapters | detailed theory + derivations + solved examples + Most Important Questions | board ready

1. Unit I-II: Electrostatics and Current Electricity

Ch	Chapter	Page
1	Electric Charges and Fields	3
2	Electrostatic Potential and Capacitance	22
3	Current Electricity	36

2. Unit III-IV: Magnetism, EMI and AC

Ch	Chapter	Page
4	Moving Charges and Magnetism	49
5	Magnetism and Matter	63
6	Electromagnetic Induction	77
7	Alternating Current	91

3. Unit V-VI: EM Waves and Optics

Ch	Chapter	Page
8	Electromagnetic Waves	101
9	Ray Optics and Optical Instruments	112
10	Wave Optics	126

4. Unit VII-VIII: Dual Nature, Atoms and Nuclei

Ch	Chapter	Page
11	Dual Nature of Radiation and Matter	138
12	Atoms	149
13	Nuclei	161

5. Unit IX: Electronic Devices

Ch	Chapter	Page
14	Semiconductor Electronics: Materials, Devices and Simple Circuits	174

6. Last-night Revision

Section	Page
Formula sheet: Ch 1-7	187
Formula sheet: Ch 8-14, SI units and constants	191
Derivations, differences, graphs, most expected questions	195
All the best from Paheli, Bujho and Nitesh Bhaiya	199

Nitesh Bhaiya ki baat

Is notebook mein poori NCERT Physics hai: har derivation step by step, diagrams, solved examples aur har chapter ke end mein Most Important Questions answers ke saath. Roz ek chapter, aur board exam bas ek aur test lagega!

Nitesh Bhaiya | Eduspark

7. Board Paper Pattern (Theory 70 + Practical 30)

Section	Type of questions	Marks
A	16 MCQ / Assertion-Reason x 1	16
B	5 very short answer x 2	10
C	7 short answer x 3	21
D	2 case-based x 4	8
E	3 long answer x 5	15
Total	**33 questions, 3 hours**	**70**

8. Unit-wise Marks

Units	Marks
I-II. Electrostatics, Current Electricity (Ch 1-3)	16
III-IV. Magnetism, EMI and AC (Ch 4-7)	17
V-VI. EM Waves and Optics (Ch 8-10)	18
VII-VIII. Dual Nature, Atoms and Nuclei (Ch 11-13)	12
IX. Electronic Devices (Ch 14)	7
Total	**70**

Is book ko kaise padhein

1. Pehle theory padho: yellow = must learn, red = keyword.
2. Har derivation khud copy mein likho, sirf padhna kaafi nahi.
3. **Yaad karo** = trick, **Exam Alert** = marks katne wali galti, **Solved example** = numerical ka tareeka.
4. Phir chapter ke **Most Important Questions** khud solve karo, answer match karo.
5. Ray diagrams aur circuits pencil se khud banao.

Electric Charges and Fields

Most asked: Coulomb's law (vector form), electric field of a dipole (axial and equatorial), torque on a dipole, Gauss's law and its three applications, field lines properties

Rub a comb in dry hair and it picks up paper bits: that is **static electricity**. This chapter builds the whole idea from **charge, Coulomb's law and electric field** to **dipoles, flux and Gauss's law.**

1. Electric Charge

Electric charge is an intrinsic property of matter due to which it experiences electric and magnetic forces. It was first noticed by **Thales of Miletus** (about 600 BC): amber rubbed with wool attracts light objects. The word electricity comes from the Greek *elektron*, meaning amber.

- → **Two kinds**: positive and negative (names given by **Benjamin Franklin**).
- → Like charges repel, unlike charges attract.
- → SI unit: **coulomb (C)**. 1 C is the charge that flows through a wire in 1 s when the current is 1 A.
- → Charge on electron = -1.6×10^{-19} C; on proton = $+1.6 \times 10^{-19}$ C.

Why does charge exist? (atomic picture)

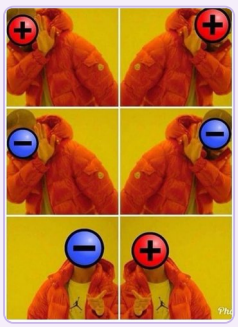
All matter is built from atoms. An atom has a tiny, heavy, positively charged **nucleus** (protons and neutrons) surrounded by light, negatively charged **electrons**. In a normal atom the number of electrons equals the number of protons, so the atom is electrically neutral. A body becomes charged only when this balance is disturbed: a body that **loses electrons** becomes positive and a body that **gains electrons** becomes negative. Protons are locked inside the nucleus and do not move during ordinary charging, so in everyday charging only electrons are transferred.

Key terms

- → **Charging by friction**: rubbing transfers electrons from one body to the other. Glass rod + silk: glass becomes +, silk becomes −.
- → **Gold-leaf electroscope**: a simple device to **detect** charge; the leaves diverge when charged.
- → **Polarity**: the property that distinguishes the two kinds of charge.

Rubbed pair	Positive	Negative
Glass rod + silk	glass rod	silk
Ebonite (plastic) rod + fur/wool	fur/wool	ebonite rod
Comb + dry hair	hair	comb

Rubbed pair	Gets positive charge	Gets negative charge
Glass rod and silk	glass rod	silk
Plastic comb and dry hair	hair	comb
Ebonite rod and wool	wool	ebonite rod
Polythene and wool	wool	polythene



Charges ki dosti: like ko like pasand nahi, unlike pe dil aa jaata hai

Asli baat: Like charges repel, unlike charges attract. Force acts along the line joining the charges (central force).

Yaad karo

- → Charging = electron transfer, proton kabhi nahi hilta.
- → Electron gaya toh positive, electron aaya toh negative.
- → Rubbing se charge banta nahi, sirf shift hota hai.

Solved example

Q. When a glass rod is rubbed with a silk cloth, charges appear on both. A similar phenomenon is observed with many other pairs of bodies. Explain how this observation is consistent with the law of conservation of charge.

Sol. Before rubbing, both bodies are neutral (total charge zero). Rubbing only **transfers electrons** from glass to silk; no charge is created or destroyed. Glass gains $+q$ and silk gains exactly $-q$, so the total remains $q + (-q) = 0$. Charges always appear in equal and opposite pairs, which is exactly what conservation of charge requires.

2. Conductors and Insulators

Conductors	*Insulators*
Allow electricity to pass easily	Offer high resistance to electricity
Have free electrons	No free electrons
Charge spreads over the whole surface	Charge stays where it is put
Metals, human body, earth	Glass, porcelain, plastic, nylon, wood

Earthing (grounding): connecting a charged conductor to the earth so its extra charge flows away. The earth acts as a huge reservoir of charge. Household appliances have a third (earth) wire for safety.

Semiconductors (Si, Ge) have conductivity between conductors and insulators.

Why do metals conduct? Free electrons explained

In a metal, the outermost electrons of each atom are loosely bound. When atoms join to form a solid, these electrons are no longer attached to any particular atom and move freely through the whole volume of the metal; they are called **free electrons**. When charge is given to a metal, these free electrons rearrange almost instantly and the charge spreads over the whole outer surface. In an insulator such as glass or rubber, every electron stays bound to its own atom, so a charge placed at one point stays there.

This explains a common observation: a metal rod held in the hand cannot be charged by rubbing, because the charge flows through the rod and the human body (a conductor) to the earth. If the metal rod has an insulating (wooden or plastic) handle, it does get charged. Semiconductors such as silicon and germanium lie in between: their ability to conduct is much less than that of metals but much more than that of insulators.

3. Charging by Induction

A charged body can charge a neutral conductor without touching it. The induced charge is always **opposite** in sign to the inducing charge.

1. Bring a negatively charged rod near a neutral metal sphere on an insulating stand. Electrons in the sphere are pushed to the far side; the near side becomes positive.
2. Connect the far side to earth. The negative (repelled) electrons flow to the ground.
3. Remove the earth connection first, keeping the rod near.
4. Now remove the rod. The positive charge spreads uniformly over the sphere.

Exam Alert

→ Order matters: **earth removed before the rod.** If the rod is removed first, electrons come back and the sphere stays neutral.

→ A charged body attracts a neutral body (by induction), but **repulsion is the sure test** of charge.

NCERT Activity: charging by rubbing

Run a dry plastic comb through dry hair and bring it near small bits of paper: they jump up to it. The comb gains electrons (negative); it induces opposite charge on the near side of each bit (polarisation) and attracts it. On a humid day the effect is weak because moisture leaks the charge away.

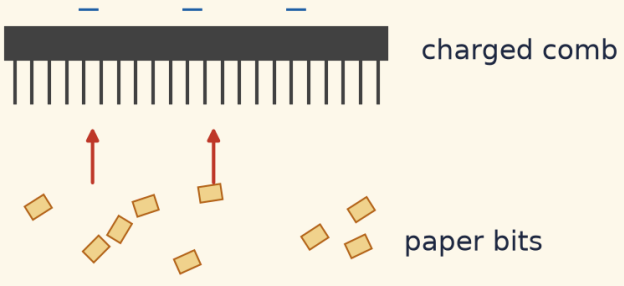


Fig. A rubbed comb attracts bits of paper

NCERT Activity: gold-leaf electroscope

Touch the metal disc with a charged rod: charge spreads to the two gold leaves, which repel and diverge. More charge, wider divergence. Bringing a charged rod near (without touching) also opens the leaves, by induction. The electroscope detects charge and its sign.

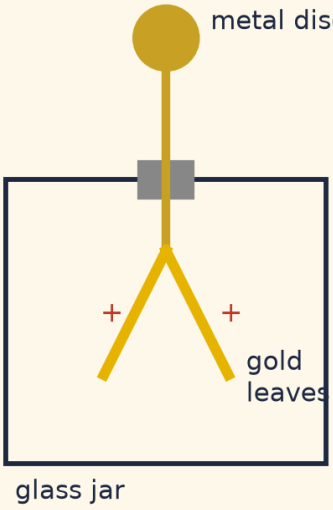
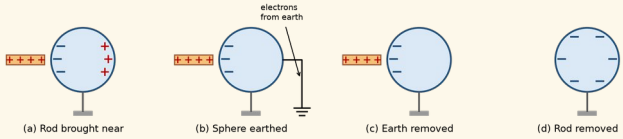


Fig. Gold-leaf electroscope

Charging by induction: step by step



Charging a metal sphere negatively by induction using a positively charged rod

Step 1: A positively charged rod is brought near an uncharged, insulated metal sphere (without touching). Free electrons in the sphere are attracted towards the rod and gather on the near side, leaving the far side positive. The sphere is still neutral overall; only its charges are separated.

Step 2: The far side of the sphere is connected to the earth with a wire. The positive charges on the far side are neutralised by electrons that flow up from the earth. The negative charge on the near side stays, held by the attraction of the rod.

Step 3: The earth connection is removed while the rod is still near. The sphere now has a net negative charge. **Step 4:** The rod is taken away. The negative charge spreads uniformly over the sphere. The sphere has acquired a charge opposite to that of the rod, and the rod has lost none of its own charge.

Why a charged body attracts a neutral one

A charged comb induces opposite charge on the near side of a paper bit and like charge on the far side. The near (opposite) charge is closer, so attraction is stronger than repulsion: net force is **attraction**. This is why attraction alone does not prove that a body is charged; **repulsion** is the sure test.

4. Basic Properties of Electric Charge

A. Additivity

Charges are **scalars** and add algebraically: total charge $q = q_1 + q_2 + q_3 + \dots$. A system with +1, +2, −3 and +5 units has net charge +5 units. (Mass is always positive; charge can be + or −.)

B. Conservation of charge

The total charge of an isolated system remains constant. Charge can neither be created nor destroyed; it can only be transferred. In pair production a γ-photon creates an electron and a positron: net charge 0 before and after.

C. Quantisation of charge

$$q = \pm ne \quad (n = 1, 2, 3, \dots) \quad e = 1.6 \times 10^{-19}\ \text{C}$$

Charge exists only in **integral multiples** of the basic unit e. It was first suggested by Faraday's laws of electrolysis and demonstrated by **Millikan's oil-drop experiment** (1912).

Quick fact

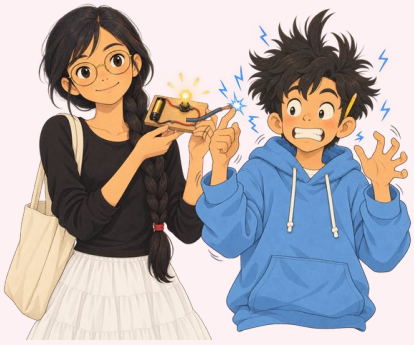
At the macroscopic level charges are huge (a charge of 1 μC has about 10^{13} electrons), so quantisation can be ignored and charge is treated as **continuous**. Quarks have charges $\pm e/3$ and $\pm 2e/3$, but they never exist free.

Solved example

Q. How many electrons make up −1 C of charge?

Ans. $n = q/e = 1/(1.6 \times 10^{-19}) =$ **6.25×10^{18} electrons.**

Paheli-Bujho



Bujho: Paheli, comb ko baal pe ragda toh comb negative kyun hua? Electron kahan se aaye?
Paheli: Baalon se! Friction mein electron hi transfer hote hain, proton nucleus mein locked rehte hain.
Bujho: Matlab baal ab positive aur thode gareeb!
Paheli: Haan, aur total charge same: conservation of charge.

Additivity, conservation and quantisation explained

Additivity: charge is a scalar, so the total charge of a system is the algebraic sum of the individual charges, with signs. For example, charges +1 C, +2 C, −3 C and +4 C give a total of +4 C. Mass is also additive, but mass is always positive, while charge can be positive or negative.

Conservation: the total charge of an isolated system never changes. Charges may be transferred, and pairs of opposite charges may be created or destroyed together (for example, in pair production a γ-ray photon produces an electron and a positron), but the net charge stays the same. In every nuclear reaction and radioactive decay, the total charge before and after is equal.

Quantisation: $q = ne$, where n is an integer. Since the smallest free charge is $e = 1.6 \times 10^{-19}$ C, one coulomb contains about 6.25×10^{18} electronic charges. In laboratory work charges are of the order of μC, so the graininess of charge is not noticed.

Solved example

Q. (a) Explain the meaning of the statement "electric charge of a body is quantised". (b) Why can one ignore quantisation of electric charge when dealing with macroscopic, i.e., large scale charges?
Sol. (a) Charge on any body is always an integral multiple of the basic charge e: $q = ne$, $n = 0, \pm1, \pm2, \ldots$ with $e = 1.6 \times 10^{-19}$ C. A charge such as 1.5e cannot exist on a free body.
(b) Macroscopic charges are of the order of μC, i.e., about 10^{13} electronic charges. A step of one e is negligibly small compared with this, so charge behaves as **continuous** and quantisation can be ignored.

Solved example

Q. A body has a charge of $+3.2 \times 10^{-19}$ C. Has it lost or gained electrons, and how many? Can a body have a charge of 2.4×10^{-19} C?
Sol. $n = q/e = (3.2 \times 10^{-19})/(1.6 \times 10^{-19}) = 2$. The charge is positive, so the body has **lost 2 electrons**. For 2.4×10^{-19} C, $n = 1.5$, which is not an integer, so **such a charge is not possible.**

5. Coulomb's Law

The force between two point charges is **directly proportional to the product of the charges** and **inversely proportional to the square of the distance** between them, acting along the line joining them.

$$F = k\,q_1 q_2 / r^2, \quad k = 1/4\pi\varepsilon_0 = 9 \times 10^9\ \text{N m}^2\ \text{C}^{-2}$$

→ ε_0 = permittivity of free space = $8.854 \times 10^{-12}\ \text{C}^2\ \text{N}^{-1}\ \text{m}^{-2}$.

→ Definition of 1 C from this law: two equal charges 1 m apart in vacuum repelling with force 9×10^9 N.

→ Valid for **point charges at rest.**

Coulomb's law in vector form

$$\vec{F}_{21} = (1/4\pi\varepsilon_0)\,(q_1 q_2/r^2)\,\hat{r}_{12}$$ force on q_2 due to q_1

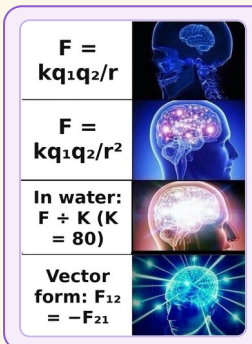
Here $\hat{r}_{12}$ is the unit vector from q_1 to q_2. Since $\hat{r}_{21} = -\hat{r}_{12}$, we get $\vec{F}_{12} = -\vec{F}_{21}$: Coulomb forces obey **Newton's third law.** The vector form holds for both like and unlike charges (the sign of $q_1 q_2$ decides the direction).

Force in a medium

$$F_{\text{medium}} = F_{\text{vacuum}} / K, \quad K = \varepsilon/\varepsilon_0 = \varepsilon_r$$

K is the **dielectric constant** (relative permittivity) of the medium. For water K = 80, so the force between ions becomes 80 times weaker: that is why salts dissolve in water. For a metal K = ∞.

Coulomb force	*Gravitational force*
Between charges	Between masses
Attractive or repulsive	Always attractive
Depends on medium	Independent of medium
Very strong (for two protons $F_e/F_g \approx 10^{36}$)	Very weak
Both: inverse-square, central, conservative	Both: inverse-square, central, conservative



Coulomb's law ka level-up: r ka square aur medium ka K mat bhoolna

Asli baat: $F \propto q_1q_2/r^2$; in a medium F becomes F/K. Water (K = 80) weakens ionic forces 80 times.

Vector form and Newton's third law

$$\vec{F}_{21} = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}\hat{r}_{21}, \quad \vec{F}_{12} = -\vec{F}_{21}$$

- → $\hat{r}_{21}$ is the unit vector from q_1 to q_2; the vector form gives the direction automatically for any signs of charges.
- → The forces are equal and opposite and act along the line joining the charges (central forces): Coulomb's law obeys Newton's third law.
- → Valid for point charges at rest; for charged bodies only when their size ≪ separation.

Definition of one coulomb

1 C is the charge that, placed 1 m from an equal charge in vacuum, repels it with a force of 9×10^9 N. (Practical charges are μC or nC.)

Solved example (NCERT)

Q. Compare the electric and gravitational forces between an electron and a proton.

Sol. $F_e/F_G = ke^2/(Gm_em_p) = (9 \times 10^9 \times 2.56 \times 10^{-38})/(6.67 \times 10^{-11} \times 9.1 \times 10^{-31} \times 1.67 \times 10^{-27}) \approx$ **2.4×10^{39}**. Gravity is negligible at atomic scale.

Solved example

Q. Two identical metal spheres carry +12 μC and −8 μC and attract with force F. They are touched and put back. Find the new force.

Sol. After contact each has (12 − 8)/2 = +2 μC. $F \propto q_1q_2$: $F'/F = (2 \times 2)/(12 \times 8) = 1/24$, now **repulsive**, $F' = F/24$.

NCERT Activity: Coulomb's torsion balance

Coulomb hung a light rod with a charged ball from a thin wire and brought another charged ball near it. The repulsion twisted the wire; the twist angle measured the force. Varying distance and charge, he found $F \propto q_1q_2/r^2$.

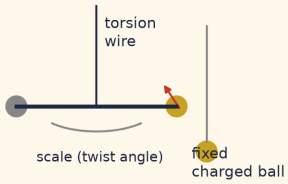


Fig. Coulomb's torsion balance (schematic)

Understanding Coulomb's law more deeply

Coulomb's law holds for **point charges**, that is, charged bodies whose size is very small compared with the distance between them. The force acts along the line joining the two charges (a central force) and obeys Newton's third law: the force on q_1 due to q_2 is equal and opposite to the force on q_2 due to q_1, even when the charges are unequal.

Effect of a medium: when the charges are placed in a medium of permittivity $\varepsilon = K\varepsilon_0$, the force becomes $F_m = q_1q_2/(4\pi K\varepsilon_0 r^2) = F_{air}/K$. Water has K ≈ 80, so the force between two ions in water is about 80 times smaller than in air. This is why salts such as NaCl dissolve easily in water.

Feature	*Electrostatic force*	*Gravitational force*
Acts between	charges	masses
Nature	attractive or repulsive	always attractive
Depends on medium	yes (F/K)	no
Strength (e and p)	about 10^{39} times stronger	very weak
Common feature	inverse square law, central, conservative	inverse square law, central, conservative

Solved example

Q. What is the force between two small charged spheres having charges 2×10^{-7} C and 3×10^{-7} C placed 30 cm apart in air?

Sol. Given $q_1 = 2 \times 10^{-7}$ C, $q_2 = 3 \times 10^{-7}$ C, r = 0.30 m.

$F = kq_1q_2/r^2 = (9 \times 10^9 \times 2 \times 10^{-7} \times 3 \times 10^{-7})/(0.30)^2 = (5.4 \times 10^{-4})/0.09$

$F = \mathbf{6 \times 10^{-3}}$ **N, repulsive** (both charges positive).

Solved example

Q. The electrostatic force on a small sphere of charge 0.4 μC due to another small sphere of charge −0.8 μC in air is 0.2 N. (a) What is the distance between the two spheres? (b) What is the force on the second sphere due to the first?

Sol. (a) $F = k|q_1q_2|/r^2 \Longrightarrow r^2 = k|q_1q_2|/F = (9 \times 10^9 \times 0.4 \times 10^{-6} \times 0.8 \times 10^{-6})/0.2 = 1.44 \times 10^{-2}$ m^2

r = **0.12 m = 12 cm**.

(b) By Newton's third law the force on the second sphere is equal and opposite: **0.2 N, attractive** (directed towards the first sphere).

Solved example

Q. Check that the ratio ke^2/Gm_em_p is dimensionless. Look up the values of the constants and find its value. What does it signify?

Sol. ke^2/r^2 and Gm_em_p/r^2 are both forces, so ke^2/Gm_em_p = (force × r^2)/(force × r^2): **dimensionless**.

Value = $(9 \times 10^9 \times (1.6 \times 10^{-19})^2)/(6.67 \times 10^{-11} \times 9.1 \times 10^{-31} \times 1.67 \times 10^{-27}) \approx \mathbf{2.3 \times 10^{39}}$.

It is the ratio of the electric force to the gravitational force between an electron and a proton at the same separation: the electric force is about 10^{39} times stronger.

Solved example

Q. (a) Two insulated charged copper spheres A and B have their centres separated by a distance of 50 cm. What is the mutual force of electrostatic repulsion if the charge on each is 6.5×10^{-7} C? The radii of A and B are negligible compared to the distance of separation. (b) What is the force of repulsion if each sphere is charged double the above amount, and the distance between them is halved?

Sol. (a) $F = kq^2/r^2 = (9 \times 10^9 \times (6.5 \times 10^{-7})^2)/(0.50)^2 = \mathbf{1.52 \times 10^{-2}}$ **N.**

(b) $F' = k(2q)(2q)/(r/2)^2 = 16\,kq^2/r^2 = 16 \times 1.52 \times 10^{-2} =$ **0.243 N.**

6. Forces between Multiple Charges

Superposition principle: the force on any charge due to a number of charges is the **vector sum** of the forces due to each charge, each taken as if the others were absent.

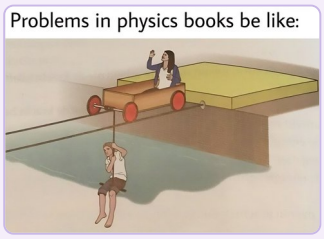
$$F_1 = F_{12} + F_{13} + \ldots + F_{1n}$$

For two forces at angle θ: $|F| = \sqrt{(F_1^2 + F_2^2 + 2F_1F_2 \cos\theta)}$.

Solved example

Q. Three equal charges q are at the corners of an equilateral triangle of side l. Find the force on a charge Q at the centroid.

Ans. The three forces are equal in magnitude and 120° apart, so their vector sum is **zero**.



Superposition ke sawaal: charges har jagah, bas dimaag shant rakho

Asli baat:

Superposition: find each force separately, then add as vectors. Symmetric equal charges give zero force at the centre.

Yaad karo

→ Equal charges, symmetric arrangement, centre pe force = 0 (triangle, square, circle).

→ Superposition = har force alag nikalo, phir vector add karo. Direction ka dhyaan rakhna.

Solved example: equilateral triangle

Q. Three charges q each sit at the corners of an equilateral triangle of side l. Find the force on a charge Q at the centroid.

Sol. Each force has magnitude $kqQ/(l/\sqrt{3})^2 = 3kqQ/l^2$; the three are 120° apart, so their vector sum is **zero** (symmetry).

Solved example: square

Q. Four charges +q, +q, −q, −q sit at the corners of a square (like charges adjacent). Find the force on +Q at the centre.

Sol. The forces due to the two +q charges point away from them; those due to −q point towards them. Pairs add along one side direction: net force $= 4\sqrt{2}\, kqQ/a^2$ ⊥ to the line of the +q charges, towards the −q side.

Yaad karo

- → Symmetric arrangement of equal charges: centre pe net force zero.
- → Vector addition ke liye components lo: $F_x = \Sigma F \cos\theta$, $F_y = \Sigma F \sin\theta$.

Superposition principle: method for numericals

When many charges act on a charge q_0, the force due to each charge is calculated as if the others were absent, and the individual forces are then added as **vectors**. Method: (1) draw the charges and the direction of each force (away for like charges, towards for unlike charges); (2) find each magnitude using Coulomb's law with the positive values of the charges; (3) resolve the forces into x and y components and add; (4) find the resultant and its direction. Symmetry often makes many components cancel.

Solved example

Q. Four point charges $q_A = 2\ \mu C$, $q_B = -5\ \mu C$, $q_C = 2\ \mu C$ and $q_D = -5\ \mu C$ are located at the corners of a square ABCD of side 10 cm. What is the force on a charge of 1 μC placed at the centre of the square?

Sol. The centre O is equidistant ($r = 5\sqrt{2}$ cm) from all corners.

A and C are opposite corners with equal charges (2 μC): their forces on 1 μC are equal in magnitude and opposite in direction, so they cancel.

B and D are opposite corners with equal charges (−5 μC): their forces also cancel.

Net force = **zero**.

Solved example

Q. Suppose the spheres A and B in the previous question have identical sizes. A third sphere of the same size but uncharged is brought in contact with the first, then brought in contact with the second, and finally removed from both. What is the new force of repulsion between A and B?

Sol. Let $q = 6.5 \times 10^{-7}$ C. Identical spheres share charge equally on contact.

C touches A: $q_A = q_C = q/2$.

C touches B: $q_B = q_C = (q + q/2)/2 = 3q/4$.

$F' = k(q/2)(3q/4)/r^2 = (3/8)\, kq^2/r^2 = (3/8) \times 1.52 \times 10^{-2} = \mathbf{5.7 \times 10^{-3}\ N}$.

7. Electric Field

The **electric field** at a point is the force experienced by a **unit positive test charge** placed at that point (the test charge being so small that it does not disturb the source).

$$E = \lim_{q_0 \to 0} F/q_0 \quad \text{unit: N C}^{-1}\ (= \text{V m}^{-1})$$

$$\text{Field of a point charge: } E = (1/4\pi\varepsilon_0)\, q/r^2\, \hat{r}$$

- → E is a **vector**: radially outward for +q, inward for −q.
- → Force on a charge q in field E: **F = qE**. A proton moves along E; an electron moves opposite to E.
- → Field of a point charge has **spherical symmetry**: same magnitude at all points on a sphere centred on the charge.

Field due to a system of charges: $E = E_1 + E_2 + \ldots$ (superposition again).

Quick fact

The field is not just a maths trick. For time-varying (moving) charges, the field carries **energy and momentum** and travels at the speed of light. This is the idea behind electromagnetic waves (Chapter 8).

Field due to a system of charges

$E = E_1 + E_2 + \ldots = (1/4\pi\varepsilon_0)\,\Sigma\,(q_i/r_i^2)\,\hat{r}_i$ superposition: vector sum

Physical significance of field

→ The field tells us the force on any charge placed there without knowing the source charges: $F = qE$.

→ In electrostatics it is a convenient tool; for time-varying situations it carries energy and momentum (electromagnetic waves), so the field is real, not just a mathematical device.

→ A test charge q_0 must be very small so it does not disturb the source charges.

Solved example: neutral point

Q. $+4q$ and $+q$ are a distance d apart. Where is $E = 0$?

Sol. Between them at x from $4q$: $4q/x^2 = q/(d - x)^2 \Longrightarrow 2/x = 1/(d - x) \Longrightarrow$ **$x = 2d/3$** from the $4q$ charge.

Solved example: electron in a field

Q. An electron falls 1.5 cm from rest in a field of 2×10^4 N/C. Find the time taken. Compare with a proton.

Sol. $a = eE/m$; $t = \sqrt{(2hm/eE)}$. For the electron $t = 2.9 \times 10^{-9}$ s; the proton is 1836 times heavier, so it takes $\sqrt{1836} \approx 43$ times longer (1.3×10^{-7} s). Gravity is negligible.

Yaad karo

→ Positive charge ka field bahar ki taraf, negative ka andar ki taraf.

→ Do opposite charges ke beech field add hota hai, do same charges ke beech cancel ho sakta hai (neutral point).

→ $F = qE$: electron pe force field ke ulta.

8. Electric Field Lines

An **electric field line** is a curve drawn such that the tangent at any point gives the direction of the field at that point. The idea was given by **Faraday**.

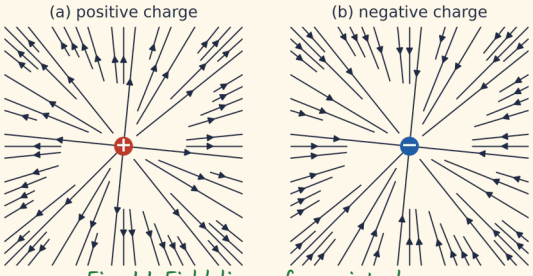


Fig. 1.1 Field lines of a point charge

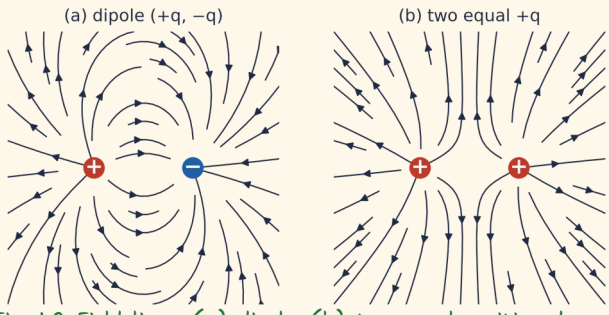


Fig. 1.2 Field lines: (a) dipole, (b) two equal positive charges

Properties of field lines

1. Start from a positive charge and end on a negative charge (or at infinity).
2. In a charge-free region they are **continuous curves** without breaks.
3. Two field lines never cross (otherwise the field would have two directions at one point).
4. Electrostatic field lines **do not form closed loops** (the field is conservative).
5. They are crowded where the field is strong and far apart where it is weak.
6. They meet the surface of a conductor **normally** (at 90°).

Field lines ka golden rule: kabhi cross nahi karti

Asli baat: Lines start on +q, end on −q, never cross, never form closed loops, and meet conductors at 90°.

Paheli-Bujho

Bujho: Paheli, do field lines cross kar jayein toh kya hoga? Traffic jam?

Paheli: Us point pe tangent do directions dega, matlab ek charge ko do taraf force. Impossible!

Bujho: Toh field lines bhi lane driving follow karti hain.

Paheli: Exactly, aur kabhi loop nahi banati.

Field aur flux samajh aa gaya toh aap bhi isi team mein ho

Asli baat: Field $E = F/q_0$ is a property of space around a charge; field lines show its direction and strength.

Solved example

Q. (a) An electrostatic field line is a continuous curve. That is, a field line cannot have sudden breaks. Why not? (b) Explain why two field lines never cross each other at any point.

Sol. (a) A field line traces the path of a free positive test charge. The field exists at every point in space around the charge, and a test charge moves continuously, so the line has **no sudden breaks**. A break would mean the field vanishes abruptly there. (b) The tangent to a field line gives the direction of E at that point. If two lines crossed, E would have **two directions at one point**, which is impossible.

Field lines give a picture of the field: where lines are crowded the field is strong, where they are far apart the field is weak. Number of lines is not a physical quantity; only their relative density matters. In a uniform field the lines are parallel and equally spaced.

9. Electric Flux

Electric flux through a surface is the number of field lines crossing it, measured as the surface integral of E:

$$\Delta\varphi = E \cdot \Delta S = E\,\Delta S\cos\theta; \quad \varphi = \oint E \cdot dS$$

→ Flux is a **scalar**. SI unit: $\mathbf{N\,m^2\,C^{-1}}$ (or V m).

→ The area vector ΔS points along the outward normal of the surface.

→ θ = 0°: flux maximum (EΔS); θ = 90°: flux zero; θ = 180°: flux negative (entering).

Solved example

Q. A uniform field $E = 3 \times 10^3\,\hat{i}$ N/C passes through a square of side 10 cm whose plane is parallel to the yz-plane. Find the flux. What if the normal makes 60° with the x-axis?

Ans. $\varphi = EA = 3 \times 10^3 \times 0.01 = \mathbf{30\ N\,m^2/C}$. At 60°: $\varphi = 30\cos 60^\circ = \mathbf{15\ N\,m^2/C}$.

Area vector

A small area ΔS is represented by a vector ΔS of magnitude ΔS directed along the outward normal (for a closed surface).

$$\Delta\Phi = E \cdot \Delta S = E\,\Delta S\cos\theta$$

→ θ = 0°: flux maximum (EΔS); θ = 90°: zero; θ > 90°: negative (lines entering a closed surface).

→ Flux is a scalar; SI unit $N\,m^2\,C^{-1}$ (or V m).

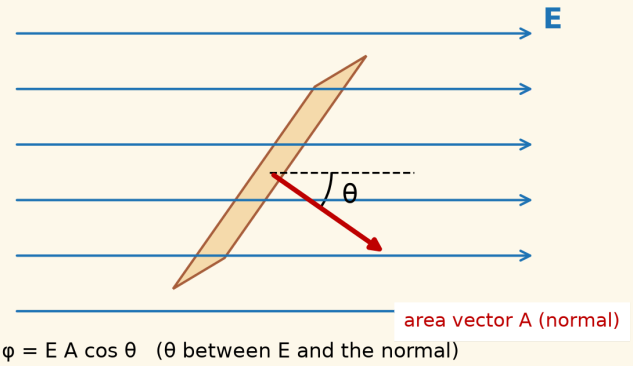
→ Flux is proportional to the number of field lines crossing the surface.

Solved example

Q. $E = 3 \times 10^3\,\hat{i}$ N/C. Find the flux through a square of side 10 cm whose plane is (a) the yz-plane, (b) inclined so its normal makes 60° with the x-axis.

Sol. (a) $\Phi = EA = 3 \times 10^3 \times 0.01 = \mathbf{30\ N\,m^2/C}$. (b) $\Phi = EA\cos 60^\circ = \mathbf{15\ N\,m^2/C}$.

Flux through an area at an angle



$\varphi = E\,A\cos\theta$ (θ between E and the normal)

Flux through a tilted surface depends on the angle between E and the area vector

Flux measures the number of field lines crossing a surface. The area is treated as a vector whose direction is along the **outward normal**. Only the component of E along the normal (E cos θ) actually crosses the surface. So flux is maximum (EA) when the surface faces the field (θ = 0°), zero when the surface is parallel to the field (θ = 90°), and negative when lines enter a closed surface (θ > 90°).

Solved example

Q. What is the net flux of the uniform electric field of the previous question through a cube of side 20 cm oriented so that its faces are parallel to the coordinate planes?

Sol. The field is uniform. Flux entering through the face perpendicular to −x equals flux leaving through the face perpendicular to +x; the other four faces are parallel to E (zero flux). Also, the cube encloses no charge, so by Gauss's law $\varphi = q_{enc}/\varepsilon_0$.
Net flux = **zero**.

10. Electric Dipole

A pair of equal and opposite charges +q and −q separated by a small distance 2a is an **electric dipole**. Its strength is the **dipole moment**:

$$p = q \times 2a \text{ vector, directed from } -q \text{ to } +q\text{; unit C m}$$

The total charge of a dipole is zero, but its field is not, because the two charges are at different points. At large distance the dipole field falls as $1/r^3$ (faster than the $1/r^2$ of a single charge).

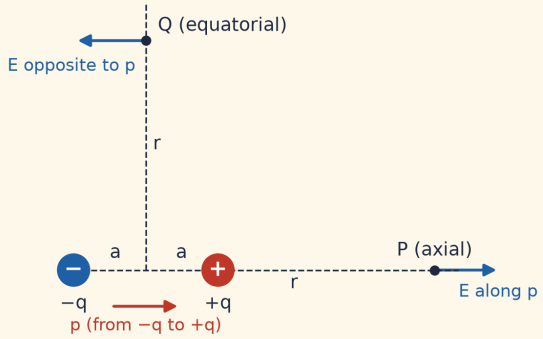


Fig. 1.3 Axial point P and equatorial point Q of a dipole

A. Field on the axial line (end-on)

At P, distance r from the centre: $E_{+q} = kq/(r-a)^2$ (away), $E_{-q} = kq/(r+a)^2$ (towards). Net:

$$E = k\,q\,[1/(r-a)^2 - 1/(r+a)^2] = k\,(4qar)\,/\,(r^2-a^2)^2$$

$$\text{For } r \gg a\text{: } E_{axial} = (1/4\pi\varepsilon_0)\,2p\,/\,r^3 \text{ along p}$$

B. Field on the equatorial line (broad-side)

At Q, both charges are at distance $\sqrt{(r^2+a^2)}$. The components perpendicular to the axis cancel; components parallel to the axis add (pointing from +q to −q):

$$E = 2\,kq\cos\theta\,/\,(r^2+a^2) = k\,(2qa)\,/\,(r^2+a^2)^{3/2}$$

$$\text{For } r \gg a\text{: } E_{equatorial} = (1/4\pi\varepsilon_0)\,p\,/\,r^3 \text{ opposite to p}$$

Yaad karo

- $E_{axial} = 2 \times E_{equatorial}$ (same r). Ratio 2 : 1.
- Axial: E p ki direction mein. Equatorial: E p ke ulta.
- Dipole field $\propto 1/r^3$, point charge $\propto 1/r^2$.

Physical significance: in most molecules the centres of positive and negative charges coincide, so p = 0 (non-polar, e.g. CO_2, CH_4). Molecules like H_2O have a permanent dipole moment (polar). In an external field even non-polar molecules develop an induced dipole moment.

Physical significance of dipoles

- **Polar molecules** (H_2O, HCl) have a permanent dipole moment: the centres of positive and negative charge do not coincide.
- **Non-polar molecules** (CO_2, CH_4) have zero moment, but an external field induces one.
- Dipole field falls as $1/r^3$, faster than a single charge ($1/r^2$), because the two charges nearly cancel far away.
- **Ideal (point) dipole**: $q \to \infty$, $2a \to 0$ with $p = q \times 2a$ finite.

Solved example

Q. Charges ±10 μC are 5 mm apart. Find E at 15 cm from the centre on (a) the axis, (b) the equatorial line.

Sol. $p = 10^{-5} \times 5 \times 10^{-3} = 5 \times 10^{-8}$ C m. (a) $E = 2kp/r^3 = 2 \times 9 \times 10^9 \times 5 \times 10^{-8}/(0.15)^3 =$ **2.67×10^5 N/C** along p. (b) $E = kp/r^3 =$ **1.33×10^5 N/C** opposite to p.

Solved example

Q. A system has two charges $q_A = 2.5 \times 10^{-7}$ C and $q_B = -2.5 \times 10^{-7}$ C located at points A (0, 0, −15 cm) and B (0, 0, +15 cm), respectively. What are the total charge and electric dipole moment of the system?

Sol. Total charge = $2.5 \times 10^{-7} - 2.5 \times 10^{-7} =$ **0**.
Separation 2a = 30 cm = 0.30 m.
$p = q \times 2a = 2.5 \times 10^{-7} \times 0.30 =$ **7.5×10^{-8} C m**.
Direction: from −q (at B, z = +15 cm) to +q (at A, z = −15 cm), i.e., along the **negative z-axis**.

11. Dipole in a Uniform External Field

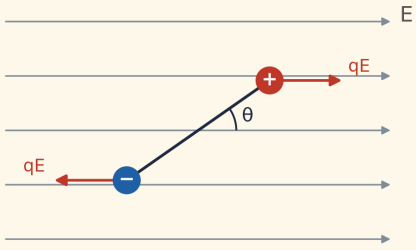


Fig. 1.4 Dipole in a uniform field E

The forces +qE and −qE are equal and opposite, so the **net force is zero**. But they act at different points and form a couple. Torque:

$$\tau = qE \times 2a \sin\theta = pE \sin\theta; \quad \boldsymbol{\tau = p \times E}$$

→ τ is maximum (pE) when θ = 90°.

→ τ = 0 at θ = 0° (stable equilibrium) and θ = 180° (unstable equilibrium).

→ The torque tries to **align p along E**.

Non-uniform field: net force is not zero. If p is parallel to E, the dipole moves towards the stronger field (direction of increasing E). That is why a charged comb attracts neutral paper bits: it **polarises** the paper (induced dipoles) and the non-uniform field pulls them in.

Exam Alert

→ Uniform field: $F_{net} = 0$, $\tau \neq 0$ (only rotation). Non-uniform field: $F_{net} \neq 0$ and τ may be ≠ 0.

→ Derivation of $\tau = pE \sin\theta$ with a labelled diagram: 3 marks, board favourite.

lly = I love you
lmy = I miss you
Brb = Be right back
pV = nRT
F = ma
KE = (1/2)mv^2
p1v1 = p2v2
E = mc^2

Physics ki apni love language: τ = p × E

Asli baat: Torque τ = pE sin θ: maximum at 90°, zero at 0° (stable) and 180° (unstable).

Dipole in a non-uniform field

In a non-uniform field the forces on +q and −q are unequal, so the dipole feels a **net force** as well as a torque. If p is along E, it moves towards the stronger field; if opposite, towards the weaker field.

Example: a comb rubbed in dry hair attracts bits of paper: the comb's non-uniform field polarises the paper (induced dipoles), and the net force pulls them towards the comb.

Work done in rotating a dipole

$$W = pE(\cos\theta_1 - \cos\theta_2) \text{ from } \theta_1 \text{ to } \theta_2$$

Solved example

Q. A dipole ($p = 4 \times 10^{-9}$ C m) makes 30° with a field of 5×10^4 N/C. Find the torque.

Sol. $\tau = pE \sin\theta = 4 \times 10^{-9} \times 5 \times 10^4 \times 0.5$ $= \mathbf{10^{-4}}$ **N m.**

Angle θ (p with E)	Torque pE sin θ	Potential energy −pE cos θ	Equilibrium
0°	0	−pE (minimum)	stable
90°	pE (maximum)	0	not in equilibrium
180°	0	+pE (maximum)	unstable

Solved example

Q. An electric dipole of moment 2×10^{-8} C m is placed in a uniform field of 4×10^4 N/C. Find the work done in rotating it from the stable position (θ = 0°) to θ = 180°.

Sol. $W = pE(\cos\theta_1 - \cos\theta_2) = 2 \times 10^{-8} \times 4 \times 10^4 \times (1 - (-1)) = \mathbf{1.6 \times 10^{-3}}$ **J.**

12. Continuous Charge Distribution

Distribution	Density	Unit	Element charge
Linear (wire)	$\lambda = \Delta q/\Delta l$	C m^{-1}	$dq = \lambda\, dl$
Surface (sheet)	$\sigma = \Delta q/\Delta S$	C m^{-2}	$dq = \sigma\, dS$
Volume (sphere)	$\rho = \Delta q/\Delta V$	C m^{-3}	$dq = \rho\, dV$

$$E = (1/4\pi\varepsilon_0) \int dq\, \hat{r} / r^2$$

$$E = (1/4\pi\varepsilon_0) \int (dq/r^2)\, \hat{r} \text{ superposition becomes an integral}$$

These densities are macroscopic averages: the volume element is small compared to the body but large enough to hold many charges.

Solved example

Q. A polythene piece rubbed with wool is found to have a negative charge of 3×10^{-7} C. (a) Estimate the number of electrons transferred (from which to which?). (b) Is there a transfer of mass from wool to polythene?

Sol. (a) $n = q/e = (3 \times 10^{-7})/(1.6 \times 10^{-19}) =$ **1.875×10^{12} electrons**, transferred **from wool to polythene** (polythene becomes negative).

(b) Yes, electrons have mass. Mass transferred $= n m_e = 1.875 \times 10^{12} \times 9.1 \times 10^{-31} \approx$ **1.7×10^{-18} kg**, which is negligible.

In practice, a charged body contains an enormous number of charges, so it is treated as a continuous distribution. The small element of charge is written as $dq = \lambda\, dl$ (line), $\sigma\, dA$ (surface) or $\rho\, dV$ (volume), and the total field is found by adding (integrating) the fields of all such elements.

13. Gauss's Law

The total electric flux through any closed surface is $1/\varepsilon_0$ times the total charge enclosed by it.

$$\varphi = \oint E \cdot dS = q_{enclosed} / \varepsilon_0$$

Proof for a point charge

Take a sphere of radius r around q. At every point $E = kq/r^2$ along the outward normal, so $\varphi = E \times 4\pi r^2 = (q/4\pi\varepsilon_0 r^2) \times 4\pi r^2 =$ **q/ε_0**.

- → Valid for **any closed surface** of any shape and size (the Gaussian surface).
- → Charges **outside** the surface contribute zero net flux (lines entering = lines leaving).
- → But E in the equation is the field due to **all** charges, inside and outside.
- → The Gaussian surface must not pass through a discrete charge.
- → Gauss's law follows from the inverse-square nature of Coulomb's law.

Solved example

Q. A point charge +10 μC is at the centre of a cube of side 20 cm. Find the flux through (a) the whole cube, (b) one face.

Ans. (a) $\varphi = q/\varepsilon_0 = 10^{-5}/(8.854 \times 10^{-12}) =$ **1.13×10^6 N m²/C**. (b) By symmetry, one-sixth: **1.88×10^5 N m²/C**.

Important points about Gauss's law

1. True for **any** closed surface of any shape and size.
2. q in the law is the net charge **enclosed**; charges outside contribute zero net flux (as many lines enter as leave).
3. But E on the surface is due to **all** charges, inside and outside.
4. The surface chosen is called a Gaussian surface; it must not pass through a discrete charge.
5. Most useful when symmetry makes E constant over the surface.
6. Based on the inverse-square law; it is equivalent to Coulomb's law.

Solved example: cube

Q. A charge q sits at the centre of a cube. Find the flux through (a) the whole cube, (b) one face. What if q is at a corner?

Sol. (a) q/ε_0. (b) By symmetry, $q/6\varepsilon_0$. At a corner, 8 cubes enclose it: flux through the cube = **$q/8\varepsilon_0$**.

Solved example

Q. The net flux through a closed surface is 8×10^3 N m²/C. Find the charge enclosed. What if the radius of the surface is doubled?

Sol. $q = \varepsilon_0 \Phi = 8.85 \times 10^{-12} \times 8 \times 10^3 =$ **7.08×10^{-8} C**. Flux stays the same: it depends only on the enclosed charge.

Using Gauss's law to find charge

Solved example

Q. Careful measurement of the electric field at the surface of a black box indicates that the net outward flux through the surface of the box is 8.0×10^3 N m²/C. (a) What is the net charge inside the box? (b) If the net outward flux through the surface of the box were zero, could you conclude that there were no charges inside the box? Why or why not?

Sol. (a) Gauss's law: $q = \varepsilon_0 \varphi = 8.854 \times 10^{-12} \times 8.0 \times 10^3 =$ **7.08×10^{-8} C ≈ 0.07 μC**.

(b) **No.** Zero flux only means the net (algebraic) charge inside is zero. The box may contain equal positive and negative charges.

Solved example

Q. A point charge causes an electric flux of -1.0×10^3 N m^2/C to pass through a spherical Gaussian surface of 10.0 cm radius centred on the charge. (a) If the radius of the Gaussian surface were doubled, how much flux would pass through the surface? (b) What is the value of the point charge?

Sol. (a) Flux depends only on the charge enclosed, so it is unchanged: $\mathbf{-1.0 \times 10^3\ N\ m^2/C}$.

(b) $q = \varepsilon_0\varphi = 8.854 \times 10^{-12} \times (-1.0 \times 10^3) = \mathbf{-8.85 \times 10^{-9}\ C \approx -8.9\ nC}$.

Cube trick

- → Charge q at the centre of a cube: flux through each face = $q/6\varepsilon_0$.
- → Charge q at a corner of a cube: total flux through the cube = $q/8\varepsilon_0$.
- → Charge q at the centre of a face: flux through the cube = $q/2\varepsilon_0$.

14. Applications of Gauss's Law

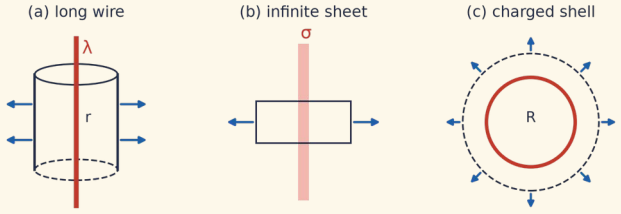


Fig. 1.5 Gaussian surfaces: (a) cylinder for a wire, (b) pill-box for a sheet, (c) sphere for a shell

A. Infinitely long straight charged wire (λ)

Take a coaxial cylinder of radius r and length l. E is radial, so flux through the flat ends is zero. Through the curved surface: $\varphi = E \times 2\pi rl$. Charge enclosed = λl.

$$E \times 2\pi rl = \lambda l/\varepsilon_0 \implies \mathbf{E = \lambda / 2\pi\varepsilon_0 r}$$ $E \propto 1/r$

B. Infinite plane sheet (σ)

Take a cylinder (pill-box) of cross-section A piercing the sheet. Flux only through the two flat faces: $\varphi = 2EA$. Charge enclosed = σA.

$$2EA = \sigma A/\varepsilon_0 \implies \mathbf{E = \sigma / 2\varepsilon_0}$$ independent of distance

E points away from a positive sheet on both sides. For two oppositely charged parallel sheets, the field between them is $\mathbf{\sigma/\varepsilon_0}$ and zero outside.

C. Uniformly charged thin spherical shell (R, σ)

- → **Outside (r > R)**: Gaussian sphere of radius r encloses q. $E \times 4\pi r^2 = q/\varepsilon_0 \implies \mathbf{E = q/4\pi\varepsilon_0 r^2}$, as if the whole charge were at the centre.
- → **On the surface (r = R)**: $E = q/4\pi\varepsilon_0 R^2 = \sigma/\varepsilon_0$ (since $q = 4\pi R^2\sigma$).
- → **Inside (r < R)**: charge enclosed = 0, so **E = 0**.

Exam Alert: Gauss's law S-markers

- → Wire: $E = \lambda/2\pi\varepsilon_0 r$. Sheet: $E = \sigma/2\varepsilon_0$. Shell: outside kq/r^2, inside 0.
- → Har derivation mein: Gaussian surface ka diagram, flux kahan zero hai ye likho, phir q_{enc}.
- → Graph of E vs r for a shell: 0 till R, then falls as $1/r^2$. Often asked.

Field between two parallel sheets

Two infinite sheets with densities +σ and −σ: between them the fields add, outside they cancel.

$E_{inside} = \sigma/\varepsilon_0$, $E_{outside} = 0$ basis of the parallel plate capacitor (Ch. 2)

If both sheets carry +σ: E = 0 between them and σ/ε_0 outside.

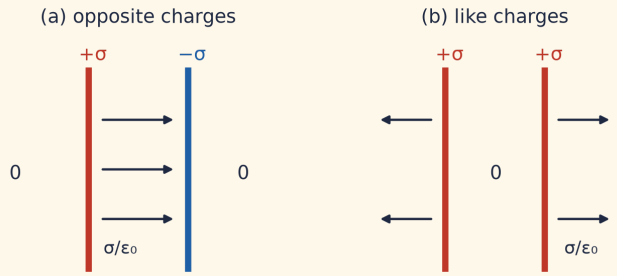


Fig. Field of two parallel charged sheets

Field of a shell vs distance

Region	E
Inside (r < R)	0
On surface (r = R)	$\sigma/\varepsilon_0 = kq/R^2$
Outside (r > R)	kq/r^2 (as if all charge at centre)

Quick fact

E = 0 inside a charged shell is why a person in a car is safe from lightning, and why sensitive instruments are kept in metal casings.

Solved example

Q. An infinite line charge produces 9×10^4 N/C at 2 cm. Find λ.

Sol. $E = 2k\lambda/r \implies \lambda = Er/2k = 9 \times 10^4 \times 0.02/(1.8 \times 10^{10}) = \mathbf{10^{-7}\ C/m}$.

Solved example

Q. A uniformly charged conducting sphere of 2.4 m diameter has a surface charge density of 80.0 $\mu C/m^2$. (a) Find the charge on the sphere. (b) What is the total electric flux leaving the surface of the sphere?

Sol. $R = 1.2$ m, $\sigma = 80.0 \times 10^{-6}$ C/m^2.

(a) $q = \sigma \times 4\pi R^2 = 80 \times 10^{-6} \times 4\pi \times (1.2)^2 = \mathbf{1.45 \times 10^{-3}}$ **C.**

(b) $\varphi = q/\varepsilon_0 = (1.45 \times 10^{-3})/(8.854 \times 10^{-12}) = \mathbf{1.6 \times 10^8}$ **N m^2/C.**

Solved example

Q. Two large, thin metal plates are parallel and close to each other. On their inner faces, the plates have surface charge densities of opposite signs and of magnitude 17.0×10^{-22} C/m^2. What is E: (a) in the outer region of the first plate, (b) in the outer region of the second plate, and (c) between the plates?

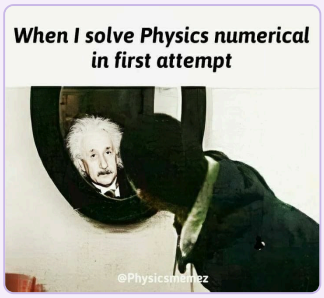
Sol. Each sheet gives $\sigma/2\varepsilon_0$. Outside the pair the two fields are opposite and cancel; between them they add.

(a) $E = \mathbf{0}$. (b) $E = \mathbf{0}$.

(c) $E = \sigma/\varepsilon_0 = (17.0 \times 10^{-22})/(8.854 \times 10^{-12}) = \mathbf{1.92 \times 10^{-10}}$ **N/C**, directed from the positive plate to the negative plate.

Charge distribution	*Field E*	*Depends on r as*
Point charge / outside a shell	$q/4\pi\varepsilon_0 r^2$	$1/r^2$
Infinite line charge	$\lambda/2\pi\varepsilon_0 r$	$1/r$
Infinite plane sheet	$\sigma/2\varepsilon_0$	independent of r
Inside a thin shell	0	zero
Just outside a conductor	σ/ε_0	independent of r
Between two opposite sheets	σ/ε_0	uniform

15. Solved Numericals (Board Pattern)



Pehli baar mein numerical sahi ho jaaye toh aisa feel hota hai

Asli baat: Steps: write given data in SI units, pick the formula, put values, check the unit of the answer.

N1. Two identical spheres carry +12 μC and −4 μC. They are touched and separated to the same distance. Ratio of new force to old force?

Sol. After contact each gets $(12 - 4)/2 = +4$ μC. $F_{old} \propto 12 \times 4 = 48$ (attractive); $F_{new} \propto 4 \times 4 = 16$ (repulsive). Ratio = **1 : 3**, and the force changes from attraction to repulsion.

N2. Find the force between two protons 10^{-15} m apart.

Sol. $F = 9 \times 10^9 \times (1.6 \times 10^{-19})^2/(10^{-15})^2 = \mathbf{230.4}$ **N.** Huge for such tiny particles, so a stronger nuclear force must hold the nucleus together.

N3. An electron falls through 1.5 cm in a uniform field of 2×10^4 N/C. Find its acceleration and time of fall.

Sol. $a = eE/m = (1.6 \times 10^{-19} \times 2 \times 10^4)/(9.1 \times 10^{-31}) = \mathbf{3.5 \times 10^{15}}$ **m/s^2**. $t = \sqrt{(2d/a)} = \sqrt{(2 \times 0.015/3.5 \times 10^{15})} = \mathbf{2.9 \times 10^{-9}}$ **s.** Gravity is negligible.

N4. A dipole (charges ±10 μC, separation 5 mm) is placed at 30° to a uniform field of 2×10^5 N/C. Find the torque.

Sol. $p = 10^{-5} \times 5 \times 10^{-3} = 5 \times 10^{-8}$ C m. $\tau = pE \sin 30°$ $= 5 \times 10^{-8} \times 2 \times 10^5 \times 0.5 = \mathbf{5 \times 10^{-3}}$ **N m.**

N5. A long wire has $\lambda = 2$ μC/m. Find the field at 10 cm.

Sol. $E = \lambda/2\pi\varepsilon_0 r = 2 \times 9 \times 10^9 \times 2 \times 10^{-6}/0.1 = \mathbf{3.6 \times 10^5}$ **N/C**, radially outward.

N6. A conducting sphere of radius 10 cm has charge such that the field 20 cm from the centre is 1.5×10^3 N/C inward. Find the charge.

Sol. Field inward $\Longrightarrow$ negative charge. $q = Er^2/k = 1.5 \times 10^3 \times 0.04/(9 \times 10^9) = \mathbf{-6.67}$ **nC.**

N7. A charge of 17.7×10^{-4} C is spread uniformly over a large sheet of area 200 m^2. Find E near it.
Sol. $\sigma = 8.85 \times 10^{-6}$ C/m^2; $E = \sigma/2\varepsilon_0 = \mathbf{5 \times 10^5}$ **N/C.**

N8. A spherical conducting shell of radius 10 cm carries 3.2×10^{-7} C. Find E at 5 cm, 10 cm and 15 cm from its centre.
Sol. 5 cm: **0**. 10 cm: $kq/R^2 = \mathbf{2.88 \times 10^5}$ **N/C**. 15 cm: $\mathbf{1.28 \times 10^5}$ **N/C.**

N9. A point charge of 2 μC is at the centre of a cubic Gaussian surface of side 9 cm. Find the net flux.
Sol. $\Phi = q/\varepsilon_0 = 2 \times 10^{-6}/8.85 \times 10^{-12} = \mathbf{2.26 \times 10^5}$ **N m^2/C** (independent of the cube size).

Yaad karo: numerical tricks

- → Touching identical spheres: charge $(q_1 + q_2)/2$ each.
- → Neutral point: like charges, beech mein (chhote charge ke paas); unlike charges, bahar (chhote charge ki taraf).
- → Field inward = negative charge.

1 minute recap

- ✓ Charge: additive, conserved, quantised ($q = ne$).
- ✓ Coulomb: $F = kq_1q_2/r^2$; medium mein F/K.
- ✓ $E = F/q_0$; point charge kq/r^2.
- ✓ Dipole: $p = q \cdot 2a$; $E_{axial} = 2kp/r^3$, $E_{eq} = kp/r^3$; $\tau = p \times E$.
- ✓ Gauss: $\varphi = q_{enc}/\varepsilon_0$; wire, sheet, shell results.

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

Questions dekh ke ghabrao mat, answers saath mein hain

Asli baat: Most asked: dipole field derivations, torque on dipole, Gauss law for wire, sheet and shell.

A. Objective and very short answer (1 mark)

Q1. IM The SI unit of electric flux is:
(a) N C^{-1}
(b) N m^2 C^{-1}
(c) N m C^{-1}
(d) C m^{-2}
Ans: (b) N m^2 C^{-1}.

Q2. IM Board fav A charge q is placed at the centre of a cube. The flux through one face is:
(a) q/ε_0
(b) $q/3\varepsilon_0$
(c) $q/6\varepsilon_0$
(d) zero
Ans: (c) $q/6\varepsilon_0$.

Q3. IM If the distance between two point charges is doubled, the force becomes:
(a) half
(b) one-fourth
(c) double
(d) four times
Ans: (b) one-fourth.

Q4. IM The electric field inside a uniformly charged thin spherical shell is:
(a) zero
(b) σ/ε_0
(c) $\sigma/2\varepsilon_0$
(d) infinite
Ans: (a) zero.

Q5. IM A dipole is placed in a uniform field with p parallel to E. It is in:
(a) stable equilibrium
(b) unstable equilibrium
(c) translational motion
(d) rotational motion
Ans: (a) stable equilibrium, net force and torque both zero.

Q6. IM The ratio of electric field on the axial line to that on the equatorial line of a short dipole at equal distance is:
(a) 1 : 1
(b) 1 : 2
(c) 2 : 1
(d) 4 : 1
Ans: (c) 2 : 1.

Q7. 1M Electric field due to an infinite line charge varies with distance r as:
(a) $1/r^2$
(b) $1/r$
(c) r
(d) independent of r
Ans: (b) $1/r$.

Q8. 1M The dielectric constant of a metal is:
(a) 0
(b) 1
(c) 80
(d) infinite
Ans: (d) infinite.

Q9. 1M Board fav Why do two field lines never intersect?
Ans: At the point of intersection the field would have two directions, which is impossible.

Q10. 1M What is the net charge on a dipole? Is its field zero?
Ans: Net charge is zero, but the field is not zero because the two charges are at different positions.

Q11. 1M Name the experiment that showed the quantisation of charge.
Ans: Millikan's oil-drop experiment.

Q12. 1M A glass rod rubbed with silk acquires a charge of $+1.6 \times 10^{-12}$ C. Is there a transfer of mass?
Ans: Yes. Electrons are transferred from glass to silk: $n = 10^7$ electrons, mass transferred $= 10^7 \times 9.1 \times 10^{-31} = 9.1 \times 10^{-24}$ kg (negligible).

Q13. 1M Can electrostatic field lines form closed loops? Why?
Ans: No. The electrostatic field is conservative; a closed loop would mean non-zero work in a closed path.

Q14. 1M Sketch the field lines for a uniform field.
Ans: Equally spaced, parallel straight lines in one direction.

B. Assertion and Reason

Options: (a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q15. 1M **A:** A charged comb attracts small bits of paper. **R:** The comb induces opposite charges on the near side of the paper (polarisation).
Ans: (a)

Q16. 1M Board fav **A:** Gauss's law is valid for any closed surface. **R:** The flux through a closed surface depends only on the charge enclosed.
Ans: (a)

Q17. 1M **A:** A dipole in a uniform field experiences a net force. **R:** The two charges of a dipole experience equal and opposite forces.
Ans: (d) A is false: net force is zero in a uniform field; R is true.

Q18. 1M **A:** Electric field at a point inside a charged shell is zero. **R:** No charge is enclosed by a Gaussian surface drawn inside the shell.
Ans: (a)

C. Short answer (2–3 marks)

Q19. 2M Board fav State Coulomb's law and write it in vector form. Show that it obeys Newton's third law.
Ans: $F = (1/4\pi\varepsilon_0)\, q_1 q_2/r^2$ along the line joining them. Vector form $\vec{F}_{21} = k(q_1 q_2/r^2)\,\hat{r}_{12}$. Since $\hat{r}_{21} = -\hat{r}_{12}$, $\vec{F}_{12} = -\vec{F}_{21}$.

Q20. 2M Define electric dipole moment. Write its SI unit and direction.
Ans: $p = q \times 2a$, a vector from $-q$ to $+q$. Unit: C m.

Q21. 3M Board fav Derive the expression for the torque on an electric dipole in a uniform field. When is it maximum?
Ans: Forces $+qE$ and $-qE$ form a couple; arm $= 2a \sin\theta$; $\tau = qE \cdot 2a \sin\theta = pE \sin\theta$, i.e. $\vec{\tau} = \vec{p} \times \vec{E}$. Maximum $(= pE)$ at $\theta = 90^\circ$.

Q22. 3M Board fav State Gauss's law. Using it, find the field due to an infinitely long straight wire of linear charge density λ.
Ans: $\varphi = q_{enc}/\varepsilon_0$. Coaxial cylinder (radius r, length l): flux through ends zero; $E \cdot 2\pi r l = \lambda l/\varepsilon_0 \Longrightarrow E = \lambda/2\pi\varepsilon_0 r$, radially outward for $\lambda > 0$.

Q23. 3M Find the field due to an infinite plane sheet of charge. What is the field between two oppositely charged parallel sheets?
Ans: Pill-box: $2EA = \sigma A/\varepsilon_0 \Longrightarrow E = \sigma/2\varepsilon_0$. Between oppositely charged sheets the fields add: σ/ε_0; outside they cancel: 0.

Q24. 2M Two point charges +4 μC and −4 μC are 6 cm apart. Find the field at the midpoint.
Ans: Each gives $E = kq/r^2 = 9 \times 10^9 \times 4 \times 10^{-6}/(0.03)^2 = 4 \times 10^7$ N/C, both towards −q. Net = **8×10^7 N/C** towards the negative charge.

Q25. 3M Two charges 5×10^{-8} C and -3×10^{-8} C are 16 cm apart. Locate the point on the line joining them where the electric field is zero.
Ans: The neutral point lies outside, beyond the smaller charge. Let x = distance from −3 nC (beyond it): $5/(16+x)^2 = 3/x^2 \Longrightarrow \sqrt{5}\,x = \sqrt{3}\,(16+x) \Longrightarrow x \approx 54.9$ cm. So about 55 cm beyond the -3×10^{-8} C charge.

Q26. 2M Distinguish between conductors and insulators (any three points).
Ans: Conductors have free electrons, insulators do not; charge spreads over a conductor, stays localised on an insulator; conductors offer low resistance, insulators very high.

Q27. 3M Why is charging by induction possible only for conductors? Explain the steps.
Ans: Conductors have free electrons that can redistribute. Steps: bring charged rod near, earth the far side, remove earth, then remove the rod; the conductor gets the opposite charge.

Q28. 2M What is electric flux? A uniform field E = 500 N/C makes 60° with the normal to a 0.02 m^2 surface. Find φ.
Ans: $\varphi = EA\cos\theta = 500 \times 0.02 \times 0.5 =$ **5 N m^2/C.**

Q29. 2M Plot E versus r for a uniformly charged spherical shell of radius R.
Ans: $E = 0$ for $r < R$; $E = q/4\pi\varepsilon_0 R^2$ at $r = R$ (maximum); then decreases as $1/r^2$ for $r > R$.

Q30. 3M Why does a dipole in a non-uniform field experience a force? Explain using the comb and paper example.
Ans: In a non-uniform field the forces on +q and −q are unequal, so net force ≠ 0. The comb polarises paper bits into dipoles; the stronger field near the comb pulls them in.

D. Long answer (5 marks)

Q31. 5M Board fav Derive the electric field of a dipole at a point on (a) the axial line, (b) the equatorial line. Show that $E_{axial} = 2E_{eq}$ for a short dipole.
Ans: (a) $E = k\cdot 4qar/(r^2-a^2)^2 \to 2kp/r^3$ for $r \gg a$, along p. (b) Perpendicular components cancel; $E = k\cdot 2qa/(r^2+a^2)^{3/2} \to kp/r^3$, opposite to p. Ratio 2 : 1.

Q32. 5M Board fav State Gauss's law. Using it, derive the field due to a uniformly charged thin spherical shell at points (i) outside, (ii) inside. Draw the E-r graph.
Ans: Outside: $E\cdot 4\pi r^2 = q/\varepsilon_0 \Longrightarrow E = q/4\pi\varepsilon_0 r^2$. Inside: $q_{enc} = 0 \Longrightarrow E = 0$. Graph: zero up to R, maximum at R, then $1/r^2$ fall.

E. Case-based

Q33. 4M Board fav Amber, rubbed with wool, picks up dust. A student charges a metal sphere on an insulating stand without touching it, using a negatively charged rod. (a) Name the process. (b) What charge does the sphere get? (c) What happens if the rod is removed before the earth? (d) Why must the stand be insulating?
Ans: (a) Charging by induction. (b) Positive. (c) Electrons return; the sphere becomes neutral. (d) Otherwise the charge leaks to the ground.

Q34. 4M Water molecules have a permanent dipole moment of 6.2×10^{-30} C m. (a) Is water polar? (b) Find the maximum torque in a field of 10^6 N/C. (c) In which orientation is the dipole in stable equilibrium? (d) Why does salt dissolve easily in water?
Ans: (a) Yes. (b) $\tau_{max} = pE = 6.2 \times 10^{-24}$ N m. (c) p parallel to E (θ = 0). (d) K = 80 weakens the ionic attraction 80 times.

Q35. 3M Two point charges q and −2q are at a distance d. Where should a third charge be placed so that it experiences no net force?
Ans: Unlike charges: the point lies outside, on the side of the smaller charge q. $k\,q/x^2 = k\,2q/(d+x)^2$ $\Longrightarrow d + x = \sqrt{2}\,x \Longrightarrow x = d/(\sqrt{2}-1) \approx 2.41d$ from q.

Q36. 2M A hollow conducting sphere carries charge Q. What is the field (a) at its centre, (b) just outside its surface?
Ans: (a) Zero. (b) $Q/4\pi\varepsilon_0 R^2 = \sigma/\varepsilon_0$, normal to the surface.

Q37. 1M Board fav If the radius of a Gaussian surface enclosing a charge is halved, how does the flux change?
Ans: No change. Flux depends only on the enclosed charge.

Q38. 2M Four charges +q, +q, −q, −q are placed at the corners of a square (like charges adjacent). Is the field at the centre zero?
Ans: No. The fields of diagonally opposite unlike charges add along each diagonal, giving a non-zero resultant parallel to the side joining the positive pair to the negative pair.

Q39. 1M An electric dipole is enclosed by a closed surface. What is the net flux?
Ans: Zero, because the net charge enclosed is +q − q = 0.

Q40. 2M Board fav Why is the electrostatic field normal to the surface of a conductor at every point?
Ans: A tangential component would move the free charges along the surface; in electrostatic equilibrium there is no such motion, so only the normal component survives.

F. Extra practice (board PYQ style)

Q41. 2M Board fav A thin metallic spherical shell of radius R carries charge Q. A point charge q is placed at its centre. Find the field at (a) $r < R$, (b) $r > R$.
Ans: (a) Only q is enclosed: $E = q/4\pi\varepsilon_0 r^2$. (b) Q + q enclosed: $E = (Q+q)/4\pi\varepsilon_0 r^2$.

Q42. 3M Board fav Two large thin metal plates are parallel and close. Their inner faces carry surface densities $+\sigma$ and $-\sigma$. Find E (a) to the left of the first plate, (b) between them, (c) to the right.
Ans: (a) 0, (b) σ/ε_0, directed from + to − plate, (c) 0.

Q43. 2M A charge Q is to be divided into q and (Q − q) so that the force between the parts at a fixed distance is maximum. Find q.
Ans: $F \propto q(Q-q)$. $dF/dq = Q - 2q = 0$ $\Longrightarrow$ **q = Q/2.**

Q44. 2M An electron and a proton are released from rest in the same uniform field. Compare their (a) forces, (b) accelerations.
Ans: (a) Equal in magnitude (eE), opposite in direction. (b) $a \propto 1/m$, so $a_e/a_p = m_p/m_e \approx 1836$.

Q45. 2M Board fav What is the flux through a cube of side a if a charge q is placed at one of its corners?
Ans: The corner charge is shared by 8 such cubes, so $\varphi = q/8\varepsilon_0$.

Q46. 2M Show that the electric field of a point charge has spherical symmetry.
Ans: $E = kq/r^2$ depends only on r, not on direction; on any sphere centred at the charge, |E| is the same and E is radial.

Q47. 2M A particle of mass 2×10^{-3} kg and charge 5 μC stays suspended in a vertical field. Find E.
Ans: $qE = mg \Longrightarrow E = mg/q = 2 \times 10^{-3} \times 9.8/(5 \times 10^{-6})$ = **3920 N/C**, upward.

Q48. 1M Define electric field intensity. Is it a scalar or vector? Give its SI unit.
Ans: Force per unit positive test charge; a vector; N/C (or V/m).

Q49. 2M Board fav Two point charges 3 μC and −3 μC are 20 cm apart. Find E at the midpoint.
Ans: Both fields point towards −q: $E = 2 \times kq/r^2 = 2 \times 9 \times 10^9 \times 3 \times 10^{-6}/0.01$ = **5.4×10^6 N/C** towards −3 μC.

Q50. 1M Why must the test charge be infinitesimally small?

Ans: So that it does not disturb the distribution of source charges, and the measured field is the true field.

Q51. 2M Sketch field lines for a positively charged conducting sphere and explain why they are normal to the surface.

Ans: Radial lines outward; if E had a tangential component, charges on the surface would move, contradicting electrostatic equilibrium.

Q52. 1M A thin wire ring of radius R carries charge Q. Find E at the centre.

Ans: By symmetry, contributions from opposite elements cancel: **E = 0.**

Electrostatics complete karke class mein entry

Asli baat: Agla chapter (Potential) isi pe based hai: $E = -dV/dr$. Field strong hai toh potential bhi aasan.

Formula sheet: Chapter 1

Quantity	Formula
Quantisation	$q = ne$
Coulomb force	$F = q_1q_2/4\pi\varepsilon_0r^2$
Field of point charge	$E = q/4\pi\varepsilon_0r^2$
Dipole moment	$p = q(2a)$
Axial / equatorial	$2kp/r^3$ / kp/r^3
Torque on dipole	$\tau = pE \sin\theta$
Flux	$\varphi = EA\cos\theta = q_{enc}/\varepsilon_0$
Wire / sheet	$\lambda/2\pi\varepsilon_0r$ / $\sigma/2\varepsilon_0$
Shell (out / in)	kq/r^2 / 0

Common mistakes

- Coulomb ke formula mein charge ka sign daal ke force ki direction mat nikalo; direction alag se socho.
- Gauss's law mein q sirf andar ka; par E sab charges se banta hai.
- Dipole: p ki direction −q se +q.
- Sheet ($\sigma/2\varepsilon_0$) aur conductor surface (σ/ε_0) mix mat karo.
- Flux mein θ area vector (normal) ke saath lo.

Quick compare: point charge vs dipole

	Point charge	Dipole
E far away	$\propto 1/r^2$	$\propto 1/r^3$
Net charge	q	zero
In uniform field	net force qE	net force 0, torque pE sin θ
Field lines	radial	from +q to −q

Paheli-Bujho

Paheli: Bujho, chapter khatam. Ek line mein Gauss's law?

Bujho: Band dabbe ka total flux = andar ka charge / ε_0. Bahar wale charge ka hisaab zero!

Paheli: Aur shell ke andar field?

Bujho: Zero! Isliye lightning mein car safe. Ab main electrostatics ka boss hoon.

Exam Alert: last check

- → Dipole field derivations (axial + equatorial) and Gauss's law applications: 5 marks pakke.
- → Units: E in N/C, φ in N m^2/C, p in C m.
- → Cube problem: centre pe q ho toh har face $q/6\varepsilon_0$.

Electrostatic Potential and Capacitance

Most asked: potential due to a dipole, potential energy of a dipole, capacitance of parallel plate capacitor with dielectric, series-parallel combinations, energy stored in a capacitor

Chapter 1 gave us the electric **field** (a vector). This chapter gives the easier scalar picture: **potential**, **potential energy**, **equipotential surfaces**, and the device that stores charge: the **capacitor**.

1. Electrostatic Potential Energy

The electrostatic force is **conservative**: work done by it depends only on the initial and final positions, not on the path. So we can define a **potential energy**.

Work done by an external force in moving a charge q from R to P against the field (without acceleration) is stored as potential energy:

$$U_P - U_R = W_{RP}$$

→ Only the **difference** in potential energy is physically meaningful.

→ We choose U = 0 at **infinity**; then U at P = work done in bringing the charge from infinity to P.

Potential energy difference between two points

Consider a charge q moved from R to P in the field of a fixed charge Q. Because the electrostatic force is conservative, the work done by the external force equals the increase in potential energy, and the same amount is returned when the charge goes back.

→ Work done by the field = $-\Delta U$; work done by the external agent = $+\Delta U$.

→ Potential energy is defined only up to an additive constant; only differences are measurable.

→ For two like charges U decreases as they separate (the field does positive work).

Why potential energy can be defined: conservative force

A force is called conservative when the work it does in moving a body between two points depends only on the initial and final positions, not on the path followed. Gravity and the electrostatic (Coulomb) force are both conservative. For such forces the work done can be stored as **potential energy** and recovered completely. For a non-conservative force such as friction, the work depends on the path and is lost as heat, so no potential energy can be defined.

An important consequence: the work done by the electrostatic field around any **closed path is zero**. If a charge is taken from A to B and back to A along any route, the field does no net work. This is why the potential at a point has a single, definite value.

Gravitational	*Electrostatic*
Mass m in field g	Charge q in field E
Work against gravity stored as mgh	Work against electric force stored as $q\Delta V$
Always attractive	Attractive or repulsive
Zero at infinity (by choice)	Zero at infinity (by choice)

2. Electrostatic Potential

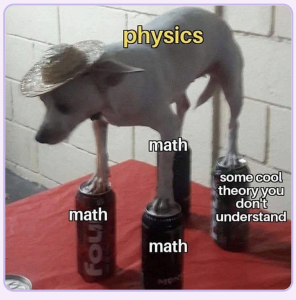
Electric potential at a point is the work done by an external force in bringing a unit positive charge from infinity to that point, without acceleration.

$$V = W/q_0 \quad \text{SI unit: volt (V)} = J\,C^{-1}$$

→ Potential is a **scalar**; it can be positive, negative or zero.

→ 1 volt: 1 joule of work is done to bring 1 coulomb from infinity to that point.

→ Potential difference $V_P - V_R = (U_P - U_R)/q$.



Potential samajhna = ek unit charge ko upar chadhana

Asli baat: Field is a vector, potential is a scalar: adding potentials of many charges is simple algebra, no angles needed.

Potential difference and work

$$V_P - V_R = W_{RP}/q$$

→ A positive charge moves naturally from high to low potential; a negative charge from low to high.

→ The work done by the field in moving charge q through a potential difference V is qV, which becomes kinetic energy if no other force acts: $\frac{1}{2}mv^2 = qV$.

Solved example

Q. An electron is accelerated from rest through 100 V. Find its kinetic energy and speed.

Sol. $K = eV = 1.6 \times 10^{-19}$ J = **100 eV**. $v = \sqrt{(2eV/m)} = \sqrt{(2 \times 1.6 \times 10^{-19}/9.1 \times 10^{-31})} =$ **5.9×10^6 m/s**.

Potential is a **scalar** and is a property of the point in the field, not of the test charge placed there. The same point has the same potential whether we place 1 μC or 1 C there; only the potential energy (qV) changes. Potential difference is what matters physically: a charge moves from high to low potential if positive and from low to high potential if negative.

Solved example

Q. How much work is done in moving a charge of 2 μC from a point at 10 V to a point at 60 V?

Sol. $W = q(V_B - V_A) = 2 \times 10^{-6} \times (60 - 10) =$ **1×10^{-4} J**.

3. Potential due to a Point Charge

Bring a unit positive charge from infinity to point P at distance r from Q. Work done against the repulsive force:

$$V = -\int_{\infty}^{r} E \cdot dr = (1/4\pi\varepsilon_0)\, Q/r$$

→ $V \propto 1/r$ (compare $E \propto 1/r^2$).

→ For $Q > 0$, V is positive; for $Q < 0$, V is negative.

→ V has the same value at all points equidistant from Q (spherical symmetry).

Derivation step by step

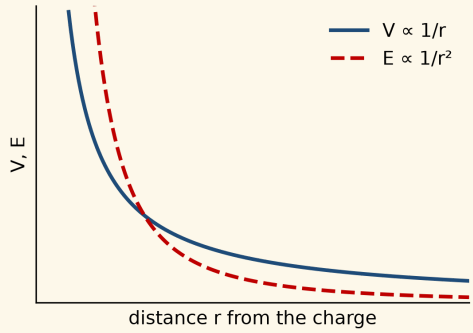
1. Electric field of Q at distance r′: $E = kQ/r'^2$, radially outward.
2. Work done by the external force against E in moving unit charge a distance dr′ towards Q: $dW = -E \cdot dr'$.
3. Integrate from ∞ to r: $V(r) = -\int_{\infty}^{r} kQ/r'^2\, dr' = kQ[1/r']_{\infty}^{r} = kQ/r$.

The result holds for any sign of Q, and the work is independent of the path taken.

Solved example (NCERT)

Q. Calculate the potential at a point P due to a charge of 4×10^{-7} C located 9 cm away. Find the work done in bringing a charge of 2×10^{-9} C from infinity to P.

Sol. $V = 9 \times 10^9 \times 4 \times 10^{-7}/0.09 =$ **4×10^4 V**. $W = qV =$ **8×10^{-5} J**.



Potential falls as 1/r while the field falls faster, as $1/r^2$

For a point charge, potential falls as 1/r while the field falls as $1/r^2$. Doubling the distance halves V but makes E one-fourth. Potential due to a positive charge is positive everywhere and that due to a negative charge is negative everywhere.

Solved example

Q. A spherical conductor of radius 12 cm has a charge of 1.6×10^{-7} C distributed uniformly on its surface. What is the electric field (a) inside the sphere, (b) just outside the sphere, (c) at a point 18 cm from the centre of the sphere?

Sol. (a) Inside a conductor **E = 0**.

(b) $E = kq/R^2 = (9 \times 10^9 \times 1.6 \times 10^{-7})/(0.12)^2 =$ **1.0×10^5 N/C**.

(c) $E = kq/r^2 = (9 \times 10^9 \times 1.6 \times 10^{-7})/(0.18)^2 =$ **4.4×10^4 N/C**.

4. Potential due to an Electric Dipole

At P at distance r from the centre, making angle θ with the dipole axis, with r_1, r_2 the distances from +q and −q:

$$V = kq\,(1/r_1 - 1/r_2), \quad r_1 \approx r - a\cos\theta,\ r_2 \approx r + a\cos\theta$$

$$V = (1/4\pi\varepsilon_0)\, p\cos\theta / r^2 \quad (r \gg a) = p\cdot\hat{r} / 4\pi\varepsilon_0 r^2$$

→ **Axial point** (θ = 0°): $V = kp/r^2$; (θ = 180°): $V = -kp/r^2$.

→ **Equatorial point** (θ = 90°): **V = 0**, though E ≠ 0 there.

→Dipole potential $\propto 1/r^2$; single charge $\propto 1/r$.

Exam Alert

→Equatorial line pe V = 0 but E = $kp/r^3 \neq 0$. Ye trick question bahut aata hai.

→Derivation of $V = kp\cos\theta/r^2$: 3 marks.

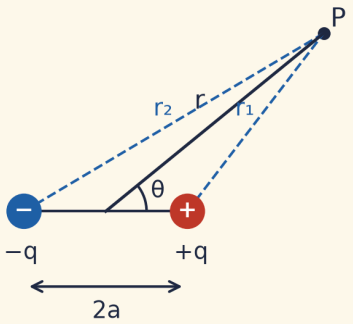


Fig. Geometry for the potential of a dipole at P

Derivation step by step

1. V at P = $kq/r_1 - kq/r_2$.
2. By geometry ($r \gg a$): $r_1^2 = r^2 + a^2 - 2ar\cos\theta \approx r^2(1 - 2a\cos\theta/r)$, so $1/r_1 \approx (1/r)(1 + a\cos\theta/r)$.
3. Similarly $1/r_2 \approx (1/r)(1 - a\cos\theta/r)$.
4. Subtracting: $V = kq \cdot 2a\cos\theta/r^2 = kp\cos\theta/r^2$.

	Single charge	Dipole
V depends on	r only	r and θ
V ∝	1/r	$1/r^2$
Equatorial value	—	zero

Point on	Potential V	Field E
Axial line	kp/r^2	$2kp/r^3$
Equatorial line	0	kp/r^3
General point (r, θ)	$kp\cos\theta/r^2$	$(kp/r^3)\sqrt{(1+3\cos^2\theta)}$

Yaad karo

→Dipole ka V ∝ $1/r^2$, point charge ka V ∝ 1/r.

→Equatorial line pe V = 0 par E zero nahi.

→Dipole ka V direction (θ) pe depend karta hai, point charge ka nahi.

Solved example

Q. A dipole has charges ±10 μC separated by 2 mm. Find the potential at a point on its axis 10 cm from the centre, and at a point on its equatorial line at the same distance.

Sol. $p = q \times 2a = 10 \times 10^{-6} \times 2 \times 10^{-3} = 2 \times 10^{-8}$ C m.

Axial: $V = kp/r^2 = (9 \times 10^9 \times 2 \times 10^{-8})/(0.1)^2 = \mathbf{1.8 \times 10^4}$ **V.**

Equatorial: **V = 0.**

5. Potential due to a System of Charges

$$V = (1/4\pi\varepsilon_0)\,\Sigma\, q_i/r_i \quad \text{algebraic sum (scalars)}$$

Uniformly charged spherical shell (R, q): outside, V = kq/r (as if all charge were at the centre). On the surface and **everywhere inside**, V = kq/R (constant), because E = 0 inside.

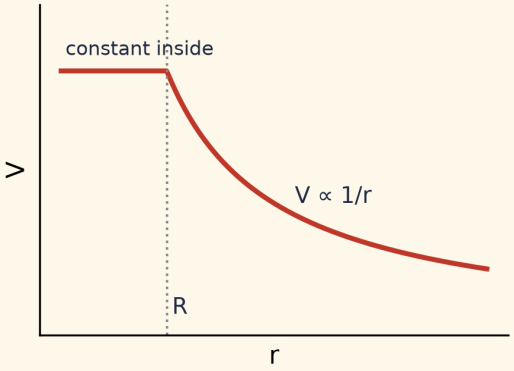


Fig. 2.1 Potential of a charged shell vs r

Solved example

Q. Charges +4 μC and −2 μC are 1 m apart. Find the point on the line joining them (between them) where V = 0.

Sol. $k(4/x - 2/(1-x)) = 0 \Longrightarrow 4(1-x) = 2x \Longrightarrow$ **x = 2/3 m** from the +4 μC charge.

Solved example

Q. Two charges 5×10^{-8} C and -3×10^{-8} C are 16 cm apart. At what points on the line joining them is V = 0?

Sol. Between: $5/x = 3/(16 - x) \Longrightarrow$ **x = 10 cm** from the positive charge. Outside (beyond the negative charge): $5/x = 3/(x - 16) \Longrightarrow$ **x = 40 cm** from the positive charge.

Solved example: square

Q. Charges q, −q, q, −q sit at corners of a square of side a. Find V at the centre and the work to bring a charge Q there.

Sol. Distances equal, charges sum to zero: **V = 0**, so W = 0.

Solved example

Q. A regular hexagon of side 10 cm has a charge 5 μC at each of its vertices. Calculate the potential at the centre of the hexagon.

Sol. Each vertex is at distance r = side = 0.10 m from the centre.

$V = 6 \times kq/r = 6 \times (9 \times 10^9 \times 5 \times 10^{-6})/0.10 =$ **2.7×10^6 V.**

Solved example

Q. Two tiny spheres carrying charges 1.5 μC and 2.5 μC are located 30 cm apart. Find the potential and electric field at the midpoint of the line joining them.

Sol. r = 0.15 m from each.

$V = k(q_1 + q_2)/r = (9 \times 10^9 \times 4 \times 10^{-6})/0.15 = \mathbf{2.4 \times 10^5\ V}$.

The fields are opposite: $E = k(q_2 - q_1)/r^2 = (9 \times 10^9 \times 1 \times 10^{-6})/(0.15)^2 = \mathbf{4 \times 10^5\ N/C}$, directed from the 2.5 μC charge towards the 1.5 μC charge.

Solved example

Q. A cube of side b has a charge q at each of its vertices. Find the potential and electric field at the centre of the cube.

Sol. Each vertex is at half the body diagonal: $r = \sqrt{3}\, b/2$.

$V = 8 \times q/(4\pi\varepsilon_0 \times \sqrt{3}\, b/2) = \mathbf{4q/(\sqrt{3}\, \pi\varepsilon_0 b)}$.

The fields of diagonally opposite charges cancel in pairs, so **E = 0**.

6. Equipotential Surfaces

A surface on which the potential is the **same at every point** is an equipotential surface.

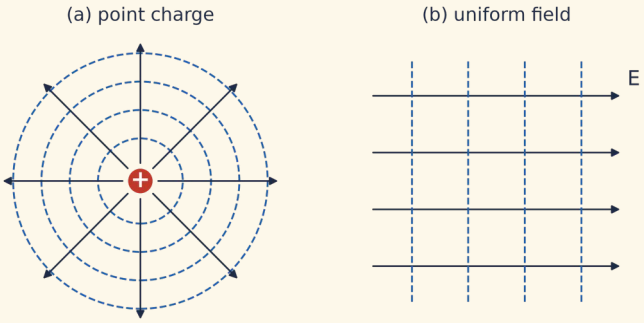


Fig. 2.2 Equipotential surfaces (dashed): (a) concentric spheres for a point charge, (b) parallel planes for a uniform field

Properties

1. No work is done in moving a charge on an equipotential surface ($\Delta V = 0$).
2. E is always normal to the equipotential surface at every point.
3. Two equipotential surfaces never intersect.
4. They are closer together where the field is strong.

Relation between field and potential

$E = -dV/dr$ E points in the direction of steepest decrease of V

$|E| = |\Delta V|/\Delta l$: the magnitude of E is the potential drop per unit displacement normal to the equipotential surface. That is why V m^{-1} is also a unit of E.

Equipotential surface pe charge ghumao, kaam zero, maza poora
Asli baat: W = qΔV = 0 on an equipotential surface, so E must be perpendicular to it.

Paheli-Bujho

Bujho: Paheli, potential aur potential energy mein kya fark? Dono same lagte hain.
Paheli: Potential har point ki property hai: per unit charge.
Energy us charge ki: U = qV.
Bujho: Matlab V ek dukaan ka rate hai aur U poora bill!
Paheli: Haha, perfect analogy, Bujho.

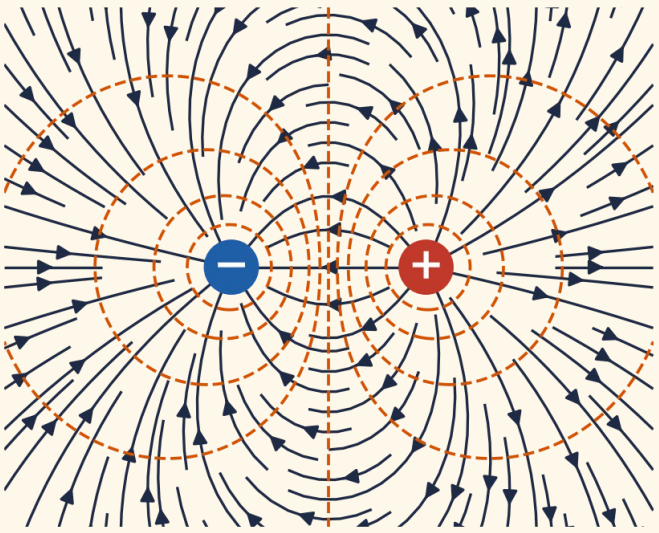


Fig. Field lines (solid) and equipotentials (dashed) of a dipole

Shapes of equipotential surfaces

Source	Equipotential surfaces
Point charge	concentric spheres
Uniform field	parallel planes ⊥ to E
Line charge	coaxial cylinders
Dipole	closed surfaces around each charge; equatorial plane is V = 0

NCERT Activity: equipotentials around a dipole

Place two oppositely charged electrodes in a shallow tray of slightly conducting water, connect a voltmeter probe and find points of equal reading. Joining them traces the dashed curves above; a compass of small charge always moves ⊥ to them, showing E ⊥ equipotentials.

Why E is perpendicular to an equipotential surface

On an equipotential surface V is the same everywhere, so moving a charge along it requires no work: $W = q\Delta V = 0$. If E had a component along the surface, work would be done in moving a charge along it. Hence E has no component along the surface: it must be **normal** to it, pointing from higher to lower potential. Also, $E = -dV/dr$: the field is strong where equipotentials are crowded and weak where they are far apart.

Field configuration	Shape of equipotential surfaces
Point charge	concentric spheres
Uniform field (along x)	planes perpendicular to x
Infinite line charge	co-axial cylinders
Dipole	closed curved surfaces; plane V = 0 at the bisector

7. Potential Energy of a System of Charges

Two charges: $U = (1/4\pi\varepsilon_0)\, q_1q_2/r_{12}$

Three charges: $U = k(q_1q_2/r_{12} + q_2q_3/r_{23} + q_1q_3/r_{13})$

→ U > 0 for like charges (work done against repulsion).

→ U < 0 for unlike charges (system is bound).

Potential energy in an external field

→ Single charge: $U = qV(r)$.

→ Two charges: $U = q_1V(r_1) + q_2V(r_2) + kq_1q_2/r_{12}$.

Potential energy of a dipole in a uniform field

Work done in rotating the dipole from θ_0 to θ_1: $W = pE(\cos\theta_0 - \cos\theta_1)$. Taking $\theta_0 = 90^\circ$ as zero:

$$U(\theta) = -pE\cos\theta = -p \cdot E$$

Orientation	θ	U	Equilibrium
p along E	0°	−pE (minimum)	stable
p ⊥ E	90°	0	–
p opposite E	180°	+pE (maximum)	unstable

Solved example

Q. Work done to rotate a dipole ($p = 10^{-29}$ C m) from stable to unstable position in $E = 10^6$ N/C?

Sol. $W = U(180^\circ) - U(0^\circ) = 2pE = \mathbf{2 \times 10^{-23}}$ J.

Electron volt dekh ke joule samajhna

Asli baat: 1 eV = 1.6×10^{-19} J: energy gained by an electron accelerated through 1 V.

Sign of potential energy

→ U positive: like charges, energy needed to bring them together (repulsion).

→ U negative: unlike charges, the system is bound; energy must be supplied to separate them.

8. Electrostatics of Conductors

1. Inside a conductor, **E = 0** (free charges move till the field vanishes).
2. At the surface, E is **normal** to the surface.
3. Net charge in the interior is zero; any excess charge resides on the **surface**.
4. Potential is **constant** throughout the volume and equal on the surface.
5. Field at the surface: $\mathbf{E} = \sigma/\varepsilon_0\, \hat{n}$.
6. Inside a cavity (with no charge), E = 0: **electrostatic shielding**.

Quick fact

During lightning, sitting inside a car is safe: the metal body acts as a shield (Faraday cage). Sensitive instruments are kept in metal boxes for the same reason.

Potential energy of a system of three charges: derivation

1. Bring q_1 from ∞: no work (no field yet).
2. Bring q_2 to distance r_{12}: $W_2 = kq_1q_2/r_{12}$.
3. Bring q_3: $W_3 = kq_3(q_1/r_{13} + q_2/r_{23})$.
4. Total U = sum of energies of all pairs, independent of the order of assembly.

Solved example (NCERT)

Q. Four charges q, −q, q, −q sit at corners of a square ABCD of side 10 cm . Find the work to put together the arrangement.

Sol. 4 adjacent pairs ($-kq^2/a$ each) and 2 diagonal pairs ($+kq^2/\sqrt{2}a$ each): $U = (kq^2/a)(-4 + \sqrt{2}) = \mathbf{-2.59\, kq^2/a}$.

Electrostatic shielding

The field inside a cavity of a conductor is zero, whatever the charges outside and whatever the shape of the cavity. This is called **electrostatic shielding**. Sensitive instruments are kept inside metal boxes, a car is a safe place during lightning, and coaxial cables have an outer metal sheath for the same reason.

Solved example

Q. A spherical conducting shell of inner radius r_1 and outer radius r_2 has a charge Q. (a) A charge q is placed at the centre of the shell. What is the surface charge density on the inner and outer surfaces? (b) Is the field inside a cavity (with no charge) zero, even if the shell is not spherical?

Sol. (a) The inner surface acquires −q and the outer surface Q + q.

$\sigma_{inner} = \mathbf{-q/4\pi r_1^2}$, $\sigma_{outer} = \mathbf{(Q+q)/4\pi r_2^2}$.

(b) **Yes.** If there were a field in the cavity, the work done around a closed path partly in the cavity and partly in the conductor would not be zero, which is impossible for an electrostatic field.

Solved example

Q. Two charged conducting spheres of radii a and b are connected to each other by a wire. What is the ratio of electric fields at the surfaces of the two spheres?

Sol. Connected spheres are at the same potential: $kQ_1/a = kQ_2/b \Longrightarrow Q_1/Q_2 = a/b$.

$E_1/E_2 = (Q_1/a^2)/(Q_2/b^2) = \mathbf{b/a}$. The smaller sphere (sharper surface) has the stronger field, which explains why charge leaks from sharp points.

Solved example

Q. A long charged cylinder of linear charge density λ is surrounded by a hollow co-axial conducting cylinder. What is the electric field in the space between the two cylinders?

Sol. Take a co-axial Gaussian cylinder of radius r and length l between them. Flux $= E(2\pi rl) = \lambda l/\varepsilon_0$. $E = \mathbf{\lambda/(2\pi\varepsilon_0 r)}$, radially outward (for positive λ). The outer conductor does not affect the field inside it.

9. Dielectrics and Polarisation

Dielectrics are non-conducting substances. In an external field E_0, their molecules develop dipole moments (**polarisation**), which produce a field opposing E_0.

Polar molecules	*Non-polar molecules*
Permanent dipole moment (centres of + and − charge separated)	No permanent dipole moment (centres coincide)
Randomly oriented; field aligns them	Field induces a dipole moment
H_2O, HCl	O_2, N_2, CO_2

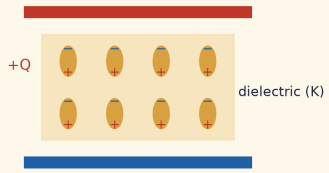


Fig. 2.3 Dielectric slab polarised in a capacitor

Polarisation $P = \chi_e E$; $K = 1 + \chi_e = \varepsilon/\varepsilon_0$

The induced surface charges reduce the field inside the dielectric: $\mathbf{E = E_0/K}$.

Why is E zero inside a conductor? Step by step

1. A conductor has free electrons. If $E \neq 0$ inside, they would drift and a current would flow.
2. In electrostatics nothing moves, so the free charges rearrange until the internal field is cancelled.
3. By Gauss's law, a surface just inside encloses zero charge, so any excess charge sits on the outer surface.

Charge density and sharp points

Surface charge density is highest where the radius of curvature is smallest. At sharp points σ (and E =

σ/ε_0) can become large enough to ionise air: **corona discharge**. This is used in lightning conductors.

NCERT Activity: Van de Graaff generator

- → A belt carries charge to a large hollow metal sphere; charge moves to the outer surface because E = 0 inside.
- → The sphere can be raised to millions of volts, used to accelerate charged particles.
- → Principle: charge given to an inner conductor always flows to the outer conductor, whatever its potential.

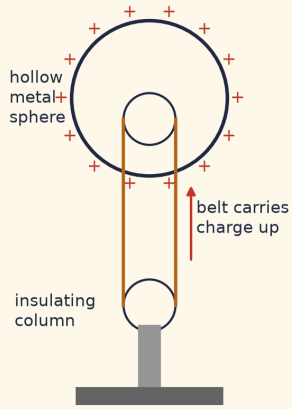


Fig. Van de Graaff generator (schematic)

Dielectrics are of two kinds. In **polar** molecules (H_2O, HCl) the centres of positive and negative charge do not coincide, so each molecule has a permanent dipole moment, randomly oriented without a field. In **non-polar** molecules (O_2, N_2, CO_2) the centres coincide; an external field displaces them and induces a dipole moment. In both cases the field produces a net dipole moment per unit volume, called **polarisation** $P = \chi_e E$, and the induced surface charges reduce the field inside the dielectric.

Polar	Non-polar
Permanent dipole moment	No permanent dipole moment
Field aligns existing dipoles	Field induces dipoles
H_2O, HCl, NH_3	O_2, N_2, H_2, CO_2

10. Capacitors and Capacitance

A **capacitor** is a system of two conductors separated by an insulator, used to store charge and energy. When the conductors carry +Q and −Q with potential difference V:

$$C = Q/V \quad \text{SI unit: farad (F)} = \text{C V}^{-1}$$

- → C depends only on the **geometry** (size, shape, separation) and the medium, not on Q or V.
- → 1 F is very large; practical units: μF (10^{-6}), nF, pF (10^{-12}).
- → Symbol: two parallel lines of equal length.

Capacitance depends only on the **geometry** (size, shape and separation of the conductors) and on the medium between them; it does not depend on the charge Q or on the potential V. Doubling Q doubles V, so the ratio C = Q/V stays the same. A large capacitance means the capacitor can hold a large charge at a small potential difference, so the field between the plates stays below the breakdown value of the medium (about 3×10^6 V/m for air).

Factor increased	Effect on C
Plate area A	C increases ($C \propto A$)
Separation d	C decreases ($C \propto 1/d$)
Dielectric constant K	C increases K times
Charge Q or voltage V	no change

11. Parallel Plate Capacitor

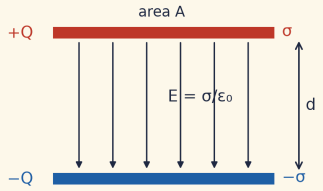


Fig. 2.4 Parallel plate capacitor

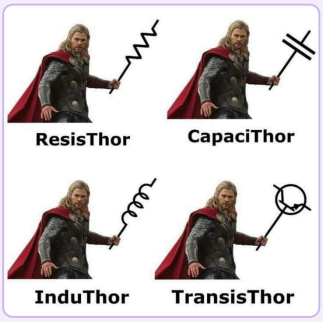
Two plates of area A, separation d ($d \ll$ size of plates), charge densities $\pm\sigma = \pm Q/A$. Between the plates the fields of the two sheets add: $E = \sigma/\varepsilon_0$; outside they cancel.

$$V = Ed = Qd/\varepsilon_0 A \implies C_0 = \varepsilon_0 A/d$$

Solved example

Q. Plates of area 1 m^2 are 1 cm apart in air. Find C.

Sol. $C = 8.85 \times 10^{-12} \times 1/0.01 = 885$ **pF**. Even a 1 m^2 capacitor is under 1 nF; that is why 1 F is huge.



CapaciThor: charge store karna iska superpower hai

Asli baat: $C = \varepsilon_0 A/d$: bigger plates and smaller gap give a bigger capacitor; a dielectric multiplies it by K.

12. Effect of Dielectric on Capacitance

With a dielectric of constant K completely filling the space, the field drops to E_0/K, so V drops and C rises:

$$C = K\,\varepsilon_0 A/d = K\,C_0$$

Quantity	Battery disconnected (Q fixed)	Battery connected (V fixed)
Charge Q	same	increases K times
Voltage V	decreases K times	same
Field E	decreases K times	same
Capacitance C	increases K times	increases K times
Energy U	decreases K times	increases K times

Partially filled (slab of thickness $t < d$)

$C = \varepsilon_0 A / [d - t + t/K]$ metal slab ($K = \infty$): $C = \varepsilon_0 A/(d - t)$

Exam Alert

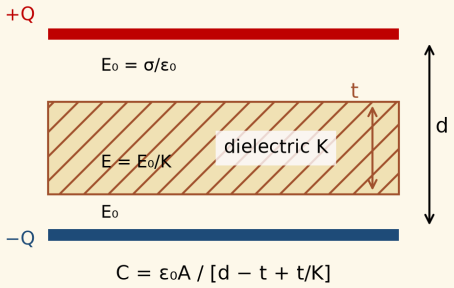
→ Battery connected vs disconnected wali table: 5-mark case question mein pakki.

→ Slab question: formula $d - t + t/K$ yaad karo; metal ke liye $t/K = 0$.

Why does the capacitance increase? (step by step)

1. Without dielectric: $E_0 = \sigma/\varepsilon_0$, $V_0 = E_0 d$.
2. Dielectric gets polarised: surface charges $\pm\sigma_p$ appear, giving an opposing field.
3. Net field $E = (\sigma - \sigma_p)/\varepsilon_0 = E_0/K$; $V = V_0/K$.
4. $C = Q/V = KQ/V_0 = KC_0$.

Dielectric	K
Vacuum	1
Air	1.0006
Paper	3.5
Mica	6
Water	80



Dielectric slab of thickness t (less than d) between the plates

If a slab of thickness t ($t < d$) is inserted, the field is E_0 in the air gap $(d - t)$ and E_0/K inside the slab. $V = E_0(d - t) + (E_0/K)t$, so $C = \varepsilon_0 A/[d - t + t/K]$. For a metal slab, $K \to \infty$ and $C = \varepsilon_0 A/(d - t)$.

13. Combination of Capacitors

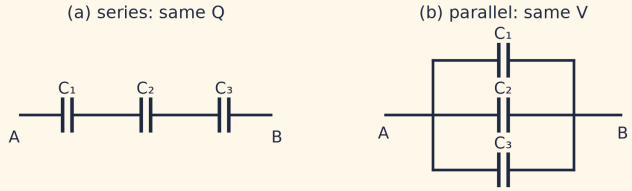


Fig. 2.5 Capacitors in (a) series, (b) parallel

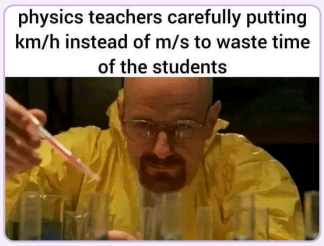
Series	Parallel
Same charge Q on each	Same voltage V across each
$V = V_1 + V_2 + V_3$	$Q = Q_1 + Q_2 + Q_3$
$1/C = 1/C_1 + 1/C_2 + 1/C_3$	$C = C_1 + C_2 + C_3$
C_{eq} less than the smallest	C_{eq} more than the largest

Note: capacitors combine like **resistors in reverse**. Two capacitors in series: $C = C_1C_2/(C_1 + C_2)$.

Solved example

Q. Three 9 pF capacitors in series across 120 V. Find C_{eq} and V on each. What if in parallel?

Sol. Series: C = 3 pF, each gets 40 V. Parallel: C = 27 pF, each 120 V, $Q = 9 \times 120 = 1080$ pC each.



Series aur parallel ka formula ulta karke yaad rakho

Asli baat: Capacitors in series add like resistors in parallel, and vice versa.

Mixed combinations: method

1. Identify groups that are clearly in series (same Q, one path) or parallel (same two end points).
2. Replace each group by its equivalent, redraw, repeat.
3. Find total $Q = C_{eq}V$, then work back to each capacitor.

Solved example (NCERT style)

Q. Four 10 μF capacitors: C_1, C_2, C_3 in series, the group in parallel with C_4, across 500 V. Find C_{eq} and the charge on each.

Sol. Series group: 10/3 μF; total = 10/3 + 10 = **13.3 μF**. $Q_4 = 10 \times 500 = 5$ mC; each in series: $(10/3) \times 500 =$ **1.67 mC**.

Solved example

Q. A network has $C_1 = 2\ \mu F$ and $C_2 = 2\ \mu F$ in parallel, the pair in series with $C_3 = 4\ \mu F$, connected to a 12 V battery. Find the equivalent capacitance and the charge on C_3.

Sol. $C_{12} = 2 + 2 = 4\ \mu F$. In series with 4 μF: $C_{eq} = (4 \times 4)/(4 + 4) = \mathbf{2\ \mu F}$.

$Q = C_{eq}V = 2 \times 12 = \mathbf{24\ \mu C}$ on C_3 (and 12 μC on each of C_1 and C_2).

14. Energy Stored in a Capacitor

Charging is done bit by bit: to move dq against the existing voltage q/C, work dW = (q/C)dq. Total work:

$$U = Q^2/2C = \tfrac{1}{2}CV^2 = \tfrac{1}{2}QV$$

Energy density

For a parallel plate capacitor, $U = \tfrac{1}{2}(\varepsilon_0 A/d)(Ed)^2 = \tfrac{1}{2}\varepsilon_0 E^2(Ad)$. Dividing by the volume Ad:

$$u = \tfrac{1}{2}\,\varepsilon_0\,E^2$$ energy per unit volume, true for any field

Yaad karo

- → Energy field mein store hoti hai, plates pe nahi.
- → Two capacitors share charge: common V = $(C_1V_1 + C_2V_2)/(C_1 + C_2)$; energy hamesha thodi lost (heat/radiation).

Paheli-Bujho

Bujho: Paheli, capacitor battery jaisa hi hai na? Dono charge rakhte hain.

Paheli: Nahi! Battery chemical energy se voltage banati hai. Capacitor sirf field mein energy store karta hai, aur ek jhatke mein de deta hai.

Bujho: Isliye camera ka flash itna tez chamakta hai!

Paheli: Bilkul, wahi capacitor discharge hai.

NCERT Activity: charging and discharging a capacitor

Connect a large capacitor (1000 μF) to a battery through a switch, then disconnect the battery and connect the capacitor across a small bulb or LED. The bulb flashes briefly: the energy $\tfrac{1}{2}CV^2$ stored in the field is released. Camera flashes work the same way.

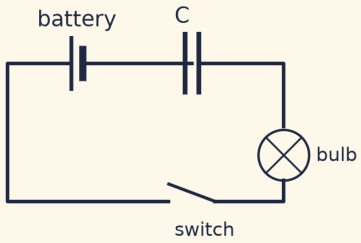


Fig. Capacitor, battery and bulb arrangement

Energy in a dielectric-filled capacitor

- → Battery disconnected: $U = Q^2/2C$ falls K times (work is done on the slab as it is pulled in).
- → Battery connected: $U = \tfrac{1}{2}CV^2$ rises K times (the battery supplies extra energy).

Solved example

Q. A 900 pF capacitor charged to 100 V is disconnected and connected to another uncharged 900 pF capacitor. Find the energy lost.

Sol. $U_i = \tfrac{1}{2} \times 900 \times 10^{-12} \times 10^4 = 4.5 \times 10^{-6}$ J. V = 50 V; $U_f = \tfrac{1}{2} \times 1800 \times 10^{-12} \times 2500 = 2.25 \times 10^{-6}$ J. Lost = $\mathbf{2.25 \times 10^{-6}}$ J.

Where is the energy stored? Between the plates, in the electric field. Energy density $u = \tfrac{1}{2}\varepsilon_0 E^2$; multiplying by the volume Ad gives $\tfrac{1}{2}\varepsilon_0 E^2 Ad = \tfrac{1}{2}CV^2$, the same result.

15. Solved Numericals (Board Pattern)

N1. Find the work done in bringing a charge of 2 μC from infinity to a point 0.3 m from a 6 μC charge.

Sol. $W = qV = 2 \times 10^{-6} \times 9 \times 10^9 \times 6 \times 10^{-6}/0.3 = \mathbf{0.36}$ J.

N2. Four charges +q, −q, +q, −q are at the corners of a square of side a (alternately). Find V at the centre.

Sol. All at distance $a/\sqrt{2}$; sum of charges = 0 ⟹ **V = 0** (E is also zero by symmetry).

N3. A 12 pF capacitor is connected to a 50 V battery. Find the energy stored.

Sol. $U = \tfrac{1}{2}CV^2 = \tfrac{1}{2} \times 12 \times 10^{-12} \times 2500 = \mathbf{1.5 \times 10^{-8}}$ J.

N4. A 600 pF capacitor charged to 200 V is connected to an uncharged 600 pF capacitor. Find the energy lost.
Sol. Initial $U = \frac{1}{2} \times 600 \times 10^{-12} \times 200^2 = 1.2 \times 10^{-5}$ J. Common V = 100 V; final $U = \frac{1}{2} \times 1200 \times 10^{-12} \times 100^2 = 6 \times 10^{-6}$ J. Lost = **6×10^{-6} J** (half).

N5. A parallel plate capacitor ($A = 6 \times 10^{-3}$ m^2, d = 3 mm) is filled with mica (K = 6) and connected to 100 V. Find C and Q.
Sol. $C = K\varepsilon_0 A/d = 6 \times 8.85 \times 10^{-12} \times 6 \times 10^{-3}/(3 \times 10^{-3})$ = **106 pF**; $Q = CV$ = **1.06×10^{-8} C**.

N6. A parallel plate capacitor with air has C = 8 pF. What will be C if the distance is halved and the space filled with K = 6?
Sol. $C = K\varepsilon_0 A/(d/2) = 2K\,C_0$ = **96 pF**.

N7. Three capacitors 2 pF, 3 pF and 4 pF are in parallel. Find C_{eq} and the charge on each across 100 V.
Sol. C = **9 pF**; Q = 200 pC, 300 pC, 400 pC.

N8. A slab of thickness 3 mm (K = 3) is placed between plates 6 mm apart. Find the ratio of new to old capacitance.
Sol. $C/C_0 = d/(d - t + t/K) = 6/(3 + 1)$ = **1.5**.

Formula sheet: Chapter 2

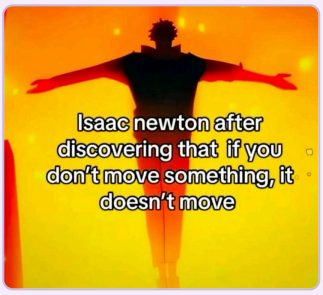
Quantity	Formula
Point charge	$V = kq/r$
Dipole	$V = kp\cos\theta/r^2$
Field-potential	$E = -dV/dr$
Two charges	$U = kq_1q_2/r$
Dipole in field	$U = -pE\cos\theta$
Parallel plate	$C = K\varepsilon_0 A/d$
Slab	$C = \varepsilon_0 A/(d - t + t/K)$
Series / parallel	$1/C = \Sigma 1/C_i$ / $C = \Sigma C_i$
Energy	$U = \frac{1}{2}CV^2$; $u = \frac{1}{2}\varepsilon_0 E^2$

1 minute recap

- ✓ V = W/q, scalar; point charge kq/r; dipole kp cos θ/r^2.
- ✓ E ⊥ equipotential surface; E = −dV/dr.
- ✓ Conductor: E = 0 inside, charge on surface, V constant.
- ✓ C = Q/V; parallel plate $\varepsilon_0 A/d$; dielectric K times.
- ✓ U = ½CV^2; series and parallel rules.

Most Important Questions

Objective, short, long and case-based, with answers



Exam hall mein capacitor ke sawaal dekh ke
Asli baat: Most asked: dipole potential, C with dielectric, series-parallel, energy stored and charge sharing.

A. Objective and very short answer (1 mark)

Q1. IM The potential at a point on the equatorial line of a dipole is:
(a) zero
(b) kp/r^2
(c) $2kp/r^2$
(d) kp/r^3
Ans: (a) zero.

Q2. IM The SI unit of capacitance is:
(a) volt
(b) coulomb
(c) farad
(d) henry
Ans: (c) farad.

Q3. IM Board fav If a dielectric of constant K fills a capacitor connected to a battery, the charge:
(a) remains same
(b) becomes K times
(c) becomes 1/K times
(d) becomes zero
Ans: (b) becomes K times.

Q4. 1M Work done in moving a charge of 5 C on an equipotential surface is:
(a) 5 J
(b) 1 J
(c) 25 J
(d) zero
Ans: (d) zero.

Q5. 1M Two capacitors of 6 μF and 3 μF in series have equivalent capacitance:
(a) 9 μF
(b) 2 μF
(c) 4.5 μF
(d) 18 μF
Ans: (b) 2 μF.

Q6. 1M Energy density in an electric field E is:
(a) $\varepsilon_0 E$
(b) $\varepsilon_0 E^2$
(c) $\frac{1}{2}\varepsilon_0 E^2$
(d) $2\varepsilon_0 E^2$
Ans: (c) $\frac{1}{2}\varepsilon_0 E^2$.

Q7. 1M Board fav Why is E normal to an equipotential surface?
Ans: If E had a component along the surface, work would be done in moving a charge on it, contradicting $\Delta V = 0$.

Q8. 1M What is the potential inside a charged hollow conducting sphere?
Ans: Constant, equal to the potential on its surface (kq/R).

Q9. 1M Can two equipotential surfaces intersect?
Ans: No. At the intersection the potential would have two values.

Q10. 1M Define dielectric constant in terms of capacitance.
Ans: $K = C$ (with dielectric)$/C_0$ (vacuum).

Q11. 1M What is the potential energy of a dipole in stable equilibrium?
Ans: $-pE$ (minimum), with p parallel to E.

Q12. 1M Give the relation between E and V.
Ans: $E = -dV/dr$.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M **A:** The field inside a conductor is zero. **R:** Free charges in a conductor move until the internal field vanishes.
Ans: (a)

Q14. 1M Board fav **A:** A dielectric slab reduces the field between the plates of a capacitor. **R:** The induced surface charges of the slab produce a field opposite to the applied field.
Ans: (a)

Q15. 1M **A:** Equivalent capacitance of capacitors in series is more than the largest one. **R:** In series the charge on each capacitor is the same.
Ans: (d) A false (it is less than the smallest); R true.

Q16. 1M **A:** Potential on the equatorial line of a dipole is zero. **R:** The electric field on the equatorial line is also zero.
Ans: (c) A true; R false ($E = kp/r^3$).

C. Short answer (2–3 marks)

Q17. 2M Derive the expression for the potential due to a point charge.
Ans: Work in bringing unit charge from ∞ to r against $E = kQ/r^2$: $V = -\int_{\infty}^{r} kQ/r^2\, dr = kQ/r$.

Q18. 3M Board fav Derive the potential due to a dipole at a general point. Hence find it on axial and equatorial lines.
Ans: $V = kq(1/r_1 - 1/r_2)$ with $r_{1,2} \approx r \mp a\cos\theta \Longrightarrow V = kp\cos\theta/r^2$. Axial: kp/r^2; equatorial: 0.

Q19. 3M Board fav Derive the expression for the potential energy of a dipole in a uniform field.
Ans: Work in rotating from θ_0 to θ: $\int pE\sin\theta\, d\theta = pE(\cos\theta_0 - \cos\theta)$. With $\theta_0 = 90^\circ$ as reference: $U = -pE\cos\theta$.

Q20. 3M Board fav Derive $C = \varepsilon_0 A/d$ for a parallel plate capacitor.
Ans: Field between plates $E = \sigma/\varepsilon_0 = Q/\varepsilon_0 A$; $V = Ed = Qd/\varepsilon_0 A$; $C = Q/V = \varepsilon_0 A/d$.

Q21. 2M Show that the energy stored in a capacitor is $\frac{1}{2}CV^2$.

Ans: $dW = (q/C)dq$; $W = \int_0^Q (q/C)dq = Q^2/2C = \frac{1}{2}CV^2$.

Q22. 3M Derive the formula for capacitors in series and in parallel.

Ans: Series: same Q, $V = \Sigma V_i = Q\Sigma 1/C_i \Longrightarrow 1/C = \Sigma 1/C_i$. Parallel: same V, $Q = \Sigma Q_i = V\Sigma C_i \Longrightarrow C = \Sigma C_i$.

Q23. 2M Distinguish between polar and non-polar molecules with examples.

Ans: Polar: permanent dipole moment (H_2O, HCl). Non-polar: no permanent dipole; moment only induced by a field (O_2, CO_2).

Q24. 2M What is electrostatic shielding? Give one application.

Ans: The field inside a cavity of a conductor is zero whatever the outside field. Application: staying in a car during lightning; sensitive instruments in metal cases.

Q25. 3M Draw equipotential surfaces for (a) a point charge, (b) a uniform field, (c) a dipole.

Ans: (a) Concentric spheres. (b) Equally spaced planes perpendicular to E. (c) Closed surfaces around each charge, with the equatorial plane as V = 0.

Q26. 3M Board fav A slab of thickness t and dielectric constant K is inserted between plates (separation d). Find C.

Ans: Field in air E_0, in slab E_0/K. $V = E_0(d - t) + E_0t/K$. $C = \varepsilon_0A/(d - t + t/K)$.

Q27. 2M A capacitor is charged and disconnected. A dielectric is inserted. What happens to Q, V, C, E and U?

Ans: Q same; V, E, U decrease K times; C increases K times.

Q28. 2M Two charges 2 μC and −2 μC are 6 cm apart. Find the potential energy of the system.

Ans: $U = kq_1q_2/r = 9 \times 10^9 \times (-4 \times 10^{-12})/0.06 = \mathbf{-0.6\ J}$.

D. Long answer (5 marks)

Q29. 5M Board fav (a) Derive C for a parallel plate capacitor partly filled with a dielectric slab. (b) Find the energy stored. (c) What happens to the energy when the slab is inserted with the battery connected?

Ans: (a) $C = \varepsilon_0A/(d - t + t/K)$. (b) $U = \frac{1}{2}CV^2$. (c) V fixed, C increases, so U increases (the battery supplies extra energy).

Q30. 5M (a) Show that energy is lost when two charged capacitors are connected. (b) Derive the energy density of an electric field.

Ans: (a) $\Delta U = C_1C_2(V_1 - V_2)^2/2(C_1 + C_2) > 0$, lost as heat. (b) $U = \frac{1}{2}\varepsilon_0E^2Ad \Longrightarrow u = \frac{1}{2}\varepsilon_0E^2$.

E. Case-based

Q31. 4M Board fav A camera flash uses a 100 μF capacitor charged to 300 V and discharged in 1 ms. (a) Energy stored? (b) Average power during the flash? (c) Why is a capacitor used instead of the battery directly? (d) How would energy change if V were doubled?

Ans: (a) $U = \frac{1}{2} \times 10^{-4} \times 9 \times 10^4 = 4.5$ J. (b) $P = 4.5/10^{-3} = 4500$ W. (c) A capacitor releases energy very quickly, a battery cannot. (d) $U \propto V^2$: four times.

Q32. 4M Inside a conductor E = 0 and charge lies on the surface. (a) What is V inside a charged conductor? (b) Where is surface charge density highest? (c) Why is a metal car safe in lightning? (d) What is E just outside a conductor?

Ans: (a) Constant, equal to the surface value. (b) At sharp points (small radius of curvature). (c) Electrostatic shielding. (d) σ/ε_0, normal to the surface.

Q33. 2M Find C_{eq} between A and B for 2 μF and 4 μF in parallel, connected in series with 3 μF.

Ans: Parallel: 6 μF; with 3 μF in series: $6 \times 3/9 = \mathbf{2\ \mu F}$.

Q34. 3M Board fav A 4 μF capacitor is charged by a 200 V supply, disconnected, and connected to an uncharged 2 μF capacitor. Find the heat lost.

Ans: Initial $U = \frac{1}{2} \times 4 \times 10^{-6} \times 4 \times 10^4 = 0.08$ J. Common $V = 800/6 = 133.3$ V; final $U = \frac{1}{2} \times 6 \times 10^{-6} \times 133.3^2 = 0.0533$ J. Lost = **0.0267 J.**

Q35. 2M Why is the electric potential inside a charged conducting shell constant?
Ans: E = 0 inside, and E = −dV/dr, so V does not change: it equals the surface value.

Q36. 2M What is the potential at the centre of a square of side a with charges +q at all four corners?
Ans: V = 4kq/(a/√2) = 4√2 kq/a.

F. Extra practice (board PYQ style)

Q37. 3M Board fav A charge of 8 mC is at the origin. Find the work done in taking a small charge of -2×10^{-9} C from P(0, 0, 3 cm) to Q(0, 4 cm, 0) via R(0, 6 cm, 9 cm).
Ans: Path does not matter. $W = q(V_Q - V_P) = -2 \times 10^{-9} \times 9 \times 10^{9} \times 8 \times 10^{-3}(1/0.04 - 1/0.03) =$ **1.2 J.**

Q38. 2M Why is a dielectric with high K preferred in capacitors?
Ans: $C = K\varepsilon_0 A/d$ grows with K, so more charge is stored for the same size and voltage; it also prevents plates touching and raises breakdown strength.

Q39. 2M The plates of a charged isolated capacitor are pulled apart. How do C, V and U change?
Ans: Q fixed; $C = \varepsilon_0 A/d$ decreases; $V = Q/C$ increases; $U = Q^2/2C$ increases (work is done in pulling).

Q40. 2M Three capacitors 2, 3 and 6 μF are in series with 10 V. Find the charge on each.
Ans: $1/C = 1/2 + 1/3 + 1/6 = 1 \Longrightarrow C = 1$ μF; Q = 10 μC on each.

Q41. 1M Why does the potential of a conductor rise when charge is given to it?
Ans: V = Q/C; for fixed C, V is proportional to Q.

Q42. 1M What is dielectric strength? Give its value for air.
Ans: The maximum field a dielectric can bear without breakdown; for air about 3×10^6 V/m.

Q43. 1M A charge of 8 mC moves between two points with a potential difference of 20 V. Find the work done.
Ans: $W = qV = 8 \times 10^{-3} \times 20 =$ **0.16 J.**

Q44. 2M Why is the electrostatic potential inside a hollow charged conductor the same everywhere?
Ans: E = 0 inside, so no work is done moving a charge inside: V equals the surface value.

Q45. 2M Board fav What is the potential energy of an electron–proton pair 0.53 Å apart (in eV)?
Ans: $U = -ke^2/r = -4.35 \times 10^{-18}$ J = **−27.2 eV.**

Q46. 2M Why does a lightning conductor have a pointed end?
Ans: σ and E are very high at sharp points, so charge leaks safely to earth (corona discharge).

Quick compare: field vs potential

Electric field E	Potential V
Vector	Scalar
N/C or V/m	volt (J/C)
Point charge: kq/r^2	kq/r
Dipole: $\propto 1/r^3$	$\propto 1/r^2$
Zero inside a conductor	Constant inside a conductor
$E = -dV/dr$	$V = -\int E \cdot dr$

Q47. 1M Can the potential at a point be zero while the field is non-zero? Give an example.
Ans: Yes: the midpoint of a dipole (or any equatorial point).

Q48. 1M Can the field be zero where the potential is non-zero? Example.
Ans: Yes: inside a charged spherical shell (E = 0, V = kq/R).

Q49. 2M Board fav A 4 μF capacitor is charged to 50 V and a 2 μF capacitor to 100 V. They are connected with like plates together. Find the common potential.
Ans: $V = (C_1V_1 + C_2V_2)/(C_1 + C_2) = (200 + 200)/6 =$ **66.7 V.**

Q50. 3M Show graphically how V and E vary with distance from the centre of a charged conducting sphere.
Ans: E: 0 inside, jumps to kq/R^2 at the surface, then falls as $1/r^2$. V: constant kq/R inside, then falls as 1/r.

Q51. 2M Board fav Two capacitors in series have C_{eq} = 2 μF; in parallel, 9 μF. Find them.
Ans: $C_1 + C_2 = 9$, $C_1C_2 = 18 \Longrightarrow$ **6 μF and 3 μF.**

Yaad karo: one-line answers

→ 1 eV = 1.6×10^{-19} J.
→ Equipotential pe kaam zero.
→ Conductor: E = 0, V constant andar.
→ Dielectric: C × K.
→ Energy: $\frac{1}{2}CV^2$; density $\frac{1}{2}\varepsilon_0 E^2$.

Capacitor ka energy formula yaad karne ka smart tareeka
Asli baat: U = ½QV, then substitute Q = CV or V = Q/C. One formula, three forms.

Common mistakes

→ Potential scalar hai: vector ki tarah angle mat lagao.
→ Equatorial line pe V = 0, E nahi.
→ Slab formula mein d − t + t/K; metal slab ke liye t/K = 0.
→ Series mein Q same, parallel mein V same.

Paheli-Bujho

Paheli: Bujho, capacitor ki energy kahan store hoti hai?
Bujho: Plates mein?
Paheli: Nahi, plates ke beech ke electric field mein: $u = \frac{1}{2}\varepsilon_0 E^2$.
Bujho: Matlab khaali jagah bhi kaam ki hai. Jaise mera dimaag!

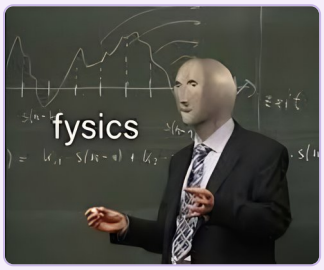
Exam Alert: last check

→ Dipole potential + energy derivations, C with slab: 5 marks pakke.
→ Battery connected vs disconnected table yaad rakho.
→ Charge sharing mein energy hamesha loss.

CURRENT ELECTRICITY

Most asked: drift velocity and its relation with current, resistivity and its temperature dependence, internal resistance, combination of cells, Kirchhoff's rules, Wheatstone bridge

Charges at rest gave us electrostatics. Now we let them **flow**. This chapter explains what current really is inside a wire, why wires resist it, and how to solve any circuit with **Kirchhoff's two rules**.



Current Electricity shuru karne se
pehle sabka mood

Asli baat: Good news: this chapter is formula-friendly. Ohm, Kirchhoff and Wheatstone cover most board questions.

1. Electric Current

Electric current is the **rate of flow of charge** through a cross-section of a conductor.

$I = \Delta q/\Delta t$ (steady), $I = dq/dt$ (instantaneous) SI unit: ampere (A) = $C\,s^{-1}$

→ Direction of conventional current: direction of flow of **positive** charge, opposite to electron flow.

→ Current is a **scalar**: it has direction but does not obey vector addition (Kirchhoff's junction rule adds currents algebraically).

→ 1 A: one coulomb of charge crosses a section per second.

Current in different media

→ **Metals**: free electrons carry current.

→ **Electrolytes**: positive and negative ions move in opposite directions; both add to the current.

→ **Gases**: ions and electrons (discharge tubes, lightning).

→ **Semiconductors**: electrons and holes.

Solved example

Q. A current of 4.8 A flows in a wire. How many electrons cross a section per second?

Sol. $n = I/e = 4.8/(1.6 \times 10^{-19}) = 3 \times 10^{19}$ **electrons/s.**

Current is a scalar even though it has a direction: currents at a junction add algebraically, not by the parallelogram law. The conventional direction is the direction of flow of positive charge, opposite to the flow of electrons in a metal. In electrolytes and gases both positive and negative ions carry current.

Solved example

Q. A current of 3.2 A flows through a wire. How many electrons pass a cross-section in 1 s?

Sol. $n = It/e = (3.2 \times 1)/(1.6 \times 10^{-19}) =$ **2×10^{19} electrons.**

Yaad karo

→ 1 A = 1 C/s: ek second mein 6.25×10^{18} electron.

→ Current scalar hai, direction hone ke bawajood.

→ Conventional current electron flow ke ulta.

2. Electric Currents in Conductors

In metals, free electrons move randomly with thermal speeds of about 10^5 m/s; the average velocity is zero, so no net current flows. When a field E is applied, electrons gain a small average velocity opposite to E.

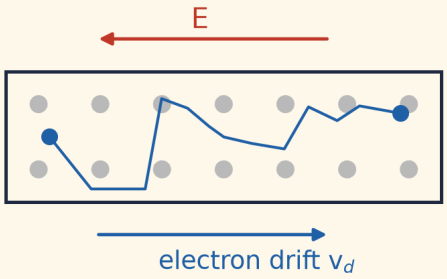


Fig. 3.1 Random motion of an electron with a slow drift opposite to E

Why does a steady field give a steady current?

Between collisions an electron accelerates; at each collision it loses its extra velocity and starts again. So instead of speeding up forever, electrons settle into an average drift, giving a steady current for a

steady field. The wire's resistance comes from these collisions.

In the absence of an electric field, free electrons in a metal move randomly with high thermal speeds (about 10^5 m/s) and collide with the fixed ions. Because the motion is random in all directions, the average velocity is zero and there is no net current. When a battery sets up a field, each electron gets a small additional velocity opposite to E between collisions; this slow, steady drift superposed on the random motion is what constitutes the current.

Random (thermal) motion	Drift motion
Exists even without a field	Only when a field is applied
Speed about 10^5 m/s	Speed about 10^{-4} m/s
Average velocity zero	Average velocity v_d opposite to E
No net current	Gives the current $I = neAv_d$

3. Drift Velocity and Relaxation Time

Relaxation time (τ): average time between two successive collisions of an electron (about 10^{-14} s).

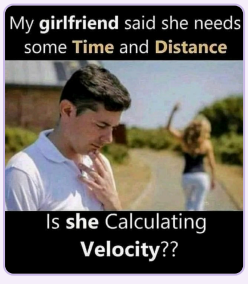
Drift velocity (v_d): average velocity with which free electrons drift opposite to E.

$$a = -eE/m \implies v_d = -eE\tau/m$$ order: 10^{-4} m/s

Relation between current and drift velocity

In time Δt, electrons within a length $v_d\Delta t$ cross the area A. Number = $nAv_d\Delta t$, charge = $neAv_d\Delta t$.

$$I = neAv_d$$ n = number density of free electrons



Drift velocity sun ke sab yahi soch rahe the

Asli baat: v_d is only about 10^{-4} m/s, yet a bulb lights instantly: the field travels through the wire at nearly the speed of light.

Mobility

$$\mu = |v_d|/E = e\tau/m$$ SI unit: $m^2\ V^{-1}\ s^{-1}$

Exam Alert

→ Derivation of $I = neAv_d$ and $\rho = m/ne^2\tau$: 3 marks, har saal aata hai.

→ Bulb turant kyun jalta hai jab v_d itni slow hai? Ye reasoning question pakka yaad rakho.

Derivation of drift velocity: step by step

1. Force on an electron: $F = -eE$; acceleration $a = -eE/m$.
2. Velocity of the i-th electron after time t_i since its last collision: $v_i = u_i - eEt_i/m$.
3. Average over all electrons: the u_i average to zero (random), and the t_i average to τ.
4. So $v_d = -eE\tau/m$.

Solved example (NCERT)

Q. Estimate v_d in a copper wire of area $1.0 \times 10^{-7}\ m^2$ carrying 1.5 A ($n = 8.5 \times 10^{28}\ m^{-3}$). Compare with thermal speed (~10^5 m/s).

Sol. $v_d = I/neA = \mathbf{1.1 \times 10^{-3}\ m/s}$, about 10^{-8} times the thermal speed.

If drift is so slow, why does a bulb glow at once?

When the switch is closed, the **electric field** is set up throughout the circuit almost at the speed of light. Free electrons everywhere in the wire, including those inside the bulb filament, start drifting at the same moment. The bulb does not wait for electrons from the switch to reach it.

Solved example

Q. A steady current flows in a metallic conductor of non-uniform cross-section. Which of these quantities is constant along the conductor: current, current density, electric field, drift speed?

Sol. Only the **current** is constant (charge is conserved). Current density $j = I/A$, electric field $E = \rho j$ and drift speed $v_d = I/neA$ all vary inversely with the area of cross-section.

Quantity	Symbol	Typical value in copper
Free electron density	n	$8.5 \times 10^{28}\ m^{-3}$
Thermal speed	u	about 10^5 m/s
Drift speed	v_d	about 10^{-4} m/s
Relaxation time	τ	about 10^{-14} s
Mobility	$\mu = e\tau/m$	about $4 \times 10^{-3}\ m^2/V\ s$

4. Ohm's Law

At constant temperature and physical conditions, the current through a conductor is proportional to the potential difference across it.

$$V = IR$$ SI unit of R: ohm (Ω)

Derivation from drift velocity

$I = neAv_d = neA(eE\tau/m)$ and $E = V/l$. So $I = (ne^2A\tau/ml)V$, giving:

$$R = ml/(ne^2\tau A) = \rho\, l/A, \quad \boldsymbol{\rho = m/(ne^2\tau)}$$

Vector form

$$\mathbf{j = \sigma E}, \quad j = I/A \text{ (current density)}, \quad \sigma = 1/\rho$$

Quantity	Symbol	SI unit
Resistance	R	Ω
Resistivity	ρ	Ω m
Conductivity	σ	S m^{-1}
Current density	j	A m^{-2}

Resistance ka asli matlab, bina formula ke

Asli baat: R = ρl/A: longer wire, more resistance; thicker wire, less. ρ depends only on the material and temperature.

NCERT Activity: verifying Ohm's law

Connect a resistance wire R in series with a battery, an ammeter and a rheostat, with a voltmeter across R. Change the rheostat setting, note V and I for 5–6 readings and plot V against I. A straight line through the origin shows V ∝ I; its slope gives R.

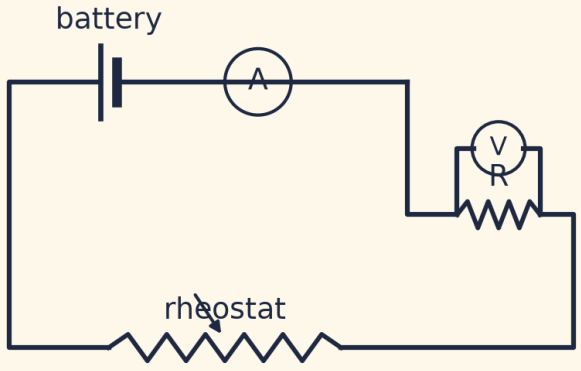


Fig. Circuit for verifying Ohm's law

Resistance depends on

Factor	Effect
Length l	R ∝ l
Area A	R ∝ 1/A
Material	through ρ
Temperature	R rises for metals

Solved example

Q. A negligibly small current is passed through a wire of length 15 m and uniform cross-section 6.0×10^{-7} m^2, and its resistance is measured to be 5.0 Ω. What is the resistivity of the material at the temperature of the experiment?

Sol. $\rho = RA/l = (5.0 \times 6.0 \times 10^{-7})/15 = \mathbf{2.0 \times 10^{-7}}$ **Ω m.**

Solved example

Q. A wire of resistance R is stretched to n times its original length. Find its new resistance.

Sol. Volume is constant: $A' = A/n$ and $l' = nl$. $R' = \rho(nl)/(A/n) = \mathbf{n^2R}$. For example, doubling the length makes the resistance four times.

Yaad karo

- → R = ρl/A: R wire pe depend karta hai, ρ sirf material aur temperature pe.
- → Stretch karne pe l badhta hai aur A ghatta hai, dono R badhate hain.
- → j = σE: Ohm's law ka microscopic form.

5. Limitations of Ohm's Law

Materials that do not obey V ∝ I are **non-ohmic**:

1. V stops being proportional to I (e.g. a bulb filament heats up).
2. The relation depends on the sign of V (diode: current flows easily one way only).
3. More than one value of V for the same I (GaAs).

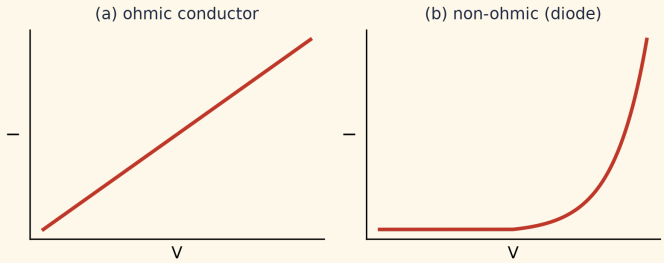


Fig. 3.2 V-I graphs: (a) ohmic, (b) non-ohmic

Solved example

Q. Is Ohm's law universally applicable for all conducting elements? If not, give examples of elements which do not obey Ohm's law.

Sol. No. It is obeyed by metallic conductors at constant temperature. Non-ohmic examples: **vacuum tubes, semiconductor diodes, thermistors, electrolytes** and GaAs, where the V-I graph is not a straight line or depends on the direction of V.

Three ways Ohm's law fails

→ V–I graph is not a straight line (filament lamp at high current, as temperature rises).
→ The relation depends on the sign of V (diode conducts in one direction only).
→ The relation is not unique: more than one V for the same I (GaAs).

6. Resistivity of Various Materials

Class	ρ (Ω m)	Examples
Conductors	10^{-8} to 10^{-6}	Ag, Cu, Al
Semiconductors	10^{-5} to 10^{6}	Si, Ge
Insulators	10^{11} to 10^{19}	glass, rubber

Colour code of carbon resistors

Colour	Digit	Colour	Digit
Black	0	Green	5
Brown	1	Blue	6
Red	2	Violet	7
Orange	3	Grey	8
Yellow	4	White	9

First two bands: digits; third: multiplier (power of 10); fourth: tolerance (gold ±5%, silver ±10%, none ±20%). Mnemonic: **B B ROY of Great Britain had a Very Good Wife.**

Solved example

Q. Bands: yellow, violet, orange, gold. Find R.
Sol. 4, 7, × 10^3: **47 kΩ ± 5%.**

Alloys such as **manganin, constantan and nichrome** are used for making standard resistors and heating elements because they have a high resistivity and a very low temperature coefficient of resistivity, so their resistance hardly changes with temperature. Copper and aluminium, with low resistivity, are used for connecting wires.

Solved example

Q. Two wires of equal length, one of aluminium and the other of copper, have the same resistance. Which of the two wires is lighter? (ρ_{Al} = 2.63 × 10^{-8} Ω m, ρ_{Cu} = 1.72 × 10^{-8} Ω m; densities 2.7 × 10^3 and 8.9 × 10^3 kg m^{-3})
Sol. Same R and l ⟹ A ∝ ρ. Mass m = (density) × A × l ∝ density × ρ.
m_{Al}/m_{Cu} = (2.7 × 2.63)/(8.9 × 1.72) = **0.46.**
Aluminium is lighter, which is why it is used for long-distance overhead power cables.

7. Temperature Dependence of Resistivity

$\rho_T = \rho_0[1 + \alpha(T - T_0)]$ α = temperature coefficient of resistivity

→ **Metals**: α positive. As T rises, ions vibrate more, τ decreases, ρ increases.
→ **Nichrome, manganin, constantan**: very weak T dependence, high ρ: used in standard resistors and heaters.
→ **Semiconductors**: α negative. n increases rapidly with T, so ρ falls.

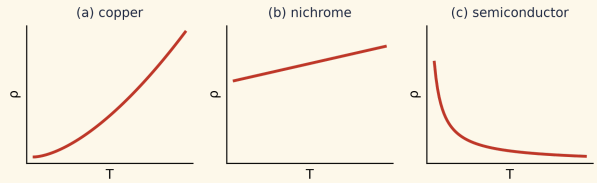


Fig. 3.3 Resistivity vs temperature

Paheli-Bujho

Bujho: Paheli, garam karne se metal ka resistance badhta hai, par semiconductor ka kam? Ulta kaise?
Paheli: Metal mein electrons pehle se bahut hain; garmi se sirf takkar badhti hai, toh τ kam. Semiconductor mein garmi naye free electrons bana deti hai, n badh jaata hai.
Bujho: Matlab metal mein traffic jam, semiconductor mein nayi gaadiyan!
Paheli: Exactly.

Solved example (NCERT)

Q. A silver wire has resistance 2.1 Ω at 27.5 °C and 2.7 Ω at 100 °C. Find α.
Sol. α = (2.7 − 2.1)/(2.1 × 72.5) = **0.0039 $°C^{-1}$.**

Solved example (NCERT)

Q. A nichrome heater draws 3.2 A initially at 27 °C and settles at 2.8 A on 230 V. Find the steady temperature ($\alpha = 1.7 \times 10^{-4}\ °C^{-1}$).

Sol. $R_{27} = 230/3.2 = 71.9\ \Omega$; $R_T = 230/2.8 = 82.1\ \Omega$. $T - 27 = (82.1 - 71.9)/(71.9 \times 1.7 \times 10^{-4}) \approx 835 \Longrightarrow$ **$T \approx 862$ °C.**

Material	Effect of rising temperature on ρ	Reason
Metals (Cu, Ag)	increases (α positive)	τ decreases: more collisions
Alloys (nichrome, manganin)	increases very slightly	very small α
Semiconductors (Si, Ge)	decreases (α negative)	n increases rapidly
Insulators	decreases	more charge carriers released

8. Electrical Energy and Power

When charge ΔQ moves through a potential drop V, energy $\Delta W = V\Delta Q = VI\Delta t$ is converted (to heat, in a resistor).

$$P = VI = I^2R = V^2/R$$ SI unit: watt (W)

Power transmission

Power lost in transmission cables of resistance R_c: $P_c = P^2R_c/V^2$. To deliver the same power P with less loss, power is sent at **very high voltage** (and low current), and stepped down near homes.

Solved example

Q. A copper wire of area $1.0 \times 10^{-7}\ m^2$ carries 1.5 A. If $n = 8.5 \times 10^{28}\ m^{-3}$, find v_d.

Sol. $v_d = I/neA = 1.5/(8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-7})$ **$= 1.1 \times 10^{-3}$ m/s.**

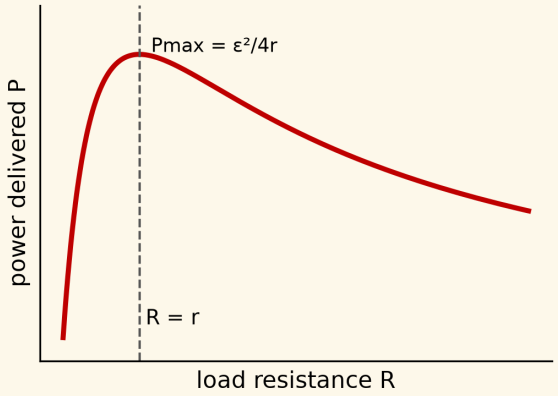
Joule's law of heating

$$H = I^2Rt$$ heat produced in time t

→ Applications: electric iron, heater, bulb filament, fuse.

→ A fuse wire has low melting point and is placed in series; excess current melts it and breaks the circuit.

→ For devices in parallel at the same V (household wiring), the one with lower R draws more power.



Power delivered to the load is maximum when R equals the internal resistance r

Maximum power transfer: a cell of emf ε and internal resistance r delivers power $P = \varepsilon^2R/(R + r)^2$ to a load R. Setting $dP/dR = 0$ gives **$R = r$**, and then $P_{max} = \varepsilon^2/4r$. At this point the efficiency is only 50%, since equal power is wasted inside the cell.

Quantity	Series (same I)	Parallel (same V)
Power formula to use	$P = I^2R$	$P = V^2/R$
Larger R	more power, glows brighter	less power, glows dimmer
Household wiring	not used	used: each appliance gets full V

Solved example

Q. Two bulbs rated 100 W, 220 V and 60 W, 220 V are connected in series across 220 V. Which glows brighter?

Sol. $R = V^2/P$: $R_{100} = 484\ \Omega$, $R_{60} = 807\ \Omega$. In series the current is the same and $P = I^2R$, so the **60 W bulb glows brighter** (higher resistance). In parallel the 100 W bulb would glow brighter.

9. Cells, EMF and Internal Resistance

EMF (ε): potential difference between the terminals of a cell when **no current** is drawn.

Internal resistance (r): resistance offered by the electrolyte inside the cell.

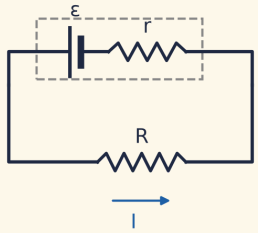


Fig. 3.4 Cell of emf ε and internal resistance r with external R

$$I = \varepsilon/(R + r),\quad \text{terminal voltage } V = \varepsilon - Ir$$

→ While discharging, $V < \varepsilon$. While charging, $V = \varepsilon + Ir$.

→ On open circuit ($I = 0$), $V = \varepsilon$. On short circuit ($R = 0$), $I_{max} = \varepsilon/r$.

→ $r = (\varepsilon/V - 1)R$: used to find internal resistance from ε and V.

Yaad karo

→ EMF force nahi hai, ye work per unit charge hai (volt).

→ r jitna kam, cell utna achha: car battery ka r bahut kam, isliye bada current deti hai.

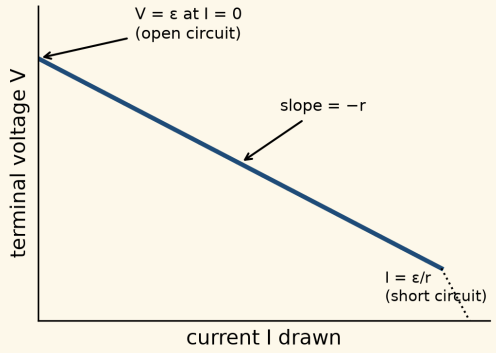
Derivation: terminal voltage

1. Current through the circuit: $I = \varepsilon/(R + r)$.
2. Potential drop inside the cell: Ir.
3. Potential difference across R (terminal voltage): $V = IR = \varepsilon - Ir$.

Solved example

Q. A cell of emf 2 V and internal resistance 0.5 Ω gives a current of 0.5 A. Find the terminal voltage and R.

Sol. $V = 2 - 0.5 \times 0.5 =$ **1.75 V**; $R = V/I =$ **3.5 Ω**.



Terminal voltage falls linearly with current: $V = \varepsilon - Ir$

The graph of terminal voltage against current drawn is a straight line with intercept ε on the V-axis and slope $-r$. A cell of low internal resistance keeps its terminal voltage nearly constant even when large currents are drawn, which is why a car battery must have a very small r.

Solved example

Q. The storage battery of a car has an emf of 12 V. If the internal resistance of the battery is 0.4 Ω, what is the maximum current that can be drawn from the battery?

Sol. Maximum current when the terminals are short-circuited ($R = 0$): $I_{max} = \varepsilon/r = 12/0.4 =$ **30 A**.

Solved example

Q. A storage battery of emf 8.0 V and internal resistance 0.5 Ω is being charged by a 120 V dc supply using a series resistor of 15.5 Ω. What is the terminal voltage of the battery during charging? What is the purpose of having a series resistor in the charging circuit?

Sol. While charging, the emfs oppose: $I = (120 - 8)/(15.5 + 0.5) = 7$ A.

$V = \varepsilon + Ir = 8 + 7 \times 0.5 =$ **11.5 V** (more than ε during charging).

The series resistor **limits the current**; without it the current would be 112/0.5 = 224 A, which would damage the battery.

Solved example

Q. A low-voltage supply from which one needs high currents must have very low internal resistance. Why? A high-tension (say 6 kV) supply must have a very large internal resistance. Why?

Sol. Maximum current = ε/r. For high current at low ε, r must be **very small**. In a high-tension supply, a large r limits the current to a safe value if the terminals are accidentally shorted.

Solved example

Q. A cell of emf 2 V and internal resistance 0.1 Ω is connected to a 3.9 Ω resistor. Find the current, the terminal voltage and the lost volts.

Sol. $I = \varepsilon/(R + r) = 2/4 =$ **0.5 A**. $V = IR = 0.5 \times 3.9 =$ **1.95 V**. Lost volts = $Ir =$ **0.05 V**.

10. Cells in Series and in Parallel

Series	Parallel
$\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$	$\varepsilon_{eq} = (\varepsilon_1 r_2 + \varepsilon_2 r_1)/(r_1 + r_2)$
$r_{eq} = r_1 + r_2$	$1/r_{eq} = 1/r_1 + 1/r_2$
Opposing cell: $\varepsilon_{eq} = \varepsilon_1 - \varepsilon_2$	n identical cells: $\varepsilon_{eq} = \varepsilon$, $r_{eq} = r/n$
n identical: $n\varepsilon$, nr	Used for large current with low R

Rule of thumb: **series** gives more voltage; **parallel** gives more current capacity with smaller internal resistance.

Combination of resistors

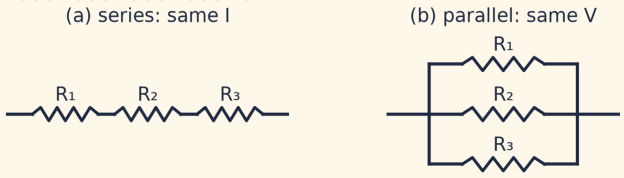


Fig. Resistors in (a) series, (b) parallel

Series	Parallel
$R = R_1 + R_2 + R_3$	$1/R = 1/R_1 + 1/R_2 + 1/R_3$
Same current	Same voltage
R_{eq} > largest	R_{eq} < smallest
V divides in ratio of R	I divides in inverse ratio of R

Derivation: parallel cells step by step

1. Same terminal voltage V across both: $V = \varepsilon_1 - I_1 r_1 = \varepsilon_2 - I_2 r_2$.
2. Total current $I = I_1 + I_2 = (\varepsilon_1 - V)/r_1 + (\varepsilon_2 - V)/r_2$.
3. Rearranging: $V = \varepsilon_{eq} - I r_{eq}$ with ε_{eq} and r_{eq} as in the table.

Solved example

Q. Six lead-acid secondary cells, each of emf 2.0 V and internal resistance 0.015 Ω, are joined in series to supply a resistance of 8.5 Ω. What are the current drawn from the supply and its terminal voltage?

Sol. $\varepsilon = 6 \times 2.0 = 12$ V; $r = 6 \times 0.015 = 0.09$ Ω.

$I = 12/(8.5 + 0.09) = \mathbf{1.4\ A}$; $V = IR = 1.397 \times 8.5 = \mathbf{11.9\ V}$.

Solved example

Q. (a) Three resistors 1 Ω, 2 Ω and 3 Ω are combined in series. What is the total resistance? (b) If the combination is connected to a battery of emf 12 V and negligible internal resistance, obtain the potential drop across each resistor.

Sol. (a) $R = 1 + 2 + 3 = \mathbf{6\ \Omega}$.

(b) $I = 12/6 = 2$ A. $V_1 = 2$ V, $V_2 = 4$ V, $V_3 = 6$ V: **2 V, 4 V, 6 V**.

Solved example

Q. (a) Three resistors 2 Ω, 4 Ω and 5 Ω are combined in parallel. What is the total resistance? (b) If the combination is connected to a battery of emf 20 V and negligible internal resistance, determine the current through each resistor and the total current.

Sol. (a) $1/R = 1/2 + 1/4 + 1/5 = 19/20 \Longrightarrow R = \mathbf{20/19\ \Omega \approx 1.05\ \Omega}$.

(b) $I_1 = 20/2 = \mathbf{10\ A}$, $I_2 = 20/4 = \mathbf{5\ A}$, $I_3 = 20/5 = \mathbf{4\ A}$; total = **19 A**.

Solved example

Q. Given n resistors each of resistance R, how will you combine them to get the maximum and minimum effective resistance? What is the ratio of the maximum to minimum resistance?

Sol. Maximum: all in series, nR. Minimum: all in parallel, R/n.

Ratio = $nR/(R/n) = \mathbf{n^2}$.

Combination	Equivalent emf	Equivalent internal resistance	Best when
n identical cells in series	nε	nr	$R \gg r$ (large external load)
m identical cells in parallel	ε	r/m	$R \ll r$ (small external load)
Mixed: n in series, m such rows	nε	nr/m	maximum current when nr/m = R

Solved example

Q. Two cells of emf 1.5 V and 2.0 V with internal resistances 1 Ω and 2 Ω are connected in parallel (like terminals together). Find the emf and internal resistance of the equivalent cell.

Sol. $\varepsilon_{eq} = (\varepsilon_1 r_2 + \varepsilon_2 r_1)/(r_1 + r_2) = (1.5 \times 2 + 2.0 \times 1)/3 = \mathbf{1.67\ V}$.

$r_{eq} = r_1 r_2/(r_1 + r_2) = 2/3 = \mathbf{0.67\ \Omega}$.

Solved example

Q. Four identical cells, each of emf 2 V and internal resistance 0.5 Ω, are connected in series to a 6 Ω resistor. Find the current and the terminal voltage across the combination. What happens if one cell is connected the wrong way round?

Sol. ε = 8 V, r = 2 Ω: I = 8/(6 + 2) = **1 A**, V = IR = **6 V**.

One cell reversed: ε = 2 + 2 + 2 − 2 = 4 V, r still 2 Ω: I = 4/8 = **0.5 A**. The current is halved.

11. Kirchhoff's Rules

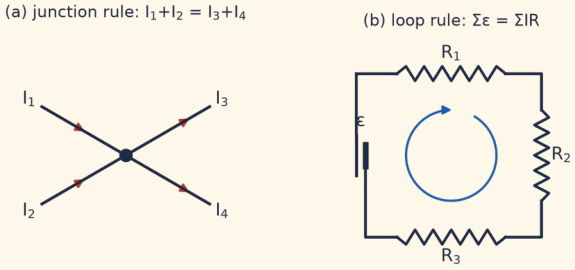


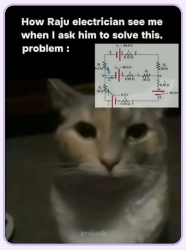
Fig. 3.5 (a) Junction rule, (b) Loop rule

1. **Junction rule (KCL)**: at any junction, the sum of currents entering = sum of currents leaving. Based on conservation of charge.
2. **Loop rule (KVL)**: the algebraic sum of changes in potential around any closed loop is zero. Based on conservation of energy.

Sign convention for the loop rule

→ Across a resistor in the direction of current: −IR; against the current: +IR.

→ Across a cell from − to + terminal: +ε; from + to −: −ε.



Kirchhoff ke rules aate hi circuit solve, Raju bhi electrician ban gaya

Asli baat: Steps: mark currents at each junction, apply the junction rule, then write one loop equation per independent loop.

Solved example (NCERT style)

Q. A battery of 10 V and negligible r is connected across a cube of 12 equal 1 Ω resistors between diagonally opposite corners. Find the equivalent resistance and current.

Sol. By symmetry, current I splits into 3, then 6, then 3 paths. Loop along one path: 10 = (I/3)(1) + (I/6)(1) + (I/3)(1) = 5I/6 ⟹ **I = 12 A**; R_{eq} = **5/6 Ω**.

Yaad karo

→ Junction pe ΣI = 0; loop mein ΣΔV = 0.

→ Symmetry dekho: equal resistance paths mein current barabar batata hai.

Kirchhoff's rules: method for numericals

(1) Mark a current with an assumed direction in every branch. (2) Use the junction rule to reduce the number of unknown currents. (3) Choose closed loops and apply the loop rule: going along the assumed current through a resistor, the change in potential is −IR; going from the negative to the positive terminal of a cell it is +ε. (4) Solve the equations. A negative answer simply means the actual current is opposite to the assumed direction.

Solved example

Q. Two cells of emf 6 V and 4 V with internal resistances 1 Ω and 2 Ω are connected in parallel (positive to positive) across an external resistance of 4 Ω. Find the current through the 4 Ω resistor.

Sol. Let the common terminal voltage be V.

Currents from the cells: (6 − V)/1 and (4 − V)/2.

Junction rule: (6 − V) + (4 − V)/2 = V/4.

Multiply by 4: 24 − 4V + 8 − 2V = V ⟹ V = 32/7 = 4.57 V.

Current through 4 Ω = V/4 = **8/7 A ≈ 1.14 A**.

Solved example

Q. In a circuit, a 10 V cell (internal resistance negligible) drives current through a 2 Ω resistor in series with a parallel pair of 3 Ω and 6 Ω. Using Kirchhoff's rules, find the current in each resistor.

Sol. 3 Ω and 6 Ω in parallel = 2 Ω; total R = 4 Ω, so I = 10/4 = **2.5 A** through the 2 Ω resistor.

Voltage across the pair = 2.5 × 2 = 5 V. I_3 = 5/3 = **1.67 A**, I_6 = 5/6 = **0.83 A**. Check (junction rule): 1.67 + 0.83 = 2.5 A.

12. Wheatstone Bridge

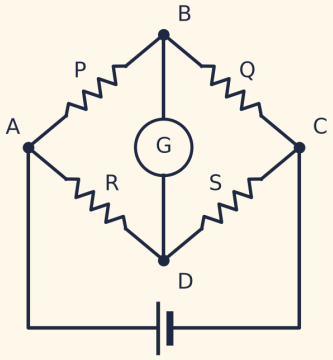


Fig. 3.6 Wheatstone bridge

Four resistors P, Q, R, S form a bridge with a galvanometer G between B and D. The bridge is **balanced** when no current flows through G ($V_B = V_D$).

> Balance condition: **P/Q = R/S**

Derivation

With $I_g = 0$, the current in P equals that in Q (I_1), and in R equals that in S (I_2). Loop ABDA: $I_1P - I_2R = 0$. Loop BCDB: $I_1Q - I_2S = 0$. Dividing: **P/Q = R/S**.

- → Used to find an unknown resistance: S = RQ/P.
- → A balanced bridge is most sensitive when all four resistances are of the same order.
- → In a balanced bridge, G (or any element between B and D) can be removed; it carries no current.

Exam Alert

- → Wheatstone derivation via Kirchhoff: 3 marks.
- → Bridge balanced dikhe toh beech wala resistor hata do, phir series-parallel se solve.

Paheli-Bujho

Bujho: Paheli, galvanometer mein current zero aaya toh circuit kharab hai kya?
Paheli: Nahi Bujho, wahi toh jeet hai! Zero matlab B aur D same potential pe hain: bridge balanced.
Bujho: Matlab yahan zero lana hi topper wali baat hai.

NCERT Activity: metre bridge

The metre bridge is a practical Wheatstone bridge. R (known) and S (unknown) are placed in the two gaps; a jockey J slides on a uniform 1 m wire until the galvanometer shows zero. Then R/S = l/(100 – l), giving S = R(100 – l)/l.

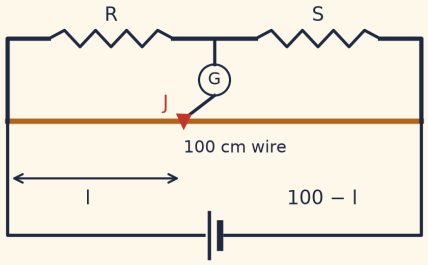


Fig. Metre bridge

Solved example

Q. In a metre bridge the balance point is at 39.5 cm with R = 12.5 Ω in the left gap. Find S.
Sol. S = 12.5 × 60.5/39.5 = **19.1 Ω.**

Solved example

Q. In a Wheatstone bridge P = 10 Ω, Q = 20 Ω, R = 15 Ω. What value of S balances the bridge? If the galvanometer and cell are interchanged, does the balance change?
Sol. P/Q = R/S ⟹ S = RQ/P = 15 × 20/10 = **30 Ω.** The balance condition is symmetric, so interchanging the galvanometer and the cell **does not disturb** the balance.

Solved example

Q. Four resistors P = 2 Ω, Q = 4 Ω, R = 6 Ω and S = 12 Ω form a Wheatstone bridge with a galvanometer between the junctions B and D. A 6 V cell of negligible internal resistance is connected across A and C. Find the current drawn from the cell.
Sol. P/Q = 2/4 = 0.5 and R/S = 6/12 = 0.5: the bridge is **balanced**, so no current flows through the galvanometer and it can be removed.
Arms in parallel: (P + Q) = 6 Ω and (R + S) = 18 Ω.
R_{eq} = (6 × 18)/24 = 4.5 Ω.
I = 6/4.5 = **1.33 A.**

13. Solved Numericals (Board Pattern)

Numericals mein units bhool gaye toh 20 mein se 2
Asli baat: Write the formula, put SI units, then the number. Half the marks in numericals are for steps.

N1. A wire of resistance 4 Ω is stretched to twice its length. Find the new resistance.
Sol. Volume constant: l → 2l, A → A/2. R = ρl/A ∝ l^2 ⟹ **16 Ω.**

N2. A battery of emf 10 V and r = 3 Ω is connected to a resistor. If I = 0.5 A, find R and the terminal voltage.
Sol. $R + r = 10/0.5 = 20 \Longrightarrow R = 17\ \Omega$. $V = IR = 8.5$ V.

N3. The resistance of a platinum wire is 2 Ω at 0 °C and 6.8 Ω at 80 °C. Find α.
Sol. $\alpha = (R_T - R_0)/(R_0 \Delta T) = 4.8/(2 \times 80) = 0.03\ °C^{-1}$.

N4. Two cells of 1.5 V (r = 0.5 Ω) and 2 V (r = 1 Ω) are in parallel. Find ε_{eq} and r_{eq}.
Sol. $\varepsilon_{eq} = (1.5 \times 1 + 2 \times 0.5)/1.5 = 1.67$ V; $r_{eq} = 0.5/1.5 = 0.33\ \Omega$.

N5. In a Wheatstone bridge P = 10 Ω, Q = 30 Ω, R = 15 Ω. Find S for balance.
Sol. $S = RQ/P = 15 \times 30/10 = 45\ \Omega$.

Formula sheet: Chapter 3

Quantity	Formula
Current	$I = dq/dt = neAv_d$
Drift velocity	$v_d = eE\tau/m$
Mobility	$\mu = v_d/E = e\tau/m$
Resistance	$R = \rho l/A$; $\rho = m/ne^2\tau$
Ohm (vector)	$j = \sigma E$
Temperature	$\rho_T = \rho_0[1 + \alpha \Delta T]$
Power	$P = VI = I^2R = V^2/R$
Cell	$V = \varepsilon - Ir$; $I = \varepsilon/(R + r)$
Wheatstone	$P/Q = R/S$

1 minute recap

- ✓ $I = neAv_d$; $v_d \approx 10^{-4}$ m/s.
- ✓ $\rho = m/ne^2\tau$; metals ρ rises with T, semiconductors falls.
- ✓ $V = \varepsilon - Ir$; series adds ε and r.
- ✓ KCL = charge conservation, KVL = energy conservation.
- ✓ Wheatstone balance: P/Q = R/S.

Most Important Questions

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. 1M The drift velocity of electrons in a conductor is of the order of:
(a) 10^8 m/s
(b) 10^5 m/s
(c) 10^{-4} m/s
(d) 10^{-12} m/s
Ans: (c) 10^{-4} m/s.

Q2. 1M Board fav Kirchhoff's junction rule is based on conservation of:
(a) energy
(b) charge
(c) momentum
(d) mass
Ans: (b) charge.

Q3. 1M The resistivity of a semiconductor with rise in temperature:
(a) increases
(b) decreases
(c) remains same
(d) first increases then decreases
Ans: (b) decreases.

Q4. 1M SI unit of mobility is:
(a) $m^2\ V^{-1}\ s^{-1}$
(b) $m\ V^{-1}\ s^{-1}$
(c) $m^2\ V\ s^{-1}$
(d) $V\ m^{-1}$
Ans: (a) $m^2\ V^{-1}\ s^{-1}$.

Q5. 1M A wire is stretched to three times its length. Its resistance becomes:
(a) 3 times
(b) 1/3
(c) 6 times
(d) 9 times
Ans: (d) 9 times.

Q6. 1M The terminal voltage of a cell is greater than its emf when:
(a) open circuit
(b) discharging
(c) it is being charged
(d) short circuited
Ans: (c) it is being charged.

Q7. 1M Board fav Why are alloys like manganin used in standard resistors?
Ans: High resistivity and very low temperature coefficient, so R hardly changes with temperature.

Q8. 1M Define relaxation time.
Ans: Average time between two successive collisions of a free electron with ions.

Q9. 1M How does drift velocity change if the length of the wire is doubled at the same V?
Ans: E = V/l halves, so $v_d = eE\tau/m$ halves.

Q10. 1M Is current a vector? Why?
Ans: No. It has direction but does not obey vector addition laws.

Q11. 1M What is the internal resistance of an ideal cell?
Ans: Zero.

Q12. 1M Why is power transmitted at high voltage?
Ans: Loss P^2R_c/V^2 is small at high V (low current).

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M Board fav **A:** A bulb glows almost instantly when switched on. **R:** Drift velocity of electrons is very high.
Ans: (c) A true; R false (the field is set up almost instantly; v_d is small).

Q14. 1M **A:** Resistivity of metals increases with temperature. **R:** Relaxation time decreases with temperature.
Ans: (a)

Q15. 1M **A:** Kirchhoff's loop rule follows from conservation of energy. **R:** The net change in potential around a closed loop is zero.
Ans: (a)

Q16. 1M **A:** Terminal voltage of a cell is always equal to its emf. **R:** Every cell has some internal resistance.
Ans: (d) A false; R true.

C. Short answer (2–3 marks)

Q17. 2M Board fav Derive the relation $I = neAv_d$.
Ans: Charge crossing A in Δt is $neAv_d\Delta t$; $I = \Delta q/\Delta t = neAv_d$.

Q18. 3M Board fav Derive $\rho = m/ne^2\tau$, and state the factors on which resistivity depends.
Ans: $I = neAv_d$ with $v_d = eV\tau/ml$ gives $R = ml/ne^2\tau A = \rho l/A$. ρ depends on n and τ, i.e. on material and temperature.

Q19. 2M Define mobility. Obtain its expression and SI unit.
Ans: $\mu = v_d/E = e\tau/m$; $m^2\ V^{-1}\ s^{-1}$.

Q20. 3M Draw ρ–T graphs for copper, nichrome and a semiconductor and explain.
Ans: Copper: ρ rises (τ falls). Nichrome: small linear rise. Semiconductor: ρ falls (n rises). See Fig. 3.3.

Q21. 2M Distinguish between emf and terminal voltage.
Ans: EMF: p.d. with no current; cause. Terminal voltage: p.d. when current flows, $V = \varepsilon - Ir$; less than ε while discharging.

Q22. 3M Board fav Derive ε_{eq} and r_{eq} for two cells in parallel.
Ans: $I = I_1 + I_2$, $V = \varepsilon_1 - I_1r_1 = \varepsilon_2 - I_2r_2 \Longrightarrow \varepsilon_{eq} = (\varepsilon_1r_2 + \varepsilon_2r_1)/(r_1 + r_2)$, $r_{eq} = r_1r_2/(r_1 + r_2)$.

Q23. 2M State Kirchhoff's rules with sign conventions.
Ans: Junction: $\Sigma I_{in} = \Sigma I_{out}$. Loop: $\Sigma\Delta V = 0$; $-IR$ along current, $+\varepsilon$ from – to +.

Q24. 3M Board fav Derive the balance condition of a Wheatstone bridge.
Ans: Using KVL in loops ABDA and BCDB with $I_g = 0$: $I_1P = I_2R$ and $I_1Q = I_2S \Longrightarrow P/Q = R/S$.

Q25. 2M Show that $j = \sigma E$.
Ans: $j = I/A = nev_d = ne(eE\tau/m) = (ne^2\tau/m)E = \sigma E$.

Q26. 2M A cell of emf 2 V and $r = 0.1\ \Omega$ is connected to a 3.9 Ω resistor. Find the terminal voltage.
Ans: $I = 2/4 = 0.5$ A; $V = 2 - 0.05 =$ **1.95 V**.

D. Long answer (5 marks)

Q27. 5M Board fav (a) Explain drift velocity and relaxation time. (b) Derive the expression for resistivity. (c) Explain why ρ of metals increases with T.
Ans: (a) definitions. (b) $\rho = m/ne^2\tau$. (c) More vibration, more collisions, τ falls, ρ rises.

Q28. 5M (a) State Kirchhoff's rules. (b) Use them to derive the Wheatstone bridge condition. (c) When is the bridge most sensitive?
Ans: (a), (b) as above. (c) When P, Q, R, S are of the same order.

E. Case-based

Q29. 4M Board fav A car battery has emf 12 V and $r = 0.05\ \Omega$. The starter motor needs 100 A. (a) Terminal voltage while starting? (b) Power lost inside the battery? (c) Why must r be small? (d) Short-circuit current?
Ans: (a) $12 - 5 = 7$ V. (b) $I^2r = 500$ W. (c) To deliver large current without large internal drop. (d) $\varepsilon/r = 240$ A.

Q30. 4M An electric heater uses nichrome. (a) Why nichrome? (b) How does ρ of nichrome vary with T? (c) A 1000 W, 220 V heater: find R. (d) Current drawn?
Ans: (a) High ρ, high melting point, does not oxidise. (b) Increases very slightly. (c) $R = V^2/P = 48.4\ \Omega$. (d) $I = 4.55$ A.

Q31. 1M Four resistors 2, 4, 6 and 12 Ω form a Wheatstone bridge (P, Q, R, S in order). Is it balanced?
Ans: $P/Q = 2/4 = 0.5$, $R/S = 6/12 = 0.5$: **balanced**.

Q32. 2M Find the equivalent resistance of n identical resistors R in series and in parallel. Ratio?
Ans: nR and R/n; ratio n^2.

Q33. 2M Why does the brightness of a bulb fall slightly when a heavy appliance is switched on?
Ans: Larger current through the line and the source resistance increases the drop, reducing V across the bulb.

Q34. 1M If the temperature of a conductor rises, what happens to the drift velocity at fixed V?
Ans: τ decreases, so $v_d = eE\tau/m$ decreases.

Q35. 3M Board fav Copper has $n = 8.5 \times 10^{28}\ m^{-3}$. How long does an electron take to drift 3 m in a wire of area $2 \times 10^{-6}\ m^2$ carrying 3 A?
Ans: $v_d = 3/(8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 2 \times 10^{-6}) = 1.1 \times 10^{-4}$ m/s; $t = 3/v_d \approx$ **2.7×10^4 s** (about 7.5 h).

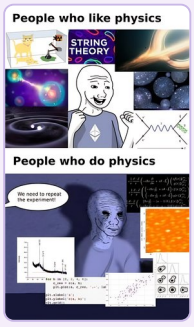
Q36. 1M What is the colour band coding significance of a fourth silver band?
Ans: Tolerance of ±10%.

Q37. 2M Two wires of the same material have lengths in ratio 1:2 and diameters in ratio 2:1. Ratio of resistances?
Ans: $R \propto l/d^2 \Longrightarrow (1/4):(2/1) =$ **1:8**.

Q38. 1M What is the maximum current that can be drawn from a cell?
Ans: When $R = 0$: $I_{max} = \varepsilon/r$.

Q39. 1M Can the terminal voltage of a cell ever be zero? When?
Ans: Yes, on short circuit: $V = \varepsilon - (\varepsilon/r)r = 0$.

Q40. 2M n identical cells (ε, r) are joined in series to an external R. Find the current. When is it nearly n times that of one cell?
Ans: $I = n\varepsilon/(R + nr)$. If $R \gg nr$, $I \approx n\varepsilon/R$: about n times the single-cell current.



Circuit diagrams dekhna vs khud solve karna
Asli baat: Practice tip: redraw every circuit neatly, mark junctions with letters, and only then apply the rules.

F. Extra practice (board PYQ style)

Q41. 2M Board fav A wire of resistance R is cut into two equal parts, which are joined in parallel. Find the new resistance.
Ans: Each part R/2; in parallel R/4.

Q42. 2M Explain why the resistance of a conductor increases with temperature in terms of electron collisions.
Ans: Ions vibrate with larger amplitude, electrons collide more often, τ falls, so $\rho = m/ne^2\tau$ rises.

Q43. 1M A potential difference V is applied across a conductor. How do v_d and I change if V is doubled?
Ans: Both double ($v_d \propto E \propto V$; $I \propto v_d$).

Q44. 2M Board fav Two bulbs of 100 W and 60 W (220 V) are in series across 220 V. Which glows brighter?
Ans: $R = V^2/P$: the 60 W bulb has more R. Same current in series, $P = I^2R$: **60 W glows brighter.**

Q45. 3M Find the current through each branch: a 6 V cell (negligible r) feeds 3 Ω in series with a parallel pair of 6 Ω and 3 Ω.
Ans: Parallel pair = 2 Ω; total 5 Ω; I = 1.2 A. Split: 6 Ω gets 0.4 A, 3 Ω gets 0.8 A.

Q46. 1M What happens to the balance point of a Wheatstone bridge if the cell and galvanometer are interchanged?
Ans: Nothing: the balance condition P/Q = R/S still holds.

Q47. 3M Board fav A cell gives 1.9 V across 3.8 Ω and 1.8 V across 1.8 Ω. Find ε and r.
Ans: $I_1 = 0.5$ A, $I_2 = 1$ A. $\varepsilon = 1.9 + 0.5r = 1.8 + r \Longrightarrow r = 0.2$ Ω, $\varepsilon =$ **2 V.**

Q48. 2M Which is more dangerous, 220 V AC through a person of 1 kΩ or 100 V? Why do birds on a wire not get shocked?
Ans: Higher V gives higher current. A bird touches one wire only: no potential difference across its body.

Common mistakes

→ KVL mein sign galat lagana: current ke saath chalo toh −IR.
→ Stretched wire mein sirf l nahi, A bhi badalta hai.
→ Parallel cells ka ε_{eq} simple average nahi hota.
→ Bulb ki brightness power se decide hoti hai, series mein zyada R wala zyada chamakta hai.

Quick revision: resistors vs cells vs capacitors

	Series	Parallel
Resistors	$R = R_1 + R_2$ (same I)	$1/R = 1/R_1 + 1/R_2$ (same V)
Cells	ε, r both add	r falls; current capacity rises
Capacitors	$1/C = 1/C_1 + 1/C_2$ (same Q)	$C = C_1 + C_2$ (same V)
Bulbs	higher R glows brighter	higher P (lower R) glows brighter

Yaad karo: one-line answers

→ Bulb turant: field turant set hota hai.
→ Manganin: low α, high ρ.
→ KCL: charge; KVL: energy.
→ Short circuit: V = 0, $I = \varepsilon/r$.
→ Wire stretched n times: R becomes n^2 R.

Exam Alert: last check

→ $I = neAv_d$, $\rho = m/ne^2\tau$, parallel cells, Wheatstone: in 4 derivations poora chapter cover.
→ Stretched wire: $R \propto l^2$ (volume constant).

Moving Charges and Magnetism

Most asked: motion of a charge in a magnetic field, Biot-Savart law and field on the axis of a loop, Ampere's circuital law (wire, solenoid), force between parallel wires, torque on a current loop, moving coil galvanometer and its conversion

In 1820 Oersted saw a compass needle turn near a current-carrying wire: **moving charges make magnetic fields**, and magnetic fields push on moving charges. This chapter builds both halves, ending with the galvanometer.

1. Magnetic Force: the Lorentz Force

A charge q moving with velocity v in electric field E and magnetic field B experiences the **Lorentz force**:

$$F = q[E + v \times B]$$

Magnetic part: $F = q(v \times B)$, $|F| = qvB \sin\theta$

- → F = 0 if the charge is at rest or moves along B (θ = 0° or 180°).
- → F is maximum (qvB) when v ⊥ B.
- → F is always ⊥ to v, so the magnetic force does **no work**: speed and kinetic energy stay constant.
- → Direction: right-hand rule for v × B (reverse it for a negative charge).

SI unit of B: tesla (T) = N s C^{-1} m^{-1}; 1 gauss = 10^{-4} T

Right hand ya left hand? Har baar ka confusion

Asli baat: Force on a moving charge: right-hand rule (or Fleming's left-hand rule for a current). Flip the direction for an electron.

Velocity selector

In crossed fields (E ⊥ B ⊥ v), the electric force qE and magnetic force qvB oppose each other. They balance when:

$v = E/B$ only particles with this speed pass undeflected

- → Used by J.J. Thomson to measure e/m of the electron.
- → Independent of charge and mass of the particle.

Field	Force on charge	Work done
Electric E	qE (even at rest)	can change KE
Magnetic B	qv × B (only when moving)	zero

Why does a magnetic force do no work?

Work done by a force is W = F · s. The magnetic force F = q(v × B) is always perpendicular to the velocity v, and the displacement ds = v dt is along v. Hence F · ds = 0 at every instant. A magnetic field can change the **direction** of motion of a charge but never its **speed** or kinetic energy. This is why a charged particle in a pure magnetic field moves with constant speed, and why magnetic fields alone cannot be used to speed up particles in an accelerator; the energy always comes from an electric field.

Direction of the magnetic force: two rules

(i) **Right-hand screw rule (cross product):** curl the fingers of the right hand from v towards B through the smaller angle; the thumb gives v × B. For a positive charge the force is along the thumb; for a negative charge (electron) it is opposite.

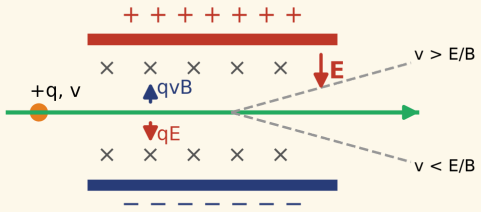
(ii) **Fleming's left-hand rule:** stretch the forefinger, middle finger and thumb of the left hand mutually perpendicular. Forefinger along B (field), middle finger along the current (direction of motion of positive charge), then the thumb gives the force (motion).

Special cases of F = qvB sin θ

Situation	Force
Charge at rest (v = 0)	Zero
v parallel or antiparallel to B (θ = 0° or 180°)	Zero: moves straight
v ⊥ B (θ = 90°)	Maximum, F = qvB
Neutral particle (q = 0)	Zero

The SI unit of B follows from F = qvB: 1 tesla is the field in which a charge of 1 C moving at 1 m s^{-1} perpendicular to the field experiences a force of 1 N,

so 1 T = 1 N s C^{-1} m^{-1} = 1 N A^{-1} m^{-1}. A smaller unit, the gauss, is also used: 1 G = 10^{-4} T. The earth's field is about 0.4 G, a laboratory magnet about 1 T, and a neutron star about 10^8 T.



B into page (×). Undeflected when qE = qvB, i.e. v = E/B

Fig. 4.7 Velocity selector: crossed E and B fields

How a velocity selector works: in Fig. 4.7 the electric force qE acts downward and the magnetic force qvB acts upward on a positive charge. Only particles with exactly v = E/B feel zero net force and pass straight through the slit. Faster particles are bent towards the magnetic force, slower ones towards the electric force. The selection does not depend on the charge or mass of the particle. J. J. Thomson used this arrangement to measure e/m of the electron, and it is used today in mass spectrometers.

Solved example

Q. In a velocity selector E = 3×10^4 V m^{-1} and B = 0.1 T. What speed is selected? Will an electron of the same speed also pass undeflected?

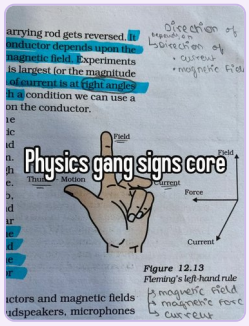
Sol. v = E/B = $3 \times 10^4 / 0.1$ = **3×10^5 m s^{-1}**. Yes: for an electron both forces reverse together, so they still cancel. The selector depends only on speed.

2. Force on a Current-Carrying Conductor

For a straight rod of length l carrying current I in a uniform field B:

$$\vec{F} = I(\vec{l} \times \vec{B}), \quad |F| = IlB \sin\theta$$

For an arbitrary shaped wire, $\vec{F} = \Sigma I\, d\vec{l} \times \vec{B}$. Direction by **Fleming's left-hand rule**: forefinger B, middle finger I, thumb F.



Fleming ka left-hand rule lagate waqt exam hall ka scene

Asli baat: First finger: field. Second finger: current. Thumb: force (motion). Keep all three mutually perpendicular.

Solved example (NCERT)

Q. A straight wire of mass 200 g and length 1.5 m carries 2 A. It is suspended in mid-air by a uniform horizontal field. Find B.

Sol. IlB = mg $\Longrightarrow$ B = 0.2 × 9.8/(2 × 1.5) = **0.65 T.**

Solved example

Q. A 3 cm wire carrying 10 A is placed inside a solenoid $\perp$ to its axis where B = 0.27 T. Find the force.

Sol. F = IlB = 10 × 0.03 × 0.27 = **8.1×10^{-2} N.**

Derivation from the Lorentz force

Consider a rod of length l and cross-section A carrying current I in a field B. Let n be the number density of free electrons and v_d their drift velocity. Total number of charge carriers = nAl. Force on each carrier = $q(\vec{v}_d \times \vec{B})$. Total force: $\vec{F} = (nAl)\, q\, (\vec{v}_d \times \vec{B})$. Since the current $I = nqAv_d$, we can write $nqAv_d\, \vec{l} = I\vec{l}$, giving

$$\vec{F} = I(\vec{l} \times \vec{B}), \quad |F| = IlB \sin\theta$$ l is along the current

For a wire of arbitrary shape, the force on each small element is $d\vec{F} = I\, d\vec{l} \times \vec{B}$ and the total force is found by adding these vectorially. An important result follows: in a **uniform** field the net force on any closed current loop is zero, because the vector sum of all the dl around a closed path is zero. The loop may still feel a torque (Section 8).

Applications of force on a conductor

- → Electric motor: the force on the coil sides produces rotation.
- → Loudspeaker: a coil in a radial field pushes the cone back and forth with the audio current.
- → Moving coil galvanometer and ammeters.
- → Rail gun and electromagnetic pumps for liquid metals in reactors.

Solved example

Q. A 0.5 m wire carrying 4 A makes an angle of 30° with a uniform field of 0.2 T. Find the force.

Sol. F = IlB sin θ = 4 × 0.5 × 0.2 × sin 30° = **0.2 N**, perpendicular to the plane containing the wire and B.

Yaad karo

- → Uniform field mein closed loop par net force zero, torque zero nahi bhi ho sakta.
- → Wire field ke parallel ho toh force zero.
- → Fleming left hand = motor rule.

3. Motion in a Magnetic Field

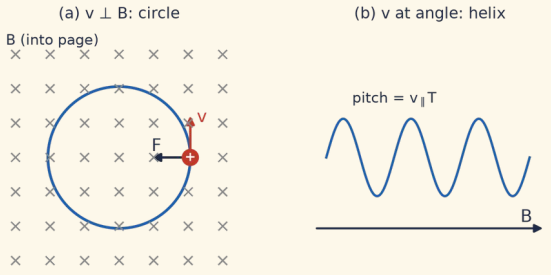


Fig. 4.1 Path of a charged particle: (a) circle when v ⊥ B, (b) helix otherwise

Case 1: v ⊥ B (circular path)

qvB provides the centripetal force mv^2/r:

$$r = mv/qB, \quad \omega = qB/m, \quad T = 2\pi m/qB, \quad \nu = qB/2\pi m$$

→ Period and frequency do **not** depend on speed or radius.

→ Faster particles move in bigger circles but take the same time.

Case 2: v at angle θ to B (helical path)

$v_{\parallel} = v\cos\theta$ is unaffected; $v_{\perp} = v\sin\theta$ gives circular motion. The combination is a **helix**.

$$r = mv\sin\theta/qB, \quad \text{pitch } p = v\cos\theta \cdot T = 2\pi mv\cos\theta/qB$$

Solved example

Q. An electron ($v = 3 \times 10^7$ m/s) moves ⊥ to B = 6×10^{-4} T. Find r and ν.

Sol. $r = mv/eB = 9.1 \times 10^{-31} \times 3 \times 10^7/(1.6 \times 10^{-19} \times 6 \times 10^{-4})$ = **0.28 m**. $\nu = v/2\pi r$ = **17 MHz**.

Exam Alert

→ Magnetic force kaam nahi karta: KE constant. MCQ aur A-R mein favourite.

→ r = mv/qB aur T = 2πm/qB: T speed pe depend nahi karta.

Frequency independent of speed: why it matters

Since T = 2πm/qB does not depend on v, all particles of the same type circle at the same frequency. This idea is used in accelerators and in mass spectrometers, where ions of different mass curve on circles of different radii.

Solved example (NCERT)

Q. An electron with speed 3×10^7 m/s moves in a plane ⊥ to B = 6×10^{-4} T. Find its radius and frequency. Its energy in keV?

Sol. r = 28 cm, ν = 17 MHz, $K = \frac{1}{2}mv^2 = 4.1 \times 10^{-16}$ J = **2.6 keV**.

Why the path is a circle: step by step

1. The magnetic force qvB is always perpendicular to v, so the speed v stays constant.
2. Since v and B are constant in magnitude and v ⊥ B, the force has constant magnitude qvB.
3. A force of constant magnitude that is always perpendicular to velocity is exactly a centripetal force.
4. Therefore $qvB = mv^2/r$, which gives r = mv/qB, and the motion is uniform circular motion.

The radius can also be written in terms of momentum p and kinetic energy K: r = p/qB = √(2mK)/qB. For a particle accelerated from rest through a potential difference V, K = qV, so r = √(2mV/q)/B. These forms are very useful in comparing different particles, for example protons, deuterons and α-particles.

Particle accelerated through same V	Radius ratio
Proton : deuteron	1 : √2
Proton : α-particle	1 : √2
Electron vs proton	Electron much smaller ($r \propto \sqrt{m}$)

Helical motion: when v makes an angle θ with B, resolve v into $v_{\parallel} = v\cos\theta$ (along B) and $v_{\perp} = v\sin\theta$. The parallel component feels no force and stays constant; the perpendicular component produces circular motion of radius $r = mv\sin\theta/qB$. The combination is a helix. The distance moved along B in one revolution is the **pitch** $p = v\cos\theta \times T = 2\pi mv\cos\theta/qB$. Charged particles from the sun spiral along the earth's field lines towards the poles in this way, producing the aurora.

Solved example

Q. A proton and an α-particle enter a uniform field perpendicularly with the same velocity. Find the ratio of the radii of their paths.

Sol. $r = mv/qB \propto m/q$. $r_{\alpha}/r_p = (4m/2e)/(m/e) = 2$. The α-particle moves in a circle of twice the radius.

Solved example

Q. An electron accelerated through 2.0 kV enters a field of 0.15 T at right angles. Find the radius of its path.

Sol. $v = \sqrt{(2eV/m)} = \sqrt{(2 \times 1.6 \times 10^{-19} \times 2000/9.1 \times 10^{-31})} = 2.65 \times 10^{7}$ m s^{-1}.

$r = mv/eB = 9.1 \times 10^{-31} \times 2.65 \times 10^{7}/(1.6 \times 10^{-19} \times 0.15) =$ **1.0 mm.**

4. Biot-Savart Law

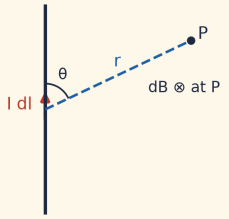


Fig. 4.2 Field dB due to a current element I dl

The magnetic field dB at P due to a current element I dl at displacement r:

$dB = (\mu_0/4\pi)\, I\, dl \times r / r^3$, $|dB| = (\mu_0/4\pi)\, I\, dl \sin\theta/r^2$ $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}

Biot-Savart (B)	Coulomb (E)
Source: current element (vector)	Source: charge (scalar)
$dB \propto 1/r^2$, depends on angle θ	$E \propto 1/r^2$, no angle
dB ⊥ both I dl and r	E along r
Zero along the line of the element	Never zero due to a single charge

Field due to a straight wire of finite length (from Biot-Savart)

$B = (\mu_0 I/4\pi r)(\sin\varphi_1 + \sin\varphi_2)$ r: perpendicular distance from the wire

For an infinitely long wire, $\varphi_1 = \varphi_2 = 90°$, giving $B = \mu_0 I/2\pi r$ (same as from Ampere's law).

Solved example (NCERT)

Q. An element $\Delta l = \Delta x\, \hat{i}$ carries I = 10 A at the origin. Find B at 0.5 m on the y-axis (Δx = 1 cm).

Sol. $|dB| = \mu_0 I \Delta l/4\pi r^2 = 10^{-7} \times 10 \times 0.01/0.25 =$ **4 × 10^{-8} T**, along +z.

Understanding each factor in the Biot-Savart law

The law $dB = (\mu_0/4\pi)\, I\, dl \sin\theta/r^2$ was obtained from experiment. Each factor has a physical meaning: (i) $dB \propto I$, a larger current gives a stronger field; (ii) $dB \propto dl$, a longer element contributes more; (iii) $dB \propto \sin\theta$, so a point lying along the line of the element ($\theta = 0°$) receives no field from it; (iv) $dB \propto 1/r^2$, the inverse-square fall-off, just like Coulomb's law. The direction of dB is along dl × r, perpendicular to the plane containing dl and r, given by the right-hand screw rule.

Coulomb's law (electric)	Biot-Savart law (magnetic)
Source: point charge q (scalar)	Source: current element I dl (vector)
Field along r (radial)	Field ⊥ to both dl and r
No angle dependence	Depends on sin θ
$1/r^2$ dependence	$1/r^2$ dependence

The constant $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1} is the permeability of free space. It is related to the permittivity ε_0 and the speed of light by $c = 1/\sqrt{(\mu_0\varepsilon_0)}$, a deep link between electricity, magnetism and light.

Solved example

Q. A wire is bent into a semicircle of radius 10 cm with long straight leads along a diameter line, and carries 5 A. Find B at the centre.

Sol. The straight leads point towards the centre (θ = 0°), so they give no field. The semicircle gives half the loop value: $B = \mu_0 I/4R = 4\pi \times 10^{-7} \times 5/(4 \times 0.1) =$ **1.57 × 10^{-5} T.**

5. Field on the Axis of a Circular Loop

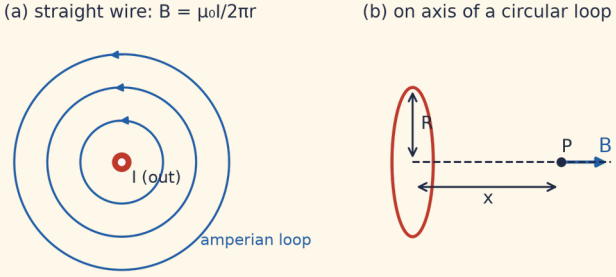


Fig. 4.3 (a) Field lines round a straight wire, (b) point P on the axis of a loop

Each element dl is ⊥ to its $r = \sqrt{(x^2 + R^2)}$. Components perpendicular to the axis cancel in pairs; components along the axis (dB cos θ, cos θ = R/r) add:

$$B = \mu_0 I R^2 / 2(x^2 + R^2)^{3/2} \text{ along the axis}$$

→ **At the centre** (x = 0): $B = \mu_0 I/2R$; for N turns, $\mu_0 NI/2R$.

→ **Far away** ($x \gg R$): $B = \mu_0 I R^2/2x^3$, like a dipole.

→ Direction: **right-hand thumb rule**: curl fingers along the current, thumb gives B.

Paheli-Bujho

Bujho: Paheli, loop ke centre pe field itni simple $\mu_0 I/2R$ kaise ho gayi?

Paheli: Centre pe har element ki distance R hai aur angle 90°. Sab dB ek hi direction mein, bas jod do: $2\pi R \times \mu_0 I/4\pi R^2$.

Bujho: Aur bahar jaate jaate cancel-cancel khelne lagte hain!

Paheli: Haan, sirf axis wala component bachta hai.

Derivation step by step (loop axis)

1. Element dl at the top of the loop; distance to P is $r = \sqrt{(x^2 + R^2)}$; dl ⊥ r, so $dB = \mu_0 I\, dl/4\pi r^2$.
2. dB has an axial component dB cos θ (cos θ = R/r) and a perpendicular one dB sin θ.
3. The perpendicular components of diametrically opposite elements cancel.
4. $B = \int dB \cos\theta = (\mu_0 I/4\pi r^2)(R/r)(2\pi R) = \mu_0 I R^2/2(x^2 + R^2)^{3/2}$.

Yaad karo

→ Loop ka face jahan se current anticlockwise dikhe, wo North pole.

→ Clockwise dikhe toh South pole (clock rule).

Special cases and the loop as a magnet

From $B = \mu_0 N I R^2/2(x^2 + R^2)^{3/2}$:

(i) **At the centre** (x = 0): $B = \mu_0 NI/2R$, the maximum value on the axis.

(ii) **Far away** ($x \gg R$): $B \approx \mu_0 N I R^2/2x^3 = (\mu_0/4\pi)\, 2m/x^3$, where $m = NI\pi R^2$ is the magnetic moment. This is exactly the form of the field on the axis of a bar magnet, which shows that a current loop behaves like a magnetic dipole.

(iii) The field decreases continuously as we move away from the centre along the axis.

Which face is north? Look at the face of the loop: if the current appears anticlockwise, that face acts as a north pole; if clockwise, it is a south pole (the clock rule). The field lines come out of the north face.

Field at the centre of arcs and loops

Shape	B at centre
Full loop, N turns	$\mu_0 NI/2R$
Semicircle	$\mu_0 I/4R$
Quarter circle	$\mu_0 I/8R$
Arc of angle θ (radian)	$\mu_0 I\theta/4\pi R$

Solved example

Q. A coil of 100 turns and radius 10 cm carries 1 A. Find B on its axis at 10 cm from the centre.

Sol. $x = R = 0.1$ m. $B = \mu_0 N I R^2/2(2R^2)^{3/2} = 4\pi \times 10^{-7} \times 100 \times 1 \times 0.01/(2 \times (0.02)^{3/2}) = \mathbf{2.2 \times 10^{-4}}$ T, about 35% of the value at the centre.

6. Ampere's Circuital Law

The line integral of B around any closed loop equals μ_0 times the net current threading the loop:

$$\oint B \cdot dl = \mu_0 I_{enclosed}$$

Sign: curl the fingers of the right hand along the loop; currents along the thumb are positive.

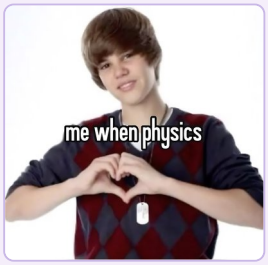
Application 1: long straight wire

By symmetry B is tangential and constant on a circle of radius r. $B(2\pi r) = \mu_0 I$:

$$B = \mu_0 I/2\pi r$$

→ Field lines are concentric circles; direction by right-hand thumb rule.

→ Inside a thick wire of radius a ($r < a$): $B = \mu_0 I r/2\pi a^2$ ($\propto r$).



Ampere's law samajh aane ke baad derivations dekh ke

Asli baat: Ampere's law works when symmetry makes B constant on the loop: straight wire, solenoid, toroid.

Application 2: long solenoid

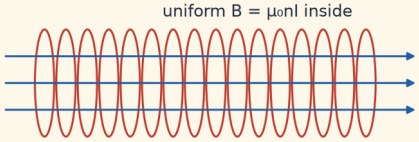


Fig. 4.4 Field of a long solenoid

Take a rectangular loop with one side (length h) inside. Only the inside side contributes;

enclosed current = nhI:

$Bh = \mu_0 nhI \implies B = \mu_0 nI$ n = turns per unit length

→ Inside: uniform and along the axis.
Outside: nearly zero.

→ At the ends: $B = \mu_0 nI/2$.

→ A solenoid behaves like a bar magnet.

Toroid

A toroid is a solenoid bent into a ring.
Ampere's law on a circle of radius r inside the windings: $B(2\pi r) = \mu_0 NI$:

$B = \mu_0 NI/2\pi r = \mu_0 nI$ zero inside the open space and outside

Solved example (NCERT)

Q. A long straight wire carries 35 A. Find B at 20 cm.

Sol. $B = \mu_0 I/2\pi r = 2 \times 10^{-7} \times 35/0.2 = \mathbf{3.5 \times 10^{-5}}$ T.

Ampere's law in words and its use

Ampere's circuital law states that the line integral of B around any closed path equals μ_0 times the net current threading the surface bounded by that path: $\oint B \cdot dl = \mu_0 I_{enclosed}$. The closed path is called an Amperian loop. The law is always true, but it gives B easily only when there is enough symmetry for B to be constant in magnitude along the loop and either parallel or perpendicular to it. It plays the same role in magnetism that Gauss's law plays in electrostatics.

Sign convention: curl the fingers of the right hand along the direction in which the loop is traversed; currents along the thumb count as positive and opposite ones as negative.

Field inside and outside a thick wire

For a long straight wire of radius a carrying current I uniformly over its cross-section:
Outside (r > a): the loop encloses all of I, so $B(2\pi r) = \mu_0 I$ and $B = \mu_0 I/2\pi r$.
Inside (r < a): the loop encloses only $I_e = I(r^2/a^2)$, so $B(2\pi r) = \mu_0 I r^2/a^2$, giving $B = \mu_0 I r/2\pi a^2$. Thus $B \propto r$ inside, is maximum at the surface, and $B \propto 1/r$ outside (Fig. 4.8).

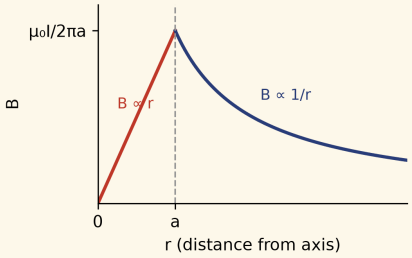


Fig. 4.8 B versus distance r for a thick straight wire

Solved example

Q. A solenoid 0.5 m long has 500 turns and carries 5 A. Find B inside it and at its ends.

Sol. $n = 500/0.5 = 1000\ m^{-1}$. $B = \mu_0 nI = 4\pi \times 10^{-7} \times 1000 \times 5 = \mathbf{6.28 \times 10^{-3}}$ T. At the ends $B = \mu_0 nI/2 = \mathbf{3.14 \times 10^{-3}}$ T.

Yaad karo

→ Solenoid ke andar B uniform, length ya radius par depend nahi karta, sirf n aur I par.

→ Toroid ke bahar aur beech ke khaali hisse mein B = 0.

→ Wire ke andar $B \propto r$, bahar $B \propto 1/r$.

7. Force between Two Parallel Currents

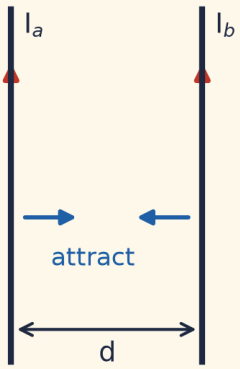


Fig. 4.5 Parallel currents attract

Field of wire a at wire b: $B_a = \mu_0 I_a/2\pi d$. Force on length L of b: $F = I_b L B_a$:

$$F/L = \mu_0 I_a I_b/2\pi d$$

→ Same direction currents **attract**; opposite currents **repel** (opposite of charges).

→ **Definition of ampere**: the current which, in two infinitely long parallel wires 1 m apart in vacuum, produces a force of 2×10^{-7} N per metre.

Yaad karo

→ Charges: like repel. Currents: like (same direction) attract. Ulta hai, yaad rakho!

→ Ampere ki definition 1 mark mein seedha aati hai.

Derivation step by step

1. Field of wire a at the location of b: $B_a = \mu_0 I_a/2\pi d$, directed $\perp$ to b.
2. Force on length L of b: $F_{ba} = I_b L B_a = \mu_0 I_a I_b L/2\pi d$.
3. By Fleming's rule, it points towards a when currents are parallel: attraction.

NCERT Activity: force between parallel currents

Hang two long thin aluminium strips side by side and pass current through both: in the same direction they swing towards each other; in opposite directions they move apart. This is how the ampere is defined.

Why parallel currents attract

Wire 1 produces a field $B_1 = \mu_0 I_1/2\pi d$ at the position of wire 2. Wire 2 carries current I_2 in this field, so it feels a force per unit length $F/l = I_2 B_1 = \mu_0 I_1 I_2/2\pi d$. Using Fleming's left-hand rule, the force points towards wire 1 when the currents are in the same direction and away from it when they are opposite. By Newton's third law, wire 1 feels an equal and opposite force. Note the contrast with charges: like charges repel, but like (parallel) currents attract.

Definition of the ampere

If $I_1 = I_2 = 1$ A and d = 1 m, then $F/l = 2 \times 10^{-7}$ N m^{-1}. This gave the classical definition: one ampere is the steady current which, flowing in two infinitely long, straight, parallel conductors of negligible cross-section placed 1 m apart in vacuum, produces a force of 2×10^{-7} N per metre of length between them. (Since 2019 the SI ampere is defined by fixing the value of the elementary charge e, but this result remains exact for practical purposes.)

Solved example

Q. Two long parallel wires 5 cm apart carry 10 A and 20 A in opposite directions. Find the force per metre and its nature.

Sol. $F/l = 2 \times 10^{-7} \times 10 \times 20/0.05 = \mathbf{8 \times 10^{-4}}$ **N m^{-1}**, repulsive (opposite currents).

Real-life note: in a high-current busbar or in the cables connecting a car battery to its starter motor, these forces can be large enough to make the conductors jump, so they are clamped firmly.

8. Torque on a Current Loop

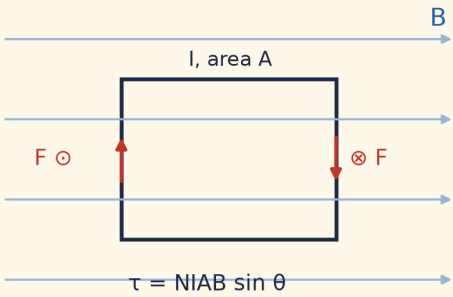


Fig. 4.6 Rectangular loop in a uniform field

In a uniform B, the net force on a loop is zero but the forces on opposite sides form a couple:

$\tau = NIAB \sin\theta = \mathbf{m} \times \mathbf{B}$ θ: angle between the normal (m) and B

Magnetic dipole moment of a loop

$\mathbf{m} = NIA\,\hat{n}$ direction by right-hand rule; SI unit A m^2

- → τ is maximum when the plane of the loop is parallel to B (θ = 90°).
- → τ = 0 when the plane is ⊥ to B (θ = 0°): stable equilibrium.
- → A current loop behaves like a magnetic dipole: compare $\tau = p \times E$.

Circulating electron as a dipole

$\mu_l = evr/2 = (e/2m_e)\,l$; $\mu_B = eh/4\pi m_e = 9.27 \times 10^{-24}$ A m^2 Bohr magneton (n = 1)

Derivation step by step (torque)

1. Rectangular loop: sides a (⊥ to B) and b; normal at angle θ to B.
2. Forces on the two sides of length b are equal and opposite, along the same line: they cancel and give no torque.
3. Forces on sides of length a: F = IbB each, opposite, separated by a sin θ: a couple.
4. $\tau = IbB \times a \sin\theta = IAB \sin\theta$; for N turns, $\tau = NIAB \sin\theta = \mathbf{m} \times \mathbf{B}$.

Torque in different positions

$\tau = NIAB \sin\theta$, where θ is the angle between the normal to the coil (direction of m) and B.

(i) θ = 90° (plane of coil parallel to B): torque is **maximum**, $\tau = NIAB$.

(ii) θ = 0° (plane perpendicular to B, m along B): τ = 0, **stable equilibrium**; a small displacement produces a restoring torque.

(iii) θ = 180° (m opposite to B): τ = 0, **unstable equilibrium**.

Potential energy of a loop in a field

The work done in rotating the dipole from θ_1 to θ_2 against the torque is $W = \int mB \sin\theta\, d\theta = mB(\cos\theta_1 - \cos\theta_2)$. Taking $U = 0$ at $\theta = 90^\circ$, the potential energy is $U = -m \cdot B = -mB\cos\theta$. It is minimum ($-mB$) in the stable position and maximum ($+mB$) in the unstable position.

Electric dipole in E	*Current loop in B*
$p = q(2a)$	$m = NIA$
$\tau = p \times E$	$\tau = m \times B$
$U = -p \cdot E$	$U = -m \cdot B$
Net force zero in uniform E	Net force zero in uniform B

Bohr magneton: for an electron in the first Bohr orbit, the magnetic moment works out to $\mu_B = eh/4\pi m_e = 9.27 \times 10^{-24}$ A m^2. This is the natural unit of atomic magnetic moments and is the origin of the magnetism of materials studied in the next chapter.

Solved example

Q. A coil of 50 turns and area 4×10^{-3} m^2 carries 2 A in a field of 0.5 T. Find (a) the maximum torque, (b) the work needed to turn it from the stable to the unstable position.

Sol. $m = NIA = 50 \times 2 \times 4 \times 10^{-3} = 0.4$ A m^2. (a) $\tau_{max} = mB =$ **0.2 N m**. (b) $W = mB(\cos 0^\circ - \cos 180^\circ) = 2mB =$ **0.4 J**.

9. Moving Coil Galvanometer

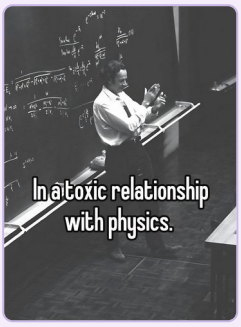
A coil of N turns, area A, pivoted in a **radial** magnetic field (concave pole pieces and a soft iron core) so the plane of the coil is always parallel to B ($\sin\theta = 1$).

Deflecting torque NIAB = restoring torque $k\varphi \Longrightarrow \varphi = (NAB/k)\, I$ Linear scale

- → **Current sensitivity:** $\varphi/I = NAB/k$.
- → **Voltage sensitivity:** $\varphi/V = NAB/kR_G$.
- → Doubling N doubles current sensitivity but also doubles R_G, so voltage sensitivity stays the same.
- → Radial field makes the scale linear; soft iron core strengthens B.

Conversion

Into ammeter	*Into voltmeter*
Low resistance **shunt** S in parallel	High resistance R in series
$S = I_g G/(I - I_g)$	$R = V/I_g - G$
Ideal ammeter: zero resistance	Ideal voltmeter: infinite resistance
Connected in series in a circuit	Connected in parallel across an element



Galvanometer aur shunt ka rishta

Asli baat: A shunt protects the galvanometer by carrying most of the current; that is how a galvanometer becomes an ammeter.

Exam Alert

- → Galvanometer principle + construction + conversion: 5 marks ka pakka sawaal.
- → N double karne se voltage sensitivity kyun nahi badhti: reasoning yaad karo.

Why is the galvanometer scale linear?

In a radial field, B is always parallel to the coil's plane, so $\sin\theta = 1$ at every deflection. Torque NIAB is balanced by the spring torque $k\varphi$, giving $\varphi \propto I$: equal steps of current give equal steps on the scale.

Solved example (NCERT)

Q. A galvanometer has $R_G = 15\ \Omega$ and gives full scale for 4 mA. How will you convert it into (a) a voltmeter of 0–10 V, (b) an ammeter of 0–6 A?

Sol. (a) $R = 10/0.004 - 15 =$ **2485 Ω** in series. (b) $S = 0.004 \times 15/5.996 =$ **0.01 Ω** in parallel.

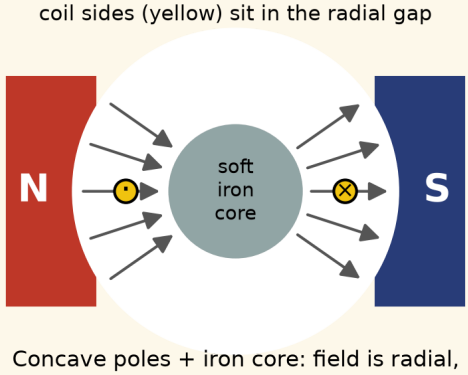


Fig. 4.9 Radial magnetic field in a moving coil galvanometer

Construction and working, step by step

1. A rectangular coil of many turns of fine insulated copper wire is wound on a light frame and pivoted (or suspended) between the concave poles of a strong permanent magnet.
2. A soft iron cylinder (core) is placed inside the coil. The concave poles and the core together make the field **radial** in the gap, and the iron also makes it stronger.
3. When current I flows, the deflecting torque is $\tau = NIAB$ ($\sin\theta = 1$ always, because the plane of the coil is always parallel to the radial field).
4. A spring (or phosphor-bronze suspension) provides a restoring torque $k\varphi$, where φ is the angular deflection.
5. In equilibrium $k\varphi = NIAB$, so $\varphi = (NAB/k)\,I$. The deflection is proportional to current and the scale is linear.

Current sensitivity $I_s = \varphi/I = NAB/k$ (deflection per unit current). **Voltage sensitivity** $V_s = \varphi/V = NAB/kR$ (deflection per unit voltage). Increasing N raises current sensitivity, but it also raises R in the same proportion, so the voltage sensitivity need not increase. This is a favourite board question.

Ammeter	Voltmeter
Galvanometer + low shunt S in parallel	Galvanometer + high R in series
$S = I_g G/(I - I_g)$	$R = V/I_g - G$
Very low total resistance	Very high total resistance
Connected in series in the circuit	Connected in parallel across the element
Ideal: zero resistance	Ideal: infinite resistance

Solved example

Q. A galvanometer of resistance 100 Ω gives full-scale deflection for 1 mA. Convert it into (a) a voltmeter of range 0–10 V, (b) an ammeter of range 0–1 A.

Sol. (a) $R = V/I_g - G = 10/10^{-3} - 100 = $ **9900 Ω in series.** (b) $S = I_g G/(I - I_g) = 10^{-3} \times 100/0.999 =$ **0.1 Ω in parallel.**

Yaad karo

→ Radial field: coil ka plane hamesha B ke parallel, isliye scale linear.
→ N badhao toh current sensitivity badhti hai, voltage sensitivity zaroori nahi.
→ Ammeter = G + chhota shunt parallel; voltmeter = G + bada R series.

10. Solved Numericals (Board Pattern)

N1. A wire carries 8 A in a field of 0.15 T at 30° to it. Force per metre?
Sol. $F/l = IB\sin\theta = 8 \times 0.15 \times 0.5 =$ **0.6 N/m.**

N2. A 100-turn coil of radius 8 cm carries 0.4 A. Field at the centre?
Sol. $B = \mu_0 NI/2R = 4\pi \times 10^{-7} \times 100 \times 0.4/0.16 =$ **3.1×10^{-4} T.**

N3. A solenoid of length 0.5 m, radius 1 cm, 500 turns, carries 5 A. Field inside?
Sol. $n = 1000/m$; $B = \mu_0 nI = 4\pi \times 10^{-7} \times 1000 \times 5 =$ **6.28×10^{-3} T.**

N4. Two parallel wires 4 cm apart carry 8 A and 5 A in opposite directions. Force on 10 cm of one wire?
Sol. $F = \mu_0 I_1 I_2 L/2\pi d = 2 \times 10^{-7} \times 40 \times 0.1/0.04 =$ **2×10^{-5} N**, repulsive.

N5. A galvanometer (G = 12 Ω) shows full scale for 3 mA. Convert it into a voltmeter of range 18 V.
Sol. $R = V/I_g - G = 18/0.003 - 12 =$ **5988 Ω** in series.

N6. The same galvanometer into an ammeter of range 6 A?
Sol. $S = I_g G/(I - I_g) = 0.003 \times 12/5.997 \approx$ **6×10^{-3} Ω** in parallel.

Oersted's discovery

NCERT Activity: Oersted's experiment

Place a magnetic compass needle below a straight wire running north-south. Switch on a current: the needle deflects. Reverse the current: it deflects the other way. Switch off: it returns. A current produces a magnetic field around it, whose direction depends on the current's direction.

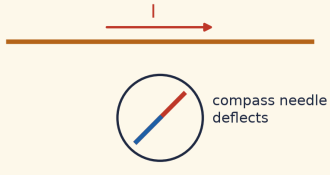


Fig. Oersted's experiment

Magnetic field: conventions

- Field out of the page: ⊙ (dot, like an arrow tip coming at you).
- Field into the page: ⊗ (cross, like the tail of an arrow).
- Magnetic field B is a vector field; its SI unit is the tesla.

Electric field vs magnetic field

Electric field E	Magnetic field B
Produced by charges (at rest or moving)	Produced by moving charges (currents) and magnets
Acts on any charge	Acts only on moving charges
Force qE along E	Force $q(\vec{v} \times \vec{B})$ ⊥ to v and B
Can change speed (does work)	Cannot change speed (no work)
Field lines begin and end on charges	Field lines are closed loops
Unit: N C^{-1} or V m^{-1}	Unit: tesla (N A^{-1} m^{-1})

Solved example (toroid)

Q. A toroid has a non-ferromagnetic core of inner radius 25 cm and outer radius 26 cm, with 3500 turns carrying 11 A. Find B inside the core, outside the toroid and in the empty space it encloses.

Sol. Mean radius r = 0.255 m. Inside the core $B = \mu_0 NI/2\pi r = 4\pi \times 10^{-7} \times 3500 \times 11/(2\pi \times 0.255)$ = **3.0×10^{-2} T**. Outside and in the enclosed space the Amperian loop encloses zero net current, so **B = 0**.

Solved example (direction)

Q. A long horizontal wire carries 50 A from north to south. Find B at a point 2.5 m east of it.

Sol. $B = 2 \times 10^{-7} \times 50/2.5 = \mathbf{4 \times 10^{-6}}$ **T**. With the right thumb pointing south, the fingers rise on the east side: B is **vertically upward**.

Solved example (rod held up by a field)

Q. A horizontal rod of length 0.45 m and mass 60 g hangs from two vertical wires and carries 5 A. What field normal to the rod makes the tension zero? What is the tension if the current is reversed?

Sol. $IlB = mg \Longrightarrow B = 0.06 \times 9.8/(5 \times 0.45)$ = **0.26 T**. On reversal the magnetic force acts downward: T = 2mg = **1.18 N**.

Solved example (power line)

Q. An overhead line carries 90 A from east to west. Find B 1.5 m below it.

Sol. $B = 2 \times 10^{-7} \times 90/1.5 = \mathbf{1.2 \times 10^{-5}}$ **T**. Thumb towards west: below the wire the field points **south**.

Solved example (meter sensitivity)

Q. Meter M_1: R = 10 Ω, N = 30, A = 3.6×10^{-3} m^2, B = 0.25 T. Meter M_2: R = 14 Ω, N = 42, A = 1.8×10^{-3} m^2, B = 0.50 T. Same springs. Compare their current and voltage sensitivities.

Sol. $I_{s2}/I_{s1} = (42 \times 1.8 \times 0.50)/(30 \times 3.6 \times 0.25)$ = **1.4**. $V_{s2}/V_{s1} = 1.4 \times 10/14$ = **1**.

Solved example (car battery cables)

Q. The cables from a car battery to the starter motor carry 300 A and are 1.5 cm apart. Find the force per metre and its nature.

Sol. $F/l = 2 \times 10^{-7} \times 300^2/0.015$ = **1.2 N m^{-1}**. The currents are opposite (to and from the motor), so the force is **repulsive**.

Formula sheet: Chapter 4

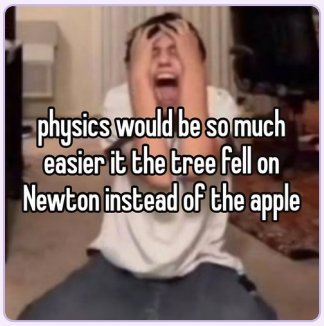
Quantity	Formula
Lorentz force	$F = q(E + v \times B)$
Wire in field	$F = I\,l \times B$
Circular motion	$r = mv/qB$; $T = 2\pi m/qB$
Biot-Savart	$dB = \mu_0 I\,dl \sin\theta/4\pi r^2$
Loop axis / centre	$\mu_0 IR^2/2(x^2+R^2)^{3/2}$ / $\mu_0 NI/2R$
Straight wire	$B = \mu_0 I/2\pi r$
Solenoid	$B = \mu_0 nI$
Parallel wires	$F/L = \mu_0 I_1 I_2/2\pi d$
Torque	$\tau = NIAB \sin\theta$
Galvanometer	$\varphi = NABI/k$; $S = I_g G/(I - I_g)$

1 minute recap

- ✓ $F = q(v \times B)$: no work, speed constant.
- ✓ $r = mv/qB$; T independent of v.
- ✓ Loop centre $\mu_0 I/2R$; wire $\mu_0 I/2\pi r$; solenoid $\mu_0 nI$.
- ✓ Parallel same-direction currents attract.
- ✓ $\tau = NIAB \sin\theta$; galvanometer: shunt → ammeter, series R → voltmeter.

Most Important Questions

Objective, short, long and case-based, with answers



Magnetism ke sawaal dekh ke students

Asli baat: Most asked: field on axis of a loop, Ampere's law for solenoid, force between wires and the galvanometer.

A. Objective and very short answer (1 mark)

Q1. IM A charged particle moves parallel to a magnetic field. The force on it is:
(a) qvB
(b) maximum
(c) zero
(d) qvB/2
Ans: (c) zero.

Q2. IM Board fav The work done by a magnetic force on a moving charge is:
(a) zero
(b) positive
(c) negative
(d) depends on speed
Ans: (a) zero.

Q3. IM The SI unit of magnetic field is:
(a) weber
(b) tesla
(c) henry
(d) gauss
Ans: (b) tesla.

Q4. IM Two parallel wires carrying currents in the same direction:
(a) attract
(b) repel
(c) no force
(d) rotate
Ans: (a) attract.

Q5. IM The field at the centre of a circular loop of radius R carrying current I is:
(a) $\mu_0 I/2\pi R$
(b) $\mu_0 I/4R$
(c) $\mu_0 I/2R$
(d) $\mu_0 IR$
Ans: (c) $\mu_0 I/2R$.

Q6. IM An ammeter is a galvanometer with:
(a) high resistance in series
(b) low resistance in parallel
(c) low resistance in series
(d) high resistance in parallel
Ans: (b) low resistance in parallel.

Q7. IM Board fav Does the period of a charged particle in a magnetic field depend on its speed?
Ans: No. $T = 2\pi m/qB$ is independent of v and r.

Q8. IM State the condition for the force on a moving charge to be maximum.
Ans: When $v \perp B$ ($\theta = 90^\circ$).

Q9. IM Define one ampere in terms of force between wires.
Ans: Current giving 2×10^{-7} N/m between two long parallel wires 1 m apart in vacuum.

Q10. IM Why is a radial field used in a galvanometer?
Ans: So that the plane of the coil is always parallel to B, $\tau = NIAB$, giving a linear scale.

Q11. IM What is the magnetic moment of a current loop?
Ans: $m = NIA$, along the normal by the right-hand rule.

Q12. IM What is the resistance of an ideal voltmeter?
Ans: Infinite.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M Board fav **A:** The kinetic energy of a charged particle in a magnetic field stays constant. **R:** The magnetic force is always perpendicular to velocity.
Ans: (a)

Q14. 1M **A:** Increasing the number of turns increases the voltage sensitivity of a galvanometer. **R:** Current sensitivity is proportional to N.
Ans: (d) A false (R_G also rises); R true.

Q15. 1M **A:** The field inside a long solenoid is uniform. **R:** The field outside a long solenoid is nearly zero.
Ans: (b)

Q16. 1M **A:** Antiparallel currents repel. **R:** Like charges repel.
Ans: (b) both true, but R does not explain A.

C. Short answer (2–3 marks)

Q17. 3M Board fav Show that a charged particle moving $\perp$ to a uniform B follows a circle. Find r and T.
Ans: $qvB = mv^2/r \Longrightarrow r = mv/qB$; $T = 2\pi r/v = 2\pi m/qB$.

Q18. 2M State Biot–Savart law in vector form.
Ans: $dB = (\mu_0/4\pi)\, I\, dl \times r/r^3$.

Q19. 3M Board fav Derive the field on the axis of a circular loop.
Ans: Perpendicular components cancel; $B = \int dB\cos\theta = \mu_0 IR^2/2(x^2+R^2)^{3/2}$.

Q20. 2M Using Ampere's law, find B due to a long straight wire.
Ans: $B(2\pi r) = \mu_0 I \Longrightarrow B = \mu_0 I/2\pi r$.

Q21. 3M Board fav Derive B inside a long solenoid using Ampere's law.
Ans: Rectangular loop: $Bh = \mu_0 nhI \Longrightarrow B = \mu_0 nI$.

Q22. 3M Derive the force per unit length between two parallel currents.
Ans: $B_a = \mu_0 I_a/2\pi d$; $F = I_b L B_a \Longrightarrow F/L = \mu_0 I_a I_b/2\pi d$.

Q23. 3M Derive the torque on a rectangular current loop in a uniform field.
Ans: Forces on sides $\perp$ to B: IbB each, arm a sin θ $\Longrightarrow \tau = IabB\sin\theta = NIAB\sin\theta$.

Q24. 2M Explain how a galvanometer is converted into an ammeter.
Ans: A small shunt $S = I_g G/(I - I_g)$ in parallel carries most current; total resistance becomes very small.

Q25. 3M Compare Biot–Savart law with Coulomb's law (any three points).
Ans: Vector source vs scalar source; angle dependence; dB $\perp$ to r vs E along r; both $\propto 1/r^2$.

Q26. 2M Find the magnetic moment of an electron revolving in an orbit of radius r with speed v.
Ans: $I = ev/2\pi r$; $\mu = IA = evr/2$.

D. Long answer (5 marks)

Q27. 5M Board fav (a) Explain the principle, construction and working of a moving coil galvanometer. (b) Define current and voltage sensitivity. (c) Why does increasing N not always increase sensitivity?
Ans: (a) Torque on coil in radial field balanced by spring: $\varphi = NABI/k$. (b) NAB/k and NAB/kR_G. (c) R_G rises with N.

Q28. 5M (a) State Ampere's circuital law. (b) Derive B inside a solenoid and a toroid. (c) Draw the field lines.
Ans: (a) $\oint B\cdot dl = \mu_0 I$. (b) Solenoid $\mu_0 nI$; toroid $\mu_0 NI/2\pi r$ inside, zero outside.

E. Case-based

Q29. 4M Board fav A proton enters a uniform field of 0.5 T at right angles with speed 2×10^6 m/s. (a) Radius? (b) Period? (c) Does its speed change? (d) What path if it entered at 60°?
Ans: (a) $r = mv/qB = 1.67\times10^{-27}\times2\times10^6/(1.6\times10^{-19}\times0.5) = 0.042$ m. (b) $T = 2\pi m/qB = 1.3\times10^{-7}$ s. (c) No. (d) Helix.

Q30. 4M Power lines in a house run parallel. (a) Do two wires carrying 10 A in opposite directions 5 cm apart attract or repel? (b) Force per metre? (c) Why is this force usually not felt? (d) Which law gives the field of each wire?
Ans: (a) Repel. (b) $2\times10^{-7}\times100/0.05 = 4\times10^{-4}$ N/m. (c) Very small. (d) Ampere's law / Biot–Savart.

Q31. 2M Board fav An electron and a proton enter the same field with the same velocity ⊥ to B. Compare their radii.
Ans: $r = mv/qB$, same q, v, B: $r_p/r_e = m_p/m_e \approx 1836$; they curve in opposite directions.

Q32. 2M A circular coil of 20 turns, radius 10 cm, carries 5 A in B = 0.1 T. Maximum torque?
Ans: $\tau = NIAB = 20 \times 5 \times \pi \times 0.01 \times 0.1 =$ **0.31 N m.**

Q33. 2M Two concentric loops of radii R and 2R carry currents I in opposite directions. Field at the centre?
Ans: $\mu_0 I/2R - \mu_0 I/4R = \mu_0 I/4R$.

Q34. 2M A galvanometer of resistance 50 Ω gives full scale for 5 mA. What resistance converts it to a 0–10 V voltmeter?
Ans: $R = 10/0.005 - 50 =$ **1950 Ω** in series.

Q35. 1M Can a magnetic field accelerate a charge at rest?
Ans: No: $F = q(v \times B) = 0$ for $v = 0$.

Q36. 1M What is the field at the centre of a semicircular wire of radius R carrying I?
Ans: Half of the loop value: $\mu_0 I/4R$.

Q37. 1M Why should an ammeter have low resistance?
Ans: It is in series; low resistance does not change the circuit current.

Q38. 2M Board fav A straight wire of mass 200 g and length 1.5 m carries 2 A and is suspended in mid-air by a horizontal field. Find B.
Ans: $IlB = mg \Longrightarrow B = 0.2 \times 9.8/(2 \times 1.5) =$ **0.65 T.**

Q39. 1M What is the pitch of the helix of a charged particle entering B at 90°?
Ans: Zero; the path is a circle.

Q40. 1M A loop is placed with its plane perpendicular to a uniform B. Torque?
Ans: Zero (m parallel to B): stable equilibrium.

F. Extra practice (board PYQ style)

Q41. 1M A long straight wire carries 35 A. Find B at 20 cm from it.
Ans: $B = \mu_0 I/2\pi r = 2 \times 10^{-7} \times 35/0.2 =$ **3.5×10^{-5} T.**

Q42. 2M Board fav An α-particle and a proton enter a field ⊥ with the same kinetic energy. Ratio of radii?
Ans: $r = \sqrt{(2mK)}/qB$: $r_\alpha/r_p = \sqrt{(4/1)}/2 =$ **1:1.**

Q43. 3M A rectangular coil (10 cm × 5 cm, 50 turns) carries 2 A in a field of 0.2 T with its plane at 30° to B. Torque?
Ans: θ (normal with B) = 60°. $\tau = 50 \times 2 \times 5 \times 10^{-3} \times 0.2 \times \sin 60° =$ **0.087 N m.**

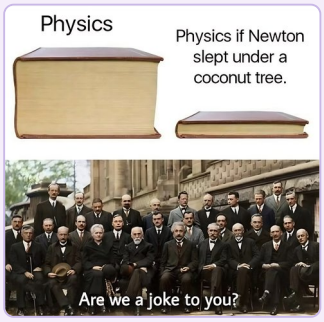
Q44. 1M Why are the pole pieces of a galvanometer concave?
Ans: To produce a radial field so the coil plane is always parallel to B.

Q45. 2M Board fav Two straight wires A and B, 10 cm apart, carry 4 A and 6 A in the same direction. Where is B zero between them?
Ans: $\mu_0 4/2\pi x = \mu_0 6/2\pi(0.1 - x) \Longrightarrow x =$ **4 cm** from A.

Q46. 2M A charged particle passes undeflected through crossed E and B fields. Find its speed.
Ans: $qE = qvB \Longrightarrow v = E/B$ (velocity selector).

Q47. 2M How can a moving coil galvanometer be made more sensitive?
Ans: Increase N, A or B; use a spring with smaller k.

Q48. 1M Name the rule to find the direction of the field due to a straight current.
Ans: Right-hand thumb rule (Maxwell's cork-screw rule).



Magnetism khatam karke apna confidence
Asli baat: Revise the formula sheet once more: magnetism numericals are mostly direct substitution.

Quick compare: field formulas

Source	B
Long straight wire	$\mu_0 I/2\pi r$
Finite wire	$(\mu_0 I/4\pi r)(\sin\varphi_1 + \sin\varphi_2)$
Centre of loop (N turns)	$\mu_0 NI/2R$
Arc of angle θ	$\mu_0 I\theta/4\pi R$
Solenoid (inside / end)	$\mu_0 nI$ / $\mu_0 nI/2$
Toroid (inside)	$\mu_0 NI/2\pi r$

Q49. 1M A circular coil of 100 turns, radius 8 cm carries 0.4 A. Find B at the centre.

Ans: $B = \mu_0 NI/2R = 4\pi \times 10^{-7} \times 100 \times 0.4/0.16 = 3.1 \times 10^{-4}$ T.

Q50. 2M Board fav A toroid has core radius 25 cm, 3500 turns, current 11 A. Find B inside the core.

Ans: $n = 3500/(2\pi \times 0.25) = 2228$ /m; $B = \mu_0 nI = 3.08 \times 10^{-2}$ T.

Q51. 1M Why is a galvanometer with a radial field preferred?

Ans: The scale becomes linear and the torque is maximum for every position.

Q52. 1M An electron moves along +x in a field along +y. Direction of force?

Ans: $F = -e(v \times B) = -e(\hat{x} \times \hat{y}) = -\hat{z}$ direction.

Q53. 2M Board fav Two long wires carry 10 A each in opposite directions, 2 m apart. Find B midway.

Ans: Fields add: $2 \times \mu_0 I/2\pi(1) = 4 \times 10^{-6}$ T.

Q54. 1M Can a moving charge be in equilibrium in E and B fields? When?

Ans: Yes, if $qE = -qv \times B$, i.e. $v = E/B$ with crossed fields.

Quick compare: Biot-Savart vs Ampere

Biot-Savart law	Ampere's law
Field of a small current element	Line integral around a closed loop
Works for any shape (with integration)	Easy only with symmetry
Loop axis, arc, finite wire	Long wire, solenoid, toroid

Fleming, Ampere aur Biot-Savart: humare liye kaam karoge?

Asli baat: Pick the tool: symmetry present? Ampere. Arc or loop? Biot-Savart. Force? Fleming's left-hand rule.

Common mistakes

- → Electron ke liye force ki direction ulti karna bhoolna.
- → θ ko loop ke plane se lena; formula mein normal ka angle chahiye.
- → Shunt series mein lagana: shunt hamesha parallel.
- → n (turns per metre) aur N (total turns) mix karna.

Paheli-Bujho

Paheli: Bujho, quick quiz: solenoid ke andar field kitni?

Bujho: $\mu_0 nI$! Aur bahar lagbhag zero.

Paheli: Aur same direction ke do currents?

Bujho: Attract! Charges se ulta. Ab main magnetism ka topper hoon.

Paheli: Pehle 48 questions solve karo, phir topper.

Q55. 2M A wire of length l is bent into a circle of one turn. Field at the centre for current I?

Ans: $R = l/2\pi$; $B = \mu_0 I/2R = \pi\mu_0 I/l$.

Q56. 2M Board fav If the same wire is bent into a coil of two turns, how does the field at the centre change?

Ans: R halves and N doubles: $B = \mu_0 NI/2R$ becomes **4 times**.

Exam Alert: last check

- → Loop axis field, solenoid, parallel wires, galvanometer: 4 derivations, poora chapter.
- → Electron ka force proton ke ulta, r mass ke hisaab se.

MAGNETISM AND MATTER

Most asked: bar magnet as an equivalent solenoid, magnetic dipole in a uniform field, Gauss's law in magnetism, magnetisation and magnetic intensity, properties of dia-, para- and ferromagnetic substances, Curie's law

Chapter 4 showed that currents make magnetic fields. Here we look at **magnets** themselves and at how different materials respond to a magnetic field: some push it away, some pull it in weakly, and iron pulls it in strongly.

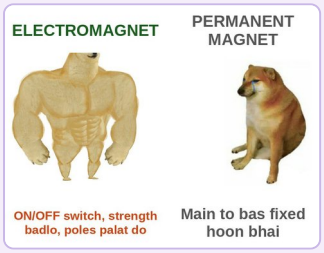
1. Basic Properties of Magnets

→ A freely suspended bar magnet points roughly **north–south**.

→ Like poles repel, unlike poles attract.

→ Poles cannot be isolated: cutting a magnet gives two smaller magnets, each with N and S (**no magnetic monopoles**).

→ Magnetic field lines form **closed loops**: outside from N to S, inside from S to N.



Magnet todne se monopole milega?
Magnet: nahi bhai

Asli baat: Every piece of a broken magnet has both N and S poles. Isolated magnetic poles (monopoles) have never been found.

NCERT Activity: iron filings and compass

Place a sheet of paper on a bar magnet and sprinkle iron filings; tap gently. The filings line up along field lines. Move a small compass around: its needle points along the tangent to the lines, from S end to N end (north pole of the needle points away from the N of the magnet).

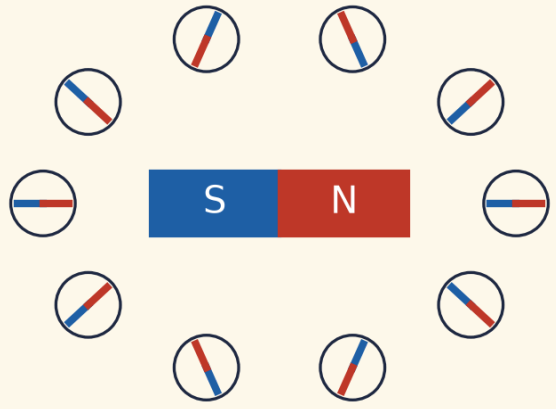


Fig. Compass needles align along the field of a bar magnet

Poles, pole strength and magnetic dipole moment

The regions near the ends of a bar magnet where the attraction of iron filings is strongest are called the **poles**. The pole that points towards geographic north when the magnet hangs freely is the **north pole (N)**; the other is the **south pole (S)**. The straight line through the two poles is the **magnetic axis**, and the line through the centre perpendicular to it is the **magnetic equator** of the magnet.

A bar magnet is the simplest example of a **magnetic dipole**. Its strength is measured by the **magnetic dipole moment m**, a vector directed from the south pole to the north pole inside the magnet. In the pole picture, if q_m is the pole strength and $2l$ the magnetic length, $m = q_m \times 2l$. Since magnetism really comes from circulating currents, the more fundamental definition is $m = NIA$ for a coil, which gives the SI unit A m^2 (equivalently J T^{-1}).

Why poles always come in pairs

Magnetism in a material arises from tiny current loops: electrons orbiting nuclei and spinning about their own axes. Every current loop has one face that behaves as N and the other as S. However finely a magnet is divided, each piece still contains complete loops, so each piece has both poles. This is why an isolated north or south pole, a **magnetic monopole**, has never been observed.

Important points

→ Direction of m: from S to N (inside the magnet).

→ SI unit of m: A m^2 = J T^{-1}.

→ Magnetic length (distance between effective poles) is about 0.84 times the geometric length.

→ A freely suspended magnet aligns along the earth's field because the torque $m \times B$ turns it until m is parallel to B.

2. Magnetic Field Lines

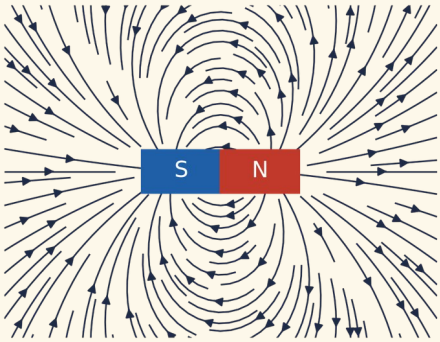


Fig. 5.1 Field lines of a bar magnet

1. Field lines are continuous **closed curves** (unlike electric lines, which start and end on charges).
2. The tangent at any point gives the direction of B.
3. Closer lines mean a stronger field.
4. Two field lines never intersect.

Magnetic field lines vs electric field lines

Magnetic	Electric
Closed loops	Start on +q, end on −q
Exist inside the magnet (S to N)	Zero inside a conductor
No monopoles	Isolated charges exist
Never intersect	Never intersect

Solved example

Q. Why do magnetic field lines not intersect?

Sol. At an intersection there would be two tangents, i.e. two directions of B at one point, which is impossible.

Properties of field lines explained

A magnetic field line is an imaginary curve drawn so that the tangent at every point gives the direction of the net field B there. Field lines are a visual way of describing a vector field: their **direction** shows the direction of B and their **density** (number of lines crossing unit area held normal to them) shows the magnitude of B. Near the poles of a bar magnet the lines crowd together, which is why the field is strongest there.

Why closed loops? Electric field lines begin on positive charges and end on negative charges. There are no magnetic charges at which a field line could begin or end, so every magnetic field line must close on itself: it leaves the N pole, curves round outside, enters the S pole and continues inside the magnet from S to N.

Why can two lines never cross? If they did, the field at the crossing point would have two directions at once, which is impossible because the net field at a point is unique.

Feature	Inside the magnet	Outside the magnet
Direction of lines	S to N	N to S
Spacing	nearly uniform, close	spreading out
Field strength	strong	decreases with distance

Applications. Field-line maps are used to design electric motors, loudspeakers and MRI magnets, where engineers need to know where the field is strong and uniform.

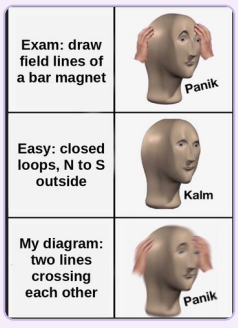
3. Bar Magnet as an Equivalent Solenoid

The field of a bar magnet and that of a current-carrying solenoid look exactly alike. Each turn of the solenoid is a small magnetic dipole; together they behave like a bar magnet of moment m = NIA.

For a solenoid of n turns per length, radius a, length 2l, the far axial field is:

$$B = (\mu_0/4\pi)\, 2m/r^3 \quad m = n(2l)I\pi a^2$$

This is the same expression as the axial field of a bar magnet, which proves the equivalence.



Field lines bahar N se S, andar S se N

Asli baat: Magnetic field lines never end: inside the magnet they return from S to N, making closed loops.

Field of a bar magnet: axial and equatorial

$B_{axial} = (\mu_0/4\pi)\, 2m/r^3$, $B_{equatorial} = -(\mu_0/4\pi)\, m/r^3$ $r \gg$ size of magnet

$B_{axial} = 2 \times B_{equatorial}$ at the same distance; the equatorial field points opposite to m.

Solved example (NCERT)

Q. What is the magnitude of the equatorial and axial fields due to a bar magnet of length 5 cm at 50 cm from its mid-point? ($m = 0.40$ A m^2)

Sol. $B_E = 10^{-7} \times 0.4/0.125 = \mathbf{3.2 \times 10^{-7}}$ T; $B_A = \mathbf{6.4 \times 10^{-7}}$ T.

Derivation: axial field of a solenoid (step by step)

1. Take a solenoid of length $2l$, radius a, n turns per unit length, carrying current I. Find B at a point P on its axis at distance r from the centre O.
2. Consider a thin element of thickness dx at distance x from O. It contains $n\,dx$ turns and behaves as a circular loop of current $nI\,dx$.
3. Field of a circular loop on its axis: $dB = \mu_0 (n\,dx) I a^2/2[(r-x)^2 + a^2]^{3/2}$.
4. Total field: $B = (\mu_0 n I a^2/2) \int_{-l}^{l} dx/[(r-x)^2 + a^2]^{3/2}$.
5. For a far point ($r \gg a$ and $r \gg l$), $[(r-x)^2 + a^2]^{3/2} \approx r^3$, so $B \approx (\mu_0 n I a^2/2r^3) \times 2l = \mu_0 n (2l) I a^2/2r^3$.
6. The magnetic moment of the solenoid is m = (total turns)(I)(area) $= n(2l)\,I\,\pi a^2$. Substituting, $B = (\mu_0/4\pi)(2m/r^3)$.

This is identical to the axial field of a bar magnet. The field lines of a solenoid and a bar magnet also look the same, and a freely suspended solenoid points north-south. Hence a bar magnet and a solenoid produce the same magnetic field: a **bar magnet behaves as an equivalent solenoid**. Cutting a solenoid in two gives two weaker solenoids, each with its own N and S ends, just as cutting a magnet does.

Equatorial field of a bar magnet (pole model)

1. Let poles of strength q_m sit at N and S, separated by $2l$, and let P lie on the equator at distance r from the centre.
2. Each pole produces a field of magnitude $(\mu_0/4\pi)\, q_m/(r^2 + l^2)$ at P.
3. The components perpendicular to the axis cancel; the components parallel to the axis add, each being the magnitude times $\cos\alpha$ where $\cos\alpha = l/(r^2 + l^2)^{1/2}$.
4. $B = (\mu_0/4\pi)\, 2q_m l/(r^2 + l^2)^{3/2} = (\mu_0/4\pi)\, m/(r^2 + l^2)^{3/2}$, directed from N to S, i.e. opposite to m.
5. For a short magnet ($r \gg l$): $B_E = -(\mu_0/4\pi)\, m/r^3$.

Point	Field (short magnet)	Direction
Axial (end-on)	$(\mu_0/4\pi)\, 2m/r^3$	along m
Equatorial (broad-on)	$(\mu_0/4\pi)\, m/r^3$	opposite to m

Solved example

Q. A short bar magnet has $m = 0.5$ A m^2. Find the magnetic field at 10 cm from its centre (a) on the axis, (b) on the equatorial line.

Sol. (a) $B_A = 10^{-7} \times 2 \times 0.5/(0.1)^3 = \mathbf{1.0 \times 10^{-4}}$ T, along m. (b) $B_E = 10^{-7} \times 0.5/(0.1)^3 = \mathbf{5.0 \times 10^{-5}}$ T, opposite to m.

Solved example

Q. A solenoid of 2000 turns and cross-sectional area 1.6×10^{-4} m^2 carries 4 A. What is its magnetic moment, and what does it behave like?

Sol. $m = NIA = 2000 \times 4 \times 1.6 \times 10^{-4} = \mathbf{1.28}$ **A m^2**, along the axis. It behaves like a bar magnet of this moment.

4. Dipole in a Uniform Magnetic Field

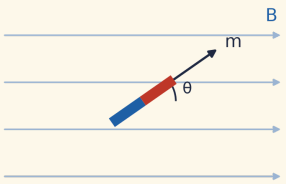


Fig. 5.2 Magnetic dipole at angle θ with B

$$\tau = m \times B, \quad |\tau| = mB \sin\theta$$

$$U = -m \cdot B = -mB\cos\theta$$

θ	U	Equilibrium
0° (m along B)	−mB (minimum)	stable
90°	0	—
180° (m opposite B)	+mB (maximum)	unstable

Small oscillations of a magnetised needle (moment of inertia I) about its stable position are simple harmonic:

$$T = 2\pi\sqrt{(I/mB)}$$

Derivation step by step (oscillations)

1. Displace the needle by small θ: restoring torque $\tau = -mB\sin\theta \approx -mB\theta$.
2. Newton's law for rotation: $I\, d^2\theta/dt^2 = -mB\theta$.
3. This is SHM with $\omega^2 = mB/I$, so $T = 2\pi\sqrt{(I/mB)}$.

Solved example (NCERT)

Q. A magnetic needle in a uniform field of 0.049 T makes 20 oscillations in 10 s; $I = 9.8 \times 10^{-6}$ kg m^2. Find m.

Sol. $T = 0.5$ s; $m = 4\pi^2 I/BT^2 = 4\pi^2 \times 9.8 \times 10^{-6}/(0.049 \times 0.25) =$ **0.032 J/T.**

Torque from the forces on the poles

Place a bar magnet of pole strength q_m and magnetic length 2l in a uniform field B at angle θ with the field. The N pole experiences a force $q_m B$ along B and the S pole an equal force opposite to B. The net force is zero, so the magnet does not move as a whole, but the two forces form a **couple**. The perpendicular distance between their lines of action is 2l sin θ, so $\tau = q_m B \times 2l \sin\theta = mB \sin\theta$, or in vector form $\boldsymbol{\tau} = \mathbf{m} \times \mathbf{B}$. The torque tends to align m with B.

Derivation: potential energy $U = -mB\cos\theta$

1. Work done by an external agent in turning the magnet slowly through dθ against the magnetic torque: $dW = \tau\, d\theta = mB \sin\theta\, d\theta$.
2. Potential energy is defined as zero at θ = 90° (m perpendicular to B).
3. $U(\theta) = \int_{90^\circ}^{\theta} mB \sin\theta\, d\theta = mB[-\cos\theta]_{90^\circ}^{\theta}$ $= -mB\cos\theta$.
4. Hence $U = -\mathbf{m}\cdot\mathbf{B}$. Work to rotate from θ_1 to θ_2: $W = mB(\cos\theta_1 - \cos\theta_2)$.

Why θ = 0° is stable: U is minimum there; any small displacement raises the energy and the torque pushes the magnet back. At θ = 180° U is maximum, and the slightest tilt makes the torque turn the magnet further away, so that equilibrium is unstable.

Oscillations: what the formula tells us

Since $T = 2\pi\sqrt{(I/mB)}$, a stronger field or a stronger magnet makes the needle oscillate faster, while a heavier needle (larger I) oscillates more slowly. Because $B \propto 1/T^2$, timing the oscillations of a compass needle is a simple way to compare two magnetic fields. This principle is used in the **vibration magnetometer**.

Solved example

Q. A bar magnet of moment 2 J T^{-1} lies along a uniform field of 0.3 T. How much work is needed to turn it through 60°?

Sol. $W = mB(\cos 0^\circ - \cos 60^\circ) = 2 \times 0.3 \times (1 - 0.5) =$ **0.3 J.**

Solved example

Q. A compass needle oscillates with a period of 2 s in a certain field. What is its period if the field is made four times stronger?

Sol. $T \propto 1/\sqrt{B}$, so $T' = 2/\sqrt{4} =$ **1 s.**

Where this is used. The torque m × B is what turns a compass needle to the north, what rotates the coil of a moving-coil galvanometer and of an electric motor, and what aligns the nuclear magnetic moments of hydrogen atoms in an MRI scanner. The energy $U = -mB\cos\theta$ explains why a compass needle, once disturbed, always settles back pointing along the field: that is its position of least energy.

Solved example

Q. A small magnet of moment 0.6 A m^2 and moment of inertia 1.2×10^{-5} kg m^2 is suspended in a uniform field of 0.04 T. Find its period of small oscillations.

Sol. $T = 2\pi\sqrt{(I/mB)} = 2\pi\sqrt{(1.2 \times 10^{-5}/(0.6 \times 0.04))}$ $= 2\pi\sqrt{(5 \times 10^{-4})} =$ **0.14 s.**

Dipole in uniform field: summary

- → Net force = 0; torque $\tau = mB\sin\theta$ (max mB at θ = 90°).
- → $U = -mB\cos\theta$: −mB (stable, 0°), 0 (90°), +mB (unstable, 180°).
- → Work from stable to unstable position = 2mB.
- → $T = 2\pi\sqrt{(I/mB)}$, so $B \propto 1/T^2$.

5. Electrostatic Analogue

Replace $p \to m$, $E \to B$ and $1/4\pi\varepsilon_0 \to \mu_0/4\pi$ to get magnetic results from electric ones:

Quantity	Electrostatics	Magnetism
Constant	$1/4\pi\varepsilon_0$	$\mu_0/4\pi$
Dipole moment	p	m
Axial field	$2p/4\pi\varepsilon_0 r^3$	$\mu_0 2m/4\pi r^3$
Equatorial field	$-p/4\pi\varepsilon_0 r^3$	$-\mu_0 m/4\pi r^3$
Torque	p × E	m × B
Energy	$-\mathbf{p}\cdot\mathbf{E}$	$-\mathbf{m}\cdot\mathbf{B}$

Exam Alert

→ Electric aur magnetic dipole ka comparison table: 2-3 marks direct.

→ Torque max at 90°, zero at 0° and 180°: MCQ favourite.

Paheli-Bujho

Bujho: Paheli, electric chapter ka sab formula yahan copy-paste ho jaata hai?

Paheli: Lagbhag! p ki jagah m, E ki jagah B, aur $1/4\pi\varepsilon_0$ ki jagah $\mu_0/4\pi$.

Bujho: Wah, ek chapter padho, do free!

Paheli: Bas ek fark yaad rakho: magnetic monopole nahi hota, isliye field lines closed loops.

Where the analogy breaks. The analogue works for fields far from a dipole, torque and energy. But electric charges can exist alone while magnetic poles cannot. So electric field lines begin and end on charges, whereas magnetic field lines are always closed; and Gauss's law gives a non-zero flux for E but zero flux for B. Inside a magnet the field points from S to N, while inside an electric dipole the field points from + to −, i.e. opposite to p.

6. Gauss's Law in Magnetism

The net magnetic flux through any closed surface is **zero**:

$$\varphi_B = \oint B \cdot dS = 0$$

→ As many field lines leave a closed surface as enter it.

→ Reason: magnetic monopoles do not exist; the simplest magnetic element is a dipole.

→ Compare: Gauss's law for E gives q/ε_0, because isolated charges exist.

Gauss ka law magnetism mein: flux hamesha zero

Asli baat: Closed surface ke andar jitni lines aati hain, utni bahar jaati hain. Net flux = 0 because there is no isolated pole.

Understanding the law

Imagine any closed surface (a Gaussian surface) drawn anywhere in a magnetic field: around one pole of a bar magnet, around the whole magnet, or around a current loop. Divide it into small area elements ΔS. The flux through each element is $B \cdot \Delta S$, positive where lines leave and negative where they enter. Because field lines are closed loops, every line that enters the surface must also leave it. So the total flux, the sum of $B \cdot \Delta S$ over the whole surface, is exactly zero.

Even a surface drawn around only the N pole of a magnet gives zero flux: the lines leaving through the outside are exactly balanced by the lines that enter from inside the magnet, travelling from S to N. In electrostatics, a surface around a charge q has flux q/ε_0; the magnetic flux is zero because there is no "magnetic charge" inside.

Gauss's law: E vs B

Electrostatics	Magnetism
$\oint E \cdot dS = q/\varepsilon_0$	$\oint B \cdot dS = 0$
Isolated charges exist	No isolated poles
Lines start and end on charges	Lines are closed loops

7. Magnetisation and Magnetic Intensity

Atoms behave like tiny current loops. In a material their net moment per unit volume is the **magnetisation**:

$$M = m_{net}/V \quad \text{SI unit: A m}^{-1}$$

Inside a long solenoid (n turns/length, current I) filled with a material, the field has two parts:

$$B = B_0 + B_m = \mu_0(H + M) \quad H = nI\text{: magnetic intensity (A m}^{-1})$$

$$M = \chi H \implies B = \mu_0(1 + \chi)H = \mu_0\mu_r H = \mu H$$

Term	Meaning
H (magnetic intensity)	field due to external currents (= nI)
M (magnetisation)	dipole moment per unit volume of the material
χ (susceptibility)	M/H, dimensionless: how easily a material is magnetised
μ_r (relative permeability)	$1 + \chi = \mu/\mu_0$

Yaad karo

→ H bahar ke current se aata hai, M material se.

→ χ negative: diamagnetic; small positive: paramagnetic; very large positive: ferromagnetic.

Origin of magnetism in atoms

Every electron revolving round the nucleus is a tiny current loop with an orbital magnetic moment; electrons also have spin moments. In most atoms these cancel in pairs; in some they do not, giving the atom a permanent moment.

Solved example (NCERT)

Q. A solenoid has a core with $\mu_r = 400$, 1000 turns over 1 m, I = 2 A. Find H, M, B and the magnetising current I_m.

Sol. $H = 2000$ A/m; $B = \mu_r\mu_0 H = 1.0$ T; $M = (\mu_r - 1)H \approx 8 \times 10^5$ A/m; $I_m = (B/\mu_0 n) - I \approx 794$ A.

Derivation: $B = \mu_0(H + M) = \mu_0(1 + \chi)H$

1. An empty long solenoid with n turns per unit length and current I has field $B_0 = \mu_0 nI$ inside.
2. Fill it with a material. The applied field magnetises the material, giving magnetisation M. The aligned atomic loops act like an extra surface current and add a field $B_m = \mu_0 M$.
3. Net field: $B = B_0 + B_m = \mu_0 nI + \mu_0 M$.
4. Define the magnetic intensity $H = B_0/\mu_0 = nI$ (it depends only on the free current). Then **$B = \mu_0(H + M)$**.
5. For most materials M is proportional to H: $M = \chi H$, where χ is the magnetic susceptibility.
6. So $B = \mu_0(H + \chi H) = \mu_0(1 + \chi)H = \mu_0\mu_r H = \mu H$, with $\mu_r = 1 + \chi$ and $\mu = \mu_0\mu_r$.

Physical meaning. H is the cause, set by the currents we control; M is the response of the material; B is the total field that actually acts on moving charges. Susceptibility χ tells how strongly the material responds, and permeability μ tells how easily field lines pass through it. M and H have the same unit, A m^{-1}, so χ and μ_r are pure numbers.

Solved example

Q. Aluminium ($\chi = 2.3 \times 10^{-5}$) is placed in a magnetic intensity H = 1000 A m^{-1}. Find M and B.

Sol. $M = \chi H = 2.3 \times 10^{-5} \times 1000 =$ **2.3×10^{-2} A m^{-1}**. $B = \mu_0(1 + \chi)H = 4\pi \times 10^{-7} \times 1.000023 \times 1000 =$ **1.257×10^{-3} T**, barely more than in vacuum, as expected for a paramagnet.

8. Magnetic Properties of Materials

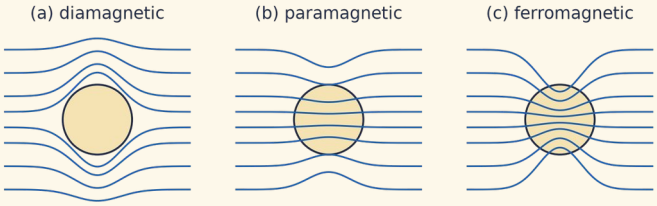


Fig. 5.3 Behaviour of field lines near (a) diamagnetic, (b) paramagnetic, (c) ferromagnetic samples

Property	Diamagnetic	Paramagnetic	Ferromagnetic
χ	small, negative (-10^{-5})	small, positive (10^{-5})	very large, positive (10^3)
μ_r	slightly < 1	slightly > 1	$\gg 1$
In non-uniform field	moves to weaker region	moves to stronger region	moves strongly to stronger region
Field lines	expelled	slightly concentrated	strongly concentrated
Temperature	no effect	$\chi \propto 1/T$ (Curie)	paramagnetic above T_c
Examples	Bi, Cu, Pb, Si, water, N_2	Al, Na, Ca, O_2	Fe, Co, Ni, Gd

Diamagnetism

Atoms have no net moment. An applied field induces orbital currents opposing it (like Lenz's law), so M is opposite to H. Present in all materials, but masked when other effects are stronger.

→ **Superconductors** are perfect diamagnets ($\chi = -1$, $\mu_r = 0$): they expel the field completely. This is the **Meissner effect**, used in maglev trains.

Paramagnetism

Atoms have permanent moments, randomly oriented by thermal motion. A field aligns them slightly. **Curie's law**:

$$\chi = C\,\mu_0/T \quad C = \text{Curie constant}$$

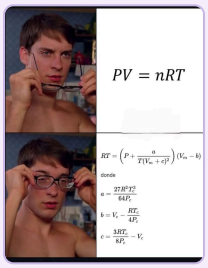
Ferromagnetism

Atomic moments align spontaneously in small regions called **domains** (about 1 mm, 10^{11} atoms). An external field grows the domains aligned with it and rotates others, producing strong magnetisation.

→ Above the **Curie temperature T_c**, a ferromagnet becomes paramagnetic: $\chi = C/(T - T_c)$.

→ T_c for iron: 1043 K; nickel: 631 K; cobalt: 1394 K.

→ **Hard ferromagnets** (alnico) keep magnetisation: permanent magnets. **Soft ferromagnets** (soft iron) lose it quickly: electromagnet and transformer cores.



Teen material, teen alag nature

Asli baat: Dia pushes the field away, para pulls it in a little, ferro pulls it in very strongly. Exam mein table seedha likho.

Exam Alert

→ Dia-para-ferro comparison (any 3 points): 3 marks, almost every year.

→ Curie's law and Curie temperature: 1–2 marks.

→ Superconductor = perfect diamagnet ($\chi = -1$).

Paheli-Bujho

Paheli: Bujho, magnet ke paas bismuth rakho toh kya hoga?

Bujho: Chipak jaayega!

Paheli: Nahi, bismuth diamagnetic hai, wo halka sa door bhaagega.

Bujho: Matlab bismuth ka magnet se breakup ho chuka hai.

Why does a dia/para/ferro sample move as it does?

→ A paramagnetic or ferromagnetic sample develops M along B and is pulled to the stronger field region (like iron to a magnet).

→ A diamagnetic sample develops M opposite to B and is pushed towards the weaker field region.

→ In a U-tube, a paramagnetic liquid rises in the arm placed between the poles; a diamagnetic liquid falls.

NCERT Activity: behaviour of a rod between poles

Suspend a small rod between the poles of a strong magnet. A paramagnetic rod (aluminium) aligns along the field; a diamagnetic rod (bismuth) sets itself ⊥ to the field.

Permanent magnets and electromagnets

Permanent magnet	*Electromagnet*
Hard ferromagnet: steel, alnico	Soft iron core in a solenoid
High retentivity and coercivity	Low retentivity, low coercivity
Magnetism stays without current	Magnetism only while current flows
Compass, speakers	Cranes, relays, bells, MRI

Retentivity: magnetisation left when the external field is removed. **Coercivity**: reverse field needed to remove it completely.

Atomic explanation in detail

Diamagnetism. In a diamagnetic atom all electrons are paired, so their orbital and spin moments cancel and the atom has no net moment. When a field is switched on, the orbital motion of each electron changes slightly: electrons whose moments point along B slow down and those whose moments point against B speed up. The result is a small induced moment opposite to B, exactly as Lenz's law predicts for any induced current. Hence χ is negative. Because no permanent moments exist to be disordered, diamagnetism hardly depends on temperature. Every substance shows this effect, but it is so weak that it is noticed only when the atoms have no permanent moment.

Paramagnetism. Atoms with unpaired electrons have a permanent moment. Without a field, thermal agitation keeps these moments randomly oriented and M = 0. A field tends to align them, while heat tends to randomise them. At ordinary temperatures only a small fraction lines up, so χ is small and positive. Lowering the temperature or raising the field increases the alignment, until at very low temperature and strong field nearly all dipoles align and M reaches its **saturation** value.

Curie's law $M = C(B_0/T)$, or $\chi = C\mu_0/T$, expresses this competition: the graph of χ against 1/T is a

straight line through the origin. It holds only while the magnetisation is well below saturation.

Ferromagnetism. In iron, cobalt, nickel and gadolinium, a quantum-mechanical exchange interaction makes neighbouring atomic moments line up spontaneously, even without any applied field. The aligned regions, called **domains**, are typically about 1 mm across and contain about 10^{11} atoms. In unmagnetised iron the domains point in random directions and cancel (Fig. 5.4a). In an applied field, domains aligned with B grow at the expense of others by the movement of domain walls, and the remaining domains rotate towards B (Fig. 5.4b). This gives a magnetisation thousands of times larger than in a paramagnet.

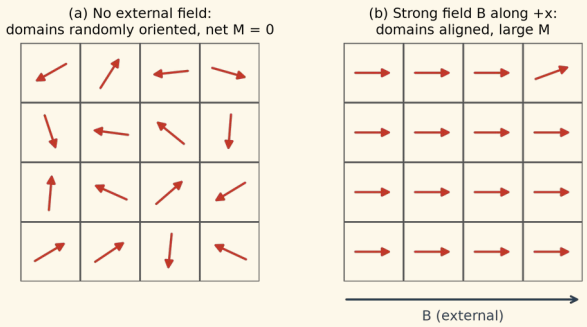


Fig. 5.4 Domains in a ferromagnet (a) without field, (b) in a strong field

Curie temperature. Heating increases thermal agitation. At the Curie temperature T_c the agitation finally destroys the spontaneous alignment inside domains, and the material becomes paramagnetic, obeying $\chi = C/(T - T_c)$ above T_c.

Substance	T_c (K)
Cobalt	1394
Iron	1043
Fe_2O_3	893
Nickel	631
Gadolinium	317

Atomic origin compared

Feature	Diamagnetic	Paramagnetic	Ferromagnetic
Electrons	all paired	some unpaired	unpaired, strongly coupled
Permanent atomic moment	none	yes, random	yes, aligned in domains
Cause of M	induced orbital change (Lenz)	partial alignment by B	growth and rotation of domains
M without field	zero	zero	can remain (retentivity)
Effect of heating	negligible	alignment decreases	para above T_c
Saturation	no	only at very low T, high B	yes, at moderate H

A useful way to remember the three classes: diamagnetism is a property of every atom but very weak; paramagnetism needs atoms that are already tiny magnets, but heat keeps them disordered; ferromagnetism needs, in addition, a strong interaction that makes neighbouring atomic magnets cooperate. That is why only a handful of elements (Fe, Co, Ni, Gd and some alloys) are ferromagnetic at room temperature, and why every ferromagnet loses this cooperation when heated above its Curie temperature.

Experimental test. When a sample is placed in a non-uniform field, a paramagnetic or ferromagnetic sample is pulled towards the region of stronger field, while a diamagnetic sample is pushed away. In a U-tube with one arm between the poles of an electromagnet, a paramagnetic solution (such as $FeCl_3$) rises in that arm, and a diamagnetic liquid (such as water) is depressed slightly. A frog or a drop of water can even be levitated in a very strong field because of its diamagnetism.

Hysteresis loop

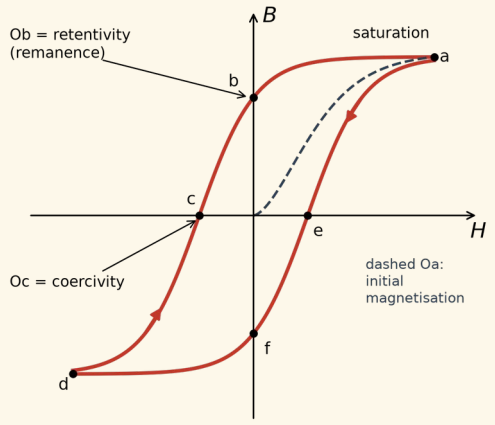


Fig. S.5 Hysteresis loop (B against H) of a ferromagnet

1. Start with unmagnetised iron and increase H: B rises along the curve Oa and finally levels off at a, where all domains are aligned (saturation).
2. Reduce H to zero: B does not return to zero but to the value Ob. This leftover field is the **retentivity** or **remanence**; the domains do not fully randomise again.
3. Apply H in the reverse direction: B becomes zero at c. The reverse intensity Oc needed is the **coercivity**.
4. Increase the reverse H further to reach reverse saturation d; then bring H back through zero (point f) and forward again through e to a.

The lagging of B behind H is called **hysteresis**. Every cycle wastes energy as heat because domain walls are dragged back and forth; this **hysteresis loss** per unit volume per cycle is proportional to the area of the loop. Materials for alternating fields must therefore have a thin, narrow loop.

Choosing a material: permanent magnet or electromagnet

Use	Required properties	Material
Permanent magnet	high retentivity, high coercivity (broad loop)	steel, alnico, cobalt steel, ticonal
Electromagnet	high permeability, low retentivity, low coercivity	soft iron
Transformer core, motor	low hysteresis loss (narrow loop), high permeability	soft iron, silicon steel

Steel has a smaller retentivity than soft iron but a much larger coercivity, so its magnetism is hard to remove: ideal for permanent magnets. Soft iron is preferred for electromagnets because its magnetism appears and disappears with the current. Permanent magnets can be made by placing a ferromagnetic rod inside a solenoid carrying current, or by stroking it repeatedly with a magnet in one direction. Electromagnets are used in electric bells, loudspeakers, telephone diaphragms, relays and the giant lifting magnets of cranes.

Solved example

Q. A paramagnetic salt has $\chi = 1.2 \times 10^{-5}$ at 27 °C. Find its susceptibility at −73 °C.

Sol. By Curie's law $\chi_2/\chi_1 = T_1/T_2 = 300/200$. $\chi_2 = 1.2 \times 10^{-5} \times 1.5 = \mathbf{1.8 \times 10^{-5}}$.

Yaad karo

- → Dia: saare electrons paired, induced moment B ke ulta (Lenz jaisa), T se farak nahi.
- → Para: unpaired electrons, garmi alignment bigaadti hai, $\chi \propto 1/T$.
- → Ferro: domains; T_c ke upar para ban jaata hai.
- → Loop ka area = har cycle mein heat loss. Transformer ke liye patla loop (soft iron).
- → Permanent magnet: high coercivity (steel); electromagnet: soft iron.

Magnetic materials in daily life

Application	Material used	Property exploited
MRI scanner, maglev train	superconducting coils	zero resistance, perfect diamagnetism
Hard disk, credit-card strip	ferromagnetic coating	domains retain the written direction
Fridge door seal, speaker	ferrite / permanent magnet	high retentivity
Transformer, motor, generator core	soft iron, silicon steel	high μ, small hysteresis loss
Scrap-lifting crane, relay, bell	soft-iron electromagnet	magnetism switches with current
Magnetic separation of ores	strong electromagnet	para/ferro particles pulled to strong field

Why is a compass needle made of a hard material but its case of a non-magnetic one? The needle must keep its magnetism for years (high coercivity), while a ferromagnetic case would concentrate the earth's field lines into itself and shield the needle. For the same reason, sensitive instruments are protected from stray fields by enclosing them in soft-iron boxes, a technique called **magnetic shielding**.

9. Solved Numericals (Board Pattern)

N1. A bar magnet of moment 0.32 J/T is placed in a field of 0.15 T at 30°. Find the torque.
Sol. $\tau = mB\sin\theta = 0.32 \times 0.15 \times 0.5 = \mathbf{0.024\ N\,m}$.

N2. For the same magnet, find the potential energy in stable and unstable positions.
Sol. Stable: $-mB = \mathbf{-0.048\ J}$; unstable: $+mB = \mathbf{+0.048\ J}$.

N3. A solenoid of 800 turns, cross-section 2.5×10^{-4} m^2, carries 3 A. Find its magnetic moment.
Sol. $m = NIA = 800 \times 3 \times 2.5 \times 10^{-4} = \mathbf{0.6\ J/T}$.

N4. A solenoid has a core with $\mu_r = 400$, $n = 1000\ m^{-1}$, $I = 2$ A. Find H, B, M.
Sol. $H = nI = \mathbf{2000\ A/m}$. $B = \mu_r\mu_0 H = 400 \times 4\pi \times 10^{-7} \times 2000 = \mathbf{1.0\ T}$. $M = (\mu_r - 1)H = \mathbf{7.98 \times 10^5\ A/m}$.

N5. A paramagnetic salt has $\chi = 2.8 \times 10^{-4}$ at 300 K. Find χ at 200 K.
Sol. $\chi \propto 1/T$: $\chi = 2.8 \times 10^{-4} \times 300/200 = \mathbf{4.2 \times 10^{-4}}$.

Extra solved examples

E1. A short bar magnet placed with its axis at 30° to a uniform field of 0.25 T experiences a torque of 4.5×10^{-2} N m. Find m.
Sol. $m = \tau/(B\sin\theta) = 0.045/(0.25 \times 0.5) = \mathbf{0.36\ J/T}$.

E2. Work needed to turn it from the most stable to the most unstable position?
Sol. $W = 2mB = 2 \times 0.36 \times 0.25 = \mathbf{0.18\ J}$.

E3. A domain in iron is a cube of side 1 μm with 8×10^{10} atoms, each of moment 9.27×10^{-24} A m^2. Find the maximum magnetisation.
Sol. $m = 8 \times 10^{10} \times 9.27 \times 10^{-24} = 7.4 \times 10^{-13}$ A m^2; $M = m/V = \mathbf{7.4 \times 10^5\ A/m}$.

Points to remember (theory)

- A bar magnet and a current-carrying solenoid produce identical external fields; $m = NIA$ for the solenoid.
- At equal distance, $B_{axial} = 2B_{equatorial}$; both fall as $1/r^3$ for a short magnet.
- In a uniform field the net force on a dipole is zero; only a torque acts. In a non-uniform field there is also a net force.
- $U = -mB\cos\theta$; work to turn a dipole from θ_1 to θ_2 is $mB(\cos\theta_1 - \cos\theta_2)$.
- $\oint B \cdot dS = 0$ for every closed surface, because magnetic monopoles do not exist.
- H depends only on free currents ($H = nI$); M depends on the material; $B = \mu_0(H + M)$.
- $\mu_r = 1 + \chi$: for diamagnets $0 \le \mu_r < 1$, for paramagnets slightly above 1, for ferromagnets of order 10^3 to 10^5.
- Curie's law $\chi = C\mu_0/T$ applies to paramagnets; above T_c a ferromagnet follows $\chi = C/(T - T_c)$.
- Area of the hysteresis loop = energy lost per unit volume per cycle; narrow loop for cores, broad loop for permanent magnets.
- Superconductors are perfect diamagnets: $\chi = -1$, $\mu_r = 0$ (Meissner effect).

Formula sheet: Chapter 5

Quantity	Formula
Torque on dipole	$\tau = m \times B$
Energy	$U = -mB\cos\theta$
Oscillation	$T = 2\pi\sqrt{(I/mB)}$
Axial / equatorial field	$\mu_0 2m/4\pi r^3$ / $\mu_0 m/4\pi r^3$
Gauss (magnetism)	$\oint B \cdot dS = 0$
Intensity	$H = nI$
Field in material	$B = \mu_0(H + M) = \mu H$
Susceptibility	$\chi = M/H$; $\mu_r = 1 + \chi$
Curie law	$\chi = C\mu_0/T$

1 minute recap

- ✓ No monopoles; field lines are closed loops.
- ✓ Bar magnet ≡ solenoid; $m = NIA$.
- ✓ $\tau = mB\sin\theta$, $U = -mB\cos\theta$.
- ✓ Net magnetic flux through a closed surface = 0.
- ✓ $B = \mu_0(H + M)$; χ sign tells dia, para or ferro.

Most Important Questions

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM The net magnetic flux through any closed surface is:
(a) positive
(b) negative
(c) zero
(d) infinite
Ans: (c) zero.

Q2. IM Board fav Susceptibility of a diamagnetic substance is:
(a) large and positive
(b) small and negative
(c) small and positive
(d) zero
Ans: (b) small and negative.

Q3. IM Above the Curie temperature, a ferromagnetic substance becomes:
(a) paramagnetic
(b) diamagnetic
(c) superconducting
(d) more ferromagnetic
Ans: (a) paramagnetic.

Q4. IM SI unit of magnetic intensity H is:
(a) tesla
(b) weber
(c) A m^2
(d) A m^{-1}
Ans: (d) A m^{-1}.

Q5. IM A superconductor has susceptibility:
(a) 0
(b) +1
(c) −1
(d) ∞
Ans: (c) −1.

Q6. IM A magnetic dipole is in stable equilibrium when the angle between m and B is:
(a) 0°
(b) 90°
(c) 180°
(d) 45°
Ans: (a) 0°.

Q7. IM Board fav Why do magnetic field lines form closed loops?
Ans: Because magnetic monopoles do not exist.

Q8. IM State Curie's law.
Ans: For a paramagnet, $\chi = C\mu_0/T$: susceptibility is inversely proportional to absolute temperature.

Q9. IM Name two diamagnetic and two paramagnetic substances.
Ans: Dia: bismuth, copper. Para: aluminium, sodium.

Q10. IM Which material is used for permanent magnets? Why?
Ans: Hard ferromagnets (alnico, steel): high retentivity and coercivity.

Q11. IM What is the Meissner effect?
Ans: Complete expulsion of magnetic field from a superconductor.

Q12. IM Define magnetic susceptibility.
Ans: $\chi = M/H$, ratio of magnetisation to magnetic intensity.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. IM Board fav **A:** Gauss's law in magnetism gives zero flux. **R:** Isolated magnetic poles do not exist.
Ans: (a)

Q14. IM **A:** A diamagnetic substance moves from a stronger to a weaker field region. **R:** Its induced magnetisation is opposite to the applied field.
Ans: (a)

Q15. IM **A:** Soft iron is used for electromagnets. **R:** Soft iron has high retentivity.
Ans: (c) A true; R false (soft iron has low retentivity and coercivity).

Q16. IM **A:** The susceptibility of a paramagnet increases with temperature. **R:** Thermal agitation disturbs the alignment of dipoles.
Ans: (d) A false (χ decreases); R true.

C. Short answer (2–3 marks)

Q17. 3M Board fav Show that a current-carrying solenoid behaves like a bar magnet.

Ans: Far axial field of a solenoid $B = \mu_0 2m/4\pi r^3$ with $m = n(2l)I\pi a^2$, same as that of a bar magnet; field line patterns are identical.

Q18. 2M Derive the potential energy of a magnetic dipole in a uniform field.

Ans: $W = \int mB \sin\theta\, d\theta$ from 90° to $\theta = -mB\cos\theta \Longrightarrow U = -\mathbf{m}\cdot\mathbf{B}$.

Q19. 2M State Gauss's law in magnetism. How does it differ from that in electrostatics?

Ans: $\oint \mathbf{B}\cdot d\mathbf{S} = 0$ (no monopoles); for E, $\oint \mathbf{E}\cdot d\mathbf{S} = q/\varepsilon_0$.

Q20. 3M Board fav Distinguish dia-, para- and ferromagnetic substances (three points).

Ans: χ: negative / small positive / large positive. μ_r: < 1 / > 1 / $\gg 1$. Behaviour in non-uniform field: weak to strong region reversed for dia.

Q21. 2M Define magnetisation and magnetic intensity. Relate them to B.

Ans: M = moment per volume; $H = nI$; $B = \mu_0(H + M)$.

Q22. 3M Explain ferromagnetism using domain theory.

Ans: Spontaneously aligned domains; external field grows aligned domains and rotates others, giving large M.

Q23. 2M What happens to a ferromagnet when heated above T_c?

Ans: Domain alignment is destroyed; it becomes paramagnetic with $\chi = C/(T - T_c)$.

Q24. 2M Give two differences between electric and magnetic field lines.

Ans: Electric lines start and end on charges, magnetic form closed loops; electric lines do not exist inside a conductor, magnetic lines pass through a magnet.

Q25. 2M A magnetic needle in a uniform field oscillates. On what factors does its period depend?

Ans: $T = 2\pi\sqrt{(I/mB)}$: moment of inertia, magnetic moment and field.

Q26. 2M Why is the susceptibility of diamagnetic substances independent of temperature?

Ans: Their moments are induced (no permanent dipoles to be disordered by heat).

D. Long answer (5 marks)

Q27. 5M Board fav (a) Define χ, μ_r, M, H and relate them. (b) Classify materials on the basis of χ with examples. (c) State Curie's law.

Ans: (a) $M = \chi H$, $B = \mu_0(1 + \chi)H$. (b) dia/para/ferro table. (c) $\chi = C\mu_0/T$.

Q28. 5M (a) Derive the torque on a magnetic dipole in a uniform field. (b) Derive its potential energy. (c) Give the conditions of stable and unstable equilibrium.

Ans: (a) $\tau = mB\sin\theta$. (b) $U = -mB\cos\theta$. (c) Stable $\theta = 0^\circ$, unstable $\theta = 180^\circ$.

E. Case-based

Q29. 4M Board fav Maglev trains float above tracks using superconducting magnets. (a) What property of superconductors is used? (b) Value of χ? (c) Is a superconductor attracted or repelled by a magnet? (d) Name the effect.

Ans: (a) Perfect diamagnetism. (b) -1. (c) Repelled. (d) Meissner effect.

Q30. 4M Transformer cores are made of soft iron. (a) Which class of material? (b) Why soft iron and not steel? (c) What happens above 1043 K? (d) What are domains?

Ans: (a) Ferromagnetic. (b) Low retentivity and coercivity, low hysteresis loss. (c) Becomes paramagnetic. (d) Regions of aligned atomic moments.

Q31. 2M Board fav A bar magnet of moment 1.5 J/T lies along a field of 0.22 T. Work done to turn it to point opposite to the field?

Ans: $W = 2mB = 2 \times 1.5 \times 0.22 =$ **0.66 J.**

Q32. 2M A solenoid with iron core ($\mu_r = 1000$), $n = 500\ m^{-1}$, $I = 0.5$ A. Find B.

Ans: $B = \mu_r\mu_0 nI = 1000 \times 4\pi \times 10^{-7} \times 250 =$ **0.31 T.**

Q33. 1M Why does a paramagnetic sample show greater magnetisation when cooled?
Ans: Less thermal agitation, so dipoles align better (Curie's law).

Q34. 2M If a bar magnet is cut into two equal halves along its length, what happens to the pole strength and moment of each?
Ans: Pole strength halves (area halves); moment halves.

Q35. 1M Why is a magnetic needle not pulled as a whole in a uniform field?
Ans: Equal and opposite forces on the poles: net force zero, only a torque acts.

Q36. 1M Which has the highest Curie temperature among Fe, Co and Ni?
Ans: Cobalt (1394 K).

Q37. 1M A sample is placed in a non-uniform field and moves to the weaker region. Name its type.
Ans: Diamagnetic.

Q38. 1M Give the relation between μ_r and χ.
Ans: $\mu_r = 1 + \chi$.

Q39. 3M Board fav A magnetic needle ($m = 6.7 \times 10^{-2}$ A m^2, $I = 7.5 \times 10^{-6}$ kg m^2) makes 10 oscillations in 6.7 s. Find B.
Ans: $T = 0.67$ s; $B = 4\pi^2 I/mT^2 = 4\pi^2 \times 7.5 \times 10^{-6}/(6.7 \times 10^{-2} \times 0.449) =$ **0.01 T**.

Q40. 2M Can a magnetic monopole exist? What would change in Gauss's law?
Ans: None has been observed. If it existed, $\oint B \cdot dS$ would equal μ_0 times the enclosed pole strength.

Magnetism and Matter khatam: chhota chapter, bade marks
Asli baat: Most asked: dia-para-ferro table, Gauss's law in magnetism, bar magnet as solenoid, Curie's law.

Q41. 2M Board fav Why is soft iron used in electromagnets and steel in permanent magnets?
Ans: Soft iron: low retentivity and coercivity, magnetises and demagnetises easily. Steel: high coercivity, keeps magnetism.

Q42. 2M A bar magnet is placed with its axis at 60° to a field of 0.2 T; $\tau = 0.06$ N m. Find m.
Ans: $m = \tau/(B \sin 60°) = 0.06/(0.2 \times 0.866) =$ **0.35 J/T**.

Q43. 1M What is the angle between m and B for maximum torque?
Ans: 90°.

Q44. 1M Why is the magnetic susceptibility of a paramagnet higher at low temperature?
Ans: Thermal agitation is less, so dipoles align more easily (Curie's law).

Q45. 2M Two identical magnets with moment m are placed at right angles with like poles touching. Net moment?
Ans: $\sqrt{2}$ m, at 45° to each.

Q46. 1M Name the property of a material that decides whether it is suitable for a transformer core.
Ans: Low hysteresis loss, high permeability, low retentivity (soft iron).

Q47. 2M Draw field lines of a current-carrying solenoid and compare with a bar magnet.
Ans: Uniform parallel lines inside, spreading outside from one end to the other: same as a bar magnet; the end with anticlockwise current is N.

Q48. 1M Why does a freely suspended magnet align north-south?
Ans: The earth behaves like a huge magnet; its field exerts a torque $m \times B$ until m is along B.

Q49. 1M What is the SI unit of magnetic moment? Give two equivalent forms.
Ans: A m^2 = J/T.

Q50. 2M Board fav If χ of a material is -4.2×10^{-6}, name its type and find μ_r.
Ans: Diamagnetic; $\mu_r = 1 + \chi =$ **0.9999958**.

Quick compare: H, B, M

Symbol	Meaning	Unit
H	magnetic intensity (from currents)	A/m
M	magnetisation (from material)	A/m
B	total field $\mu_0(H + M)$	T

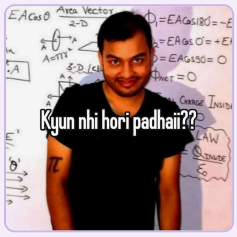
Paheli-Bujho

Paheli: Bujho, ek line mein: dia, para, ferro?

Bujho: Dia door bhaage, para thoda paas aaye,
ferro chipak jaaye!

Paheli: Aur Curie temperature ke upar ferro?

Bujho: Para ban jaata hai. Garmi mein sab dheele pad jaate hain.



Magnetism and Matter: kyu nahi ho rahi padhai? Kyunki chapter chhota hai

Asli baat: Short chapter, sure marks: revise the dia-para-ferro table and Gauss's law twice.

Common mistakes

→ Paramagnetic ka χ temperature badhne pe kam hota hai, badhta nahi.

→ Field lines magnet ke andar S se N jaati hain.

→ Soft iron: low retentivity (electromagnet); steel: high (permanent magnet).

→ Superconductor ka $\mu_r = 0$, $\chi = -1$.

Exam Alert: last check

→ Dia-para-ferro table, Curie law, Meissner effect: sabse zyada aate hain.

→ Electric vs magnetic dipole comparison table bhi pakka.

Electromagnetic Induction

Most asked: Faraday's and Lenz's laws, motional emf, self inductance of a solenoid, mutual inductance of two coaxial solenoids, energy stored in an inductor, AC generator

Chapter 4 showed that current makes a magnetic field. Faraday asked the reverse: **can a magnetic field make a current?** Yes, but only when the magnetic flux **changes**. Every generator and transformer runs on this idea.

Faraday: magnet hilao, current paao.
Guarantee deta hai?
Asli baat: Yes: any change in magnetic flux through a coil induces an emf. No change, no emf, however strong the magnet.

1. Experiments of Faraday and Henry

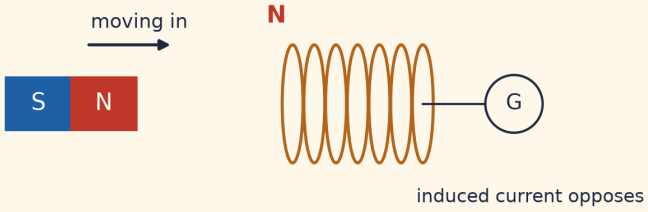


Fig. 6.1 A magnet moved towards a coil induces a current, shown by the galvanometer

1. Moving a magnet towards or away from a coil deflects a galvanometer; no deflection when the magnet is at rest.
2. Faster motion gives a larger deflection.
3. Switching the current ON or OFF in a nearby coil induces a current in the second coil (with a steady current, none).
4. An iron rod in the coils increases the effect.

Conclusion: an emf is induced whenever the magnetic flux linked with a coil changes.

NCERT Activity: Faraday's two-coil experiment

Wind two coils C_1 and C_2 close to each other. Connect C_1 to a battery through a key and C_2 to a galvanometer. On pressing the key, G deflects momentarily; on releasing it, G deflects the other way; with the key held down (steady current), there is no deflection. A changing current in one coil induces a current in the other.

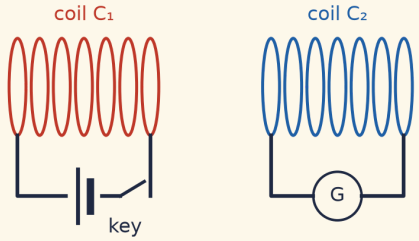


Fig. Faraday's experiment with two coils

NCERT Activity: magnet and coil

Connect a coil to a galvanometer and push the N pole of a bar magnet into it: G deflects. Pull it out: G deflects the opposite way. Hold it still: no deflection. Move the coil instead of the magnet: same result. Only relative motion (a changing flux) matters.

Why these experiments matter

Before 1830, electricity and magnetism were linked in only one direction: Oersted had shown that an electric current produces a magnetic field. Michael Faraday in England and Joseph Henry in the USA, working independently, showed the reverse link. Their experiments proved that a magnetic field can drive a current, but **only while something is changing**. A strong magnet resting inside a coil produces no current at all, while a weak magnet moved quickly produces a clear deflection. This single observation is the foundation of electrical power generation, transformers, induction cooktops and wireless chargers.

In each experiment the galvanometer shows a current although the second circuit contains no battery. The current is called the **induced current**, and the emf that drives it is the **induced emf**. The phenomenon is called **electromagnetic induction**. Three common features run through all the experiments: (i) there is relative motion between a magnet and a coil, or a changing current in a neighbouring coil; (ii) the deflection lasts only as

long as the change lasts; (iii) the direction of deflection reverses when the change reverses (approach versus recession, switch ON versus switch OFF).

Experiment	What changes	Observation
Magnet pushed into coil	B through the coil increases	G deflects one way
Magnet pulled out	B through the coil decreases	G deflects the other way
Magnet held at rest	nothing	no deflection
Key of primary pressed	current in C_1 rises, so B in C_2 rises	momentary deflection
Key held down	steady current, no change	no deflection
Key released	current falls to zero	opposite momentary deflection

Key idea

It is not the magnetic field, but the **change** in the magnetic field linked with a circuit, that induces an emf. Relative motion is what matters: moving the coil towards a fixed magnet gives exactly the same result as moving the magnet towards a fixed coil.

2. Magnetic Flux

$$\Phi_B = B \cdot A = BA\cos\theta \quad \text{SI unit: weber (Wb)} = \text{T m}^2$$

→ θ is the angle between B and the normal to the area.

→ Flux is a scalar. For a non-uniform field, $\Phi = \int B \cdot dA$.

→ Flux can be changed by changing B, A, or θ.

Solved example

Q. A square loop of side 10 cm lies in the xy-plane in B = 0.3 T along z. Find the flux. What if B makes 60° with z?

Sol. $\Phi = BA = 0.3 \times 0.01 = \mathbf{3 \times 10^{-3}\ Wb}$; at 60°, $\Phi = BA\cos 60° = \mathbf{1.5 \times 10^{-3}\ Wb}$.

Understanding flux physically

Magnetic flux measures how many magnetic field lines pass through a surface. If field lines are pictured as threads, flux is the number of threads crossing the loop. It is largest when the loop faces the field squarely (θ = 0°, Φ = BA) and zero when the loop is turned edge-on to the field (θ = 90°), because then no line crosses its face. For a coil of N identical turns, each turn links the same flux, so the total flux linkage is $N\Phi = NBA\cos\theta$.

Flux is a scalar quantity although both B and A are vectors: it is their dot product. The area vector A has magnitude equal to the area and direction along the normal to the surface. Since B is measured in tesla, 1 Wb = 1 T m^2; equivalently 1 T = 1 Wb m^{-2}, which is why B is also called the magnetic flux density.

Angle θ (B with normal)	Flux Φ	Loop orientation
0°	BA (maximum)	plane of loop ⊥ B
60°	BA/2	tilted
90°	zero	plane of loop ∥ B
180°	−BA	loop flipped over

Yaad karo

→ θ hamesha B aur area ke **normal** ke beech, plane ke beech nahi.

→ Loop ko 180° palatne par flux BA se −BA: change 2BA hota hai, BA nahi.

3. Faraday's Law of Induction

The magnitude of the induced emf equals the rate of change of magnetic flux through the circuit:

$$\varepsilon = -\frac{d\Phi_B}{dt}; \quad \text{for N turns: } \varepsilon = -N\frac{d\Phi_B}{dt}$$

Flux change nahi hua toh emf bhi nahi, jaise dimaag bina chai ke

Asli baat: ε depends on how fast flux changes, not on how much flux there is.

Ways to induce an emf

1. Change B (move a magnet, change a nearby current).
2. Change the area A (stretch or shrink the loop, slide a rod).
3. Change the angle θ (rotate the coil: AC generator).

Solved example (NCERT)

Q. A 20-turn coil of area 5×10^{-3} m^2 is in a field that drops from 0.1 T to 0 in 0.05 s. Find ε.

Sol. $\varepsilon = NA\Delta B/\Delta t = 20 \times 5 \times 10^{-3} \times 0.1/0.05 =$ **0.2 V**.

Meaning of each term

Faraday summarised his experiments in a simple law: **the magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit**. The emf depends on how fast the flux changes, not on the value of the flux itself. A coil can link a huge flux and still have zero emf if that flux is constant; on the other hand a small flux switched off very quickly can give a large emf. For a coil of N closely wound turns, the emfs of the individual turns are in series and add up, giving $\varepsilon = -N\, d\Phi/dt$. The negative sign expresses Lenz's law, discussed next.

If the circuit has resistance R, the induced current is $I = \varepsilon/R$. The charge that flows during a flux change from Φ_1 to Φ_2 is obtained by integrating $I\,dt$:

$q = \int I\,dt = (N/R)\int d\Phi = N\,\Delta\Phi / R$ independent of the rate of change

This is a useful result: moving a magnet slowly or quickly gives different emfs and currents, but the same total charge through the galvanometer.

Solved example

Q. A circular coil of 50 turns and radius 10 cm lies with its plane perpendicular to a uniform field of 0.2 T. It is turned through 180° in 0.1 s. Find the average induced emf.

Sol. $A = \pi(0.1)^2 = 0.0314$ m^2. Initial flux linkage NBA, final −NBA, so $|\Delta\Phi| = 2NBA = 2 \times 50 \times 0.2 \times 0.0314 = 0.628$ Wb. $\varepsilon = \Delta\Phi/\Delta t = 0.628/0.1 =$ **6.28 V**.

Solved example

Q. A coil of 200 turns, area 4×10^{-3} m^2 and resistance 10 Ω is held perpendicular to a field of 0.5 T. The field is switched off. Find the charge that flows through the coil.

Sol. $q = N\Delta\Phi/R = NBA/R = 200 \times 0.5 \times 4 \times 10^{-3}/10 =$ **0.04 C**, whatever the time taken for the field to fall.

4. Lenz's Law and Conservation of Energy

The **polarity** of the induced emf is such that the current it produces **opposes the change** in flux that caused it. That is the minus sign in Faraday's law.

- → North pole approaching a coil: the near face becomes N (repels).
- → North pole receding: the near face becomes S (attracts back).
- → Lenz's law follows from conservation of energy: if the induced current helped the change, energy would be created from nothing.

Magnet paas laaya aur coil ne ulta push kar diya

Asli baat: Lenz's law: nature always opposes the change in flux. The work you do against this opposition becomes electrical energy.

Exam Alert

- → Lenz's law energy conservation se kaise: 2 marks pakka.
- → Coil ke face ka N/S batao: approaching magnet ka same pole, receding ka opposite.

Using Lenz's law: examples

Situation	*Induced current*
N pole approaching coil	near face becomes N, repels
N pole moving away	near face becomes S, attracts
Current in nearby loop switched ON	opposite direction (opposes increase)
Current switched OFF	same direction (opposes decrease)

Eddy currents

When a bulk metal piece moves in (or sits in) a changing field, circulating currents called **eddy currents** are induced inside it. They oppose the motion (Lenz) and heat the metal.

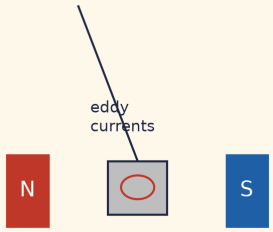


Fig. A metal plate swinging between poles is damped by eddy currents

→ **Uses**: electromagnetic braking in trains, induction furnace, electric power meters, dead-beat galvanometers.

→ **Reducing them**: laminated cores, slots in metal plates.

NCERT Activity: damped pendulum

Swing a copper plate between the poles of a strong magnet: it stops quickly. Cut slots into the plate and repeat: it swings much longer, because slots break the eddy current paths.

Lenz's law explained step by step

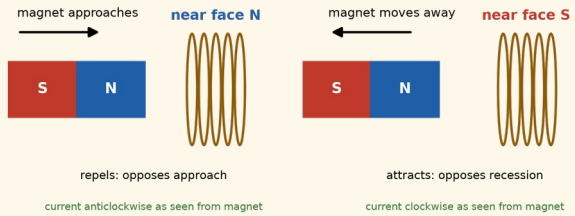


Fig. 6.5 Lenz's law: the near face of the coil always opposes the motion of the magnet

Lenz's law (1834) gives the direction of the induced current: **the induced current flows in such a direction that it opposes the change in magnetic flux that produces it**. To apply it, follow three steps: (1) find whether the flux through the coil is increasing or decreasing, and in which direction; (2) the induced current must set up its own field opposite to the original field if the flux is increasing, and along it if the flux is decreasing; (3) use the right-hand thumb rule to get the direction of current that produces this field.

When the N pole of a magnet approaches a coil, the flux towards the coil increases. The coil opposes this by making its near face a north pole, which repels the magnet; seen from the magnet, the current is anticlockwise. When the N pole is withdrawn, the flux decreases; the coil tries to keep it by making its near face a south pole, which attracts the receding magnet; the current is now clockwise. In both cases the coil resists the motion.

Why Lenz's law must be true: energy conservation

Suppose the induced current helped the change instead of opposing it. An approaching N pole would then create a south pole on the near face, which would pull the magnet in faster. The faster motion would increase the induced current, which would pull harder still. The magnet would accelerate and the current would grow without any work being done from outside, producing kinetic energy and electrical energy from nothing. This violates the law of conservation of energy, so it cannot happen. Because the induced current opposes the motion, we must do mechanical work to push the magnet in or pull it out, and **this work is exactly what appears as electrical energy**, finally dissipated as Joule heat in the coil.

Important notes on Lenz's law

→ Lenz's law opposes the **change** in flux, not the flux itself.

→ If the coil is open (no closed path), an emf is still induced but no current flows, so there is no opposing force.

→ The minus sign in $\varepsilon = -N\, d\Phi/dt$ is the mathematical form of Lenz's law.

Eddy currents: why they form

Eddy currents were discovered by Foucault (1895), so they are also called Foucault currents. A solid metal plate can be thought of as made of countless closed conducting loops. When the flux through these loops changes, because the plate moves in a field or the field around it varies, an emf is induced in every loop and currents swirl around inside the metal like eddies in water. By Lenz's law these currents oppose the cause: a moving plate is slowed down, and the energy of motion is converted into heat. In a transformer or motor core the swirling currents waste energy as heat, so the core is made of thin **laminations** separated by an insulating varnish. The laminations are placed so that the insulation lies across the path of the eddy currents, which reduces their size and hence the loss.

Application	How eddy currents help
Magnetic braking in trains	strong electromagnets near the rails induce eddy currents that oppose motion; smooth braking with no wear
Electromagnetic damping	a coil wound on a metal frame in a galvanometer stops quickly (dead-beat)
Induction furnace	high-frequency AC induces large eddy currents that melt metals
Electric power meter	a shiny aluminium disc rotates due to eddy currents and counts energy used

Yaad karo

→ Magnet aata hai: near face same pole (repel). Magnet jaata hai: near face opposite pole (attract).

→ Eddy current: fayda braking, furnace, meter, damping; nuksaan heat loss, isliye laminated core.

5. Motional Electromotive Force

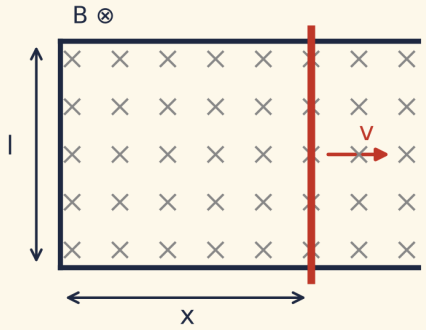


Fig. 6.2 Rod of length l moving on rails in a uniform field

A rod PQ of length l moves with speed v on frictionless rails in a field B ⊥ to the plane. Flux enclosed $\Phi = Blx$, and $dx/dt = v$:

$$\varepsilon = Blv$$

Lorentz force explanation

Each free charge q in the rod feels qvB along the rod. Work done in moving it through l is qvBl, so emf = work per unit charge = Blv.

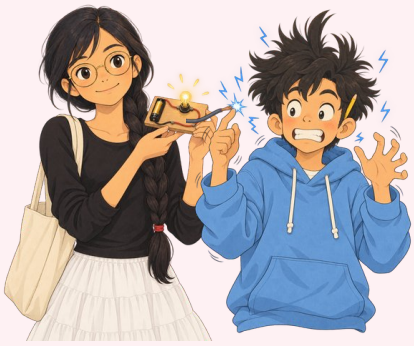
→ Current $I = Blv/R$; force needed to keep the rod moving $F = IlB = B^2l^2v/R$.

→ Power supplied $P = Fv = B^2l^2v^2/R = I^2R$ (all of it appears as heat): energy is conserved.

Rotating rod

$\varepsilon = \frac{1}{2} B\omega l^2$ rod of length l rotating about one end with angular speed ω

Paheli-Bujho



Bujho: Paheli, rod chal rahi hai, koi battery nahi, phir bhi current?

Paheli: Haan! Rod ke electrons B mein chal rahe hain, unpe qvB lagta hai. Yahi motional emf hai.

Bujho: Matlab rod khud battery ban gayi.

Paheli: Aur usko chalane ke liye jo mehnat karoge, wahi bijli banti hai.

Derivation step by step (motional emf via flux)

1. Area enclosed by the rod and rails: A = lx.
2. Flux: $\Phi = Blx$.
3. Rod moves: $dx/dt = -v$ (area decreasing), so $\varepsilon = -d\Phi/dt = Blv$.

Solved example (NCERT)

Q. A 1 m metal rod rotates at 400 rad/s about an axis through one end, ⊥ to it, in a field of 0.5 T. Find the emf between its ends.

Sol. $\varepsilon = \frac{1}{2}B\omega l^2 = \frac{1}{2} \times 0.5 \times 400 \times 1 =$ **100 V.**

Origin of motional emf: Lorentz force in detail

Motional emf is the emf induced in a conductor that moves through a magnetic field. Consider the rod PQ moving to the left with velocity v in a field B directed into the plane. Every free electron in the rod moves with the rod, so it feels a Lorentz force $F = q(v \times B)$ directed along the length of the rod. The free charges are pushed towards one end, making one end positive and the other negative. The rod therefore behaves exactly like a cell: it has an emf even though there is no chemical reaction. The work done by the magnetic force on a charge q moved from one end to the other is W = qvBl, so the emf, which is work done per unit charge, is $\varepsilon = W/q = Blv$. This derivation shows that motional emf does not need the idea of flux at all; the flux method and the Lorentz-force method give the same answer.

Energy consideration: force, power and heat

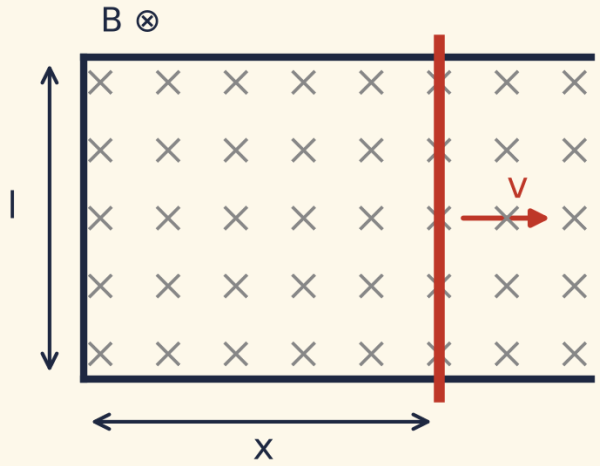


Fig. 6.6 Sliding rod: the induced current I in the field B gives a retarding force F = IlB

Let the rod of length l have resistance r and the rest of the circuit be ideal. The induced current is $I = Blv/r$. Since this current flows through the rod lying in the field, the rod feels a force $F = IlB = B^2l^2v/r$. By Lenz's law this force is directed **opposite to the velocity**: it tries to stop the rod. To keep the rod moving at constant speed an external agent must apply an equal and opposite force, and the power it supplies is

$P = Fv = B^2l^2v^2/r$ Joule heat per second: $I^2r = (Blv/r)^2 r = B^2l^2v^2/r$

The mechanical power supplied is exactly equal to the electrical power dissipated as heat. Mechanical energy is first converted into electrical energy and then into thermal energy, which is a direct check of Lenz's law and energy conservation. If the external force is removed, the retarding force slows the rod down until it stops.

Solved example

Q. A rod of length 0.5 m slides at 10 m/s on rails perpendicular to a field of 0.4 T. The total resistance of the circuit is 2 Ω. Find the emf, current, force needed for uniform motion and the power dissipated.

Sol. $\varepsilon = Blv = 0.4 \times 0.5 \times 10 = \mathbf{2\ V}$; $I = 2/2 = \mathbf{1\ A}$; $F = IlB = 1 \times 0.5 \times 0.4 = \mathbf{0.2\ N}$; $P = Fv = 0.2 \times 10 = \mathbf{2\ W}$, which equals $I^2R = 1 \times 2 = 2$ W.

Rotating rod: derivation of $\varepsilon = \tfrac{1}{2}B\omega l^2$

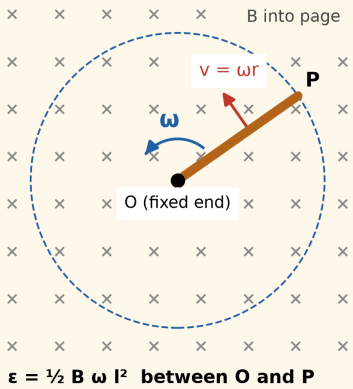


Fig. 6.7 A rod rotating about one end in a uniform field

A rod OP of length l rotates with angular speed ω about an axis through O, perpendicular to the rod and parallel to B. Different points of the rod move with different speeds, so we cannot use $\varepsilon = Blv$ directly. Take a small element dr at distance r from O. Its speed is $v = \omega r$, so the emf across it is $d\varepsilon = Bv\,dr = B\omega r\,dr$. All elements are in series, so

$\varepsilon = \int_0^l B\omega r\,dr = \tfrac{1}{2}B\omega l^2$ equivalently, Blv with v = average speed $\omega l/2$

The same result follows from flux: in one rotation (time $2\pi/\omega$) the rod sweeps area πl^2, so the rate of sweeping flux is $B\pi l^2\omega/2\pi = \tfrac{1}{2}B\omega l^2$. For a wheel with many metal spokes, all spokes are in parallel between the axle and the rim, so the emf equals that of one spoke and is **independent of the number of spokes.**

Solved example

Q. A wheel with 10 metal spokes, each 0.5 m long, rotates at 120 rev/min in a plane normal to the vertical component of the Earth's field, 0.4 G. Find the emf between the axle and the rim.

Sol. $\omega = 2\pi \times 2 = 4\pi$ rad/s; $B = 0.4 \times 10^{-4}$ T. $\varepsilon = \tfrac{1}{2}B\omega l^2 = \tfrac{1}{2} \times 0.4 \times 10^{-4} \times 4\pi \times 0.25 = \mathbf{6.28 \times 10^{-5}\ V}$, the same for any number of spokes.

6. Inductance

The flux through a coil is proportional to the current producing it: $N\Phi \propto I$. The constant of proportionality is the **inductance.**

Self inductance

$N\Phi_B = LI$, $\varepsilon = -L\,dI/dt$ SI unit: henry (H) = Wb A^{-1}

For a long solenoid (n turns/length, length l, area A): $B = \mu_0 nI$, $N\Phi = (nl)(\mu_0 nI)A$:

$L = \mu_0 n^2 A l$; with a core: $L = \mu_r \mu_0 n^2 A l$

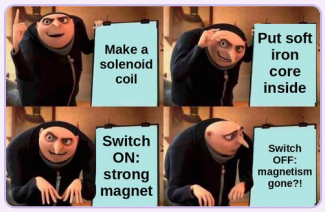
→ L depends only on geometry and the medium.

→ Self-induced emf opposes the change in current (back emf): inductance is the **electrical inertia** of a circuit.

Energy stored in an inductor

Work done against back emf while current grows from 0 to I: $W = \int LI\, dI$:

$U = \frac{1}{2} LI^2$ compare KE = $\frac{1}{2}mv^2$: L acts like mass



Inductor ka plan: current badhao toh roko, ghatao toh bhi roko

Asli baat: Self inductance opposes any change in current, like inertia opposes change in velocity.

Mutual inductance

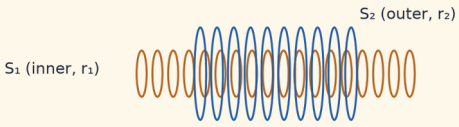


Fig. 6.3 Two long coaxial solenoids

$N_1\Phi_1 = M_{12} I_2, \quad \varepsilon_1 = -M\, dI_2/dt$

For coaxial solenoids (inner radius r_1, turns n_1, n_2 per length, length l):

$$M_{12} = M_{21} = \mu_0 n_1 n_2 \pi r_1^2 l$$

→ M depends on geometry, separation, orientation and medium.

→ With an iron core, $M = \mu_r \mu_0 n_1 n_2 \pi r_1^2 l$.

→ For tightly coupled coils, $M = \sqrt{(L_1 L_2)}$.

Yaad karo

→ $L = \mu_0 n^2 A l$ aur $M = \mu_0 n_1 n_2 \pi r_1^2 l$: dono derivations 3-mark wale.

→ Inductor ki energy $\frac{1}{2}LI^2$, capacitor ki $\frac{1}{2}CV^2$.

Inductor in a circuit

→ On switching ON a circuit with an inductor, the current grows gradually (back emf).

→ On switching OFF suddenly, a large emf L dI/dt appears and can cause a spark across the switch.

→ A choke (inductor) is used in old tube lights to provide the high starting voltage.

Solved example

Q. Two coaxial solenoids, each 50 cm long: inner radius 2 cm, 400 turns; outer 1000 turns. Find M.

Sol. $n_1 = 800/m$, $n_2 = 2000/m$; $M = \mu_0 n_1 n_2 \pi r_1^2 l = 4\pi \times 10^{-7} \times 800 \times 2000 \times \pi \times 4 \times 10^{-4} \times 0.5 =$ **1.26 mH.**

Mutual inductance of coaxial solenoids: derivation

1. Let the outer solenoid S_2 (n_2 turns per unit length) carry current I_2. Its field inside is $B_2 = \mu_0 n_2 I_2$, uniform over the inner solenoid.
2. The inner solenoid S_1 has radius r_1, length l and $n_1 l$ turns. Flux linkage: $N_1\Phi_1 = (n_1 l)(\pi r_1^2)(\mu_0 n_2 I_2)$.
3. Comparing with $N_1\Phi_1 = M_{12} I_2$: $M_{12} = \mu_0 n_1 n_2 \pi r_1^2 l$.
4. Now pass I_1 in S_1. Its field $\mu_0 n_1 I_1$ exists only inside S_1 (area πr_1^2), so the flux linkage with S_2 is $(n_2 l)(\pi r_1^2)(\mu_0 n_1 I_1)$, giving $M_{21} = \mu_0 n_1 n_2 \pi r_1^2 l = M_{12}$.

The equality $M_{12} = M_{21}$ holds for any pair of coils, not just solenoids. Physically, mutual inductance measures how well two coils share magnetic flux. It is larger when the coils are close, coaxial and wound on a common iron core, and small when they are far apart or perpendicular to each other.

Self inductance of a solenoid: derivation

1. A long solenoid has length l, cross-section A, n turns per unit length (total N = nl) and carries current I.
2. Field inside: $B = \mu_0 nI$. Flux through one turn: $BA = \mu_0 nIA$.
3. Total flux linkage: $N\Phi = (nl)(\mu_0 nIA) = \mu_0 n^2 AlI$.
4. Since $N\Phi = LI$: $L = \mu_0 n^2 Al = \mu_0 N^2 A/l$.

Self inductance depends on the square of the number of turns. Doubling the turns on the same length doubles the field and also doubles the number of turns that link it, so L becomes four times. A soft-iron core of relative permeability μ_r multiplies L by μ_r, which can be several thousand.

Energy stored in an inductor: step by step

1. While the current grows, the inductor produces a back emf $\varepsilon = L\, dI/dt$ opposing the growth.
2. The source must do work against it. In time dt, charge $dq = I\, dt$ moves through the back emf, so $dW = \varepsilon I\, dt = LI\, dI$.

3. Total work as current rises from 0 to I: $W = \int_0^I LI\,dI = \frac{1}{2}LI^2$.
4. This work is stored as magnetic energy in the field of the inductor and is returned when the current falls.

For a solenoid, $U = \frac{1}{2}LI^2 = \frac{1}{2}(\mu_0 n^2 Al)I^2 = (B^2/2\mu_0) \times Al$, since $B = \mu_0 nI$. Hence the **magnetic energy density** is $u_B = B^2/2\mu_0$, the magnetic counterpart of the electric energy density $\frac{1}{2}\varepsilon_0 E^2$.

Self inductance	Mutual inductance
emf in a coil due to change of current in the same coil	emf in a coil due to change of current in a neighbouring coil
$N\Phi = LI$; $\varepsilon = -L\,dI/dt$	$N_1\Phi_1 = MI_2$; $\varepsilon_1 = -M\,dI_2/dt$
$L = \mu_0 n^2 Al$	$M = \mu_0 n_1 n_2 \pi r_1^2 l$
depends on one coil's geometry and core	depends on both coils, separation, orientation and core
example: choke, back emf	example: transformer, wireless charger

Mechanics	Inductor
mass m (inertia)	inductance L
velocity v	current I
$KE = \frac{1}{2}mv^2$	$U = \frac{1}{2}LI^2$
force = m dv/dt	emf = L dI/dt

Solved example

Q. A solenoid of 1000 turns, length 0.5 m and cross-section 10 cm^2 carries 2 A. Find its self inductance and the energy stored.

Sol. $L = \mu_0 N^2 A/l = 4\pi \times 10^{-7} \times 10^6 \times 10^{-3}/0.5 =$ **2.51 mH**. $U = \frac{1}{2}LI^2 = \frac{1}{2} \times 2.51 \times 10^{-3} \times 4 =$ **5.03 mJ**.

Solved example

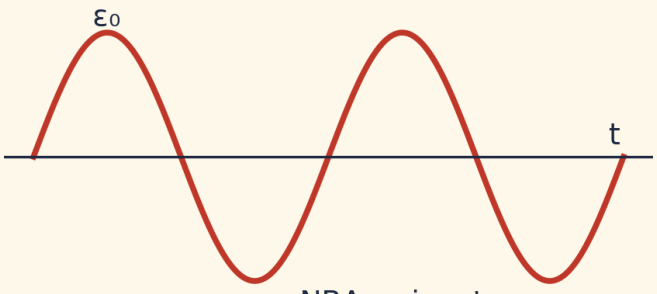
Q. A pair of coils has M = 1.5 H. The current in one coil rises from 0 to 20 A in 0.5 s. Find the emf in the other coil and its change in flux linkage.

Sol. $\varepsilon = M\,\Delta I/\Delta t = 1.5 \times 20/0.5 =$ **60 V**; $\Delta(N\Phi) = M\Delta I = 1.5 \times 20 =$ **30 Wb**.

7. AC Generator

A coil of N turns, area A rotates with angular speed ω in a uniform field B. At time t, $\theta = \omega t$ and $\Phi = NBA\cos\omega t$:

$$\varepsilon = NBA\omega \sin\omega t = \varepsilon_0 \sin\omega t \quad \varepsilon_0 = NBA\omega$$



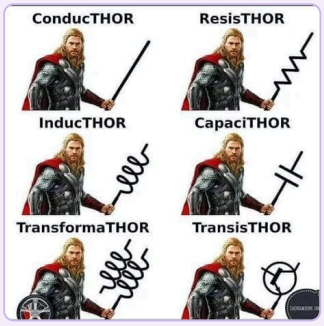
$\varepsilon = NBA\omega \sin\omega t$

Fig. 6.4 Alternating emf of a generator

→ emf is maximum when the plane of the coil is **parallel** to B (flux zero, but changing fastest).

→ emf is zero when the plane is ⊥ to B (flux maximum).

→ Mechanical energy (turbine, water, wind) → electrical energy. India: 50 Hz.



EMI ke Thor: Resis-Thor, Induc-Thor, Transis-Thor

Asli baat: Inductor (L), capacitor (C) and resistor (R) together build every AC circuit of the next chapter.

Exam Alert

→ AC generator: principle, $\varepsilon = NBA\omega \sin\omega t$ derivation, kab max kab zero: 3-5 marks.

→ Rod on rails ($\varepsilon = Blv$), force and power: case question favourite.

Derivation step by step (AC generator)

1. Coil of N turns, area A, rotating with angular velocity ω; angle between normal and B: $\theta = \omega t$.
2. Flux: $\Phi = NBA\cos\omega t$.
3. $\varepsilon = -d\Phi/dt = NBA\omega \sin\omega t$; maximum $\varepsilon_0 = NBA\omega$ when $\sin\omega t = 1$.

Coil position	Flux	emf
Plane ⊥ B ($\theta = 0^\circ$)	maximum	zero
Plane ∥ B ($\theta = 90^\circ$)	zero	maximum
$\theta = 180^\circ$	maximum (reversed)	zero
$\theta = 270^\circ$	zero	maximum (reversed)

Paheli-Bujho

Bujho: Paheli, motor aur generator mein kya fark?

Paheli: Ek hi machine hai, ulta chalao. Current do toh ghoomega (motor), ghumao toh current dega (generator).

Bujho: Matlab dono jaise main aur mera alarm: ek uthata hai, ek sulata hai!

Construction and working of the AC generator

An AC generator converts mechanical energy into electrical energy using electromagnetic induction. Its main parts are: (i) the **armature**, a rectangular coil of many turns wound on a soft-iron core; (ii) strong **field magnets** that provide a uniform radial field; (iii) two **slip rings** attached to the ends of the coil, which rotate with it; and (iv) two carbon **brushes** that press against the slip rings and connect the rotating coil to the external circuit. The coil is turned by a turbine driven by falling water (hydroelectric), steam from burning coal or nuclear fuel (thermal), or wind.

As the coil rotates, the angle between its normal and the field changes continuously, so the flux linked with it changes, and an emf is induced. During one half of the rotation one side of the coil moves up through the field and the other moves down; in the next half their roles swap, so the direction of the emf reverses every half rotation. The output is therefore alternating and sinusoidal, with frequency $\nu = \omega/2\pi$ equal to the number of rotations per second.

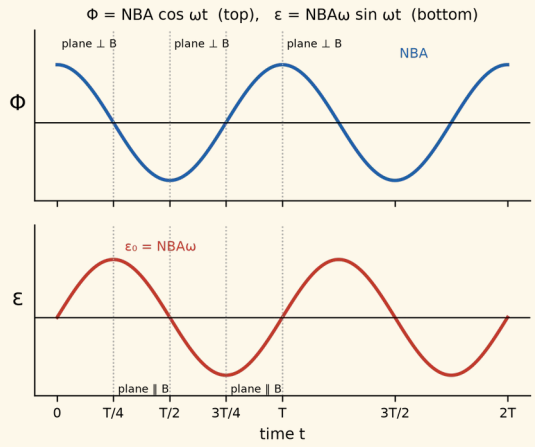


Fig. 6.8 Flux and emf of a rotating coil: emf is maximum when flux is zero

The graph shows a key point often asked in exams: the emf is not proportional to the flux but to its **rate of change**. When the plane of the coil is perpendicular to B, the flux is maximum but momentarily constant, so $\varepsilon = 0$. When the plane is parallel to B, the flux is zero but changing fastest, so ε is maximum. Φ and ε are 90° (T/4) out of phase.

Solved example

Q. A coil of 100 turns and area 0.05 m^2 rotates at 3000 rev/min in a field of 0.2 T. Find the peak emf and the emf when the plane of the coil makes 30° with the field.

Sol. $\omega = 2\pi \times 50 = 314$ rad/s. $\varepsilon_0 = NBA\omega = 100 \times 0.2 \times 0.05 \times 314 =$ **314 V**. Plane at 30° with B means the normal is at 60°, so $\varepsilon = \varepsilon_0 \sin 60° =$ **272 V**.

Generator: points to remember

- → Peak emf $\varepsilon_0 = NBA\omega$ increases with turns, field, area and speed of rotation.
- → Slip rings keep the output alternating; a split-ring commutator would make it unidirectional (DC dynamo).
- → In India the mains frequency is 50 Hz: the coil (or field system) completes 50 cycles per second.

8. Solved Numericals (Board Pattern)

N1. A coil of 500 turns has its flux changed from 2×10^{-3} Wb to 0 in 0.1 s. Find the emf.

Sol. $\varepsilon = N\Delta\Phi/\Delta t = 500 \times 2 \times 10^{-3}/0.1 =$ **10 V**.

N2. A 1 m rod moves at 5 m/s ⊥ to a field of 0.2 T. Find the emf. If R = 2 Ω, find I and the force needed.

Sol. $\varepsilon = Blv =$ **1 V**; $I =$ **0.5 A**; $F = IlB =$ **0.1 N**.

N3. A solenoid has 500 turns, length 0.5 m, area 4×10^{-4} m^2. Find L.

Sol. $L = \mu_0 N^2 A/l = 4\pi \times 10^{-7} \times 250000 \times 4 \times 10^{-4}/0.5 =$ **0.25 mH**.

N4. Current in a coil of L = 0.5 H falls from 4 A to 0 in 0.02 s. Find the emf and the initial energy.

Sol. $\varepsilon = L\Delta I/\Delta t =$ **100 V**; $U = \frac{1}{2}LI^2 =$ **4 J**.

N5. A 100-turn coil of area 0.1 m^2 rotates at 50 rev/s in 0.01 T. Find ε_0.

Sol. $\omega = 100\pi$ rad/s; $\varepsilon_0 = NBA\omega = 100 \times 0.01 \times 0.1 \times 314 =$ **31.4 V**.

Chapter at a glance: every way to induce an emf

Method	What changes	emf formula	Example
Moving magnet or changing current nearby	B	$\varepsilon = -N\,d\Phi/dt$	Faraday's experiments, transformer
Sliding rod on rails	area A	$\varepsilon = Blv$	rod-rail circuit
Rod rotating about one end	area swept	$\varepsilon = \tfrac{1}{2}B\omega l^2$	spoked wheel, Faraday disc
Coil rotating in field	angle θ	$\varepsilon = NBA\omega \sin \omega t$	AC generator
Changing current in the same coil	own flux	$\varepsilon = -L\,dI/dt$	choke, spark at switch
Changing current in nearby coil	shared flux	$\varepsilon = -M\,dI/dt$	transformer, wireless charger

All the results of this chapter are applications of one law, $\varepsilon = -d\Phi/dt$, with Lenz's law fixing the direction. Whenever you meet a new situation, first ask which of B, A or θ is changing, write the flux as a function of time, and differentiate. The energy that appears as electrical energy always comes from the work done by some external agent against the opposing force predicted by Lenz's law.

Solved example

Q. A square loop of side 20 cm and resistance 0.5 Ω is pulled out of a region of uniform field 0.5 T at a constant speed of 2 cm/s, with one side kept perpendicular to the direction of motion. Find the induced current and the time for which it flows.

Sol. While the loop is leaving, $\varepsilon = Blv = 0.5 \times 0.2 \times 0.02 = 2 \times 10^{-3}$ V, so $I = \varepsilon/R =$ **4 mA**. The current flows only while the flux is changing, that is for $t = \text{side}/v = 0.2/0.02 =$ **10 s**.

Yaad karo

- → Har sawaal mein pehle dekho: B badla, A badla, ya θ badla.
- → Flux maximum par emf zero, flux zero par emf maximum (generator).
- → Pura loop field ke andar chal raha hai to flux constant, emf zero.

Inductors in practice

- → Sudden switch-off of an inductive circuit (motor, electromagnet) gives a large emf L dI/dt, so a spark appears at the switch; this is why heavy machines are switched off gradually.
- → A choke is preferred over a resistor to limit AC because it controls current with very little heat loss.
- → Wireless phone chargers and induction cooktops work by mutual induction between a coil in the base and a coil (or metal pan) above it.

Formula sheet: Chapter 6

Quantity	Formula
Flux	$\Phi = BA\cos\theta$
Faraday	$\varepsilon = -N\,d\Phi/dt$
Motional emf	$\varepsilon = Blv$; rotating rod $\tfrac{1}{2}B\omega l^2$
Self inductance	$L = \mu_0 n^2 Al$; $\varepsilon = -L\,dI/dt$
Mutual inductance	$M = \mu_0 n_1 n_2 \pi r_1^2 l$
Energy	$U = \tfrac{1}{2}LI^2$
Generator	$\varepsilon = NBA\omega \sin \omega t$

1 minute recap

- ✓ Changing flux induces emf: $\varepsilon = -N\,d\Phi/dt$.
- ✓ Lenz: induced current opposes the change (energy conservation).
- ✓ $\varepsilon = Blv$ for a moving rod.
- ✓ $L = \mu_0 n^2 Al$, $U = \tfrac{1}{2}LI^2$; $M = \mu_0 n_1 n_2 \pi r_1^2 l$.
- ✓ Generator: $\varepsilon_0 = NBA\omega$.

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

EMI padhne se pehle vs baad

Asli baat: Most asked: Lenz with energy conservation, $\varepsilon = Blv$, L and M of solenoids, AC generator.

A. Objective and very short answer (1 mark)

Q1. [1M] The SI unit of magnetic flux is:

(a) tesla
(b) weber
(c) henry
(d) gauss

Ans: (b) weber.

Q2. IM Board fav Lenz's law is a consequence of conservation of:
(a) charge
(b) momentum
(c) energy
(d) mass
Ans: (c) energy.

Q3. IM Self inductance of a solenoid is proportional to:
(a) n^2
(b) n
(c) 1/n
(d) √n
Ans: (a) n^2.

Q4. IM The emf induced in a rod of length l moving with v ⊥ to B is:
(a) Bl/v
(b) Bv/l
(c) B/lv
(d) Blv
Ans: (d) Blv.

Q5. IM In an AC generator the emf is maximum when the plane of the coil is:
(a) perpendicular to B
(b) parallel to B
(c) at 45°
(d) at 60°
Ans: (b) parallel to B.

Q6. IM Energy stored in an inductor carrying current I is:
(a) LI
(b) LI^2
(c) $\frac{1}{2}LI^2$
(d) $2LI^2$
Ans: (c) $\frac{1}{2}LI^2$.

Q7. IM Board fav Why is inductance called electrical inertia?
Ans: It opposes any change in current, as mass opposes change in velocity.

Q8. IM State Faraday's law.
Ans: Induced emf equals the negative rate of change of magnetic flux: $\varepsilon = -d\Phi/dt$.

Q9. IM A magnet falls through a copper ring. Is its acceleration g?
Ans: No, less than g: the induced current opposes its motion (Lenz).

Q10. IM Define one henry.
Ans: Self inductance of a coil in which 1 V is induced when current changes at 1 A/s.

Q11. IM On what factors does mutual inductance depend?
Ans: Number of turns, area, separation, orientation and the medium (core).

Q12. IM What is the frequency of AC in India?
Ans: 50 Hz.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. IM **A:** An emf is induced in a coil when a magnet is held stationary inside it. **R:** Induced emf depends on flux.
Ans: (d) A false (flux is not changing); R partly true but emf depends on rate of change.

Q14. IM Board fav **A:** Lenz's law is consistent with energy conservation. **R:** Work done against the opposing induced current appears as electrical energy.
Ans: (a)

Q15. IM **A:** Self inductance of a coil increases with an iron core. **R:** Iron has high relative permeability.
Ans: (a)

Q16. IM **A:** In a generator emf is zero when flux is maximum. **R:** ε depends on the rate of change of flux, which is zero there.
Ans: (a)

C. Short answer (2–3 marks)

Q17. 2M Board fav Derive $\varepsilon = Blv$ for a rod moving on rails.
Ans: $\Phi = Blx$; $\varepsilon = d\Phi/dt = Bl\,dx/dt = Blv$.

Q18. 2M Board fav Show that Lenz's law is a consequence of energy conservation.
Ans: If induced current aided the change, the magnet would accelerate without external work, creating energy. Opposition makes us do work, which becomes electrical energy.

Q19. 3M Board fav Derive the self inductance of a long solenoid.
Ans: $B = \mu_0 nI$; $N\Phi = nl \cdot \mu_0 nIA = LI \Longrightarrow L = \mu_0 n^2 Al$.

Q20. 3M Board fav Derive the mutual inductance of two coaxial solenoids.
Ans: Field of S_2: $\mu_0 n_2 I_2$; flux through S_1: $n_1 l\ \pi r_1^2\ \mu_0 n_2 I_2 \Longrightarrow M = \mu_0 n_1 n_2 \pi r_1^2 l$.

Q21. 2M Show that energy stored in an inductor is $\frac{1}{2}LI^2$.
Ans: $dW = \varepsilon\, I\, dt = LI\, dI$; $W = \int_0^I LI\, dI = \frac{1}{2}LI^2$.

Q22. 2M Derive the emf of a rod of length l rotating about one end in a field B.
Ans: Element dr at r: $d\varepsilon = B\omega r\, dr$; $\varepsilon = \int_0^l B\omega r\, dr = \frac{1}{2}B\omega l^2$.

Q23. 3M Explain the AC generator and derive its emf.
Ans: Coil rotating in B: $\Phi = NBA\cos\omega t$; $\varepsilon = NBA\omega \sin\omega t$.

Q24. 2M Predict the polarity of a capacitor connected to a loop when a magnet's N pole moves towards the loop.
Ans: The near face becomes N; current flows anticlockwise as seen from the magnet, charging the corresponding plate positive.

Q25. 2M Find the force needed to move a rod at constant speed v on rails with resistance R.
Ans: $I = Blv/R$; $F = IlB = B^2l^2v/R$.

Q26. 2M Why does the current in a coil not reach its maximum instantly when switched on?
Ans: Self-induced back emf opposes the growth of current.

D. Long answer (5 marks)

Q27. 5M Board fav (a) State Faraday's and Lenz's laws. (b) A rod moves on rails: derive ε, I, force and power. (c) Show power is dissipated as heat.
Ans: (b) $\varepsilon = Blv$, $I = Blv/R$, $F = B^2l^2v/R$, $P = B^2l^2v^2/R$. (c) I^2R = same value.

Q28. 5M (a) Define self and mutual inductance. (b) Derive L of a solenoid and M of two coaxial solenoids. (c) Energy stored in an inductor.
Ans: As derived above; $U = \frac{1}{2}LI^2$.

E. Case-based

Q29. 4M Board fav An aircraft with wingspan 25 m flies horizontally at 180 km/h where the vertical component of earth's field is 5×10^{-4} T. (a) emf across wings? (b) Which law? (c) Can it light a bulb? (d) Would it change if the plane flew faster?
Ans: (a) $\varepsilon = Blv = 5 \times 10^{-4} \times 25 \times 50 = 0.625$ V. (b) Motional emf (Faraday). (c) No closed circuit through the bulb outside, and the emf is tiny. (d) Yes, $\varepsilon \propto v$.

Q30. 4M A metal plate swings between the poles of a magnet and quickly stops. (a) Why? (b) Name the currents. (c) How do slots reduce them? (d) One application.
Ans: (a) Eddy currents oppose motion (Lenz). (b) Eddy currents. (c) Slots break current paths. (d) Electromagnetic braking, induction furnace.

Q31. 2M Board fav A coil of area 0.04 m^2, 200 turns, is placed ⊥ to B = 0.5 T. The field drops to zero in 0.2 s. Find the average emf.
Ans: $\varepsilon = NBA/t = 200 \times 0.5 \times 0.04/0.2 =$ **20 V**.

Q32. 2M Two coils have M = 1.5 H. Current in one changes from 0 to 20 A in 0.5 s. Emf in the other?
Ans: $\varepsilon = M\, dI/dt = 1.5 \times 40 =$ **60 V**.

Q33. 2M A rod of length 1 m rotates at 400 rad/s about one end in B = 0.5 T. Find the emf.
Ans: $\varepsilon = \frac{1}{2}B\omega l^2 = \frac{1}{2} \times 0.5 \times 400 \times 1 =$ **100 V**.

Q34. 1M Why are transformer cores laminated?
Ans: To reduce eddy current losses.

Q35. 1M The current in a coil grows at 2 A/s and induces 4 V. Find L.
Ans: $L = \varepsilon/(dI/dt) =$ **2 H**.

Q36. 1M What happens to the self inductance if the number of turns is doubled (same length)?
Ans: $L \propto N^2$: becomes 4 times.

Q37. 2M A bar magnet is dropped through a long vertical copper tube. Describe its motion.
Ans: It soon falls at a slow constant (terminal) speed because eddy currents oppose its motion.

Q38. 2M Give two applications of eddy currents.
Ans: Electromagnetic brakes in trains, induction furnace, speedometers, energy meters.

F. Extra practice (board PYQ style)

Q39. 2M Board fav A square loop of side 10 cm and resistance 0.5 Ω is placed ⊥ to a field that changes at 0.1 T/s. Find the induced current.
Ans: $\varepsilon = A\,dB/dt = 0.01 \times 0.1 = 10^{-3}$ V; I = **2 mA.**

Q40. 1M A circular coil is pulled out of a field region. In which direction does the induced current flow?
Ans: So as to oppose the decrease of flux: it tries to keep the original field (Lenz).

Q41. 2M Board fav Show that the induced charge does not depend on the rate of change of flux.
Ans: $q = \int I\,dt = \int (\varepsilon/R)\,dt = N\Delta\Phi/R$: depends only on the total change.

Q42. 3M A 0.5 m long solenoid of 500 turns has radius 2 cm. A 10-turn coil is wound tightly over its middle. Find M.
Ans: $M = \mu_0 n_1 N_2 A = 4\pi \times 10^{-7} \times 1000 \times 10 \times \pi(0.02)^2$ = **1.6×10^{-5} H.**

Q43. 2M Two identical loops, one copper and one aluminium, are rotated in the same field at the same speed. Compare the emf and current.
Ans: Same emf (depends on geometry and motion); current larger in copper (lower resistance).

Q44. 1M What is the direction of induced emf in a rod moving east in a vertically downward field?
Ans: v × B: east × down = north. Positive charges are pushed north, so the north end is at higher potential.

EMI ke 6 formulas aur Lenz ka logic
clear: presentation done
Asli baat: Revise: $\Phi = BA\cos\theta$, $\varepsilon = -N\,d\Phi/dt$, Blv, L, M, $\frac{1}{2}LI^2$, $NBA\omega$.

Q45. 2M Board fav A coil of 100 turns and area 0.01 m^2 rotates at 50 rev/s in B = 0.1 T. Find ε_0 and the rms emf.
Ans: $\omega = 314$ rad/s; $\varepsilon_0 = NBA\omega$ = **31.4 V**; ε_{rms} = **22.2 V.**

Q46. 1M Current in a coil of L = 40 mH changes from 11 A to 0 in 0.1 s. Find the average emf.
Ans: $\varepsilon = L\,\Delta I/\Delta t = 0.04 \times 11/0.1$ = **4.4 V.**

Q47. 1M Why is the emf of a coil zero when the flux through it is maximum?
Ans: emf depends on $d\Phi/dt$; at maximum flux the rate of change is zero.

Q48. 2M A 0.5 m rod moves at 2 m/s ⊥ to B = 0.4 T. Find ε and the end at higher potential if v is east and B is downward.
Ans: $\varepsilon = Blv$ = **0.4 V**; v × B points north, so the north end is at higher potential.

Q49. 1M State two factors on which the self inductance of a coil depends.
Ans: Number of turns ($L \propto N^2$), area, length, and the permeability of the core.

Q50. 1M Give the energy stored in a 2 H inductor carrying 3 A.
Ans: $U = \frac{1}{2}LI^2$ = **9 J.**

Quick compare: generator vs motor

AC generator	Electric motor
Mechanical → electrical	Electrical → mechanical
Works on EMI (Faraday)	Works on force on current (Fleming left)
Fleming's right-hand rule	Fleming's left-hand rule

Common mistakes

→ θ ko plane se lena: $\Phi = BA\cos\theta$ mein θ normal ke saath.

→ Flux zero ho toh emf zero mat likhna: emf flux ke change pe depend karta hai.

→ $L \propto N^2$, $M \propto N_1N_2$.

→ Generator: emf max jab coil ka plane B ke parallel.

Quick compare: self vs mutual inductance

Self (L)	Mutual (M)
Emf in the same coil	Emf in a neighbouring coil
$\varepsilon = -L\, dI/dt$	$\varepsilon_2 = -M\, dI_1/dt$
$L = \mu_0 n^2 Al$	$M = \mu_0 n_1 n_2 \pi r_1^2 l$
Opposes change in own current	Basis of the transformer

Paheli-Bujho

Paheli: Bujho, ek line mein EMI batao.

Bujho: Flux badlo, emf paao; Lenz bolega: mujhe change pasand nahi!

Paheli: Perfect. Ab yahi formula sheet do baar revise karo.

Exam Alert: last check

→ Blv rod, L of solenoid, M of coaxial solenoids, AC generator: 4 derivations.

→ Lenz + energy conservation reasoning pakka.

ALTERNATING CURRENT

Most asked: rms values, AC through R, L and C (phase relations), reactance and impedance, phasor analysis of series LCR circuit, resonance and quality factor, power factor and wattless current, transformer

The supply at home is not steady DC: its voltage swings 50 times a second. This chapter shows how resistors, inductors and capacitors behave with such **alternating current**, using a simple tool called the **phasor**.

AC ka chapter: phasor samajh liya toh aaj maza aa jaayega

Asli baat: Every AC question becomes a right-angled triangle once you draw the phasor diagram.

1. AC Voltage and Current

$$v = v_m \sin \omega t, \quad \omega = 2\pi\nu$$

The average of a sine wave over a full cycle is **zero**, so we use the **root mean square (rms)** value: the steady DC that would produce the same heat in a resistor.

$$I = i_m/\sqrt{2} = 0.707\, i_m, \quad V = v_m/\sqrt{2}$$

- → Household 220 V is the rms value; peak $v_m = 220\sqrt{2} \approx 311$ V.
- → AC meters read rms values.
- → Average power in a resistor: $P = I^2R = V^2/R$ (rms values).

Why rms? (derivation)

1. Instantaneous power in R: $p = i^2R = i_m^2R \sin^2 \omega t$.
2. Average of $\sin^2 \omega t$ over a cycle = ½.
3. So average power = $\frac{1}{2} i_m^2R = I^2R$ with $I = i_m/\sqrt{2}$.

Solved example (NCERT)

Q. A light bulb is rated 100 W for 220 V. Find (a) its resistance, (b) peak voltage, (c) rms current.

Sol. (a) $R = V^2/P = 484\ \Omega$. (b) $v_m = 220\sqrt{2} = 311$ V. (c) $I = P/V = 0.454$ A.

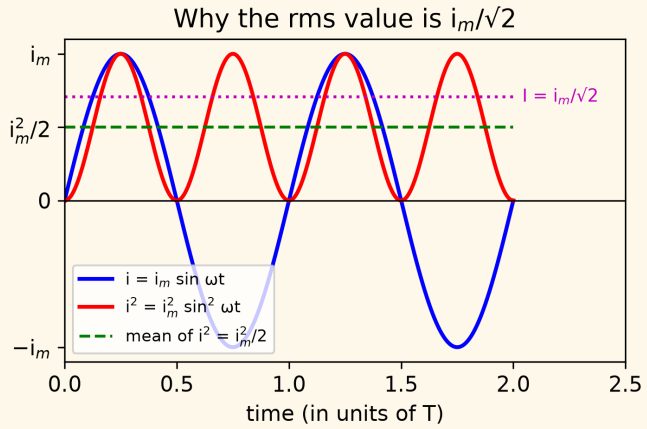
Why do we use AC at all?

Almost all electrical energy sold by power companies is AC. The main reason is that AC voltage can be **stepped up or stepped down easily and efficiently** by a transformer. Power is generated at about 11 kV, raised to hundreds of kV for long-distance transmission (so that the current, and hence the I^2R loss in the lines, is small) and finally lowered to 220 V for our homes. DC voltage cannot be changed by a simple, static device like a transformer.

- → AC generators (alternators) are simpler and cheaper to build in large sizes than DC generators.
- → AC can be converted to DC easily with rectifiers when DC is needed (charging, electronics).
- → AC motors are robust, cheap and need little maintenance.
- → **Drawback:** at the same rms voltage, the peak of AC (311 V for 220 V supply) is higher, so an AC shock is more dangerous; AC also has skin effect at high frequency.

Terms used for an AC

Term	Meaning
Amplitude (peak) i_m	Maximum value reached in a cycle
Time period T	Time for one complete cycle; $T = 1/\nu$
Frequency ν	Cycles per second; 50 Hz in India
Angular frequency ω	$\omega = 2\pi\nu = 314$ rad/s for 50 Hz
Phase	The angle $(\omega t + \varphi)$ that tells the state of the AC at time t
rms value	$i_m/\sqrt{2}$: equivalent steady (DC) current for heating



The square of a sinusoidal current always stays positive; its average is $i_m^2/2$, which gives the rms value $i_m/\sqrt{2}$.

Mean value over half a cycle: although the average over a full cycle is zero, the average over a positive half cycle is not. Integrating $i_m \sin \omega t$ from 0 to T/2 and dividing by T/2 gives $i_{av} = 2i_m/\pi = 0.637\, i_m$. Hence for a sine wave: peak : rms : half-cycle mean = 1 : 0.707 : 0.637.

Yaad karo

→ Full cycle average = 0, half cycle average = $0.637\, i_m$, rms = $0.707\, i_m$.

→ Meter jo bhi AC padhta hai, wo rms hota hai.

→ 220 V ka peak 311 V: isliye AC ka jhatka zyada khatarnak.

Solved example

An AC is given by $i = 5\sqrt{2} \sin(100\pi t)$ A. Find (a) rms current, (b) frequency, (c) heat produced in a 10 Ω resistor in 1 minute.

(a) $i_m = 5\sqrt{2}$ A, so $I = i_m/\sqrt{2} = \mathbf{5\ A}$. (b) $\omega = 100\pi = 2\pi\nu$, so $\nu = \mathbf{50\ Hz}$. (c) $H = I^2Rt = 25 \times 10 \times 60 = \mathbf{15000\ J}$.

2. AC Voltage Applied to R, L and C

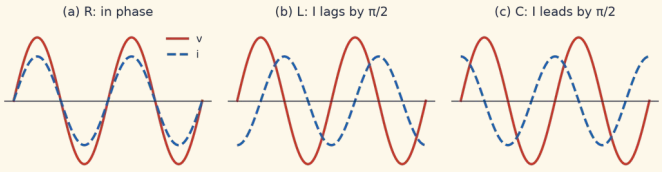


Fig. 7.1 Voltage (solid) and current (dashed) for pure R, L and C

Element	Opposition	Phase of i w.r.t. v	Average power
Resistor R	R	in phase	V I
Inductor L	$X_L = \omega L$	lags by π/2	zero
Capacitor C	$X_C = 1/\omega C$	leads by π/2	zero

Inductor

$v = L\, di/dt \Longrightarrow i = -(v_m/\omega L) \cos \omega t = i_m \sin(\omega t - \pi/2)$. **Inductive reactance $X_L = \omega L$** grows with frequency (DC passes freely, ω = 0).

Capacitor

$q = Cv \Longrightarrow i = dq/dt = \omega C v_m \cos \omega t = i_m \sin(\omega t + \pi/2)$. **Capacitive reactance $X_C = 1/\omega C$** falls with frequency (DC is blocked, $X_C = \infty$).

Capacitor mein current voltage se pehle aa jaata hai

Asli baat: Mnemonic CIVIL: in C, I leads V; in L, V leads I (I lags).

Exam Alert

→ CIVIL yaad karo: phase relation MCQ har saal.

→ Inductor DC ko pass karta hai, capacitor DC ko block: reasoning question.

Graphs of reactance vs frequency

→ $X_L = 2\pi\nu L$: straight line through the origin.

→ $X_C = 1/2\pi\nu C$: falls as 1/ν (rectangular hyperbola).

→ They are equal at ν_0, the resonant frequency.

Solved example

Q. A 15 μF capacitor is connected to 220 V, 50 Hz. Find X_C and I. What if ν is doubled?

Sol. $X_C = 1/(2\pi \times 50 \times 15 \times 10^{-6}) = \mathbf{212\ \Omega}$; $I = \mathbf{1.04\ A}$. Doubling ν halves X_C and doubles I.

Inductive reactance: physical meaning

When AC flows in a coil, the changing current produces a changing magnetic flux through the coil itself. By Lenz's law, the self-induced emf opposes this change. This opposition is the **inductive reactance** $X_L = \omega L$. The faster the current changes (higher ω) and the larger the inductance, the larger the back emf, so the larger X_L. For DC (ω = 0) the current does not change, so a pure inductor offers **no** opposition.

Capacitive reactance: physical meaning

A capacitor does not let charges cross its dielectric. In an AC circuit, however, the plates are charged, discharged and charged in the opposite sense again and again, so current flows in the connecting wires.

At a higher frequency the plates are charged and discharged more often in each second, so for the same voltage a larger current flows: $X_C = 1/\omega C$ decreases with ω. For DC, after the capacitor is fully charged the current stops, so X_C is infinite.

Property	Resistor R	Inductor L	Capacitor C
Opposition	R	$X_L = \omega L$	$X_C = 1/\omega C$
Depends on ν?	No	$\propto \nu$	$\propto 1/\nu$
For DC	R	Zero	Infinite
Phase of i w.r.t. v	In phase	Lags by $\pi/2$	Leads by $\pi/2$
Energy	Dissipated as heat	Stored in magnetic field	Stored in electric field

Important note

Energy stored in an inductor ($\frac{1}{2}Li^2$) and in a capacitor ($\frac{1}{2}Cv^2$) is returned to the source in the next quarter cycle. That is why the **average power** in a pure L or pure C is zero, even though a large current flows.

Solved example

A 0.5 H inductor and a 20 μF capacitor are each connected to a 220 V, 50 Hz supply. Find the rms current in each.

$X_L = 2\pi \times 50 \times 0.5 = 157\ \Omega$, so $I_L = 220/157 =$ **1.40 A**.
$X_C = 1/(2\pi \times 50 \times 20 \times 10^{-6}) = 159\ \Omega$, so $I_C =$ $220/159 =$ **1.38 A**.

3. Phasors

A **phasor** is a vector rotating anticlockwise with angular speed ω; its projection on the vertical axis gives the instantaneous value. Phase differences become angles between arrows.

Why phasors? In a series circuit the voltages across R, L and C are not in phase, so their rms values cannot be added like numbers ($V \neq V_R + V_L + V_C$). A phasor is a rotating arrow whose length is the peak value and whose projection on the vertical axis gives the instantaneous value. Adding phasors like vectors automatically takes care of phase differences.

→ All phasors rotate anticlockwise with the same angular speed ω, so the angles between them stay fixed.

→ The current phasor is taken as reference in a series circuit (same current through all elements).

→ V_R is along I, V_L is 90° ahead of I and V_C is 90° behind I.

4. Series LCR Circuit

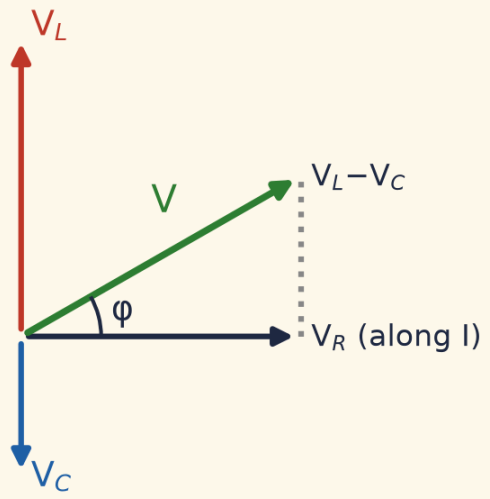


Fig. 7.2 Phasor diagram of a series LCR circuit ($X_L > X_C$)

The same current i flows through all three. V_R is along I, V_L leads by 90°, V_C lags by 90°. Adding as vectors:

$$V^2 = V_R^2 + (V_L - V_C)^2$$

$$Z = \sqrt{(R^2 + (X_L - X_C)^2)}, \quad i_m = v_m/Z$$

$$\tan\varphi = (X_L - X_C)/R$$

→ $X_L > X_C$: circuit is inductive, current lags.

→ $X_C > X_L$: capacitive, current leads.

→ $X_L = X_C$: resistive, in phase (resonance).

Impedance triangle

Since $V_R = IR$ and $V_L - V_C = I(X_L - X_C)$, dividing the phasor triangle by I gives a right triangle with sides R, $X_L - X_C$ and hypotenuse Z: the angle φ appears directly.

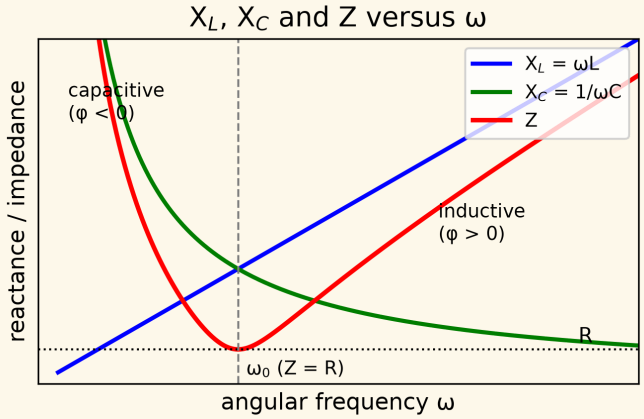
Solved example (NCERT)

Q. A resistor of 200 Ω and a capacitor of 15 μF are in series with 220 V, 50 Hz. Find the current and voltages across R and C.

Sol. $X_C = 212\ \Omega$; $Z = \sqrt{(200^2 + 212^2)} = 291.5\ \Omega$; $I =$ **0.755 A**; $V_R =$ **151 V**, $V_C =$ **160 V** (sum is more than 220 V because they are 90° apart).

Three cases of a series LCR circuit

Case	Nature	Phase φ	Current vs voltage
$X_L > X_C$	Inductive	Positive	Current lags voltage
$X_L < X_C$	Capacitive	Negative	Current leads voltage
$X_L = X_C$	Resistive (resonance)	Zero	In phase, $Z = R$



Variation of X_L, X_C and Z with ω. Z is minimum (= R) at ω_0; below ω_0 the circuit is capacitive, above it inductive.

Note that the voltage across L or C alone can be **larger than the source voltage**, because V_L and V_C are opposite in phase and partly cancel each other. This is not a violation of energy conservation.

5. Resonance

At $\omega = \omega_0$, $X_L = X_C$, so $Z = R$ is minimum and current is **maximum**:

$$\omega_0 L = 1/\omega_0 C \implies \omega_0 = 1/\sqrt{LC}, \quad \nu_0 = 1/2\pi\sqrt{LC}$$

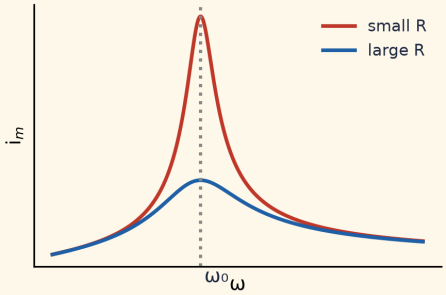


Fig. 7.3 Current vs frequency: sharper peak for smaller R

Sharpness and quality factor

$$Q = \omega_0 L/R = 1/(\omega_0 CR) = \omega_0/2\Delta\omega$$

→ Larger Q (smaller R) means a sharper, more selective resonance.

→ Used in radio and TV tuning: the circuit picks one station's frequency.

→ Resonance is possible only when both L and C are present.

Radio ki frequency resonance pe na ho toh kya takleef hai

Asli baat: Tuning a radio changes C until $\omega_0 = 1/\sqrt{LC}$ matches the station's frequency.

Paheli-Bujho

Bujho: Paheli, resonance pe inductor aur capacitor dono gayab ho jaate hain kya?

Paheli: Gayab nahi, unke voltage barabar aur ulte hote hain, isliye cancel. Circuit ko sirf R dikhta hai.

Bujho: Matlab dono ek doosre ko neutral kar dete hain, jaise main aur Monday.

Paheli: Haha, aur current maximum ho jaata hai.

LC oscillations

A charged capacitor connected to an inductor makes charge slosh back and forth: energy passes between the electric field of C and the magnetic field of L.

$$\omega_0 = 1/\sqrt{LC}; \quad q^2/2C + \tfrac{1}{2}Li^2 = \text{constant}$$

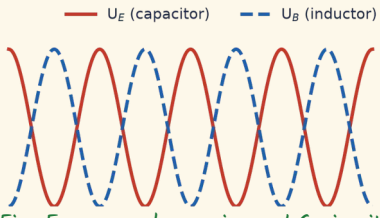


Fig. Energy exchange in an LC circuit

LC circuit	Spring-mass
Charge q	Displacement x
Current i	Velocity v
Inductance L	Mass m
1/C	Spring constant k
Electric energy	Potential energy
Magnetic energy	Kinetic energy

Applications of resonance

Tuning of radio and TV: the antenna picks up signals of many frequencies. The tuning circuit is an LC (LCR) circuit whose capacitance is varied. When its resonant frequency equals the frequency of a station, the current for that station is maximum and the station is heard clearly. A high Q (small R)

gives a sharp resonance, so neighbouring stations are rejected (good selectivity).

Condition for resonance: resonance is possible only if both L and C are present. In an RL or RC circuit, X_L and X_C cannot cancel, so there is no resonance.

Bandwidth

The two frequencies ω_1 and ω_2 at which the current falls to $i_{max}/\sqrt{2}$ (power falls to half) are called half-power frequencies. The bandwidth is $2\Delta\omega = \omega_2 - \omega_1 = R/L$, and $Q = \omega_0/2\Delta\omega = \omega_0 L/R = (1/R)\sqrt{(L/C)}$.

Yaad karo

- → Resonance ke liye L aur C dono chahiye.
- → Q zyada = resonance sharp = tuning achhi.
- → R kam karo, Q badhega.
- → Resonance pe $V_L = V_C$, aur ye V se bada bhi ho sakta hai.

Solved example

A series LCR circuit has L = 2.0 H, C = 32 μF and R = 10 Ω. Find the resonant angular frequency and the Q factor.

$\omega_0 = 1/\sqrt{(LC)} = 1/\sqrt{(2 \times 32 \times 10^{-6})} = 1/(8 \times 10^{-3}) =$ **125 rad/s**. $Q = \omega_0 L/R = 125 \times 2/10 =$ **25**.

6. Power in AC Circuits: Power Factor

$P = V I \cos\varphi$ $\cos\varphi = R/Z$: power factor

Circuit	φ	cos φ	Power
Pure R	0	1	maximum, VI
Pure L or C	±π/2	0	zero
LCR at resonance	0	1	maximum
General LCR	between	R/Z	VI cos φ

Wattless current

The current component I sin φ (⊥ to V) consumes no power; it is called **wattless current**. A low power factor means large current for the same power, so large I^2R losses in transmission. Capacitors are added in parallel to raise the power factor.

Yaad karo

- → Pure L ya C mein power zero: current hai, kaam nahi (wattless).
- → Power factor = R/Z; resonance pe 1.

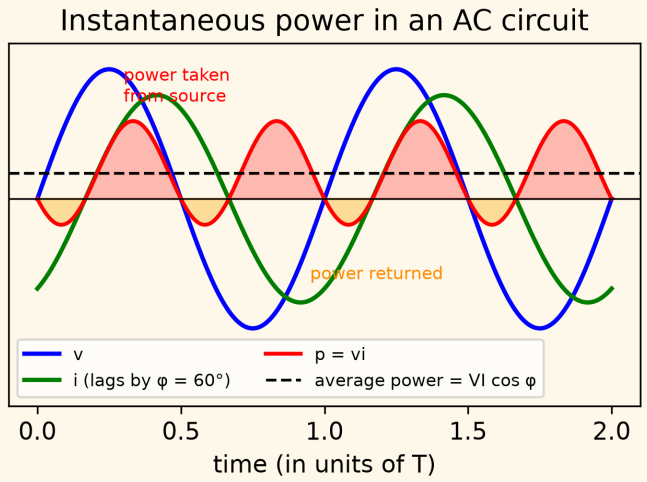
Derivation step by step (average power)

1. $v = v_m \sin\omega t$, $i = i_m \sin(\omega t + \varphi)$.
2. $p = vi = v_m i_m \sin\omega t \sin(\omega t + \varphi) = \frac{1}{2} v_m i_m [\cos\varphi - \cos(2\omega t + \varphi)]$.
3. The second term averages to zero over a cycle: $P = \frac{1}{2} v_m i_m \cos\varphi = VI \cos\varphi$.

Solved example

Q. A series LCR circuit (L = 0.12 H, C = 480 nF, R = 23 Ω) is on 230 V. Find the source frequency for maximum current and the power at that frequency.

Sol. $\omega_0 = 1/\sqrt{(0.12 \times 4.8 \times 10^{-7})} =$ **4167 rad/s** (663 Hz). $P = V^2/R = 230^2/23 =$ **2300 W**.



Instantaneous power p = vi. When v and i have opposite signs, power is returned to the source; the average over a cycle is VI cos φ.

Power factor in different circuits

Circuit	φ	cos φ	Average power
Pure R	0	1	VI (maximum)
Pure L	π/2	0	Zero
Pure C	π/2	0	Zero
Series LCR	$\tan^{-1}[(X_L - X_C)/R]$	R/Z	VI cos φ
LCR at resonance	0	1	V^2/R

Why a low power factor is undesirable

For a given power $P = VI \cos\varphi$ at a given voltage, the current needed is $I = P/(V \cos\varphi)$. If cos φ is small, a large current must flow, and the line loss I^2R becomes large. Industries with many motors (inductive loads) therefore connect **capacitors in parallel** to supply a leading current that cancels the lagging part, bringing cos φ close to 1. This is called power factor correction.

Choke coil

To reduce the current in an AC circuit (for example in a tube light), a resistor would waste energy as heat. Instead an inductor of large L and very small R, called a **choke coil**, is used. Its power factor $\cos\varphi = R/Z \approx 0$, so it limits the current while consuming almost no power.

Solved example

A choke coil of L = 0.7 H and negligible resistance is connected to a 220 V, 50 Hz supply. Find the current and the average power consumed.

$X_L = 2\pi \times 50 \times 0.7 = 220\ \Omega$, so $I = 220/220 = \mathbf{1\ A}$.
Since $\varphi = 90^\circ$, $P = VI\cos 90^\circ = \mathbf{0\ W}$.

7. Transformers

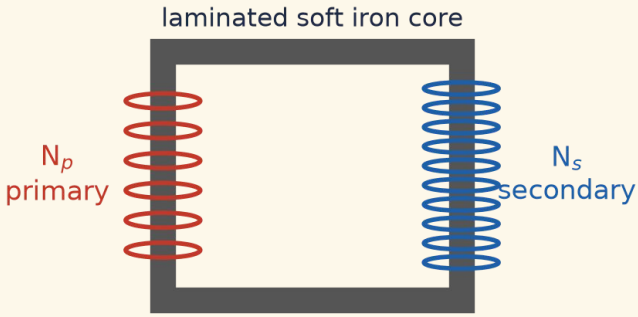


Fig. 7.4 Transformer: two coils on a common laminated core

Works on **mutual induction**. An alternating current in the primary creates changing flux in the core, which induces an emf in the secondary. For an ideal transformer (no losses):

$$V_s/V_p = N_s/N_p = I_p/I_s$$

→ **Step-up**: $N_s > N_p$; voltage up, current down.
→ **Step-down**: $N_s < N_p$.
→ Works only with AC (DC gives no changing flux).

Energy losses

Loss	Cause	Remedy
Flux leakage	not all flux links the secondary	wind coils over each other
Copper loss	I^2R heating of windings	thick copper wire
Eddy current loss	currents in the core	laminated core
Hysteresis loss	repeated magnetisation	soft iron (low hysteresis)

Transformer DC pe kaam karega? Acha!?
Asli baat: No: a transformer needs changing flux, so it works only with AC.

Exam Alert

→ LCR: Z, phase angle, resonance, Q-factor: 5 marks.
→ Transformer principle, ratio aur 4 losses: 3 marks.

NCERT Activity: step-up and step-down

Wind a primary coil of few turns and a secondary of many turns on a soft iron ring. Connect the primary to a low-voltage AC supply and a small bulb across the secondary: the bulb glows brighter as secondary turns increase. With DC in the primary, the bulb glows only for an instant when switched on or off.

Transformers in power transmission

At the power station a step-up transformer raises the voltage (e.g. 11 kV to 220 kV) so the current and I^2R losses in long lines are small. Near towns, step-down transformers bring it down to 220 V for homes.

Solved example

Q. An ideal transformer has 100 primary and 2000 secondary turns. 220 V AC is applied to the primary and 2 A flows in it. Find V_s, I_s and the power.
Sol. $V_s = \mathbf{4400\ V}$; $I_s = \mathbf{0.1\ A}$; $P = 220 \times 2 = \mathbf{440\ W}$ on both sides.

Efficiency of a transformer

Efficiency η = (output power/input power) × 100%. Practical transformers have efficiencies of 95–99%, because there are no moving parts. The small loss comes from copper loss (I^2R in windings), eddy current loss (reduced by laminated core), hysteresis loss (reduced by using soft iron) and flux leakage (reduced by winding the coils one over the other).

Step-up transformer	Step-down transformer
$N_s > N_p$	$N_s < N_p$
$V_s > V_p$, $I_s < I_p$	$V_s < V_p$, $I_s > I_p$
Used at power stations	Used at substations, chargers
Secondary wire thinner	Secondary wire thicker

Solved example (NCERT type)

A small town needs 800 kW at 220 V. It is 15 km from a plant generating at 440 V. The two-wire line has resistance 0.5 Ω/km, and the town gets power through a 4000–220 V step-down transformer. Find the line loss and the power the plant must supply.

Line resistance = 30 km × 0.5 = 15 Ω. Line current $I = 800 \times 10^3/4000 = 200$ A. Loss = $I^2R = 200^2 \times 15 =$ **600 kW**. Plant must supply 800 + 600 = **1400 kW**. Voltage drop on line = 200 × 15 = 3000 V, so the plant needs a 440 V to 7000 V step-up transformer. Such a large loss shows why transmission is done at much higher voltages.

8. Solved Numericals (Board Pattern)

N1. A 100 Ω resistor is connected to 220 V, 50 Hz. Find the rms and peak current.
Sol. $I = 2.2$ A; $i_m = 2.2\sqrt{2} =$ **3.11 A**.

N2. A 44 mH inductor is connected to 220 V, 50 Hz. Find X_L and I.
Sol. $X_L = 2\pi \times 50 \times 0.044 =$ **13.8 Ω**; $I =$ 220/13.8 = **15.9 A**.

N3. A 60 μF capacitor on 110 V, 60 Hz. Find X_C and I.
Sol. $X_C = 1/(2\pi \times 60 \times 60 \times 10^{-6}) =$ **44.2 Ω**; $I =$ **2.49 A**.

N4. L = 5 H, C = 80 μF, R = 40 Ω in series on 230 V. Find ω_0, the current at resonance and Q.
Sol. $\omega_0 = 1/\sqrt{(5 \times 8 \times 10^{-5})} =$ **50 rad/s**; $I = 230/40 =$ **5.75 A**; $Q = \omega_0 L/R =$ **6.25**.

N5. A transformer converts 220 V to 22 V. Primary has 3000 turns. Find N_s; if $I_s = 5$ A, find I_p.
Sol. $N_s =$ **300**; $I_p = 5 \times 22/220 =$ **0.5 A**.

Formula sheet: Chapter 7

Quantity	Formula
rms	$I = i_m/\sqrt{2}$
Reactances	$X_L = \omega L$; $X_C = 1/\omega C$
Impedance	$Z = \sqrt{(R^2 + (X_L - X_C)^2)}$
Phase	$\tan\varphi = (X_L - X_C)/R$
Resonance	$\omega_0 = 1/\sqrt{(LC)}$
Q-factor	$Q = \omega_0 L/R$
Power	$P = VI\cos\varphi$; $\cos\varphi = R/Z$
Transformer	$V_s/V_p = N_s/N_p = I_p/I_s$

1 minute recap

- ✓ rms = peak/√2; meters read rms.
- ✓ R in phase, L lags ($X_L = \omega L$), C leads ($X_C = 1/\omega C$).
- ✓ $Z = \sqrt{(R^2 + (X_L - X_C)^2)}$; resonance $\omega_0 = 1/\sqrt{(LC)}$.
- ✓ $P = VI\cos\varphi$; pure L, C: wattless.
- ✓ Transformer: mutual induction, AC only.

Most Important Questions

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM The rms value of an AC of peak 10 A is:
(a) 10 A
(b) 7.07 A
(c) 5 A
(d) 14.1 A
Ans: (b) 7.07 A.

Q2. IM Board fav In a pure inductor, current:
(a) is in phase
(b) leads by π/2
(c) lags voltage by π/2
(d) lags by π
Ans: (c) lags voltage by π/2.

Q3. IM At resonance in a series LCR circuit, the impedance is:
(a) R
(b) zero
(c) maximum
(d) X_L
Ans: (a) R.

Q4. 1M The power factor of a pure capacitor is:
(a) 1
(b) 0.5
(c) 0.707
(d) zero
Ans: (d) zero.

Q5. 1M A transformer works on:
(a) self induction
(b) mutual induction
(c) eddy currents
(d) Ohm's law
Ans: (b) mutual induction.

Q6. 1M Capacitive reactance with increasing frequency:
(a) increases
(b) decreases
(c) stays same
(d) first rises then falls
Ans: (b) decreases.

Q7. 1M Board fav Why does a capacitor block DC?
Ans: For DC, $\omega = 0$ so $X_C = 1/\omega C = \infty$.

Q8. 1M What is wattless current?
Ans: Current component $I \sin\varphi$ perpendicular to voltage; it consumes no power.

Q9. 1M Define quality factor.
Ans: $Q = \omega_0 L/R$: measures the sharpness of resonance.

Q10. 1M Why can't a transformer step up DC?
Ans: DC produces no changing flux, so no emf is induced.

Q11. 1M Give the resonant frequency of an LC circuit.
Ans: $\nu_0 = 1/2\pi\sqrt{(LC)}$.

Q12. 1M Why is the core of a transformer laminated?
Ans: To reduce eddy current losses.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M Board fav **A:** Average power in a pure inductor is zero. **R:** The phase difference between current and voltage is $\pi/2$.
Ans: (a)

Q14. 1M **A:** At resonance, current in a series LCR circuit is maximum. **R:** At resonance $X_L = X_C$ and $Z = R$.
Ans: (a)

Q15. 1M **A:** A step-up transformer increases power. **R:** It increases voltage.
Ans: (d) A false (power stays same, ideally); R true.

Q16. 1M **A:** AC is measured by its rms value. **R:** The average value of AC over a cycle is zero.
Ans: (a)

C. Short answer (2–3 marks)

Q17. 3M Board fav Show that the current through a pure inductor lags the voltage by $\pi/2$. Find X_L.
Ans: $v_m \sin\omega t = L\, di/dt \Longrightarrow i = i_m \sin(\omega t - \pi/2)$, $i_m = v_m/\omega L \Longrightarrow X_L = \omega L$.

Q18. 3M Board fav Show that the current through a capacitor leads the voltage by $\pi/2$. Find X_C.
Ans: $i = C\, dv/dt = \omega C v_m \cos\omega t = i_m \sin(\omega t + \pi/2)$; $X_C = 1/\omega C$.

Q19. 3M Board fav Draw the phasor diagram of a series LCR circuit and derive Z and φ.
Ans: $V^2 = V_R^2 + (V_L - V_C)^2 \Longrightarrow Z = \sqrt{(R^2 + (X_L - X_C)^2)}$, $\tan\varphi = (X_L - X_C)/R$.

Q20. 2M Derive the condition for resonance and the resonant frequency.
Ans: Current max when Z min, $X_L = X_C \Longrightarrow \omega_0 = 1/\sqrt{(LC)}$.

Q21. 3M Derive $P = VI\cos\varphi$ for an AC circuit.
Ans: $p = vi = v_m i_m \sin\omega t \sin(\omega t + \varphi)$; average over a cycle $= \tfrac{1}{2} v_m i_m \cos\varphi = VI\cos\varphi$.

Q22. 3M State the principle of a transformer and derive the turns ratio.
Ans: $\varepsilon_s = -N_s\, d\Phi/dt$, $\varepsilon_p = -N_p\, d\Phi/dt \Longrightarrow V_s/V_p = N_s/N_p$; power equal $\Longrightarrow I_p/I_s$.

Q23. 2M List four energy losses in a transformer and how to reduce them.
Ans: Flux leakage, copper loss, eddy current loss, hysteresis loss (see table).

Q24. 2M Why is power transmitted at high voltage?
Ans: Same P at lower I means smaller I^2R line losses.

Q25. 2M Define power factor. When is it maximum and minimum?
Ans: $\cos\varphi = R/Z$; maximum 1 (pure R or resonance), minimum 0 (pure L or C).

Q26. 2M A bulb and a capacitor are in series with AC. What happens to the brightness if a dielectric is inserted in the capacitor?
Ans: C rises, X_C falls, current rises: bulb glows brighter.

D. Long answer (5 marks)

Q27. 5M Board fav (a) Derive Z and φ for a series LCR circuit using phasors. (b) Derive the resonant frequency. (c) Define Q-factor and its significance.
Ans: As above; $Q = \omega_0 L/R$: larger Q means sharper tuning.

Q28. 5M (a) Explain the working of a transformer. (b) Derive the turns ratio. (c) Explain energy losses.
Ans: As above.

E. Case-based

Q29. 4M Board fav A radio receiver uses an LC circuit with L = 1 μH. (a) Value of C to tune to 100 MHz? (b) What happens if R is large? (c) Which quantity is maximum at resonance? (d) Name the phenomenon.
Ans: (a) $C = 1/(4\pi^2\nu^2L) = 2.5$ pF. (b) Flatter peak, poor selectivity. (c) Current. (d) Resonance.

Q30. 4M A power station sends 800 kW to a town through lines of resistance 15 Ω. (a) Line loss at 40 kV? (b) At 8 kV? (c) Which is better? (d) Which device changes voltage?
Ans: (a) I = 20 A; loss = 6 kW. (b) I = 100 A; loss = 150 kW. (c) High voltage. (d) Transformer.

Q31. 3M Board fav An LCR circuit has R = 30 Ω, X_L = 80 Ω, X_C = 40 Ω on 200 V. Find Z, I and cos φ.
Ans: $Z = \sqrt{(900 + 1600)} = 50$ Ω; I = 4 A; $\cos\varphi = 30/50 = 0.6$.

Q32. 1M At what frequency will a 1 H inductor have a reactance of 314 Ω?
Ans: $\nu = X_L/2\pi L =$ **50 Hz.**

Q33. 3M A 100 μF capacitor in series with 40 Ω across 110 V, 60 Hz. Find the maximum current.
Ans: $X_C = 26.5$ Ω; Z = 48 Ω; I = 2.29 A; $i_m =$ **3.24 A.**

Q34. 1M What is the average power in a series LCR circuit at resonance?
Ans: $P = V^2/R$ (power factor 1).

Q35. 2M Why is a choke coil preferred over a resistor to control AC?
Ans: It controls current with almost no power loss (wattless).

Q36. 2M Can the voltage across L in a series LCR circuit be greater than the supply voltage?
Ans: Yes, at resonance $V_L = QV$ can exceed V; V_L and V_C cancel each other.

Q37. 2M Give one difference between an LC oscillation and a mechanical spring oscillation analogy.
Ans: L ↔ mass, 1/C ↔ spring constant, q ↔ displacement, i ↔ velocity.

Q38. 2M In an LC circuit, energy oscillates between which two forms?
Ans: Electric energy of the capacitor ($q^2/2C$) and magnetic energy of the inductor ($\frac{1}{2}Li^2$).

Q39. 1M An ideal step-down transformer has N_p:N_s = 20:1. If 2200 V is applied, find V_s.
Ans: $V_s = 2200/20 =$ **110 V.**

Q40. 2M Why do we use AC for domestic supply rather than DC?
Ans: AC can be stepped up/down easily by transformers, allowing efficient transmission.

F. Extra practice (board PYQ style)

Q41. 2M Board fav An AC source v = 200 sin 314t is applied to a 50 Ω resistor. Find the rms current and frequency.

Ans: v_m = 200 V, V = 141.4 V; I = **2.83 A**; ν = 314/2π = **50 Hz.**

Q42. 2M Board fav In a series LCR circuit, V_R = 40 V, V_L = 70 V, V_C = 40 V. Find the supply voltage.

Ans: $V = \sqrt{(40^2 + 30^2)}$ = **50 V.**

Q43. 2M Plot X_L and X_C against frequency.

Ans: X_L: straight line through origin; X_C: rectangular hyperbola; they cross at ν_0.

Q44. 1M The power factor of a circuit is 0.5. What is the phase difference between V and I?

Ans: cos φ = 0.5 ⟹ φ = **60°.**

Q45. 2M A 15 μF capacitor and a 3 H inductor: find the resonant frequency.

Ans: $\omega_0 = 1/\sqrt{(45 \times 10^{-6})}$ = 149 rad/s; ν_0 = **23.7 Hz.**

Q46. 1M Why does a transformer hum?

Ans: Magnetostriction: the core vibrates as it is magnetised back and forth 100 times a second.

Quick compare: R, L, C in AC

	R	L	C
Opposition	R	ωL	1/ωC
DC (ω = 0)	R	0 (passes)	∞ (blocks)
High ω	R	very large	very small
Phase of I	in phase	lags 90°	leads 90°
Power	I^2R	zero	zero

Paheli-Bujho

Paheli: Bujho, inductor high frequency ko rokta hai ya capacitor?

Bujho: Inductor! X_L = ωL, ω badha toh rukawat badhi.

Paheli: Aur capacitor?

Bujho: Wo high frequency ka dost hai, X_C chhota ho jaata hai. CIVIL yaad hai!

Q47. 3M Board fav A coil has R = 8 Ω and X_L = 6 Ω on 200 V AC. Find I, power factor and power.

Ans: Z = 10 Ω; I = 20 A; cos φ = 0.8; P = 200 × 20 × 0.8 = **3200 W.**

Q48. 1M Two LCR circuits have the same ω_0; Q of the first is twice the second. Which is better for tuning?

Ans: The first: higher Q, sharper resonance, better selectivity.

Q49. 1M What is the phase difference between V_L and V_C in a series LCR circuit?

Ans: π (180°): they are opposite.

Yaad karo: one-line answers

- → Choke coil: low power loss.
- → Capacitor: DC block, AC pass.
- → Resonance: Z = R, I max, cos φ = 1.
- → Transformer: AC only, power same.
- → Peak = rms × √2.

Phasor diagram ko dhyan se padhte hue

Asli baat: Draw V_R along I, V_L up, V_C down: the rest is Pythagoras.

Common mistakes

- → rms aur peak mix karna: 220 V rms hai.
- → X_L = ωL mein ω = 2πν, sirf ν nahi.
- → Transformer power badhata nahi, sirf V-I badalta hai.
- → Resonance pe Z = R, zero nahi.

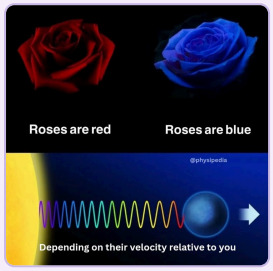
Exam Alert: last check

- → LCR phasor + resonance + Q, transformer + losses: 2 badi derivations.
- → CIVIL aur power factor table pakka.

ELECTROMAGNETIC WAVES

Most asked: displacement current and Ampere-Maxwell law, properties of EM waves, relation $c = 1/\sqrt{(\mu_0\varepsilon_0)}$ and $E_0/B_0 = c$, electromagnetic spectrum with sources and uses

A changing magnetic field makes an electric field (Faraday). Maxwell asked: does a changing electric field make a magnetic field? Yes. Together they form a self-sustaining wave that travels at the speed of light: **light itself is an electromagnetic wave.**



Roses are red, violets are blue: wavelength ka khel hai sab

Asli baat: Colour is just wavelength. Red has longer λ than violet; both are EM waves travelling at c in vacuum.

1. Introduction

Ampere's circuital law $\oint B \cdot dl = \mu_0 I$ works for steady currents. Maxwell found that it fails for a charging capacitor and added a new term, the **displacement current**. With it, the equations of electricity and magnetism predicted waves of electric and magnetic fields travelling at 3×10^8 m/s, exactly the measured speed of light.

Year	Scientist	Contribution
1864	Maxwell	Predicted EM waves; light is an EM wave
1887	Hertz	Produced and detected EM waves ($\lambda \approx 6$ m)
1895	J.C. Bose	Produced waves of 5–25 mm
1899	Marconi	Wireless transmission over long distances

Background: what Maxwell noticed

By the middle of the 19th century three facts were known: (i) electric charges produce electric fields (Gauss's law), (ii) electric currents produce magnetic fields (Ampere's law), and (iii) a changing magnetic field produces an electric field (Faraday's law). Maxwell asked the reverse question: **does a changing electric field produce a magnetic field?** While checking Ampere's law for a charging capacitor he found that the answer must be yes, otherwise the law gives two different answers for the same loop.

This single idea completed the laws of electricity and magnetism. It predicted that changing electric and magnetic fields can sustain each other and travel through empty space as a wave, with a speed exactly equal to the measured speed of light. Light was thus shown to be an electromagnetic wave, unifying electricity, magnetism and optics.

Key idea

A changing magnetic field gives an electric field (Faraday). A changing electric field gives a magnetic field (Maxwell). Together these two make electromagnetic waves possible.

2. Displacement Current

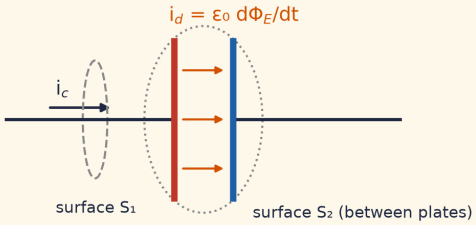


Fig. 8.1 Charging capacitor: conduction current in the wires, displacement current between the plates

The inconsistency in Ampere's law

1. Take a point P near the wire charging a capacitor and a circle around it.
2. For a flat surface S_1 bounded by the circle, the wire pierces it: $\oint B \cdot dl = \mu_0 i$.
3. For a pot-shaped surface S_2 with the same boundary but passing between the plates, no current crosses it: $\oint B \cdot dl = 0$.
4. The same loop cannot give two answers. Something crosses S_2: the changing electric field between the plates.

Maxwell's correction

Between the plates $E = q/\varepsilon_0 A$, so the electric flux $\Phi_E = EA = q/\varepsilon_0$. Then $dq/dt = \varepsilon_0 \, d\Phi_E/dt$, which equals the current in the wires. This term is called the **displacement current**:

$$i_d = \varepsilon_0 \, d\Phi_E/dt$$

$$\oint B \cdot dl = \mu_0(i_c + \varepsilon_0 \, d\Phi_E/dt)$$ Ampere-Maxwell law

→ Conduction current i_c: flow of charges. Displacement current i_d: a changing electric field. Both produce magnetic fields.

→ Total current ($i_c + i_d$) is continuous through the circuit, including the gap.

→ Outside the capacitor plates $i_d = 0$; inside, $i_c = 0$.

→ For steady fields (DC after charging) $i_d = 0$.

Exam Alert

→ Displacement current ka formula aur reason (capacitor charging): 2-3 marks har saal.

→ $i_d = i_c$ during charging: numerical mein seedha use hota hai.

Solved example (NCERT)

Q. A parallel plate capacitor with circular plates of radius 12 cm, separation 5 mm, is charged by a current of 0.15 A. Find the rate of change of potential difference and the displacement current.

Sol. $C = \varepsilon_0\pi r^2/d = 8.85 \times 10^{-12} \times \pi \times 0.0144/0.005 = 80.1$ pF. $dV/dt = I/C = 0.15/80.1 \times 10^{-12} = \mathbf{1.87 \times 10^9}$ **V/s.** $i_d = i_c = \mathbf{0.15}$ **A.**

Maxwell's equations (for reference)

Equation	Name	Meaning
$\oint E \cdot dA = Q/\varepsilon_0$	Gauss (electric)	charges produce E
$\oint B \cdot dA = 0$	Gauss (magnetic)	no monopoles
$\oint E \cdot dl = -d\Phi_B/dt$	Faraday	changing B produces E
$\oint B \cdot dl = \mu_0 i_c + \mu_0\varepsilon_0 d\Phi_E/dt$	Ampere-Maxwell	current and changing E produce B

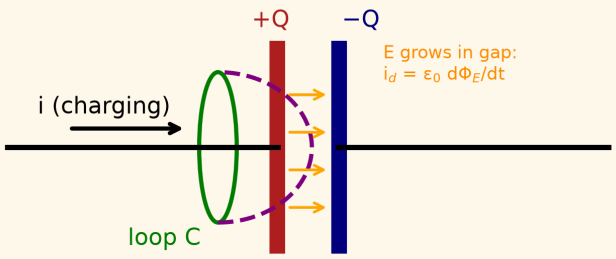
Paheli-Bujho

Bujho: Paheli, plates ke beech toh koi wire nahi, phir current kaise?

Paheli: Wahan charge nahi chalta, par electric field badalta hai. Maxwell ne kaha: badalta field bhi current jaisa kaam karta hai.

Bujho: Matlab bina wire ka current! Wireless ka asli baap.

Paheli: Bilkul, isi se EM waves aur radio bane.



Surface 1 (flat disc on C): cut by wire, $\oint B \cdot dl = \mu_0 i$
Surface 2 (pot-shaped, dashed): passes between plates; no conduction current, so Ampere's law needs i_d

Ampere's law applied to two surfaces bounded by the same loop C around the wire of a charging capacitor. Only the displacement current makes the two answers agree.

Inconsistency in Ampere's law: step by step

1. Take a loop C of radius r around the wire, just before the capacitor plate. By symmetry B is tangential and the same at all points of the loop.
2. Surface 1 (flat disc on C) is crossed by the wire carrying current i. Ampere's law gives $B(2\pi r) = \mu_0 i$, so $B = \mu_0 i/2\pi r$.
3. Surface 2 (a pot-shaped surface on the same loop C) passes through the gap between the plates. No charge flows across it, so Ampere's law gives $B(2\pi r) = 0$, i.e. $B = 0$.
4. The same loop cannot have two values of B. So Ampere's law, as originally stated, is incomplete.
5. Between the plates the electric field $E = Q/\varepsilon_0 A$ is changing. Its flux is $\Phi_E = EA = Q/\varepsilon_0$, so $\varepsilon_0 \, d\Phi_E/dt = dQ/dt = i$.
6. Maxwell called $\varepsilon_0 \, d\Phi_E/dt$ the displacement current i_d. Adding it to the conduction current, both surfaces give the same B.

$$\oint B \cdot dl = \mu_0(i_c + \varepsilon_0 \, d\Phi_E/dt)$$ Ampere-Maxwell law

Important features of displacement current

→ It is not a flow of charge; it exists wherever the electric flux changes, even in vacuum.

→ It produces a magnetic field exactly like a conduction current.

→ Total current ($i_c + i_d$) is continuous: the circuit current is i_c in the wires and an equal i_d between the plates.

→ In steady (DC) conditions, after the capacitor is charged, E is constant and $i_d = 0$.

→ Inside a good conductor i_c dominates; in an insulator with a rapidly changing field, i_d dominates.

Yaad karo

→ $i_d = \varepsilon_0 \, d\Phi_E/dt$: ye charge ka flow nahi hai.

→ Capacitor charge ho raha hai toh plates ke beech i_d = wire ka i.

→ Fully charged capacitor: E constant, $i_d = 0$.

Solved example

A parallel plate capacitor of capacitance 2 μF is connected to a source whose voltage changes at 5×10^4 V/s. Find the displacement current between the plates.

$i_d = i_c = C \, dV/dt = 2 \times 10^{-6} \times 5 \times 10^4 =$ **0.1 A.**

Solved example (NCERT type)

A capacitor with circular plates of radius 12 cm and separation 5.0 mm is charged by a constant current of 0.15 A. Find (a) its capacitance, (b) the rate of change of potential difference, (c) the displacement current across the plates.

(a) $C = \varepsilon_0 \pi r^2/d = 8.85 \times 10^{-12} \times \pi \times (0.12)^2/(5 \times 10^{-3}) =$ **80.1 pF**. (b) $dV/dt = i/C = 0.15/(80.1 \times 10^{-12}) =$ **1.87×10^9 V/s**. (c) $i_d = i =$ **0.15 A**; Kirchhoff's first rule holds at each plate if displacement current is included.

3. Electromagnetic Waves

Source of EM waves

A charge at rest produces only E; a charge moving uniformly produces a steady B as well. An **accelerated charge** produces changing E and B, which regenerate each other and travel outward: an EM wave. An oscillating charge of frequency ν radiates EM waves of the same frequency.

→ Oscillating charges in an antenna produce radio waves.

→ Electrons jumping between atomic levels produce light.

→ Fast electrons hitting a metal target produce X-rays.

NCERT Activity: Hertz's experiment

Hertz connected two metal balls with a small gap to an induction coil; sparks across the gap made charges oscillate and radiate. A nearby open ring with a small gap (the receiver) also showed tiny sparks, proving the waves crossed the room. He showed they reflect, refract and interfere like light.

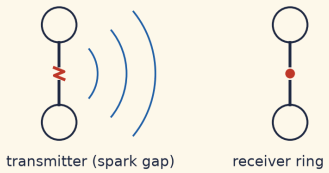


Fig. 8.2 Hertz's transmitter and receiver

Nature of EM waves

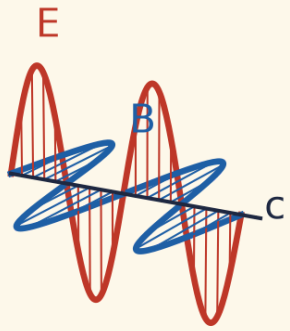


Fig. 8.3 Plane EM wave: E (red) along y, B (blue) along z, travelling along x

$$E_y = E_0 \sin(kx - \omega t), \quad B_z = B_0 \sin(kx - \omega t)$$

$$k = 2\pi/\lambda, \quad \omega = 2\pi\nu, \quad c = \omega/k = \nu\lambda$$

Properties of EM waves

1. E and B are ⊥ to each other and to the direction of propagation: EM waves are **transverse**.
2. E and B oscillate **in phase** (reach maximum together).

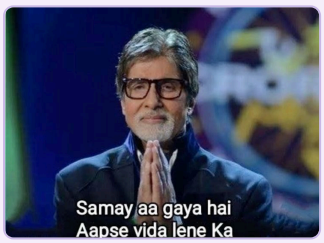
3. Need no medium: they travel through vacuum.
4. Speed in vacuum: $c = 1/\sqrt{(\mu_0\varepsilon_0)} = 3 \times 10^8$ m/s.
5. Ratio of amplitudes: $E_0/B_0 = c$.
6. In a medium: $v = 1/\sqrt{(\mu\varepsilon)} = c/n$, where $n = \sqrt{(\mu_r\varepsilon_r)}$ is the refractive index.
7. They carry **energy and momentum**, and exert pressure (radiation pressure).
8. The direction of propagation is along E × B.

Yaad karo

→ E aur B same phase, aapas mein aur direction se ⊥.

→ $c = 1/\sqrt{(\mu_0\varepsilon_0)}$: ye calculation se light ki speed aayi, isliye light = EM wave.

→ $E_0 = cB_0$, isliye E bada dikhta hai par energy dono mein barabar.



Maxwell ke equations ke baad light ko pata chala wo EM wave hai

Asli baat: From four equations, the speed $1/\sqrt{(\mu_0\varepsilon_0)}$ came out equal to the measured speed of light.

Energy and momentum

$u = \frac{1}{2}\varepsilon_0E^2 + B^2/2\mu_0$ energy density; the two parts are equal on average

Momentum delivered on complete absorption: $p = U/c$ on complete reflection: $2U/c$

→ Average energy density: $u_{av} = \frac{1}{2}\varepsilon_0E_0^2$.

→ Intensity $I = u_{av}c$ (power per unit area).

→ Sunlight on earth: about 1.3 kW/m^2; its radiation pressure is tiny ($\approx 7 \times 10^{-6}$ N/m^2) but pushes comet tails away from the sun.

Solved example (NCERT)

Q. A plane EM wave of frequency 25 MHz travels in vacuum along x. At a point E = 6.3 ĵ V/m. Find B there.

Sol. $B = E/c = 6.3/(3 \times 10^8) = \mathbf{2.1 \times 10^{-8}}$ **T**. E along y, propagation along x, so B along z (E × B ∥ x̂).

Solved example (NCERT)

Q. The magnetic field of a plane EM wave is $B_y = 2 \times 10^{-7} \sin(0.5 \times 10^3x + 1.5 \times 10^{11}t)$ T. Find λ, ν and E_0.

Sol. $k = 500$ m$^{-1} \Longrightarrow \lambda = 2\pi/k =$ **1.26 cm**. $\omega = 1.5 \times 10^{11} \Longrightarrow \nu =$ **23.9 GHz**. $E_0 = cB_0 =$ **60 V/m**; the wave travels along −x.

Solved example

Q. Light of intensity 18 W/cm^2 falls normally on a non-reflecting surface of area 20 cm^2. Find the average force.

Sol. Power = 18 × 20 = 360 W; $F = P/c = 360/(3 \times 10^8) = \mathbf{1.2 \times 10^{-6}}$ **N**.

Mathematical description of a plane EM wave

For a plane EM wave travelling along +z, with E along x and B along y:

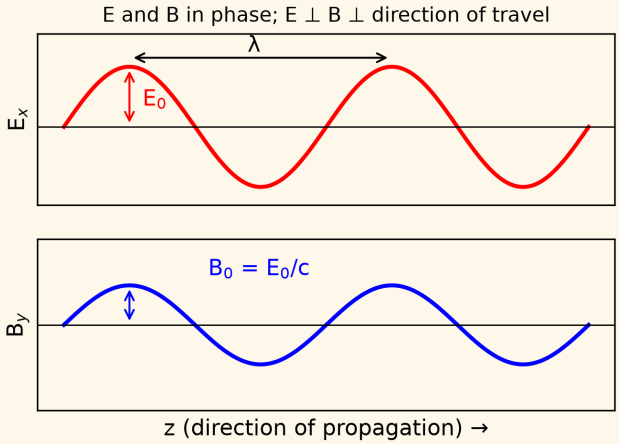
$$E_x = E_0 \sin(kz - \omega t), \quad B_y = B_0 \sin(kz - \omega t)$$

→ Wave number $k = 2\pi/\lambda$; angular frequency $\omega = 2\pi\nu$.

→ Speed $c = \omega/k = \nu\lambda = 1/\sqrt{(\mu_0\varepsilon_0)} = 3 \times 10^8$ m/s.

→ $E_0/B_0 = c$, and E and B reach their maxima at the same place and time (in phase).

→ The direction of propagation is along E × B.



Electric and magnetic fields of a plane EM wave plotted along the direction of travel: same λ, same phase, $E_0 = cB_0$.

How to read a given wave equation

1. The variable inside the sine or cosine (x, y or z) gives the axis of propagation.
2. If the argument is (kz − ωt), the wave moves along +z; if (kz + ωt), along −z.
3. $k = 2\pi/\lambda$ gives λ; $\omega = 2\pi\nu$ gives ν; check that $\omega/k = c$.
4. $B_0 = E_0/c$, and the direction of B is chosen so that E × B is along the direction of travel.

Solved example (NCERT type)

The electric field of an EM wave in vacuum is $\vec{E} = (3.1\ \text{N/C}) \cos[(1.8\ \text{rad/m})\, y + (5.4 \times 10^6\ \text{rad/s})\, t]\, \hat{i}$. Find (a) direction of propagation, (b) λ, (c) ν, (d) B_0, (e) the expression for $\vec{B}$.

(a) The argument is $(ky + \omega t)$, so the wave moves along **$-y$**. (b) $\lambda = 2\pi/k = 2\pi/1.8 =$ **3.49 m**. (c) $\nu = \omega/2\pi = 5.4 \times 10^6/2\pi =$ **8.6×10^5 Hz**. (d) $B_0 = E_0/c = 3.1/(3 \times 10^8) =$ **1.03×10^{-8} T**. (e) $\vec{E}$ is along $\hat{i}$ and travel is along $-\hat{j}$; since $\hat{i} \times \hat{k} = -\hat{j}$, $\vec{B}$ is along $\hat{k}$: **$\vec{B} = (1.03 \times 10^{-8}\ \text{T}) \cos[1.8y + 5.4 \times 10^6\, t]\, \hat{k}$**.

Energy density and intensity

The energy of an EM wave is shared equally between its electric and magnetic fields. The average energy densities are $u_E = \frac{1}{4}\varepsilon_0 E_0^2$ and $u_B = B_0^2/4\mu_0$; using $B_0 = E_0/c$ and $c^2 = 1/\mu_0\varepsilon_0$, these are equal. The total average energy density is $u = \frac{1}{2}\varepsilon_0 E_0^2$, and the intensity (energy per unit area per second) is $I = uc$.

Solved example (NCERT type)

About 5% of the power of a 100 W bulb is converted into visible light. Find the average intensity of visible radiation at (a) 1 m, (b) 10 m from the bulb (treat it as a point source).

Visible power = 5 W. $I = P/4\pi r^2$. (a) $I = 5/(4\pi \times 1^2) =$ **0.40 W/m^2**. (b) $I = 5/(4\pi \times 10^2) =$ **4.0×10^{-3} W/m^2**.

4. Electromagnetic Spectrum

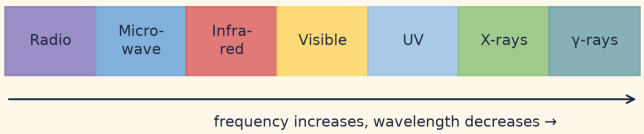


Fig. 8.4 The electromagnetic spectrum

Type	λ	Source	Uses
Radio	> 0.1 m	antennas	radio, TV, mobile
Micro-wave	1 mm–0.1 m	magnetron, klystron	radar, ovens, satellites
Infra-red	700 nm–1 mm	hot bodies	remotes, night vision
Visible	400–700 nm	atoms (electron jumps)	vision, photos
UV	1–400 nm	sun, arc lamps	sterilising, LASIK
X-rays	10^{-3}–1 nm	X-ray tube	imaging, crystals
γ-rays	< 10^{-3} nm	nuclei	cancer therapy

Type	Detected by
Radio	receiver aerials
Microwave	point-contact diodes
Infrared	thermopile, bolometer
Visible	eye, photocell
UV, X, γ	photocell, film, Geiger tube

Important facts about each band

- **Radio**: AM band 530–1710 kHz; FM 88–108 MHz; TV 54–890 MHz; mobile phones in the UHF band.
- **Microwaves**: ovens use frequencies that match the resonant frequency of water molecules, so food heats up while the container stays cool.
- **Infrared**: called heat waves; absorbed by water molecules, raising temperature. Earth re-radiates IR, which greenhouse gases trap.
- **Visible**: the only band our eyes detect; snakes can sense IR, bees can see UV.
- **Ultraviolet**: ozone layer absorbs most UV from the sun; welders wear special goggles because UV harms the eyes.
- **X-rays**: strongly penetrating; over-exposure is harmful.
- **Gamma rays**: most energetic, from nuclear reactions.

Spectrum ka order yaad nahi ho raha? Kyu nahi ho rahi padhai

Asli baat: Mnemonic for increasing frequency: Raju Made Interesting Videos Using X-ray Glasses (Radio, Micro, IR, Visible, UV, X, Gamma).

Exam Alert

→ Spectrum table: kis wave ka use kya (1–2 marks, MCQ). Microwave oven, LASIK (UV), remote (IR), radar (microwave) pakke.

→ Order of frequency/wavelength yaad rakho.

Paheli-Bujho

Paheli: Bujho, remote control ki light dikhti kyun nahi?

Bujho: Kyunki wo shy hai?

Paheli: Haha, nahi! Wo infrared hai, humari aankh sirf 400–700 nm dekhti hai.

Bujho: Phone camera se dekho toh dikhti hai, maine try kiya tha!

Paheli: Sahi, camera sensor IR pakad leta hai.

NCERT Activity: see infrared with a phone camera

Point a TV remote at a phone camera and press a button. The eye sees nothing, but the camera screen shows the LED flashing: the camera sensor detects infrared that our eyes cannot.

Why each band behaves differently

All EM waves obey the same laws and travel at the same speed in vacuum; they differ only in frequency (and wavelength). Their interaction with matter depends strongly on frequency: long radio waves are reflected by the ionosphere, microwaves set water molecules rotating (heating food), infrared makes molecules vibrate (felt as heat), visible light excites outer electrons (vision, photosynthesis), UV can break chemical bonds, and X-rays and gamma rays are energetic enough to ionise atoms and penetrate the body.

Band	Wavelength	Frequency (approx.)	Photon energy
Radio	> 0.1 m	< 10^{9} Hz	< 10^{-5} eV
Microwave	1 mm to 0.1 m	10^{9} to 10^{11} Hz	10^{-5} to 10^{-3} eV
Infrared	700 nm to 1 mm	10^{11} to 4 × 10^{14} Hz	10^{-3} to 1.8 eV
Visible	400 to 700 nm	4 to 7.5 × 10^{14} Hz	1.8 to 3.1 eV
Ultraviolet	1 to 400 nm	10^{15} to 10^{17} Hz	3 eV to 1 keV
X-rays	1 nm to 10^{-3} nm	10^{17} to 10^{20} Hz	1 keV to 1 MeV
Gamma rays	< 10^{-3} nm	> 10^{20} Hz	> 1 MeV

EM waves and our planet

→ **Ozone layer:** absorbs most harmful UV from the Sun; without it, UV would damage skin, eyes and plant life.

→ **Greenhouse effect:** the Earth radiates infrared; gases such as CO_2 and water vapour trap part of it and keep the surface warm. Without an atmosphere the average surface temperature would be much lower.

→ **Ionosphere:** reflects short-wave radio signals, making long-distance radio broadcasts possible. Higher frequencies (TV, satellite) pass through it, so satellites are needed for them.

→ **X-ray astronomy** is done from satellites because the atmosphere absorbs X-rays, while visible and radio waves reach the ground.

Yaad karo

→ Ozone: UV ko rokta hai.

→ Greenhouse: IR ko phansata hai.

→ Ionosphere: short wave radio ko reflect karta hai.

→ X-ray astronomy: satellite se, kyunki atmosphere X-ray absorb kar leta hai.

Solved example

Identify the part of the spectrum: (a) 21 cm line of hydrogen, (b) 5890 Å sodium line, (c) 14.4 keV photon from ^{57}Fe.

(a) λ = 21 cm, so $\nu = c/\lambda = 1.43 \times 10^9$ Hz: **microwave (short radio)**. (b) 589 nm: **visible (yellow)**. (c) $\lambda = hc/E$ = 1240 eV nm/14400 eV = 0.086 nm: **X-ray / gamma region** (it is a nuclear gamma ray).

EM waves compared with mechanical waves

EM waves	Mechanical (e.g. sound) waves
Need no medium; travel in vacuum	Need a material medium
Always transverse	Transverse or longitudinal (sound in air is longitudinal)
Speed 3×10^8 m/s in vacuum	Sound: about 340 m/s in air
Oscillating E and B fields	Oscillating particles of the medium
Can be polarised	Longitudinal waves cannot be polarised

Production of EM waves: why acceleration is needed

A charge at rest produces only an electric field. A charge moving with constant velocity produces a steady current, i.e. a steady magnetic field, but no wave. Only an **accelerated (oscillating) charge** produces fields that change with time; the changing E produces changing B and vice versa, and the disturbance travels outward as an EM wave. The frequency of the wave equals the frequency of oscillation of the charge. In an antenna, electrons are made to oscillate by an AC source; in an atom, electrons jumping between energy levels emit light; in a nucleus, de-excitation emits gamma rays.

Historical milestones

- → **1864:** Maxwell predicts EM waves and calculates their speed as $1/\sqrt{(\mu_0\varepsilon_0)} = c$.
- → **1887:** Hertz produces and detects EM waves (wavelength a few metres) in the laboratory.
- → **1895:** Jagadish Chandra Bose produces waves of wavelength 5 mm to 25 mm (microwaves) in Kolkata.
- → **1896–1899:** Marconi transmits EM waves over long distances; radio communication begins.

Important note

Speed of an EM wave in a medium: $v = 1/\sqrt{(\mu\varepsilon)} = c/n$, where $n = \sqrt{(\mu_r\varepsilon_r)}$. Frequency stays the same when the wave enters a new medium; wavelength becomes λ/n.

Solved example

Light of wavelength 600 nm in vacuum enters glass of refractive index 1.5. Find its speed, frequency and wavelength in glass.

$v = c/n$ = **2×10^8 m/s**. $\nu = c/\lambda = 3 \times 10^8/600 \times 10^{-9}$ = **5×10^{14} Hz** (unchanged). λ_{glass} = 600/1.5 = **400 nm**.

5. Solved Numericals (Board Pattern)

N1. Find the wavelength of an FM signal of 100 MHz.
Sol. $\lambda = c/\nu = 3 \times 10^8/10^8$ = **3 m**.

N2. The amplitude of B in an EM wave is 510 nT. Find the amplitude of E.
Sol. $E_0 = cB_0 = 3 \times 10^8 \times 510 \times 10^{-9}$ = **153 V/m**.

N3. Show that $c = 1/\sqrt{(\mu_0\varepsilon_0)}$ gives 3×10^8 m/s.
Sol. $\mu_0\varepsilon_0 = 4\pi \times 10^{-7} \times 8.85 \times 10^{-12} = 1.11 \times 10^{-17}$; $\sqrt{} = 3.33 \times 10^{-9}$; c = **3×10^8 m/s**.

N4. A radio can tune from 7.5 MHz to 12 MHz. Find the wavelength band.
Sol. $\lambda = c/\nu$: **40 m to 25 m**.

N5. The refractive index of a medium is 1.5 and μ_r = 1. Find the speed of EM waves and ε_r.
Sol. $v = c/n$ = **2×10^8 m/s**; $n = \sqrt{(\varepsilon_r)} \Longrightarrow \varepsilon_r$ = **2.25**.

N6. A capacitor is charged by 1 A. Find the displacement current between the plates.
Sol. $i_d = i_c$ = **1 A**.

More solved examples (NCERT type)

1. A parallel plate capacitor with circular plates of radius 6.0 cm has C = 100 pF and is connected to a 230 V AC supply of angular frequency 300 rad/s. Find (a) the rms conduction current, (b) whether the conduction current equals the displacement current, (c) the amplitude of B at a point 3.0 cm from the axis between the plates.

(a) $I_{rms} = V_{rms}\omega C = 230 \times 300 \times 100 \times 10^{-12}$ = **6.9 μA**. (b) **Yes**, $i_d = i_c$ at every instant. (c) Inside the plates $B = \mu_0 r\, i_0/2\pi R^2$ with peak current $i_0 = \sqrt{2} \times 6.9$ μA = 9.76 μA: $B_0 = 4\pi \times 10^{-7} \times 0.03 \times 9.76 \times 10^{-6}/(2\pi \times 0.06^2)$ = **1.63×10^{-11} T**.

2. A plane EM wave of frequency 25 MHz travels in free space along x. At a point, $E = 6.3\,\hat{j}$ V/m. Find B at this point.

$B = E/c = 6.3/(3 \times 10^8) = 2.1 \times 10^{-8}$ T. E is along $\hat{j}$ and travel along $\hat{i}$; since $\hat{j} \times \hat{k} = \hat{i}$, $\mathbf{B = 2.1 \times 10^{-8}\,\hat{k}\ T}$.

3. The electric field of a plane EM wave oscillates at 2.0×10^{10} Hz with amplitude 48 V/m. Find (a) λ, (b) B_0, (c) show that the average energy density of E equals that of B.

(a) $\lambda = c/\nu = 3 \times 10^8/2 \times 10^{10} = \mathbf{1.5\ cm}$. (b) $B_0 = 48/(3 \times 10^8) = \mathbf{1.6 \times 10^{-7}\ T}$. (c) $u_B = B_0^2/4\mu_0 = E_0^2/(4\mu_0 c^2) = \frac{1}{4}\varepsilon_0 E_0^2 = u_E$, since $1/\mu_0 c^2 = \varepsilon_0$.

4. A radio can tune to any station in the 7.5 MHz to 12 MHz band. Find the wavelength band.

$\lambda = c/\nu$: $3 \times 10^8/7.5 \times 10^6 = 40$ m and $3 \times 10^8/12 \times 10^6 = 25$ m. Band: **25 m to 40 m** (short-wave radio).

5. A charged particle oscillates about its mean position with a frequency of 10^9 Hz. What is the frequency of the EM waves it produces? What is their wavelength?

The waves have the same frequency as the oscillating charge: $\mathbf{10^9}$ **Hz**. $\lambda = c/\nu = \mathbf{0.3}$ **m** (microwave/radio region).

6. Light of intensity 6.4×10^4 W/m^2 falls normally on a perfectly absorbing surface of area 1.0 cm^2. Find the energy received in 30 minutes and the momentum delivered.

$U = IAt = 6.4 \times 10^4 \times 10^{-4} \times 1800 = \mathbf{1.15 \times 10^4\ J}$. $p = U/c = 1.15 \times 10^4/(3 \times 10^8) = \mathbf{3.84 \times 10^{-5}\ kg\ m/s}$ (twice this for a perfect reflector).

Formula sheet: Chapter 8

Quantity	Formula
Displacement current	$i_d = \varepsilon_0\, d\Phi_E/dt$
Ampere-Maxwell	$\oint B \cdot dl = \mu_0(i_c + i_d)$
Speed in vacuum	$c = 1/\sqrt{(\mu_0\varepsilon_0)}$
Speed in medium	$v = 1/\sqrt{(\mu\varepsilon)} = c/n$
Amplitudes	$E_0 = cB_0$
Wave relations	$c = \nu\lambda = \omega/k$
Energy density	$u_{av} = \frac{1}{4}\varepsilon_0 E_0^2$
Momentum	$p = U/c$ (absorbed)

1 minute recap

- ✓ Changing E makes B: displacement current $\varepsilon_0 d\Phi_E/dt$.
- ✓ Accelerated charges radiate EM waves.
- ✓ Transverse, $E \perp B \perp$ direction, in phase, no medium needed.
- ✓ $c = 1/\sqrt{(\mu_0\varepsilon_0)}$, $E_0/B_0 = c$.
- ✓ Spectrum: radio, micro, IR, visible, UV, X, γ (frequency up).

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM EM waves are produced by:
(a) static charges
(b) charges in uniform motion
(c) accelerated charges
(d) neutral particles

Ans: (c) accelerated charges.

Q2. IM Board fav The waves used in LASIK eye surgery are:
(a) infrared
(b) ultraviolet
(c) X-rays
(d) microwaves

Ans: (b) ultraviolet.

Q3. IM The ratio E_0/B_0 for an EM wave in vacuum is:
(a) c
(b) 1/c
(c) c^2
(d) 1

Ans: (a) c.

Q4. IM Which has the longest wavelength?
(a) X-rays
(b) UV
(c) infrared
(d) radio waves

Ans: (d) radio waves.

Q5. IM Displacement current is due to:
(a) flow of electrons
(b) changing electric field
(c) changing magnetic field
(d) flow of ions

Ans: (b) changing electric field.

Q6. 1M The waves used in radar are:
(a) radio
(b) infrared
(c) microwaves
(d) γ-rays
Ans: (c) microwaves.

Q7. 1M Board fav Which EM waves are called heat waves? Why?
Ans: Infrared: they are readily absorbed by water molecules and raise temperature.

Q8. 1M What is the phase difference between E and B in an EM wave?
Ans: Zero: they are in phase.

Q9. 1M Name the EM waves used to kill germs in water purifiers.
Ans: Ultraviolet rays.

Q10. 1M Write the expression for the speed of EM waves in a medium.
Ans: $v = 1/\sqrt{(\mu\varepsilon)}$.

Q11. 1M Which part of the spectrum does a TV remote use?
Ans: Infrared.

Q12. 1M Why are microwaves used in ovens?
Ans: Their frequency matches the resonant frequency of water molecules, which absorb the energy and heat the food.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M Board fav **A:** EM waves are transverse. **R:** E and B are perpendicular to each other and to the direction of propagation.
Ans: (a)

Q14. 1M **A:** EM waves need a medium to travel. **R:** They are mechanical waves.
Ans: Both A and R are false: EM waves travel in vacuum and are not mechanical.

Q15. 1M **A:** Displacement current is present between the plates of a charging capacitor. **R:** The electric field between the plates changes with time.
Ans: (a)

Q16. 1M **A:** X-rays are used to detect fractures. **R:** X-rays pass through flesh but are absorbed more by bones.
Ans: (a)

C. Short answer (2–3 marks)

Q17. 2M Board fav What is displacement current? Why was it introduced?
Ans: $i_d = \varepsilon_0 d\Phi_E/dt$; to remove the inconsistency of Ampere's law for a charging capacitor and make total current continuous.

Q18. 2M Write the Ampere-Maxwell law and explain each term.
Ans: $\oint B \cdot dl = \mu_0(i_c + \varepsilon_0 d\Phi_E/dt)$: conduction current plus displacement current produce B.

Q19. 2M Board fav List any four properties of EM waves.
Ans: Transverse; no medium needed; speed c in vacuum; $E_0/B_0 = c$; carry energy and momentum.

Q20. 2M Draw a labelled diagram of a plane EM wave travelling along z.
Ans: E along x, B along y, sinusoidal and in phase (see Fig. 8.3 with axes relabelled).

Q21. 2M Arrange in increasing order of frequency: X-rays, microwaves, UV, radio, IR, γ-rays, visible.
Ans: Radio < microwave < IR < visible < UV < X-rays < γ-rays.

Q22. 2M Board fav Name the EM waves used for (a) cancer treatment, (b) satellite communication, (c) night vision, (d) sterilisation.
Ans: (a) γ-rays / X-rays; (b) microwaves; (c) infrared; (d) ultraviolet.

Q23. 2M How are X-rays produced? Give two uses.
Ans: By striking a metal target with high-energy electrons; medical imaging, study of crystal structure.

Q24. 1M Why is the ozone layer important?
Ans: It absorbs most harmful UV radiation from the sun.

Q25. 3M Show that the average energy density of E equals that of B in an EM wave.
Ans: $u_E = \frac{1}{2}\varepsilon_0 E^2$, $u_B = B^2/2\mu_0 = (E/c)^2/2\mu_0 = \frac{1}{2}\varepsilon_0 E^2$ (using $c^2 = 1/\mu_0\varepsilon_0$).

Q26. 2M Board fav A capacitor of plate area A is being charged. Show that $i_d = i_c$.
Ans: $E = q/\varepsilon_0 A$, $\Phi_E = q/\varepsilon_0$; $i_d = \varepsilon_0 d\Phi_E/dt = dq/dt = i_c$.

D. Long answer (5 marks)

Q27. 5M Board fav (a) Explain the need for displacement current with a charging capacitor. (b) Write Maxwell's generalised Ampere law. (c) Show $i_d = i_c$.
Ans: (a) Two surfaces with the same boundary give different results. (b) $\oint B \cdot dl = \mu_0(i_c + i_d)$. (c) As in Q26.

Q28. 5M (a) List the main parts of the EM spectrum in order of wavelength. (b) Give the source and one use of each.
Ans: See the spectrum table.

E. Case-based

Q29. 4M Board fav Mobile phones use waves of about 900 MHz to 2 GHz. (a) Which band? (b) Wavelength at 900 MHz? (c) Do they need a medium? (d) Speed in air?
Ans: (a) Microwave/UHF radio. (b) $\lambda = 0.33$ m. (c) No. (d) $\approx 3 \times 10^8$ m/s.

Q30. 4M Comet tails point away from the sun. (a) What pushes them? (b) Do EM waves carry momentum? (c) Momentum for energy U absorbed? (d) For reflection?
Ans: (a) Radiation pressure of sunlight (and solar wind). (b) Yes. (c) U/c. (d) 2U/c.

Q31. 1M An EM wave has $E_0 = 48$ V/m. Find B_0.
Ans: $B_0 = E_0/c = 1.6 \times 10^{-7}$ T.

Q32. 1M Find the frequency of EM waves of wavelength 600 nm.
Ans: $\nu = c/\lambda = 5 \times 10^{14}$ Hz.

Q33. 2M Board fav The electric field of an EM wave is $E = 30 \sin(2 \times 10^8 t - kx)$ V/m. Find k and λ.
Ans: $k = \omega/c = 2/3$ m^{-1}; $\lambda = 2\pi/k = 9.42$ m.

Q34. 1M Name the scientist who first produced EM waves in the laboratory.
Ans: Heinrich Hertz (1887).

Q35. 1M Why do welders wear special glasses?
Ans: To protect the eyes from intense UV produced during welding.

Q36. 1M What is the wavelength range of visible light?
Ans: About 400 nm to 700 nm.

Q37. 1M Write the expression relating c, μ_0 and ε_0.
Ans: $c = 1/\sqrt{(\mu_0\varepsilon_0)}$.

Q38. 1M Why can we not see infrared?
Ans: The eye's receptors are sensitive only to 400–700 nm.

Q39. 1M Which EM waves are used in the treatment of muscular strain?
Ans: Infrared (physiotherapy).

Q40. 2M Board fav A plane EM wave travels along +y with E along +z. What is the direction of B?
Ans: $E \times B$ must be along +y: $\hat{z} \times B \parallel \hat{y}$ $\Longrightarrow$ B along −x.

F. Extra practice (board PYQ style)

Q41. 3M Board fav Suppose the electric field amplitude of an EM wave is 120 N/C and $\nu = 50$ MHz. Find B_0, ω, k and λ.
Ans: $B_0 = 400$ nT; $\omega = 3.14 \times 10^8$ rad/s; $k = 1.05$ rad/m; $\lambda = 6$ m.

Q42. 1M Which physical quantity is the same for X-rays of 1 Å, red light of 6800 Å and radio waves of 500 m?
Ans: Their speed in vacuum, c.

Q43. 1M The charging current of a capacitor is 0.25 A. What is the displacement current across its plates?
Ans: 0.25 A ($i_d = i_c$).

Q44. 2M How does a microwave oven heat food without heating the plate much?

Ans: Water molecules in food absorb microwave energy (resonance); dry glass or ceramic plates absorb little.

Q45. 2M Why is the sky blue but sunsets red? (EM-wave view)

Ans: Shorter wavelengths (blue) scatter more from air molecules; at sunset light travels a longer path, blue is scattered away, red remains.

Q46. 2M Give the order of magnitude of the wavelength of (a) γ-rays, (b) microwaves, (c) visible light.

Ans: (a) < 10^{-12} m; (b) 10^{-3} to 10^{-1} m; (c) 10^{-7} m.

Q47. 1M What is the source of energy of EM waves from an antenna?

Ans: Oscillating (accelerated) charges in the antenna, driven by an AC source.

Q48. 2M Board fav Name the EM radiation (a) produced by klystron, (b) used in greenhouses, (c) that causes sun-tan.

Ans: (a) Microwaves; (b) infrared; (c) ultraviolet.

Quick compare: conduction vs displacement current

Conduction current	Displacement current
Flow of charges	Changing electric field
In conductors (wires)	In gaps/dielectrics (capacitor plates)
Exists for DC and AC	Exists only when E changes
Produces magnetic field	Also produces magnetic field

Yaad karo: one-line answers

→ Radar: microwave. Remote: IR. LASIK: UV. Fracture: X-ray. Cancer: γ-ray.
→ EM wave = accelerated charge ka gift.
→ E × B = direction of propagation.
→ In medium: v = c/n.

EM waves itna aasan chapter, beta padh le

Asli baat: Short chapter, sure marks: displacement current + properties + spectrum uses.

Common mistakes

→ Displacement current charge ka flow nahi hai.
→ $E_0 = cB_0$, $B_0 = cE_0$ nahi.
→ Spectrum mein UV aur IR ki position ulta mat karna.
→ Microwave aur radio ke uses mix mat karo.

Exam Alert: last check

→ Displacement current derivation, properties, spectrum uses: 90% paper isi se.
→ Direction of propagation = E × B.

Ray Optics and Optical Instruments

Most asked: mirror and lens formula derivations, refraction at a spherical surface, lens maker's formula, total internal reflection, prism (minimum deviation), power and combination of lenses, compound microscope and telescope

Light travels in straight lines (rays) when its wavelength is tiny compared with the objects it meets. Using just two laws, reflection and refraction, we can explain mirrors, lenses, prisms, rainbows, optical fibres, microscopes and telescopes.

1. Reflection by Spherical Mirrors

Laws of reflection

1. Angle of incidence = angle of reflection ($i = r$).
2. Incident ray, reflected ray and the normal at the point of incidence lie in the same plane.

Sign convention (Cartesian)

- → All distances are measured from the pole P (or optical centre O).
- → Distances in the direction of incident light are positive; against it, negative.
- → Heights above the principal axis are positive; below, negative.
- → Object is usually placed on the left, so u is negative.

Term	Meaning
Pole P	centre of the mirror surface
Centre of curvature C	centre of the sphere
Radius R	distance PC
Focus F	point where paraxial rays parallel to the axis meet (or appear to diverge from)
Focal length f	PF = R/2

Derivation step by step: f = R/2

1. A ray parallel to the axis strikes the mirror at M and reflects through F. CM is the normal, so $\angle\theta$ (incidence) = $\angle\theta$ (reflection).
2. In triangle MCF, $\angle MCF = \theta$ and $\angle MFP = 2\theta$.
3. For paraxial rays: $\tan\theta \approx \theta = MD/CD$ and $\tan 2\theta \approx 2\theta = MD/FD$.
4. So $MD/FD = 2MD/CD \Longrightarrow FD = CD/2$. With D ≈ P: **f = R/2.**

Rules for drawing ray diagrams (mirrors)

- → A ray parallel to the principal axis passes through F after reflection (concave) or appears to come from F (convex).
- → A ray through (or towards) F becomes parallel to the axis.
- → A ray through (or towards) C retraces its path.
- → A ray striking the pole P reflects symmetrically about the axis.

Mirror equation

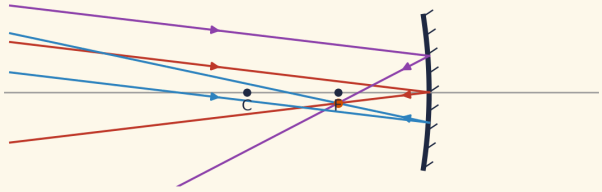
$$\mathbf{1/v + 1/u = 1/f} \quad m = h'/h = -v/u$$

Derivation step by step (mirror formula)

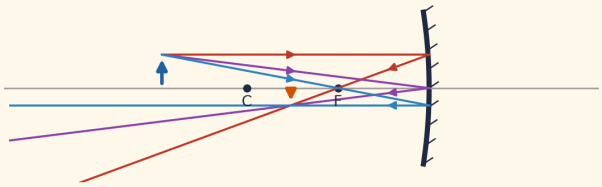
1. Object AB at distance u, image A′B′ at distance v; ray from A parallel to axis reflects through F; ray through P reflects symmetrically.
2. Triangles A′B′F and MPF are similar: $B'A'/PM = B'F/FP$; since PM = BA, $B'A'/BA = B'F/FP$.
3. Triangles A′B′P and ABP are similar: $B'A'/BA = B'P/BP$.
4. Equate: $B'F/FP = B'P/BP$, i.e. $(B'P - FP)/FP = B'P/BP$.
5. With signs: $B'P = -v$, $FP = -f$, $BP = -u \Longrightarrow (-v + f)/(-f) = -v/(-u) \Longrightarrow$ **1/v + 1/u = 1/f.**

Object position (concave)	Image	Nature
At infinity	at F	real, inverted, highly diminished
Beyond C	between C and F	real, inverted, diminished
At C	at C	real, inverted, same size
Between C and F	beyond C	real, inverted, enlarged
At F	at infinity	real, inverted, highly enlarged
Between F and P	behind mirror	virtual, erect, enlarged

(i) Object at infinity: image at F, real, point-sized



(ii) Beyond C: between F and C, real, diminished



(iii) At C: at C, real, inverted, same size

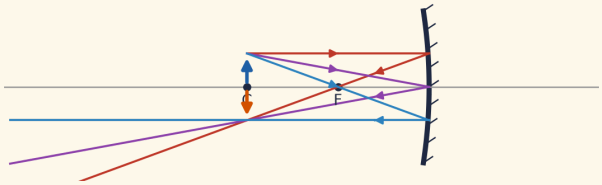
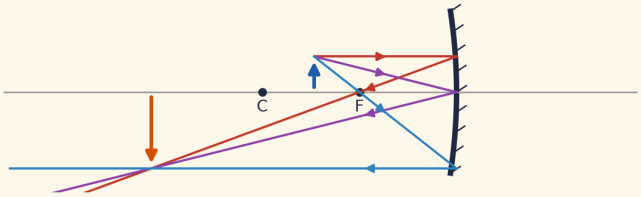
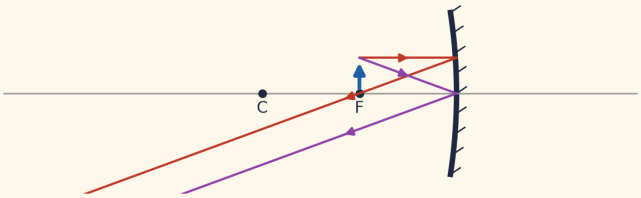


Fig. 9.1(a) Concave mirror, cases (i)–(iii). Blue = object, orange = real image, green = virtual image, dashed = rays extended back

(iv) Between C and F: beyond C, real, enlarged



(v) At F: at infinity, highly enlarged



(vi) Between F and P: behind mirror, virtual, erect, enlarged

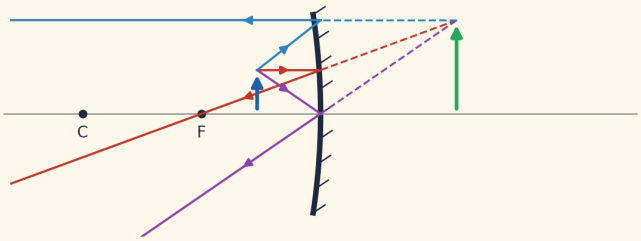


Fig. 9.1(b) Concave mirror, cases (iv)–(vi)

Image formation by a convex mirror

Convex mirror: virtual, erect, diminished

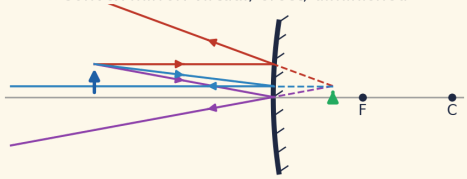


Fig. 9.2 Convex mirror: for any object position the image is virtual, erect, diminished, between P and F

Object (convex mirror)	Image	Nature
At infinity	at F (behind)	virtual, erect, point-sized
Anywhere in front	between P and F	virtual, erect, diminished

→ **Uses of concave mirrors**: shaving and make-up mirrors (object within F), dentist's mirror, torch and headlight reflectors (bulb at F), solar furnaces.

→ **Uses of convex mirrors**: rear-view mirrors in vehicles (wide field of view, always erect), security mirrors in shops, sharp turns on hill roads.

Spoon dekho toh concave mirror bhi, convex mirror bhi

Asli baat: The inside of a spoon is a concave mirror (inverted image far away); the back is convex (small erect image).

Solved example (NCERT)

Q. An object is placed at (i) 10 cm, (ii) 5 cm in front of a concave mirror of R = 15 cm. Find the image position and nature.

Sol. f = −7.5 cm. (i) u = −10: 1/v = −1/7.5 + 1/10 ⟹ v = **−30 cm**, real, m = −3 (enlarged, inverted). (ii) u = −5: 1/v = −1/7.5 + 1/5 ⟹ v = **+15 cm**, virtual behind the mirror, m = +3.

2. Refraction

Snell's law

$$n_{21} = \sin i / \sin r = n_2/n_1 = v_1/v_2$$

→ Light bends towards the normal entering a denser medium ($n_2 > n_1$).

→ Frequency stays the same; speed and wavelength change.

→ $n_{12} = 1/n_{21}$; $n_{32} = n_{31} \times n_{12}$.

→ A glass slab shifts a ray sideways (lateral shift) without changing its direction.

Apparent depth

real depth / apparent depth = n apparent depth = h/n (viewed normally)

NCERT Activity: pencil in water

Dip a pencil partly in a glass of water at an angle: it appears bent at the surface. A coin at the bottom of a bucket appears raised. Both happen because rays from the submerged part bend away from the normal as they leave water.

Solved example

Q. A tank of water (n = 1.33) is 12.5 cm deep. A needle at the bottom appears raised by how much? If the water is replaced by a liquid of n = 1.63?

Sol. Apparent depth = 12.5/1.33 = 9.4 cm ⟹ raised by **3.1 cm**. For n = 1.63: 7.67 cm ⟹ raised by **4.83 cm**.

Laws of refraction and their meaning

1. The incident ray, the refracted ray and the normal at the point of incidence lie in the same plane.
2. Snell's law: $\sin i/\sin r = n_{21} = n_2/n_1$, a constant for a given pair of media and a given colour of light.

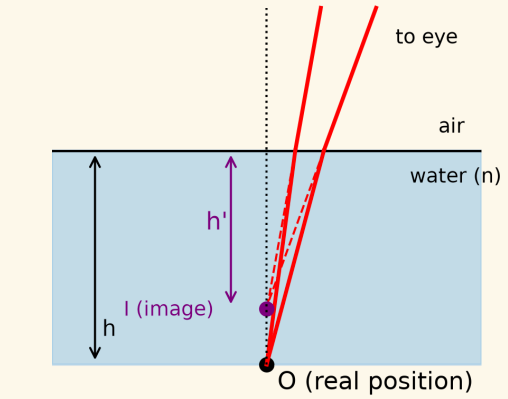
Refraction happens because light travels with different speeds in different media: $n = c/v$. When light enters a denser medium (larger n) it slows down and bends **towards** the normal; when it enters a rarer medium it speeds up and bends **away** from the normal. The frequency does not change during refraction; the wavelength changes in the same ratio as the speed ($\lambda_{medium} = \lambda_{vacuum}/n$).

→ $n_{21} = 1/n_{12}$, and $n_{32} = n_{31} \times n_{12}$.

→ Refractive index of a medium is larger for violet than for red light, so violet bends more.

→ A ray falling along the normal (i = 0) passes undeviated.

Apparent depth: h' = h/n (near-normal viewing)



Apparent depth: rays from O bend away from the normal at the water surface and appear to come from I, which is nearer to the surface.

Apparent depth: derivation step by step

1. An object O is at real depth h below the surface of water of refractive index n, viewed from directly above.
2. A ray from O meets the surface at a small distance x from the normal, making angle r with the normal in water; in air it emerges at angle i (i > r).
3. For near-normal viewing, sin ≈ tan: $\sin i/\sin r = n$ gives $(x/h')/(x/h) = n$, where h' is the depth of the image I.
4. So $h' = h/n$, and the object appears raised by $h - h' = h(1 - 1/n)$.

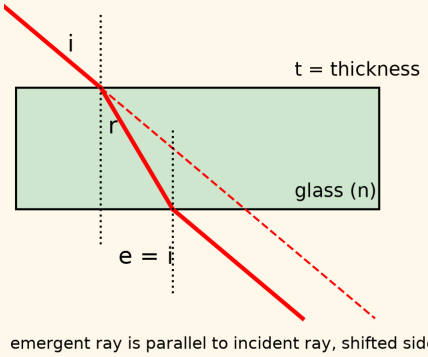
Yaad karo

→ Paani ka swimming pool hamesha kam gehra dikhta hai: $h' = h/n$.

→ Raised by $h(1 - 1/n)$: slab ke liye bhi same, $t(1 - 1/n)$.

→ Frequency kabhi nahi badalti, speed aur λ badalte hain.

Lateral shift $d = t \sin(i - r)/\cos r$



emergent ray is parallel to incident ray, shifted sideways by d

Refraction through a rectangular glass slab: the emergent ray is parallel to the incident ray but laterally shifted.

Glass slab: since the two faces are parallel, the angle of emergence equals the angle of incidence (e = i). The ray suffers no deviation but a lateral shift d = t sin(i − r)/cos r, which increases with thickness t and with i.

Solved example (NCERT type)

A tank is filled with water to a height of 12.5 cm. The apparent depth of a needle lying at the bottom, measured by a microscope, is 9.4 cm. (a) Find the refractive index of water. (b) If water is replaced by a liquid of n = 1.63 up to the same height, by what distance must the microscope be moved to focus on the needle again?

(a) n = real/apparent = 12.5/9.4 = **1.33**. (b) New apparent depth = 12.5/1.63 = 7.67 cm. The image rises by 9.4 − 7.67 = 1.73 cm, so the microscope must be moved **up by 1.73 cm**.

Solved example

A pin is viewed from above through a glass slab of thickness 15 cm and n = 1.5. By how much does it appear raised? Does this depend on where the slab is placed?

Shift = t(1 − 1/n) = 15(1 − 1/1.5) = **5 cm**. The shift does not depend on the position of the slab, only on t and n.

3. Total Internal Reflection

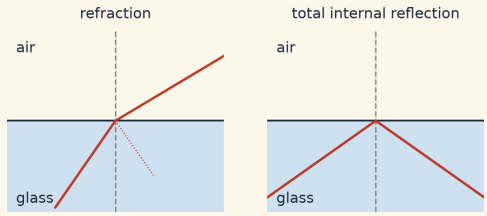


Fig. 9.2 Above the critical angle, all light is reflected back into the denser medium

When light goes from a denser to a rarer medium, the refracted ray bends away from the normal. At the **critical angle** i_c the refracted ray grazes the surface (r = 90°). For $i > i_c$, no refraction happens: all light is reflected. This is **total internal reflection (TIR)**.

$$\sin i_c = 1/n \text{ (denser medium to air)}$$

→ **Conditions:** light must go from denser to rarer, and $i > i_c$.

→ Critical angle: water 48.75°, crown glass 41°, diamond 24.4°.

Applications of TIR

→ **Mirage:** air near hot ground is rarer; light from the sky bends and is totally reflected, looking like water.

→ **Diamond:** i_c = 24.4°, so light reflects many times inside a cut diamond and it sparkles.

→ **Prisms:** right-angled prisms bend light by 90° or 180° (periscopes, binoculars).

→ **Optical fibres:** glass core (high n) with cladding (lower n); light travels by repeated TIR with little loss. Used in communication and endoscopy.

NCERT Activity: TIR with a laser

Shine a laser pointer through the side of a glass of water containing a few drops of milk (to make the beam visible), aiming upwards at the water surface. Increase the angle slowly: at a particular angle the beam stops escaping and reflects fully back into the water.

Sign convention bhool gaye?
Ab answer ulta
Asli baat: Measure from the pole; along incident light positive, against it negative. Most wrong answers in optics are sign errors.

Critical angle: derivation and dependence

When light goes from a denser medium (n_1) to a rarer medium (n_2), the refracted ray bends away from the normal. As i increases, r reaches 90° at a particular angle of incidence called the **critical angle** C. Putting r = 90° in Snell's law: $n_1 \sin C = n_2 \sin 90°$, so $\sin C = n_2/n_1$; for a denser medium in air, $\sin C = 1/n$.

Medium	n	Critical angle
Water	1.33	48.8°
Crown glass	1.52	41.1°
Dense flint glass	1.62	38.1°
Diamond	2.42	24.4°

→ Two conditions for TIR: light must go from denser to rarer, and i must exceed C.

→ Larger n means smaller C, so TIR occurs more easily (diamond sparkles because C is only 24.4°).

→ C is larger for red light than for violet, since $n_{red} < n_{violet}$.

Solved example (NCERT type)

A small bulb is placed at the bottom of a tank containing water (n = 1.33) to a depth of 80 cm. Find the area of the water surface through which light from the bulb can emerge.

sin C = 1/1.33, so C = 48.8° and tan C = 1.14. Light emerges through a circle of radius r = h tan C = 0.80 × 1.14 = 0.91 m. Area = $\pi r^2 = \pi \times (0.91)^2$ = **2.6 m²**.

Solved example (NCERT type)

An optical fibre has core of n = 1.68 and cladding of n = 1.44. Find the range of angles of incidence (with the axis, at the end face) for which light is totally reflected inside.

At core-cladding surface: sin C = 1.44/1.68 = 0.857, C = 59°. For TIR the angle with the axis inside must be at most 90° − 59° = 31°. At the end face: sin i = 1.68 sin 31° = 0.865, so i = 60°. Hence TIR for **0 < i < 60°**.

4. Refraction at Spherical Surfaces and Lenses

Derivation step by step (refraction at a convex surface)

1. Object O in medium n_1, image I in medium n_2; surface with centre C, radius R; ray hits at N, with normal NC.
2. Snell (small angles): $n_1 i = n_2 r$.
3. Angles: i = ∠NOM + ∠NCM, r = ∠NCM − ∠NIM; for small angles ∠NOM = MN/OM, ∠NCM = MN/MC, ∠NIM = MN/MI.
4. Substitute: $n_1/OM + n_2/MI = (n_2 - n_1)/MC$.
5. With signs (OM = −u, MI = +v, MC = +R):

$$n_2/v - n_1/u = (n_2 - n_1)/R$$

Lens maker's formula (derivation)

1. First surface (radius R_1) forms an image I_1 at v_1: $n_2/v_1 - n_1/u = (n_2 - n_1)/R_1$.
2. I_1 acts as object for the second surface (radius R_2): $n_1/v - n_2/v_1 = (n_1 - n_2)/R_2$.
3. Add (thin lens): $n_1(1/v - 1/u) = (n_2 - n_1)(1/R_1 - 1/R_2)$.
4. For u = ∞, v = f:

$$1/f = (n_{21} - 1)(1/R_1 - 1/R_2)$$ lens maker's formula

$$1/v - 1/u = 1/f$$ thin lens formula; m = v/u

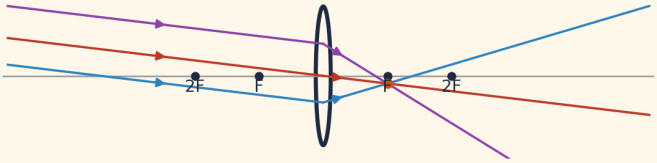
Rules for drawing ray diagrams (lenses)

→ A ray parallel to the axis passes through F_2 (convex) or appears to come from F_1 (concave).

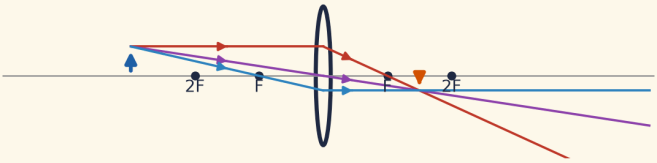
→ A ray through the optical centre O goes straight, undeviated.

→ A ray through F_1 (convex) or aimed at F_2 (concave) emerges parallel to the axis.

(i) At infinity: at F₂, real, point-sized



(ii) Beyond 2F₁: between F₂ and 2F₂, diminished



(iii) At 2F₁: at 2F₂, real, same size

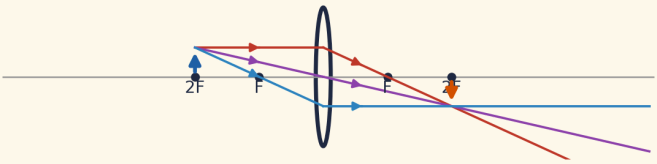
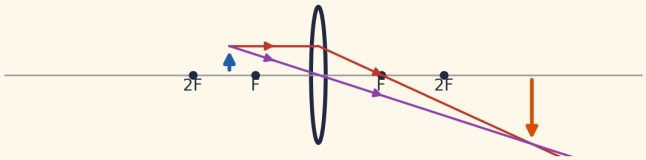
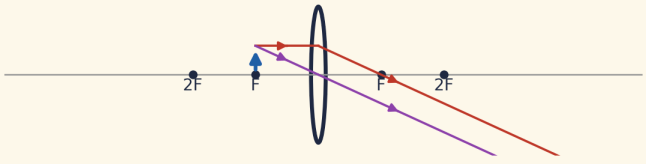


Fig. 9.4(a) Convex lens, cases (i)–(iii)

(iv) Between F₁ and 2F₁: beyond 2F₂, enlarged



(v) At F₁: at infinity, highly enlarged



(vi) Between F₁ and O: same side, virtual, erect, enlarged

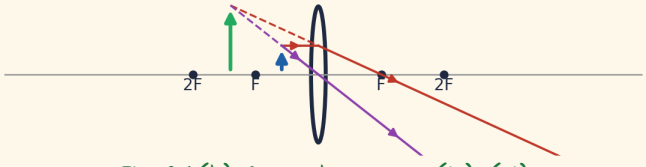


Fig. 9.4(b) Convex lens, cases (iv)–(vi)

Object (convex lens)	Image	Nature
At infinity	at F_2	real, inverted, point-sized
Beyond $2F_1$	between F_2 and $2F_2$	real, inverted, diminished
At $2F_1$	at $2F_2$	real, inverted, same size
Between F_1 and $2F_1$	beyond $2F_2$	real, inverted, enlarged
At F_1	at infinity	real, inverted, very large
Between F_1 and O	same side as object	virtual, erect, enlarged

Image formation by a concave lens

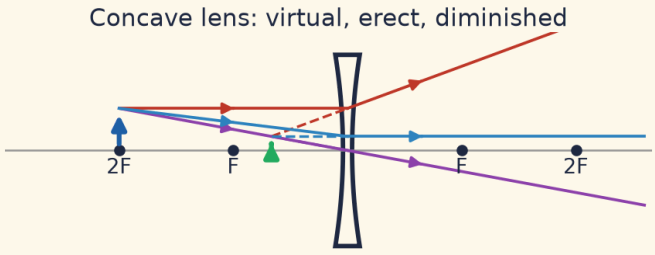


Fig. 9.5 Concave lens: image always virtual, erect, diminished, between O and F_1

Object (concave lens)	Image	Nature
At infinity	at F_1	virtual, erect, point-sized
Anywhere in front	between O and F_1	virtual, erect, diminished

NCERT Activity: image by a convex lens

Hold a convex lens in front of a white wall facing a window. Move it until a sharp, inverted, small image of the window forms on the wall. The lens-to-wall distance is roughly the focal length. Bring a candle closer to the lens and watch the image grow and move away.

Power of a lens

P = 1/f (f in metres); unit dioptre (D) convex: +ve, concave: −ve

Combination of thin lenses in contact

1. First lens: $1/v_1 - 1/u = 1/f_1$.
2. Its image is the object for the second lens: $1/v - 1/v_1 = 1/f_2$.
3. Add: $1/v - 1/u = 1/f_1 + 1/f_2$.

$$1/f = 1/f_1 + 1/f_2;\ P = P_1 + P_2;\ m = m_1 m_2$$

Convex lens, real image... virtual bhi?! PANIK

Asli baat: Object inside F of a convex lens gives a virtual, erect, enlarged image: that is how a magnifying glass works.

Solved example (NCERT)

Q. A double convex lens has radii 20 cm each and n = 1.55. Find f.

Sol. $R_1 = +20$, $R_2 = -20$: $1/f = 0.55(1/20 + 1/20) = 0.055 \Longrightarrow$ **f = 18.2 cm.**

Solved example (NCERT)

Q. A magician holds a lens (n = 1.47) in a trough of liquid and it vanishes. What is the liquid's n? Could it be water?

Sol. The lens disappears when $n_{liquid} = n_{lens} =$ **1.47** (no refraction at its surfaces). Water (1.33) will not work; glycerine (1.47) will.

Solved example

Q. Find the power of a combination of a convex lens of 25 cm and a concave lens of 20 cm in contact. Is it converging?

Sol. $P = 1/0.25 - 1/0.20 = 4 - 5 =$ **−1 D**: diverging, f = −100 cm.

Lens maker's formula: what it tells us

- → Focal length depends on the material (n) and the curvature of both faces.
- → A lens behaves differently in different media: $1/f = (n_{lens}/n_{medium} - 1)(1/R_1 - 1/R_2)$.
- → A convex lens made of glass (n = 1.5) placed in water (n = 1.33) becomes weaker: its focal length increases about four times.
- → If $n_{lens} = n_{medium}$, the lens becomes invisible and has no focusing effect ($f = \infty$).
- → If $n_{lens} < n_{medium}$, a convex lens behaves as a diverging lens (an air bubble in water acts as a concave lens).

Solved example (NCERT type)

A double convex lens of glass (n = 1.55) has both faces of the same radius. Find the radius required for a focal length of 20 cm.

$1/f = (n - 1)(1/R + 1/R) = 0.55 \times 2/R$. So $R = 1.1 \times 20 =$ **22 cm.**

Solved example (NCERT type)

A beam of light converges at a point P. A lens is placed in its path 12 cm from P. Where does the beam converge if the lens is (a) convex with f = 20 cm, (b) concave with f = 16 cm?

Here P acts as a virtual object: u = +12 cm. (a) 1/v = 1/f + 1/u = 1/20 + 1/12, v = **7.5 cm** (converges closer). (b) 1/v = −1/16 + 1/12 = 1/48, v = **48 cm** (converges farther away).

5. Refraction through a Prism

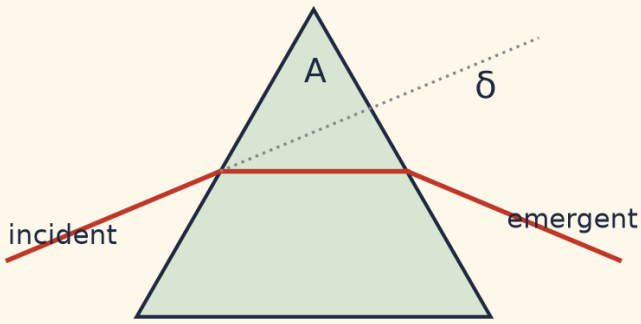


Fig. 9.4 Deviation δ by a prism of angle A

Derivation step by step (prism formula)

1. At the first face: angles i and r_1; at the second face: r_2 and e.
2. The normals and the faces give $r_1 + r_2 = A$.
3. Total deviation: $\delta = (i - r_1) + (e - r_2) = i + e - A$.
4. At minimum deviation D_m: i = e and $r_1 = r_2 = A/2$; $i = (A + D_m)/2$.
5. Snell at first face: $n = \sin i / \sin r_1$:

$$n = \sin[(A + D_m)/2] / \sin(A/2)$$ thin prism: $\delta = (n - 1)A$

→ At minimum deviation, the ray inside the prism is parallel to the base.

→ A graph of δ against i falls to D_m and then rises.

→ Different colours have different n, so violet deviates most and red least: dispersion.

Solved example

Q. A prism of angle 60° gives minimum deviation 30°. Find n.

Sol. n = sin 45°/sin 30° = 0.707/0.5 = **1.414**.

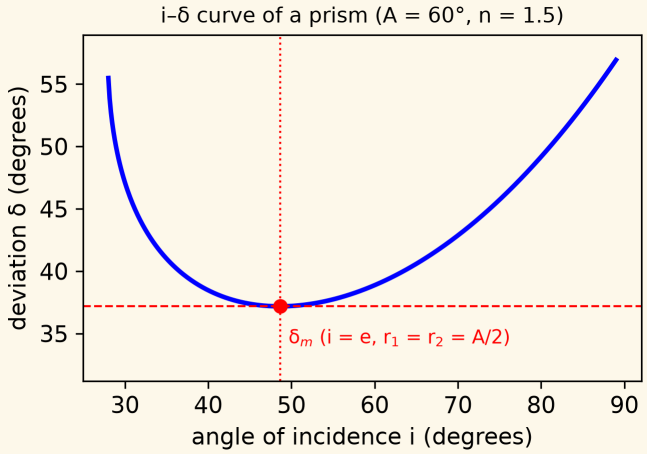
Paheli-Bujho

Bujho: Paheli, rainbow mein violet niche aur red upar kyun?

Paheli: Paani ki boondein prism jaisi hain. Violet zyada mudta hai, red kam, toh aankh tak alag angle se aate hain.

Bujho: Matlab barish ki boondein bhi optics padhti hain!

Paheli: Bilkul, aur dispersion ka exam mein question pakka.



Graph of deviation δ against angle of incidence i for a prism. Deviation is minimum when the ray passes symmetrically (i = e).

Minimum deviation: as i increases, the deviation first decreases, reaches a minimum value δ_m, and then increases. At minimum deviation the ray inside the prism is parallel to the base, i = e and $r_1 = r_2 = A/2$. For any value of δ other than δ_m, there are two values of i (because the path of light is reversible).

Thin prism and dispersion

For a prism of small angle A, sin ≈ angle, so δ = (n − 1)A. Since n depends on colour, violet (larger n) deviates more than red; white light therefore splits into its colours. This splitting is called **dispersion**, and the band of colours is the spectrum (VIBGYOR, red deviating least).

Solved example (NCERT type)

A prism of angle 60° gives a minimum deviation of 40°. Find its refractive index. What would be the minimum deviation if the prism were placed in water (n = 1.33)?

$n = \sin[(A + \delta_m)/2]/\sin(A/2) = \sin 50°/\sin 30° =$ **1.532**. In water $n_{rel} = 1.532/1.33 = 1.152$, so $\sin[(60 + \delta)/2] = 1.152 \times 0.5 = 0.576$, giving $(60 + \delta)/2 = 35.2°$ and $\delta_m =$ **10.3°**.

6. Optical Instruments

The eye

→ The eye lens focuses images on the retina; ciliary muscles change its focal length (accommodation).

→ Near point (least distance of distinct vision) D = 25 cm; far point = infinity for a normal eye.

→ Myopia (near-sightedness): corrected by a concave lens. Hypermetropia: convex lens. Astigmatism: cylindrical lens. Presbyopia: bifocal lens.

Simple microscope (magnifying glass)

$m = 1 + D/f$ (image at D); $m = D/f$ (image at infinity)

Compound microscope

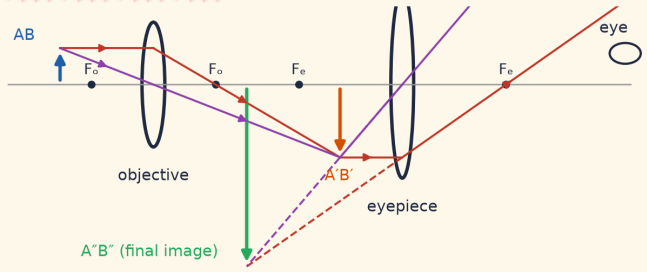


Fig. 9.7 Ray diagram of a compound microscope (final image at least distance of distinct vision)

Derivation step by step (magnifying power)

1. Objective: object AB just outside F_o gives real, inverted, enlarged image A'B'; $m_o = v_o/u_o = h'/h$.
2. Using similar triangles at the objective: $m_o = L/f_o$, where L (tube length) is the distance between the second focus of the objective and the first focus of the eyepiece.
3. Eyepiece acts as a simple microscope on A'B': $m_e = 1 + D/f_e$ (image at D) or D/f_e (image at infinity).
4. Total: $m = m_o \times m_e$.

→ Objective of short focal length forms a real, enlarged image; eyepiece acts as a simple microscope to magnify it further.

→ The final image is virtual, inverted and highly enlarged.

$m = m_o \times m_e$; $m \approx (L/f_o)(D/f_e)$ image at infinity; L = tube length

Telescope (refracting)

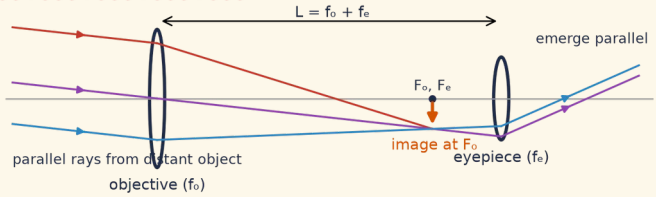


Fig. 9.8 Astronomical telescope in normal adjustment: final image at infinity

Derivation step by step (magnifying power)

1. Rays from a distant object make angle α with the axis; the objective forms a real, inverted image of height h at its focus: $\tan\alpha \approx \alpha = h/f_o$.
2. This image lies at the first focus of the eyepiece (normal adjustment), so rays emerge parallel making angle β: $\beta \approx h/f_e$.
3. Magnifying power $m = \beta/\alpha = f_o/f_e$; tube length $L = f_o + f_e$.
4. If the final image is at D: $m = (f_o/f_e)(1 + f_e/D)$.

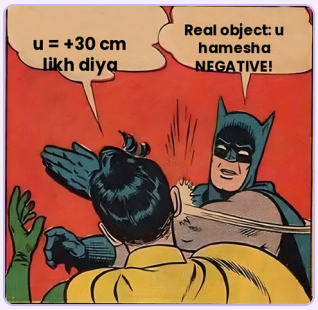
→ Objective of large focal length and aperture collects light from a distant object; eyepiece of short focal length magnifies the image.

→ Normal adjustment: final image at infinity.

$m = f_o/f_e$; length $L = f_o + f_e$ normal adjustment

→ **Reflecting telescope (Cassegrain)**: uses a concave mirror as objective. No chromatic aberration, less spherical aberration, a mirror is lighter and easier to support than a huge lens.

	Microscope	Telescope
Objective focal length	small	large
Objective aperture	small	large
Used for	tiny nearby objects	distant objects
Magnification	$(L/f_o)(D/f_e)$	f_o/f_e



u ka sign positive daala? Thappad

Asli baat: Real objects in front of a mirror or lens: u is always negative in the Cartesian convention.

Solved example (NCERT)

Q. A telescope has an objective of focal length 140 cm and an eyepiece of 5 cm. Find the magnifying power in normal adjustment and the tube length.

Sol. $m = 140/5 = \mathbf{28}$; $L = 140 + 5 = \mathbf{145\ cm}$.

Solved example (NCERT)

Q. A compound microscope has an objective of 2.0 cm and an eyepiece of 6.25 cm, 15 cm apart. Find the object distance for final image at 25 cm.

Sol. Eyepiece: $v_e = -25 \Longrightarrow 1/u_e = -1/25 - 1/6.25 \Longrightarrow u_e = -5$ cm. Objective image at $15 - 5 = 10$ cm: $1/u_o = 1/10 - 1/2 \Longrightarrow u_o = \mathbf{-2.5\ cm}$. $m_o = -4$, $m_e = 5 \Longrightarrow m = \mathbf{-20}$.

Natural phenomena due to sunlight

- **Rainbow**: sunlight is refracted, dispersed and internally reflected inside raindrops. The primary rainbow (one internal reflection) has red outside and violet inside, seen at about 40°–42° from the antisolar point; the secondary (two reflections) is fainter with colours reversed.
- **Scattering**: small air molecules scatter light with intensity $\propto 1/\lambda^4$ (Rayleigh). Blue is scattered most, so the sky looks blue; at sunrise and sunset light travels a long path, blue is removed, and the sun looks red.
- **Clouds**: large water droplets scatter all colours nearly equally, so clouds look white.

Defect	Cause	Correction
Myopia	eyeball too long / lens too strong	concave lens
Hypermetropia	eyeball too short / lens too weak	convex lens
Presbyopia	loss of accommodation with age	bifocal lens
Astigmatism	cornea not spherical	cylindrical lens

Solved example

Q. A myopic person has a far point of 80 cm. What lens is needed?

Sol. The lens must form an image of a distant object at 80 cm: $1/v - 1/u = 1/f$ with $u = \infty$, $v = -80 \Longrightarrow f = \mathbf{-80\ cm}$, $P = \mathbf{-1.25\ D}$ (concave).

Solved example

Q. A person needs to read at 25 cm but his near point is 50 cm. Find the lens.

Sol. $u = -25$, $v = -50$: $1/f = -1/50 + 1/25 = 1/50 \Longrightarrow f = \mathbf{+50\ cm}$, $P = \mathbf{+2\ D}$ (convex).

Solved example

Q. A small bulb is at the bottom of a tank of water ($n = 1.33$) 80 cm deep. Find the area of the water surface through which light can emerge.

Sol. $\sin i_c = 1/1.33 \Longrightarrow i_c = 48.75°$; radius $r = h \tan i_c = 80 \times 1.14 = 91$ cm; area $= \pi r^2 \approx \mathbf{2.6\ m^2}$.

Solved example

Q. An object is 20 cm in front of a convex lens of $f = 10$ cm; a concave lens of $f = 10$ cm is 5 cm behind the convex lens. Where is the final image?

Sol. Convex: $v_1 = 20$ cm. For the concave lens this image is 15 cm beyond it (virtual object): $u = +15 \Longrightarrow 1/v = -1/10 + 1/15 \Longrightarrow v = \mathbf{-30\ cm}$, i.e. 30 cm to the left of the concave lens (virtual).

NCERT Activity: find f of a concave mirror

Focus the image of a distant tree or window on a screen using a concave mirror. The distance between the mirror and the sharp image is its focal length. Then place a candle at different positions and check the image positions against the mirror table.

Solved example

A convex lens of focal length 20 cm made of glass ($n = 1.5$) is immersed in water ($n = 1.33$). Find its new focal length.

In air: $1/20 = (1.5 - 1)K$. In water: $1/f_w = (1.5/1.33 - 1)K = 0.128K$. Dividing, $f_w = 20 \times 0.5/0.128 = \mathbf{78\ cm}$ (about four times). The lens becomes weaker in water.

Solved example

Find the power of a combination of a convex lens of focal length 25 cm and a concave lens of focal length 40 cm placed in contact.

$P_1 = 100/25 = +4$ D, $P_2 = 100/(-40) = -2.5$ D. $P = P_1 + P_2 = \mathbf{+1.5\ D}$, so $F = 100/1.5 = \mathbf{66.7\ cm}$ (converging).

Solved example

A magnifying glass has a focal length of 5 cm. Find its magnifying power when the image is formed (a) at the near point (25 cm), (b) at infinity.

(a) $m = 1 + D/f = 1 + 25/5 = \mathbf{6}$. (b) $m = D/f = 25/5 = \mathbf{5}$.

Solved example

A ray of light is incident at 60° on a glass slab of thickness 6 cm and refractive index $\sqrt{3}$. Find the angle of refraction and the lateral shift.

$\sin r = \sin 60°/\sqrt{3} = 0.5$, so $r = \mathbf{30°}$. $d = t\sin(i - r)/\cos r = 6 \times \sin 30°/\cos 30° = 6 \times 0.5/0.866 = \mathbf{3.46\ cm}$.

7. Solved Numericals (Board Pattern)

N1. A convex lens of f = 20 cm forms an image of an object at 30 cm. Find v and m.
Sol. $1/v = 1/20 - 1/30 \Longrightarrow v = \mathbf{60\ cm}$; $m = v/u = \mathbf{-2}$.

N2. Find the critical angle for glass of n = 1.5.
Sol. $\sin i_c = 1/1.5 \Longrightarrow i_c = \mathbf{41.8°}$.

N3. A concave mirror of f = 15 cm forms an erect image 3 times the object. Find u.
Sol. $m = -v/u = 3 \Longrightarrow v = -3u$; $1/(-3u) + 1/u = -1/15 \Longrightarrow 2/(3u) = -1/15 \Longrightarrow u = \mathbf{-10\ cm}$.

N4. Power of a lens is −2.5 D. Find f and type.
Sol. $f = 1/P = \mathbf{-40\ cm}$, concave.

N5. A glass lens (n = 1.5) has f = 20 cm in air. Find f in water (n = 1.33).
Sol. $f_w/f_a = (1.5 - 1)/(1.5/1.33 - 1) = 0.5/0.128 \Longrightarrow f_w \approx \mathbf{78\ cm}$.

N6. Light enters a prism of A = 60°, n = $\sqrt{2}$ at minimum deviation. Find D_m and i.
Sol. $\sin[(60 + D)/2] = \sqrt{2}\sin 30° = 0.707 \Longrightarrow (60 + D)/2 = 45° \Longrightarrow D_m = \mathbf{30°}$, $i = \mathbf{45°}$.

More solved examples (NCERT type)

1. An object 3.0 cm high is placed 21 cm in front of a concave lens of focal length 21 cm. Describe the image. What happens as the object is moved farther away?

$u = -21$, $f = -21$: $1/v = 1/f + 1/u = -1/21 - 1/21$, so $v = \mathbf{-10.5\ cm}$ (virtual, same side). $m = v/u = 0.5$, height = **1.5 cm**, erect and diminished. As the object moves away, the image moves towards the focus and becomes smaller, but never goes beyond F.

2. A convex lens of focal length 30 cm is in contact with a concave lens of focal length 20 cm. Is the combination converging or diverging? Find its focal length.

$1/F = 1/30 - 1/20 = -1/60$, so $F = \mathbf{-60\ cm}$: the combination is **diverging**.

3. A compound microscope has an objective of focal length 2.0 cm and an eyepiece of focal length 6.25 cm, 15 cm apart. Where should the object be placed if the final image is at (a) 25 cm, (b) infinity? Find the magnifying power in each case.

(a) Eyepiece: $v_e = -25$, so $1/u_e = -1/25 - 1/6.25$ and $u_e = -5$ cm. Objective image at $v_o = 15 - 5 = 10$ cm; $1/u_o = 1/10 - 1/2$, $u_o = \mathbf{-2.5\ cm}$. $M = (v_o/|u_o|)(1 + D/f_e) = 4 \times 5 = \mathbf{20}$. (b) $u_e = -6.25$ cm, $v_o = 8.75$ cm; $1/u_o = 1/8.75 - 1/2$, $u_o = \mathbf{-2.59\ cm}$. $M = (8.75/2.59)(25/6.25) = \mathbf{13.5}$.

4. A person with near point 25 cm uses a compound microscope with objective f = 8.0 mm and eyepiece f = 2.5 cm, and sees an object placed 9.0 mm from the objective in sharp focus. Find the separation of the lenses and the magnifying power.

Objective: $1/v_o = 1/8 - 1/9$ (mm), $v_o = 72$ mm = 7.2 cm. Eyepiece (image at 25 cm): $1/u_e = -1/25 - 1/2.5$, $u_e = -2.27$ cm. Separation = $7.2 + 2.27 = \mathbf{9.47\ cm}$. $M = (72/9)(1 + 25/2.5) = 8 \times 11 = \mathbf{88}$.

5. A small telescope has an objective of focal length 144 cm and an eyepiece of focal length 6.0 cm. Find its magnifying power and the separation between the lenses (normal adjustment).

$m = f_o/f_e = 144/6 = \mathbf{24}$; $L = f_o + f_e = \mathbf{150\ cm}$.

6. A large refracting telescope has an objective of focal length 15 m. (a) With an eyepiece of focal length 1.0 cm, what is its angular magnification? (b) Find the diameter of the image of the Moon formed by the objective (Moon diameter 3.48×10^6 m, orbit radius 3.8×10^8 m).

(a) $m = 1500/1.0 = \mathbf{1500}$. (b) The Moon subtends $\alpha = 3.48 \times 10^6/3.8 \times 10^8$ rad; image size $= f_o\alpha = 15 \times 9.16 \times 10^{-3} = \mathbf{13.7\ cm}$.

7. An object and a screen are 90 cm apart. A convex lens forms a sharp image on the screen at two positions 20 cm apart. Find the focal length.

By the displacement method, $f = (D^2 - d^2)/4D = (8100 - 400)/360 = \mathbf{21.4\ cm}$.

8. At what angle should a ray be incident on one face of an equilateral prism ($n = 1.524$) so that it just suffers total internal reflection at the other face?

$\sin C = 1/1.524$, $C = 41.0^\circ$. For grazing at the second face $r_2 = C$, so $r_1 = 60^\circ - 41.0^\circ = 19.0^\circ$. $\sin i = 1.524 \times \sin 19.0^\circ = 0.496$, so $i = \mathbf{29.7^\circ}$.

Formula sheet: Chapter 9

Quantity	Formula
Mirror	$1/v + 1/u = 1/f$, $m = -v/u$, $f = R/2$
Snell	$n_1 \sin i = n_2 \sin r$
Critical angle	$\sin i_c = 1/n$
Spherical surface	$n_2/v - n_1/u = (n_2 - n_1)/R$
Lens maker	$1/f = (n - 1)(1/R_1 - 1/R_2)$
Lens	$1/v - 1/u = 1/f$, $m = v/u$
Power, combination	$P = 1/f$, $P = P_1 + P_2$
Prism	$n = \sin[(A + D_m)/2]/\sin(A/2)$
Microscope	$m \approx (L/f_o)(D/f_e)$
Telescope	$m = f_o/f_e$, $L = f_o + f_e$

1 minute recap

- ✓ Mirror: $1/v + 1/u = 1/f$; lens: $1/v - 1/u = 1/f$.
- ✓ TIR: denser to rarer and $i > i_c$; fibres, diamond, mirage.
- ✓ Lens maker and spherical-surface formulas: derive step by step.
- ✓ Prism: n from A and D_m; violet deviates most.
- ✓ Microscope: small f_o; telescope: large f_o.

Most Important Questions

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. 1M The focal length of a concave mirror of radius 30 cm is:
(a) 30 cm
(b) −15 cm
(c) 15 cm
(d) −30 cm
Ans: (b) −15 cm.

Q2. 1M Board fav A diamond sparkles because of:
(a) dispersion
(b) reflection
(c) total internal reflection
(d) refraction only
Ans: (c) total internal reflection.

Q3. 1M Power of a lens of focal length −50 cm:
(a) −2 D
(b) +2 D
(c) −0.5 D
(d) +0.5 D
Ans: (a) −2 D.

Q4. 1M In minimum deviation through a prism, the ray inside is:
(a) along the normal
(b) along the face
(c) perpendicular to the base
(d) parallel to the base
Ans: (d) parallel to the base.

Q5. 1M Which colour deviates most in a prism?
(a) red
(b) violet
(c) green
(d) yellow
Ans: (b) violet.

Q6. 1M A convex lens is placed in a liquid of the same n. Its focal length becomes:
(a) zero
(b) doubled
(c) infinite
(d) unchanged
Ans: (c) infinite.

Q7. 1M Board fav State two conditions for total internal reflection.
Ans: Light from denser to rarer medium; angle of incidence greater than the critical angle.

Q8. 1M Which mirror is used as a rear-view mirror? Why?
Ans: Convex: it gives an erect, diminished image and a wide field of view.

Q9. 1M What happens to the frequency of light on refraction?
Ans: It remains unchanged.

Q10. 1M Define one dioptre.
Ans: Power of a lens of focal length 1 m.

Q11. 1M Why are the objective lenses of telescopes large?
Ans: To gather more light and improve resolving power.

Q12. 1M What is the magnification of a plane mirror?
Ans: +1.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M Board fav **A:** Optical fibres carry light with little loss. **R:** Light travels by repeated total internal reflection.
Ans: (a)

Q14. 1M **A:** A convex mirror always forms a virtual image. **R:** Its focus lies behind the mirror.
Ans: (a)

Q15. 1M **A:** The focal length of a lens increases in water. **R:** The relative refractive index of glass decreases in water.
Ans: (a)

Q16. 1M **A:** A telescope has a large-f objective. **R:** Magnifying power = f_o/f_e.
Ans: (a)

C. Short answer (2–3 marks)

Q17. 2M Board fav Derive f = R/2 for a concave mirror.
Ans: See the step-by-step derivation in Section 9.1.

Q18. 3M Board fav Derive the mirror formula using a concave mirror.
Ans: Similar triangles A′B′F ~ MPF and A′B′P ~ ABP; with signs, $1/v + 1/u = 1/f$.

Q19. 2M Explain the formation of a mirage.
Ans: Air near hot ground is rarer; light from the sky bends progressively away from the normal and finally suffers TIR, appearing to come from the ground like a reflection in water.

Q20. 2M Derive the relation between critical angle and refractive index.
Ans: At $i = i_c$, $r = 90^\circ$: $n \sin i_c = 1 \times \sin 90^\circ \Longrightarrow \sin i_c = 1/n$.

Q21. 2M Board fav Derive $1/f = 1/f_1 + 1/f_2$ for two thin lenses in contact.
Ans: See Section 9.4.

Q22. 3M Board fav Draw a labelled ray diagram of a compound microscope and write its magnifying power.
Ans: Objective forms real enlarged image within focus of eyepiece; final virtual image. $m \approx (L/f_o)(D/f_e)$.

Q23. 2M Give two advantages of a reflecting telescope over a refracting telescope.
Ans: No chromatic aberration; less spherical aberration with parabolic mirror; mirror is lighter and easier to support.

Q24. 2M A lens behaves as converging in air and diverging in water. What can you say about its n?
Ans: n_{lens} lies between 1 and n_{water} = 1.33.

Q25. 2M Why does a prism produce dispersion but a glass slab does not?
Ans: In a slab the two faces are parallel, so colours emerge parallel and overlap; in a prism they emerge at different angles.

Q26. 2M Draw the graph of δ against i for a prism and mark D_m.
Ans: U-shaped curve; lowest point gives D_m, where i = e.

D. Long answer (5 marks)

Q27. 5M Board fav (a) Derive $n_2/v - n_1/u = (n_2 - n_1)/R$. (b) Use it to obtain the lens maker's formula.
Ans: See Section 9.4.

Q28. 5M Board fav (a) Derive the prism formula $n = \sin[(A + D_m)/2]/\sin(A/2)$. (b) A prism of A = 60° has $n = \sqrt{3}$. Find D_m.
Ans: (b) $\sin[(60 + D)/2] = \sqrt{3} \times 0.5 = 0.866 \Longrightarrow (60 + D)/2 = 60° \Longrightarrow D_m = 60°$.

Q29. 5M (a) Draw a ray diagram of a refracting telescope in normal adjustment. (b) Derive $m = f_o/f_e$. (c) Why is f_o large?
Ans: (b) $m = \beta/\alpha = (h/f_e)/(h/f_o) = f_o/f_e$. (c) For high magnification and light-gathering.

E. Case-based

Q30. 4M Board fav Optical fibres are used in endoscopes and internet cables. (a) Which phenomenon? (b) Core or cladding has higher n? (c) Find i_c for core n = 1.68, cladding 1.44. (d) One advantage.
Ans: (a) TIR. (b) Core. (c) $\sin i_c = 1.44/1.68 \Longrightarrow i_c \approx 59°$. (d) Low loss, high bandwidth, immune to electrical interference.

Q31. 4M A student uses a convex lens of f = 10 cm as a magnifier. (a) Max magnification (image at 25 cm)? (b) For image at infinity? (c) Where is the object in (a)?
Ans: (a) 1 + 25/10 = **3.5**. (b) 2.5. (c) $1/u = 1/(-25) - 1/10 \Longrightarrow u = -7.14$ cm.

Q32. 2M Board fav An object 2 cm tall is placed 15 cm from a concave lens of f = 10 cm. Find v and image size.
Ans: $1/v = -1/10 + 1/(-15) \Longrightarrow$ **v = −6 cm**; m = 0.4 $\Longrightarrow$ **0.8 cm**, erect, virtual.

Q33. 1M Find the apparent depth of a coin under 20 cm of water (n = 4/3).
Ans: **15 cm**.

Q34. 1M Two lenses of power +4 D and −2 D are in contact. Find f.
Ans: P = +2 D $\Longrightarrow$ **f = 50 cm**.

Q35. 2M Light travels from glass (n = 1.5) to water (n = 1.33). Find the critical angle.
Ans: $\sin i_c = 1.33/1.5 = 0.887 \Longrightarrow i_c \approx$ **62.5°**.

Q36. 2M Why does a star twinkle but a planet does not?
Ans: Light from a point-like star bends variably through moving air layers; a planet is an extended source, so variations average out.

Q37. 2M Board fav A telescope has magnifying power 10 and length 44 cm. Find f_o and f_e.
Ans: $f_o = 10f_e$; $11f_e = 44 \Longrightarrow$ **f_e = 4 cm**, **f_o = 40 cm**.

Q38. 1M Which defect of vision is corrected by a concave lens?
Ans: Myopia (near-sightedness).

Q39. 2M The image formed by a convex mirror is 1/3 the size of the object placed 30 cm away. Find f.
Ans: $m = -v/u = 1/3 \Longrightarrow v = 10$ cm; $1/f = 1/10 - 1/30 \Longrightarrow$ **f = 15 cm**.

Q40. 1M Why is the sky seen as blue? Which phenomenon?
Ans: Scattering: blue light (shorter λ) is scattered more by air molecules ($\propto 1/\lambda^4$).

F. Extra practice (board PYQ style)

Q41. 2M Board fav An equiconvex lens (n = 1.5) has f = 20 cm. Find the radius of curvature.
Ans: $1/20 = 0.5 \times 2/R \Longrightarrow$ **R = 20 cm**.

Q42. 2M A ray is incident normally on one face of an equilateral prism of n = 1.5. Will it emerge from the second face?
Ans: At the second face i = 60° > i_c = 41.8°: TIR, it does not emerge there.

Q43. 2M A concave mirror forms a real image twice the object size at 30 cm from the mirror. Find f.
Ans: $m = -2 \Longrightarrow u = -15$ cm, v = −30 cm; $1/f = -1/30 - 1/15 \Longrightarrow$ **f = −10 cm**.

Q44. 1M Why are danger signals red?
Ans: Red has the longest wavelength in the visible band, so it is scattered least and is seen from far.

Q45. 2M A plano-convex lens (n = 1.5) has R = 10 cm. Find its power.

Ans: $1/f = 0.5 \times (1/10) \Longrightarrow f = 20$ cm $\Longrightarrow P = +5$ D.

Q46. 1M How does the focal length of a convex lens change for violet light compared with red?

Ans: n is larger for violet, so f is smaller for violet.

Yaad karo: sign quick-check

- Concave mirror f −ve, convex mirror f +ve.
- Convex lens f +ve, concave lens f −ve.
- Real image: mirror mein v −ve, lens mein v +ve.
- m −ve = real, inverted; m +ve = virtual, erect.

Image at infinity? Pikachu face

Asli baat: Object at F of a convex lens or concave mirror sends parallel rays out: the image forms at infinity.

Quick compare: mirror vs lens

Mirror	Lens
$1/v + 1/u = 1/f$	$1/v - 1/u = 1/f$
$m = -v/u$	$m = v/u$
$f = R/2$ (any medium)	f depends on medium (lens maker)
Real image on same side as object	Real image on other side

Common mistakes

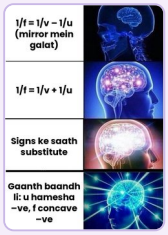
- Mirror aur lens formula mix mat karo: mirror mein +, lens mein −.
- u hamesha negative (real object).
- Lens maker mein R_2 ka sign double convex ke liye negative.
- TIR sirf denser se rarer mein.
- Telescope length = $f_o + f_e$, microscope ka nahi.

Paheli-Bujho

Paheli: Bujho, mirror formula aur lens formula mein fark kaise yaad rakhoge?

Bujho: Mirror mein sab jodo, lens mein ek minus... jaise mere marks!

Paheli: Haha. Mirror: 1/v + 1/u, lens: 1/v − 1/u. Aur sign convention pole se.



Formula rattne se lekar derivation samajhne tak

Asli baat: Learn to derive mirror, lens maker and prism formulas: boards ask derivations every year.

Exam Alert: last check

- Derivations: mirror formula, lens maker, refraction at spherical surface, prism, lenses in contact.
- Ray diagrams: microscope and telescope (5 marks).
- TIR applications and fibre case study.

WAVE OPTICS

Most asked: Huygens principle and proof of laws of reflection and refraction, Young's double slit experiment (fringe width, intensity), coherent sources, single slit diffraction, resolving power, polarisation and Malus' law, Brewster's law

Ray optics cannot explain why light bends around tiny obstacles or makes bright and dark bands. Treating light as a wave, Huygens, Young and Fresnel explained interference, diffraction and polarisation, the clearest proof that light is a wave.

1. Huygens Principle

Wavefront

A **wavefront** is the locus of all points vibrating in the same phase. Rays are always perpendicular to wavefronts, and the wavefront moves outward with the speed of the wave.

Source	Wavefront
Point source	spherical
Line source (slit)	cylindrical
Very distant source (sun, star)	plane

Huygens principle

1. Every point on a wavefront acts as a source of secondary wavelets that spread out in the forward direction with the speed of the wave.
2. The new wavefront at a later time is the forward envelope (common tangent) of these secondary wavelets.

Proof of the law of refraction (step by step)

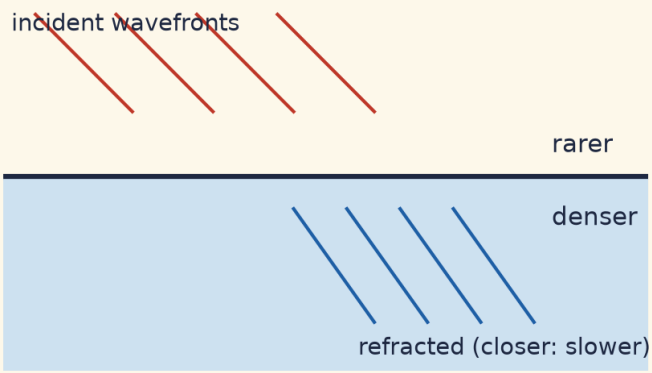


Fig. 10.1 Plane wavefronts entering a denser medium crowd together because the speed drops

1. Plane wavefront AB strikes the surface at A at angle i. While point B travels BC = $v_1\tau$ to the surface, a wavelet from A spreads AE = $v_2\tau$ into medium 2.
2. The tangent CE from C to this wavelet is the refracted wavefront; angle r with the surface.
3. From triangles ABC and AEC: sin i = BC/AC = $v_1\tau$/AC; sin r = AE/AC = $v_2\tau$/AC.
4. Divide: sin i/sin r = $v_1/v_2 = n_2/n_1$, which is **Snell's law**.

→ Since $v = \nu\lambda$ and ν does not change: $\lambda_2 = \lambda_1/n_{21}$, light has a shorter wavelength in a denser medium.

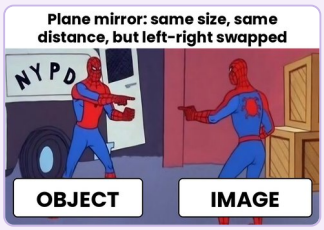
→ Reflection: the same construction with both wavelets in one medium gives AE = BC, hence i = r.

Doppler effect for light

$$\Delta\nu/\nu = -v_{radial}/c \quad (v \ll c)$$

→ Source moving away: frequency decreases, wavelength increases: **red shift** (galaxies moving away, expanding universe).

→ Source approaching: **blue shift**.



Do bilkul same sources? Coherent hona zaroori hai

Asli baat: Coherent = same frequency + constant phase difference. Only then is the pattern steady.

Wavefront: a closer look

A **wavefront** is the locus of all points of a medium that vibrate in the same phase at a given instant. Since every point on a wavefront has the same phase,

the wavefront is a surface of constant phase. The energy of the wave travels outward perpendicular to the wavefront, so the direction of a ray is always **normal to the wavefront**. This single idea connects wave optics with ray optics: a bundle of rays is simply a set of normals to a moving wavefront.

The shape of the wavefront depends on the source and on the distance from it. Close to a point source the disturbance spreads equally in all directions, so the wavefront is a sphere. A long slit or a tube light acts as a line source and gives a cylindrical wavefront. Far away from any source, a small portion of a very large sphere is practically flat, so light from the sun or a star reaches us as a **plane wavefront**.

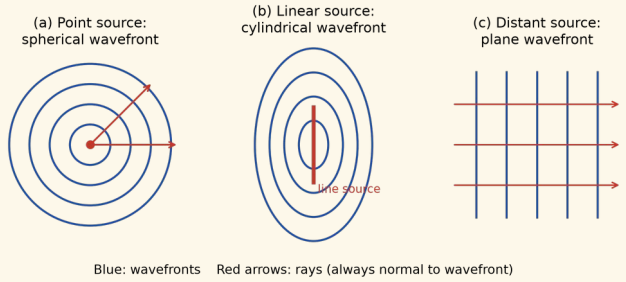


Fig. 10.A Spherical, cylindrical and plane wavefronts; rays are always normal to the wavefront

Source	Wavefront	Intensity varies as
Point source	Spherical	$I \propto 1/r^2$
Line source (slit)	Cylindrical	$I \propto 1/r$
Source at infinity	Plane	constant

Proof of the law of reflection (step by step)

1. Let a plane wavefront AB be incident on a reflecting surface MN at angle i. Point A touches the surface first.
2. In the time t that the disturbance at B takes to reach C on the surface, BC = vt.
3. In the same time, the secondary wavelet from A spreads into a hemisphere of radius AE = vt (same medium, same speed).
4. Draw a tangent from C to this hemisphere; CE is the reflected wavefront.
5. Triangles ABC and CEA are right-angled, share the hypotenuse AC and have BC = AE = vt, so they are congruent.
6. Hence $\angle BAC = \angle ECA$, which gives **i = r**. The incident ray, the reflected ray and the normal also lie in one plane, the plane of the figure.

Why frequency stays the same

Frequency is decided by the source. At the boundary, the secondary wavelets are produced by charges of the medium oscillating at the frequency of the incident light. Hence on reflection or refraction **frequency does not change**; speed and wavelength change in the ratio $v_2/v_1 = \lambda_2/\lambda_1$. The colour we see depends on frequency, which is why a red ball still looks red under water.

Refraction at a rarer medium and the critical angle

When a wavefront goes from a denser medium (speed v_1) into a rarer medium (speed $v_2 > v_1$), the secondary wavelets in the second medium spread faster, so the refracted wavefront bends away from the normal and r > i. As i increases, r reaches 90° at the critical angle i_c, given by $\sin i_c = v_1/v_2 = n_2/n_1$. Beyond this the tangent from the wavelets cannot be drawn in the second medium, and the light is totally internally reflected.

Plane wave through a prism, a lens and a mirror

- → **Prism:** the part of the wavefront passing through the thicker base travels more distance in glass and is delayed, so the emerging wavefront is tilted towards the base.
- → **Convex lens:** the centre of the wavefront passes through more glass and is delayed most; the emerging wavefront becomes a converging spherical wave that meets at the focus.
- → **Concave mirror:** the central part of the incident plane wave is reflected later; the reflected wavefront is a converging sphere centred at the focus.

In all three cases the **time taken** from any point of the object to the corresponding point of the image is the same along every ray. This is why a lens forms a sharp image even though rays travel different geometric paths: the extra path in air is exactly balanced by the shorter path in glass.

Solved example (NCERT)

Q. Light of wavelength 589 nm in air falls on a water surface. Find the wavelength, frequency and speed of (a) reflected light, (b) refracted light. Refractive index of water = 1.33.

Sol. (a) Reflected light stays in air: λ = **589 nm**, $\nu = c/\lambda = 3 \times 10^8/589 \times 10^{-9}$ = **5.09×10^{14} Hz**, v = **3×10^8 m/s**.

(b) Frequency unchanged: ν = **5.09×10^{14} Hz**; $v = c/n = 3 \times 10^8/1.33$ = **2.26×10^8 m/s**; λ = 589/1.33 = **443 nm**.

Solved example (NCERT)

Q. What is the shape of the wavefront in each case: (a) light diverging from a point source, (b) light emerging from a convex lens with a point source at its focus, (c) the portion of the wavefront of light from a distant star intercepted by the earth?

Sol. (a) **Spherical**. (b) **Plane** (rays come out parallel). (c) **Plane** (a small part of a huge sphere).

Yaad karo

- → Ray hamesha wavefront ke perpendicular.
- → Medium badla: v aur λ badalte hain, ν kabhi nahi.
- → Denser medium mein light slow hoti hai: Huygens ne yahi predict kiya, Newton ki corpuscular theory ne ulta (fast) bola, aur experiment ne Huygens ko sahi saabit kiya.

2. Coherent Sources and Interference

When two waves overlap, the resultant displacement is the sum of the individual displacements (superposition). If the phase difference stays constant, the intensity pattern is steady: **interference**.

Derivation step by step (resultant intensity)

1. Two waves at a point: $y_1 = a \cos \omega t$, $y_2 = a \cos(\omega t + \varphi)$.
2. Sum: $y = 2a \cos(\varphi/2) \cos(\omega t + \varphi/2)$, amplitude $2a \cos(\varphi/2)$.
3. Intensity $\propto$ amplitude^2: $I = 4I_0 \cos^2(\varphi/2)$.
4. For unequal intensities: $I = I_1 + I_2 + 2\sqrt{(I_1 I_2)} \cos \varphi$.

Condition	Phase difference φ	Path difference	Intensity
Constructive	0, 2π, 4π, ...	$n\lambda$	$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$
Destructive	π, 3π, 5π, ...	$(n + \frac{1}{2})\lambda$	$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$

- → Phase difference = $(2\pi/\lambda) \times$ path difference.
- → Two independent bulbs are incoherent: their phase changes randomly every 10^{-8} s, so intensities simply add ($I = I_1 + I_2$) and no pattern is seen.
- → Energy is not destroyed in interference: it is redistributed from dark to bright fringes.

Exam Alert

- → $I = 4I_0 \cos^2(\varphi/2)$ aur I_{max}/I_{min} ratio: numerical har saal.
- → Coherent sources kya hain aur do bulb se pattern kyun nahi: 2 marks.

Constructive and destructive interference: physical picture

When two waves arrive at a point with their crests together (phase difference 0, 2π, 4π, ...), their displacements add and the point is bright. When a crest of one meets a trough of the other (phase difference π, 3π, ...), they cancel and the point is dark. Energy is not destroyed at the dark points; it is **redistributed** to the bright points, so the average intensity over the pattern is still the sum of the two intensities.

Condition	Phase difference φ	Path difference Δ	Intensity
Constructive (bright)	$2n\pi$	$n\lambda$	$(\sqrt{I_1} + \sqrt{I_2})^2$
Destructive (dark)	$(2n - 1)\pi$	$(2n - 1)\lambda/2$	$(\sqrt{I_1} - \sqrt{I_2})^2$

$\varphi = (2\pi/\lambda) \times \Delta$ relation between phase difference and path difference

Why two bulbs never show interference

In an ordinary source, light is emitted by billions of atoms independently, and each atom emits a wave train lasting only about 10^{-8} s. The phase of the light therefore changes randomly many millions of times per second. With two independent bulbs, the phase difference at a point keeps changing, so a bright spot becomes dark and bright again faster than the eye can follow. The eye sees only the average, $I = I_1 + I_2$, a uniform illumination. To get a steady pattern the two

sources must be **coherent**: their phase difference must stay constant with time. Practically, coherent sources are obtained by dividing light from a single source into two parts, as in Young's experiment, or by using a laser.

Coherent sources	*Incoherent sources*
Constant phase difference	Phase difference changes randomly
Intensities do not simply add: $I = I_1 + I_2 + 2\sqrt{(I_1 I_2)}\cos\varphi$	Intensities simply add: $I = I_1 + I_2$
Sustained interference pattern	No observable pattern
Example: two slits lit by one laser	Example: two bulbs, two candles

Solved example

Q. Two coherent waves of intensities I and $4I$ superpose. Find the maximum and minimum intensities.

Sol. $I_{max} = (\sqrt{I} + \sqrt{4I})^2 = (3\sqrt{I})^2 = 9I$; $I_{min} = (2\sqrt{I} - \sqrt{I})^2 = I$.

3. Young's Double Slit Experiment

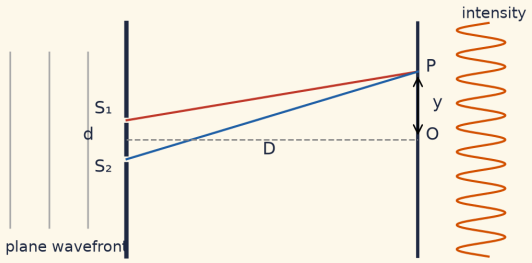


Fig. 10.2 YDSE: slits S_1, S_2 separated by d; screen at distance D

Young passed light from one slit S through two narrow slits S_1 and S_2. Since both get light from the same wavefront, they are coherent. Bright and dark bands (**fringes**) appear on the screen.

Derivation step by step (fringe width)

1. For a point P at height y on the screen: path difference $\Delta = S_2P - S_1P$.
2. $(S_2P)^2 - (S_1P)^2 = [D^2 + (y + d/2)^2] - [D^2 + (y - d/2)^2] = 2yd$.
3. Since $S_2P + S_1P \approx 2D$ ($D \gg d, y$): $\Delta = yd/D$.
4. Bright fringe: $yd/D = n\lambda \Longrightarrow y_n = n\lambda D/d$. Dark fringe: $y_n = (n + \tfrac{1}{2})\lambda D/d$.
5. Spacing between consecutive bright (or dark) fringes:

$$\beta = \lambda D/d$$ fringe width; angular width $\theta = \lambda/d$

→ All fringes have equal width; β increases with λ and D, decreases with d.

→ Central fringe ($y = 0$) is bright: path difference zero.

→ White light: central fringe white, others coloured (violet inner, red outer), only a few visible.

→ Whole apparatus in water: λ becomes λ/n, so β decreases by factor n.

→ One slit covered with a thin sheet (thickness t, index μ): pattern shifts by $(\mu - 1)tD/d$ towards the covered slit, β unchanged.

NCERT Activity: double slit with a laser

Scratch two fine parallel lines very close together on a soot-coated glass slide with a blade. Shine a laser pointer through them onto a wall a few metres away in a dark room: a row of equally spaced bright and dark spots appears. Move the wall further away and the spots spread out ($\beta \propto D$).

Fringe width ka formula: $\beta = \lambda D/d$, bas itna hi?

Asli baat: Yes. Fringe width increases with λ and D and decreases with slit separation d.

Solved example (NCERT)

Q. In YDSE the slits are 0.28 mm apart and the screen is 1.4 m away. The distance between the central bright fringe and the fourth bright fringe is 1.2 cm. Find λ.

Sol. $y_4 = 4\lambda D/d \Longrightarrow \lambda = y\,d/4D = 1.2 \times 10^{-2} \times 0.28 \times 10^{-3}/(4 \times 1.4) =$ **600 nm**.

Solved example (NCERT)

Q. In YDSE with monochromatic light of wavelength λ, the intensity at a point where the path difference is λ is K. What is it where the path difference is $\lambda/3$?

Sol. Path $\lambda \Longrightarrow \varphi = 2\pi \Longrightarrow I = 4I_0 = K$. Path $\lambda/3 \Longrightarrow \varphi = 2\pi/3 \Longrightarrow I = 4I_0\cos^2 60^\circ =$ **K/4**.

Solved example

Q. Two slits give intensities in the ratio 9 : 1. Find $I_{max} : I_{min}$.

Sol. Amplitude ratio $3 : 1 \Longrightarrow I_{max}/I_{min} = (3 + 1)^2/(3 - 1)^2 =$ **4 : 1**.

Paheli-Bujho

Bujho: Paheli, do torch jala ke deewar pe fringes kyun nahi bante?

Paheli: Kyunki dono torch coherent nahi hain. Unka phase har pal randomly badalta hai, pattern ban ke turant mit jaata hai.

Bujho: Toh Young ne kya jugaad kiya?

Paheli: Ek hi source ki light do slit se nikali. Dono ka phase saath badalta hai, toh pattern steady.

Intensity distribution in YDSE

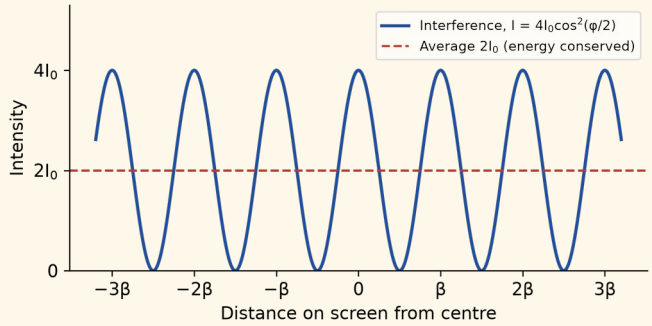


Fig. 10.B Intensity on the screen in YDSE: equally spaced, equally bright fringes; the average stays $2I_0$

For two identical slits each giving intensity I_0, $I = 4I_0\cos^2(\varphi/2)$. The bright fringes have intensity $4I_0$ and the dark fringes zero. All bright fringes are equally bright and equally spaced (for small angles), which distinguishes interference from diffraction, where the side maxima become rapidly weaker.

Angular width of a fringe

The angular separation between two consecutive bright fringes, as seen from the slits, is $\theta = \beta/D = \lambda/d$. It does not depend on the distance of the screen. If the experiment is immersed in a liquid of refractive index n, λ becomes λ/n and the angular width also becomes 1/n times.

Solved example (NCERT)

Q. The angular width of a fringe in YDSE with light of 600 nm is 0.2°. What is the angular width if the whole apparatus is immersed in water (n = 4/3)?

Sol. $\theta_w = \theta/n = 0.2^\circ \times 3/4 = \mathbf{0.15^\circ}$.

What happens to the fringes if ...

Change made	Effect on pattern	Reason
Slit separation d increased	Fringes come closer (β decreases)	$\beta = \lambda D/d$
Screen moved away (D increased)	Fringes spread out	$\beta \propto D$
Red light replaced by blue	Fringes come closer	$\lambda_{blue} < \lambda_{red}$
Whole apparatus put in water (n)	β becomes β/n	λ becomes λ/n
White light used	Central fringe white, others coloured; few fringes visible	each λ has its own β
One slit covered	Interference disappears; only diffraction of one slit	no second source
Source slit widened	Fringes become less sharp and finally vanish	different parts of the slit act as incoherent sources
Thin glass sheet over one slit	Whole pattern shifts towards the covered slit, β unchanged	extra optical path $(n-1)t$

Solved example (NCERT)

Q. Light containing 650 nm and 520 nm is used in YDSE with d = 2 mm and D = 1.2 m. Find (a) the distance of the third bright fringe from the centre for 650 nm, (b) the least distance from the centre where bright fringes of both wavelengths coincide.

Sol. (a) $y_3 = 3\lambda D/d = 3 \times 650 \times 10^{-9} \times 1.2/(2 \times 10^{-3})$ $= \mathbf{1.17\ mm}$.

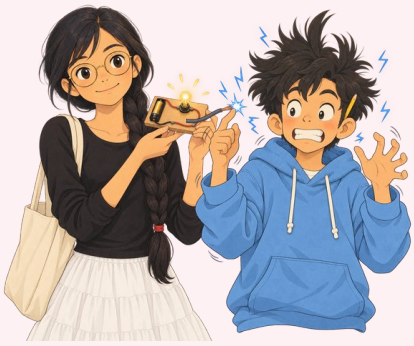
(b) Coincidence: $n_1 \times 650 = n_2 \times 520 \Longrightarrow n_1/n_2 = 4/5$. Least: $n_1 = 4$: $y = 4 \times 650 \times 10^{-9} \times 600 = \mathbf{1.56\ mm}$.

Solved example (NCERT)

Q. In YDSE with 600 nm light, the angular width of a fringe on a distant screen is 0.1°. Find the slit separation.

Sol. $\theta = \lambda/d \Longrightarrow d = \lambda/\theta = 600 \times 10^{-9}/(0.1 \times \pi/180) = \mathbf{3.44 \times 10^{-4}\ m \approx 0.34\ mm}$.

Paheli-Bujho



Bujho: Paheli, dark fringe pe light gayab kahan ho gayi? Energy toh destroy nahi hoti na?

Paheli: Bilkul nahi hoti. Jo energy dark band se hati, wahi bright band mein chali gayi. Average dekho toh $2I_0$ hi hai.

Bujho: Matlab light ne bas seating arrangement badla, total log utne hi hain!

4. Diffraction

The bending of light around the edges of an obstacle or aperture and spreading into the geometrical shadow is called **diffraction**. It is noticeable when the size of the aperture is comparable to λ.

Single slit diffraction (step by step)

1. Light of wavelength λ falls on a slit of width a; we look at angle θ on a distant screen.
2. Divide the slit into two halves. Waves from corresponding points of the halves differ in path by $(a/2)\sin\theta$.
3. If $(a/2)\sin\theta = \lambda/2$, they cancel in pairs: first minimum at $a\sin\theta = \lambda$.
4. In general minima at $a\sin\theta = n\lambda$; secondary maxima roughly at $a\sin\theta = (n + \tfrac{1}{2})\lambda$, getting weaker.

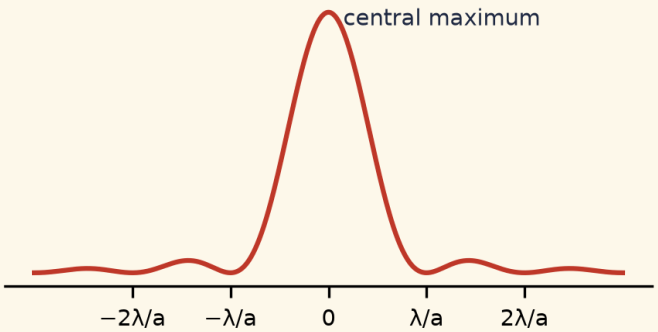


Fig. 10.3 Single slit pattern: a broad central maximum and weak secondary maxima

Width of central maximum = $2\lambda D/a$ angular width $2\lambda/a$

Interference	Diffraction
Superposition of waves from two coherent sources	Superposition of wavelets from different parts of the same wavefront
Fringes of equal width	Central maximum twice as wide as others
All bright fringes equally intense	Intensity falls rapidly for higher orders
Dark fringes nearly perfectly dark	Minima not perfectly dark

NCERT Activity: diffraction through fingers

Hold two fingers very close together so a thin gap remains, and look through it at a bright tube light. You will see faint dark and bright lines parallel to the gap. Similarly, look at a distant street light through a fine cloth or a feather: coloured patterns appear due to diffraction.

Resolving power and the validity of ray optics

- → Due to diffraction, a lens or telescope forms a small disc (Airy disc), not a point, of angular radius $\approx 1.22\lambda/D$ (D = aperture diameter).
- → Two stars can be seen separately only if their angular separation exceeds this: a bigger aperture gives better resolution.
- → Microscope: resolving power increases with refractive index of the medium (oil immersion) and smaller λ.
- → Fresnel distance $z_F = a^2/\lambda$: up to this distance ray optics is valid for an aperture a.

Solved example (NCERT)

Q. For what distance is ray optics a good approximation for an aperture of 3 mm and λ = 500 nm?

Sol. $z_F = a^2/\lambda = (3\times10^{-3})^2/(5\times10^{-7}) =$ **18 m.**

Solved example

Q. Light of 600 nm falls on a slit of width 0.2 mm; screen at 1 m. Find the width of the central maximum.

Sol. $2\lambda D/a = 2\times6\times10^{-7}\times1/(2\times10^{-4}) =$ **6 mm.**

Slit chhota karo toh central maximum aur bada

Asli baat: Width of the central maximum is $2\lambda D/a$: a narrower slit spreads light more.

Why the side maxima are weaker

Divide the slit into three equal parts at the angle where $a \sin\theta = 3\lambda/2$. Two of the parts cancel each other completely, and only the third part contributes, so the first secondary maximum has roughly one-ninth of the central intensity. At higher orders an even smaller fraction of the slit survives, so the secondary maxima become rapidly weaker. The central maximum, where all wavelets from the slit arrive nearly in phase, is the brightest and twice as wide as any secondary maximum.

Feature	Interference (YDSE)	Diffraction (single slit)
Cause	superposition of waves from two coherent sources	superposition of wavelets from different parts of the same wavefront
Fringe width	all fringes equal	central maximum twice as wide as others
Intensity	all bright fringes equally bright	intensity falls rapidly for higher orders
Dark fringes	perfectly dark	not perfectly dark
Condition for minima	path difference $(2n - 1)\lambda/2$	$a \sin\theta = n\lambda$

Seeing the single slit pattern: everyday diffraction

- → Look at a distant street light through a narrow gap between two fingers or two razor blades: bright and dark bands appear parallel to the slit.
- → The coloured rings seen around the sun or moon through a misty window, and the colours on a CD, involve diffraction of light by small particles or fine tracks.
- → Sound bends easily round a wall (λ about 1 m), so we can hear someone behind it; light ($\lambda \sim 500$ nm) does not, so we cannot see the person. Diffraction is noticeable only when the obstacle or aperture is comparable to λ.

Interference and diffraction together in YDSE

Each slit in YDSE has a finite width, so each one produces its own single slit diffraction pattern. The two-slit interference fringes are seen inside the broad central maximum of this diffraction envelope. Since d is much larger than a, many narrow interference fringes fit inside one diffraction maximum. Making the slits narrower widens the envelope and more interference fringes become visible.

Fresnel distance (step by step)

1. A beam through an aperture of size a spreads by diffraction at an angle $\theta \approx \lambda/a$.
2. After travelling a distance z, the spread is $z\lambda/a$.
3. Spreading becomes comparable to the aperture itself when $z\lambda/a \approx a$.
4. This gives the Fresnel distance $z_F = a^2/\lambda$.

For distances much less than z_F the beam travels in straight lines and ray optics is valid; beyond it diffraction dominates. Ray optics is therefore the limit of wave optics when $\lambda \to 0$.

Solved example (NCERT)

Q. A parallel beam of 500 nm falls on a narrow slit and the first minimum is 2.5 mm from the centre on a screen 1 m away. Find the slit width.

Sol. First minimum: $a \sin\theta = \lambda$, with $\sin\theta \approx x/D$. So $a = \lambda D/x = 500 \times 10^{-9} \times 1/2.5 \times 10^{-3} = \mathbf{2 \times 10^{-4}}$ **m = 0.2 mm.**

Solved example

Q. Two towers on hills 40 km apart are joined by a line that passes 50 m above a hill halfway between them. Find the longest wavelength of radio waves that can be sent without appreciable diffraction.

Sol. Here a = 50 m and the distance to the hill z = 20 km. For no appreciable spreading, $z \le a^2/\lambda \Longrightarrow \lambda \le a^2/z = 2500/20000 =$ **0.125 m = 12.5 cm.**

Yaad karo

- → Central max = $2\lambda D/a$, baaki saare = $\lambda D/a$.
- → Slit width aadhi karo: central maximum doguna chauda.
- → Diffraction dikhne ke liye $a \sim \lambda$ chahiye; isliye sound deewar ke peeche jaati hai, light nahi.

5. Polarisation

Light is a transverse wave: E vibrates perpendicular to the direction of travel. In ordinary (unpolarised) light E vibrates in all such directions randomly. If E is confined to a single plane, the light is **plane polarised**. Only transverse waves can be polarised, so polarisation proves that light is transverse.

Polaroids and Malus' law

→ A polaroid transmits only the component of E along its pass axis.

→ Unpolarised light through one polaroid: intensity drops to $I_0/2$ whatever its orientation.

→ Second polaroid (analyser) at angle θ to the first:

$$I = I_0 \cos^2 \theta$$ Malus' law

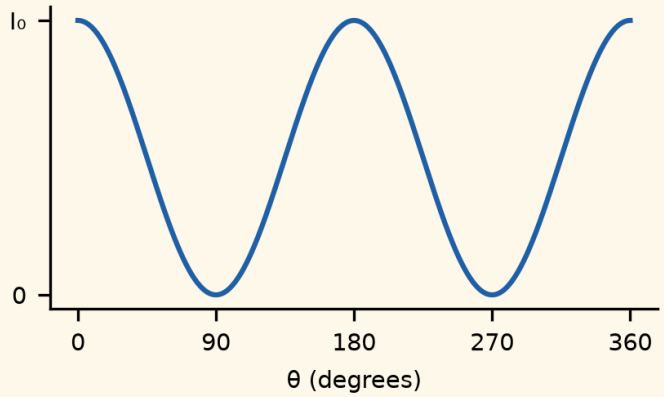


Fig. 10.4 Transmitted intensity vs angle between polaroids

Polarisation by scattering and reflection

→ Sunlight scattered at 90° by air molecules is polarised: bees use it to navigate.

→ At a particular angle of incidence (Brewster angle i_B), reflected light is completely plane polarised and the reflected and refracted rays are perpendicular.

$$\tan i_B = n$$ Brewster's law; for glass $i_B \approx 57°$

→ **Uses of polaroids**: sunglasses (cut glare from reflections), 3D movie glasses, LCD screens, photographic filters, window panes in trains.

NCERT Activity: polaroid sunglasses

Look at a laptop or phone screen through polaroid sunglasses and slowly rotate the glasses: the screen becomes dark at one angle. LCD screens emit polarised light, and the glasses act as an analyser. Two polaroid sheets held together and rotated show the same darkening.

Solved example (NCERT)

Q. Unpolarised light of intensity I_0 passes through polaroids P_1 and P_3 crossed; a third P_2 is placed between at 45°. Find the final intensity.

Sol. After P_1: $I_0/2$. After P_2: $(I_0/2)\cos^2 45° = I_0/4$. After P_3: $(I_0/4)\cos^2 45° = \mathbf{I_0/8}$.

Solved example

Q. Find the Brewster angle for water (n = 1.33).

Sol. $i_B = \tan^{-1}(1.33) = \mathbf{53.1°}$.

Light ek ray hai... ek wave hai... transverse wave hai

Asli baat: Interference and diffraction show light is a wave; polarisation shows it is a transverse wave.

Transverse nature of light: the proof

Polarisation is the property that proves light is a **transverse wave**. In a longitudinal wave (like sound in air) the vibrations are along the direction of travel, so every direction perpendicular to the ray looks the same and nothing can block one orientation. When unpolarised light passes through a polaroid P_1, the intensity becomes half whatever the orientation of P_1; but when a second polaroid P_2 is rotated, the intensity varies from maximum to zero twice in one full rotation. Such behaviour is possible only if the electric field vibrates perpendicular to the direction of propagation.

Unpolarised light	*Plane polarised light*
E vibrates in all directions perpendicular to the ray	E vibrates in a single direction perpendicular to the ray
Intensity through a rotating polaroid stays constant $(I_0/2)$	Intensity through a rotating polaroid varies as $\cos^2\theta$
Examples: sunlight, bulb	Light after a polaroid, reflected at Brewster angle

Brewster's law (step by step)

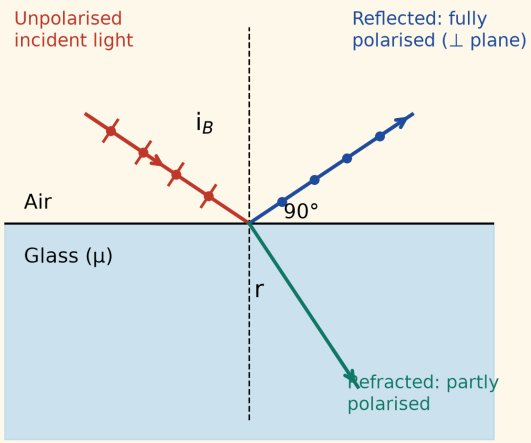


Fig. 10.C At the Brewster angle the reflected and refracted rays are perpendicular; reflected light is fully polarised

1. Experiment shows that the reflected light is completely polarised when the reflected and refracted rays are perpendicular: $i_B + r = 90^\circ$.
2. Snell's law: $\mu = \sin i_B / \sin r$.
3. Since $r = 90^\circ - i_B$, $\sin r = \cos i_B$.
4. Therefore $\mu = \sin i_B / \cos i_B$.

$$\mu = \tan i_B$$ Brewster's law

The reflected light vibrates perpendicular to the plane of incidence (shown by dots). Physically, the electrons in glass oscillate along the direction of the refracted ray's field; they cannot radiate along their own direction of oscillation, so no light with E in the plane of incidence goes into the reflected direction.

Uses of polaroids

- → Polaroid sunglasses cut the glare reflected from water and roads, since that light is partially polarised horizontally.
- → Window panes of trains and aeroplanes to control the light entering.
- → 3D movies: the two pictures are projected with perpendicular polarisations and each eye sees only one.
- → Photographic filters, LCD screens of calculators and phones, and stress analysis of glass and plastic.

Solved example (NCERT)

Q. What is the Brewster angle for air to glass transition? (μ of glass = 1.5)

Sol. $\tan i_B = 1.5 \Longrightarrow i_B = \tan^{-1}(1.5) = \mathbf{56.3^\circ}$.

Solved example

Q. Unpolarised light of intensity 32 W/m^2 passes through three polaroids. The axis of the second makes 30° and the third 90° with the first. Find the emergent intensity.

Sol. After P_1: 16 W/m^2. After P_2: $16\cos^2 30^\circ = 12$ W/m^2. P_3 is at 60° to P_2: $12\cos^2 60^\circ = \mathbf{3\ W/m^2}$.

Solved example (NCERT)

Q. Light of wavelength 5000 Å falls on a plane reflecting surface. Find the wavelength and frequency of the reflected light. For what angle of incidence is the reflected ray normal to the incident ray?

Sol. $\lambda = \mathbf{5000\ Å}$; $\nu = 3 \times 10^8 / 5 \times 10^{-7} = \mathbf{6 \times 10^{14}\ Hz}$. Reflected ray ⊥ incident ray: $i + r = 90^\circ$ and $i = r$ $\Longrightarrow \mathbf{i = 45^\circ}$.

Solved example

Q. Two polaroids are placed with their pass axes at 60° to each other. Unpolarised light of intensity I_0 falls on the first. Find the intensity of the emergent light. At what angle should the second be set to transmit $I_0/4$?

Sol. After P_1: $I_0/2$. After P_2: $(I_0/2)\cos^2 60^\circ = \mathbf{I_0/8}$. For $I_0/4$: $\cos^2\theta = 1/2 \Longrightarrow \mathbf{\theta = 45^\circ}$.

Solved example

Q. Newton's corpuscular theory predicts that light travels faster in water than in air. Explain how it reaches this result and whether experiment supports it.

Sol. In the corpuscular model the water surface attracts the light particles, increasing the normal component of velocity while the parallel component is unchanged; this bends the ray towards the normal and gives $n = v_{water}/v_{air} > 1$, i.e. **faster in water**. Foucault's experiment (1850) showed light is **slower** in water, as Huygens' wave theory predicts ($n = v_{air}/v_{water}$). The prediction is therefore **not confirmed**.

6. Solved Numericals (Board Pattern)

N1. In YDSE, d = 1 mm, D = 1 m, λ = 500 nm. Find β.

Sol. $\beta = \lambda D/d = \mathbf{0.5\ mm}$.

N2. The YDSE apparatus is immersed in water ($n = 4/3$). The fringe width in air is 0.6 mm. Find the new width.
Sol. $\beta' = \beta/n = \mathbf{0.45\ mm}$.

N3. Light of 5000 Å in air enters glass ($n = 1.5$). Find its wavelength, frequency and speed in glass.
Sol. $\lambda = \mathbf{3333\ Å}$; $\nu = c/\lambda_{air} = \mathbf{6 \times 10^{14}\ Hz}$ (same); $v = \mathbf{2 \times 10^8\ m/s}$.

N4. Two coherent waves of intensity I_0 each meet with phase difference $\pi/2$. Find the resultant intensity.
Sol. $I = 4I_0\cos^2(\pi/4) = \mathbf{2I_0}$.

N5. In a single slit, the first minimum for red light (660 nm) coincides with the first maximum for another λ'. Find λ'.
Sol. $a \sin\theta = 660\ nm = (3/2)\lambda' \Longrightarrow \lambda' = \mathbf{440\ nm}$.

N6. Polarised light of intensity 32 units passes through a polaroid at 60°. Find the transmitted intensity.
Sol. $I = 32\cos^2 60^\circ = \mathbf{8\ units}$.

Formula sheet: Chapter 10

Quantity	Formula
Snell (wave)	$\sin i/\sin r = v_1/v_2$
Intensity	$I = I_1 + I_2 + 2\sqrt{(I_1 I_2)}\cos\varphi$
Equal sources	$I = 4I_0\cos^2(\varphi/2)$
Phase-path	$\varphi = (2\pi/\lambda)\Delta$
Fringe width	$\beta = \lambda D/d$
Bright / dark	$y = n\lambda D/d$, $(n + ½)\lambda D/d$
Single slit minima	$a \sin\theta = n\lambda$
Central max width	$2\lambda D/a$
Malus	$I = I_0\cos^2\theta$
Brewster	$\tan i_B = n$
Fresnel distance	$z_F = a^2/\lambda$

1 minute recap

- ✓ Huygens: every point is a source of wavelets; proves Snell's law.
- ✓ Coherent sources needed for steady interference.
- ✓ YDSE: $\beta = \lambda D/d$, central fringe bright.
- ✓ Single slit: minima at $a \sin\theta = n\lambda$, central max $2\lambda D/a$.
- ✓ Polarisation proves light is transverse: Malus and Brewster.

Most Important Questions

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM The wavefront from a distant star is:
(a) spherical
(b) cylindrical
(c) plane
(d) elliptical
Ans: (c) plane.

Q2. IM Board fav In YDSE, if d is doubled, fringe width:
(a) doubles
(b) halves
(c) unchanged
(d) becomes four times
Ans: (b) halves.

Q3. IM Polarisation shows that light is:
(a) longitudinal
(b) particle-like
(c) mechanical
(d) transverse
Ans: (d) transverse.

Q4. IM Unpolarised light passes through a polaroid. Intensity becomes:
(a) $I_0/2$
(b) I_0
(c) $I_0/4$
(d) zero
Ans: (a) $I_0/2$.

Q5. IM Which property does not change when light goes from air to glass?
(a) speed
(b) wavelength
(c) frequency
(d) direction
Ans: (c) frequency.

Q6. IM The central fringe in YDSE with white light is:
(a) dark
(b) white
(c) red
(d) violet
Ans: (b) white.

Q7. 1M Board fav Define a wavefront.
Ans: The locus of all points vibrating in the same phase.

Q8. 1M What are coherent sources?
Ans: Sources of the same frequency with a constant phase difference.

Q9. 1M Why are two sodium lamps not coherent?
Ans: Their phases change randomly and independently.

Q10. 1M State Brewster's law.
Ans: $\tan i_B = n$; at i_B reflected light is fully polarised.

Q11. 1M What is red shift?
Ans: Increase in wavelength of light from a source moving away.

Q12. 1M How does fringe width change if the apparatus is put in water?
Ans: It decreases (β/n).

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M Board fav **A:** No interference pattern is seen with two independent bulbs. **R:** They are not coherent.
Ans: (a)

Q14. 1M **A:** Sound waves cannot be polarised. **R:** Sound waves in air are longitudinal.
Ans: (a)

Q15. 1M **A:** In interference, energy is destroyed at dark fringes. **R:** Intensity is zero there.
Ans: (d): energy is redistributed, not destroyed.

Q16. 1M **A:** Diffraction of sound is more common than of light. **R:** Wavelength of sound is comparable to everyday obstacles.
Ans: (a)

C. Short answer (2–3 marks)

Q17. 3M Board fav State Huygens principle and use it to prove Snell's law.
Ans: See Section 10.1.

Q18. 3M Board fav Derive the expression for fringe width in YDSE.
Ans: Path difference yd/D; bright at $n\lambda$; $\beta = \lambda D/d$.

Q19. 2M Derive $I = 4I_0\cos^2(\varphi/2)$ for two coherent sources.
Ans: See Section 10.2.

Q20. 3M Board fav Give three differences between interference and diffraction.
Ans: See the comparison table.

Q21. 3M What happens to the YDSE pattern if (a) one slit is closed, (b) white light is used, (c) the source slit is widened?
Ans: (a) Single slit diffraction pattern. (b) Coloured fringes with white centre. (c) Fringes become less sharp and finally vanish.

Q22. 2M Board fav State Malus' law. Unpolarised light passes two polaroids at 30°. Find the fraction transmitted.
Ans: $I = I_0\cos^2\theta$; $(1/2)(3/4) = \mathbf{3/8}$.

Q23. 2M Show that at the Brewster angle, the reflected and refracted rays are perpendicular.
Ans: $\sin i_B/\sin r = n = \tan i_B \Longrightarrow \sin r = \cos i_B \Longrightarrow i_B + r = 90°$.

Q24. 1M Give two uses of polaroids.
Ans: Sunglasses, LCD screens, 3D glasses, camera filters.

Q25. 2M How does the width of the central maximum change if (a) slit width is halved, (b) red light replaces blue?
Ans: (a) Doubles. (b) Increases (larger λ).

Q26. 2M What is the Fresnel distance? Give its significance.
Ans: $z_F = a^2/\lambda$; ray optics holds up to this distance.

D. Long answer (5 marks)

Q27. 5M Board fav (a) Describe YDSE and derive the fringe width. (b) The intensity ratio of maxima and minima is 25 : 9. Find the ratio of slit widths.
Ans: (b) $(a_1 + a_2)/(a_1 - a_2) = 5/3 \Longrightarrow a_1/a_2 = 4 \Longrightarrow$ width (intensity) ratio **16 : 1**.

Q28. 5M (a) Explain single slit diffraction and the condition for minima. (b) Draw the intensity graph. (c) Why must the slit be narrow?

Ans: (a) Pairs of points half the slit apart cancel when $a \sin\theta = n\lambda$. (c) Spreading $\propto \lambda/a$ is visible only when a is comparable to λ.

E. Case-based

Q29. 4M Board fav A student sees bright and dark bands from a laser and two slits 0.5 mm apart, screen 2 m away, λ = 650 nm. (a) β? (b) Position of 3rd dark fringe? (c) If D is halved? (d) If green laser used?

Ans: (a) β = 2.6 mm. (b) 2.5β = 6.5 mm. (c) β halves. (d) β decreases.

Q30. 4M Polaroid sunglasses cut glare from roads and water. (a) Why is glare polarised? (b) Brewster angle for water? (c) Which law gives the transmitted intensity? (d) Intensity after two crossed polaroids?

Ans: (a) Reflected light is partly polarised (fully at i_B). (b) 53°. (c) Malus' law. (d) Zero.

Q31. 1M Find the phase difference for a path difference of $\lambda/4$.

Ans: $\varphi = (2\pi/\lambda)(\lambda/4) = \pi/2$.

Q32. 1M In YDSE, the 10th bright fringe is 3 cm from the centre. Find β.

Ans: 3 mm.

Q33. 1M Light of 500 nm passes a 0.1 mm slit. Find the angular width of the central maximum.

Ans: $2\lambda/a$ = 0.01 rad.

Q34. 2M Board fav A thin sheet (μ = 1.5, t = 6 μm) covers one slit; λ = 600 nm. How many fringes shift?

Ans: $(\mu - 1)t/\lambda = 0.5 \times 6 \times 10^{-6}/6 \times 10^{-7}$ = 5 fringes.

Q35. 1M What is the shape of wavefront from a point source near a convex lens (at its focus) after refraction?

Ans: Plane.

Q36. 2M A galaxy line of 656 nm appears at 706 nm. Find its speed.

Ans: $v = c\Delta\lambda/\lambda = 3 \times 10^8 \times 50/656 \approx 2.3 \times 10^7$ m/s, moving away.

Q37. 2M Why is the sky polarised when seen at 90° to the sun?

Ans: Light scattered at 90° by air molecules has E vibrating only perpendicular to the scattering plane.

Q38. 1M Can two light waves of different frequencies form a steady interference pattern?

Ans: No: phase difference keeps changing.

Q39. 1M State one condition for sustained interference.

Ans: Coherent sources, nearly equal amplitudes, narrow and close slits.

Q40. 1M Find I_{max}/I_{min} if amplitudes are in ratio 2 : 1.

Ans: $(3/1)^2$ = 9 : 1.

Wave optics ke numericals mein bas teen formula: β, I, Malus

Asli baat: Most board numericals use $\beta = \lambda D/d$, $I = 4I_0\cos^2(\varphi/2)$ or $I = I_0\cos^2\theta$.

Common mistakes

- → Fringe width d ke inversely proportional hai, directly nahi.
- → Unpolarised light pehle polaroid ke baad $I_0/2$, $\cos^2$ nahi lagta.
- → Central maximum ki width $2\lambda D/a$ hai, $\lambda D/a$ nahi.
- → Frequency refraction mein nahi badalti.

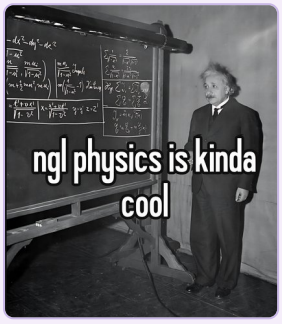
Exam Alert: last check

- → Huygens proof of Snell, YDSE fringe width derivation: 3-5 marks pakka.
- → Interference vs diffraction table aur Malus numerical.

Dual Nature of Radiation and Matter

Most asked: photoelectric effect experiments and graphs, failure of wave theory, Einstein's photoelectric equation, photon properties, de Broglie wavelength of electrons and other particles

Interference and diffraction proved light is a wave. Yet when light knocks electrons out of a metal it behaves like a stream of particles. And electrons, which we call particles, show wave behaviour. This chapter is about that double personality of both radiation and matter.



Einstein ko Nobel relativity ke liye nahi, photoelectric effect ke liye mila

Asli baat: Einstein won the 1921 Nobel Prize for explaining the photoelectric effect with light quanta (photons).

1. Electron Emission

Free electrons in a metal are held inside by the attraction of positive ions. The minimum energy needed to pull an electron out of the metal surface is called the **work function** φ_0, usually given in electron volt (1 eV = 1.6×10^{-19} J).

Type of emission	*Energy supplied by*
Thermionic emission	heat (hot filament)
Field emission	strong electric field (~10^8 V/m)
Photoelectric emission	light of suitable frequency

Metal	φ_0 (eV)
Cesium	2.14
Potassium	2.30
Sodium	2.75
Calcium	3.20
Copper	4.65
Platinum	5.65

Why electrons do not simply leave a metal

A metal contains a large number of free electrons that move randomly through the lattice of positive ions. Inside the metal each electron is pulled equally in all directions, so the net force is nearly zero. At the surface, however, there are positive ions only on the inner side, so an electron trying to leave is pulled back. The surface behaves like a potential barrier. An electron can escape only if it receives energy at least equal to the work function.

The work function depends on the nature of the metal and on the condition of its surface (impurities, oxide layers, coating). Alkali metals such as cesium, potassium and sodium have low work functions, which is why they emit electrons even with visible light and are used in photocells. Metals such as platinum and copper have high work functions and need ultraviolet light.

The electron volt

1 eV is the kinetic energy gained by an electron when it is accelerated through a potential difference of 1 volt. Energy = charge × potential = 1.6×10^{-19} C × 1 V, so **1 eV = 1.6×10^{-19} J**. Atomic and photon energies are a few eV, which is why this unit is used throughout this chapter.

Solved example

Q. The work function of sodium is 2.75 eV. Express it in joule and find the threshold wavelength.

Sol. $\varphi_0 = 2.75 \times 1.6 \times 10^{-19} = \mathbf{4.4 \times 10^{-19}}$ **J**. $\lambda_0 = hc/\varphi_0 = 1240/2.75$ nm = **451 nm** (blue light).

2. Photoelectric Effect

Discovery

→ **Hertz (1887)**: sparks across a gap were stronger when the gap was lit by UV light.

→ **Hallwachs and Lenard (1886–1902)**: a negatively charged zinc plate lost its charge under UV; a neutral plate became positive. Light was knocking out negative charges, later shown to be electrons (photoelectrons).

→ Alkali metals (Li, Na, K, Cs) show the effect even with visible light; zinc, cadmium, magnesium need UV.

NCERT Activity: Hallwachs's zinc plate

Place a clean zinc plate on the cap of a gold-leaf electroscope and charge it negatively: the leaves diverge. Shine UV light (from an arc lamp) on the plate: the leaves collapse as the plate loses electrons. Repeat with a positively charged plate: nothing happens, since UV can only remove negative charge.

Experimental setup

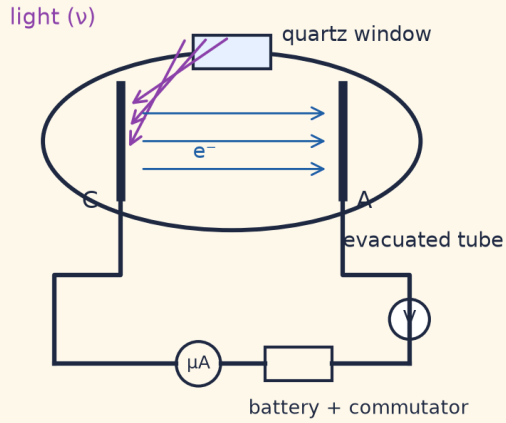


Fig. 11.1 Apparatus for studying the photoelectric effect: light falls on emitter C, electrons are collected at A

Monochromatic light falls through a quartz window on the photosensitive plate C (emitter). Photoelectrons travel to plate A (collector). The potential of A relative to C can be made positive or negative using the commutator; the microammeter reads the photocurrent.

Effect of intensity

Fig. 11.2 Photocurrent is directly proportional to intensity

→ For a fixed frequency above threshold and fixed potential, photocurrent ∝ intensity: number of photoelectrons per second ∝ intensity.

Effect of potential

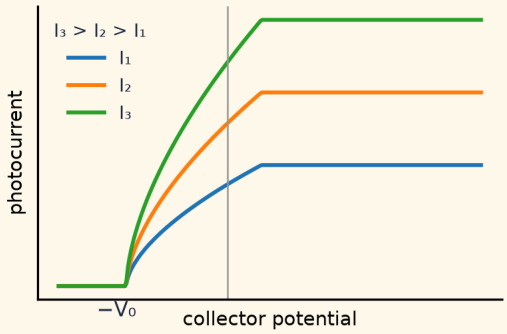


Fig. 11.3 Photocurrent vs collector potential for different intensities (same frequency)

→ As A is made more positive, the current rises and then saturates (all emitted electrons collected).

→ Making A negative reduces the current; at the **stopping potential** $-V_0$ the current becomes zero.

→ V_0 is the same for all intensities at a given frequency:

$$K_{max} = \tfrac{1}{2}mv_{max}^2 = eV_0$$

Effect of frequency

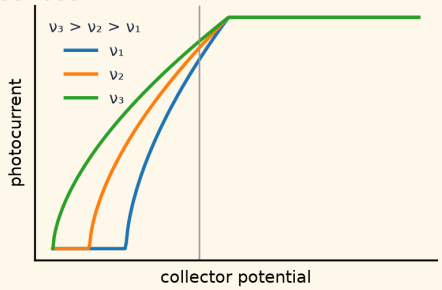


Fig. 11.4 Higher frequency gives larger stopping potential; saturation current same for equal intensity

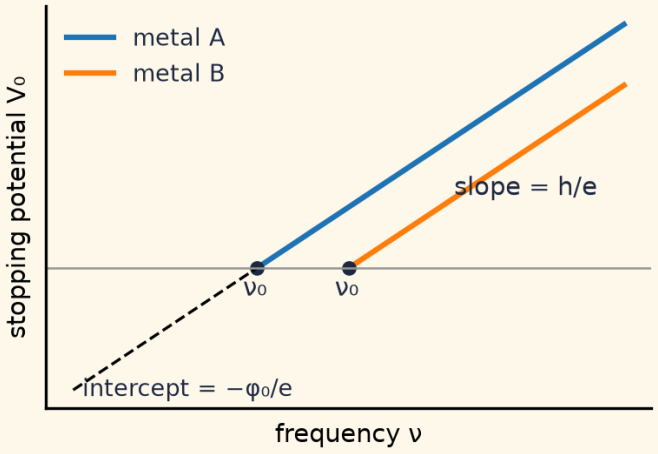


Fig. 11.5 Stopping potential vs frequency for two metals: parallel straight lines

→ V_0 increases linearly with ν; the graphs for all metals are parallel (same slope h/e).

→ Below a minimum **threshold frequency** ν_0, no emission occurs, however intense the light.

→ Emission is **instantaneous** (within 10^{-9} s) even for very dim light.

Yaad karo: 4 laws of photoelectric emission

1. Photocurrent ∝ intensity (frequency fixed).
2. Saturation current ∝ intensity; V_0 independent of intensity.
3. Threshold frequency hota hai; uske neeche koi emission nahi.
4. $K_{max} \propto (\nu - \nu_0)$, intensity se independent; emission instant.

Jab samajh aaya ki intensity badhane se K_{max} nahi badhta

Asli baat: Intensity changes only the number of electrons; maximum kinetic energy depends on frequency.

Reading the photoelectric graphs: what each one tells us

Graph	Shape	Physical meaning
Photocurrent vs intensity	straight line through origin	number of electrons emitted per second ∝ number of photons per second
Photocurrent vs collector potential (different intensities)	same stopping potential, different saturation currents	K_{max} does not depend on intensity
Photocurrent vs collector potential (different frequencies)	larger V_0 for larger ν, same saturation current	K_{max} increases with frequency
V_0 vs ν	straight line, slope h/e, intercept ν_0	gives Planck's constant and the threshold frequency
K_{max} vs ν	straight line, slope h	same slope for every metal

Why current flows even at zero collector potential

Photoelectrons leave the emitter with kinetic energies ranging from zero up to K_{max}. Some of them have enough energy to reach the collector even when no accelerating potential is applied, so a small current flows at V = 0. A positive collector potential pulls more of them in until every emitted electron is collected (saturation). A negative potential turns back the slower electrons first; at $V = -V_0$ even the fastest ones are stopped.

Why the emitted electrons have a range of energies

The work function is the minimum energy needed by the electrons that are least tightly bound and nearest the surface. Electrons deeper inside lose energy in collisions before reaching the surface, so they come out with less than K_{max}. That is why we speak only of the **maximum** kinetic energy in Einstein's equation.

Solved example (NCERT)

Q. The photoelectric cut-off voltage in an experiment is 1.5 V. What is the maximum kinetic energy of the emitted photoelectrons?

Sol. $K_{max} = eV_0 = 1.6 \times 10^{-19} \times 1.5 = \mathbf{2.4 \times 10^{-19}}$ **J (= 1.5 eV).**

Solved example (NCERT)

Q. In a photoelectric experiment the slope of the cut-off voltage versus frequency graph is 4.12×10^{-15} V s. Calculate Planck's constant.

Sol. Slope = h/e $\Longrightarrow$ h = $4.12 \times 10^{-15} \times 1.6 \times 10^{-19}$ = $\mathbf{6.59 \times 10^{-34}}$ **J s.**

3. Wave Theory Fails

Observation	Wave theory predicts	Experiment shows
K_{max} and intensity	more intensity, more K_{max}	K_{max} independent of intensity
Threshold frequency	any frequency works if intense enough	no emission below ν_0
Time lag	dim light needs hours to build energy	emission is instantaneous

Exam Alert

→ "Wave theory photoelectric effect explain kyun nahi kar paayi?" teen points: 3 marks, bahut common.

→ Graphs: I vs V, V_0 vs ν, I vs intensity: draw-and-label questions.

According to the classical wave picture, light energy is spread uniformly and continuously over the whole wavefront. A single electron intercepts only the energy falling on an area about the size of an atom ($\sim 10^{-20}$ m^2). For ordinary light intensities,

calculation shows an electron would need hours to collect enough energy to escape. Experimentally, emission begins within 10^{-9} s even with very feeble light. Similarly, a wave of any frequency should eventually give an electron enough energy if it is intense enough, yet no emission is observed below the threshold frequency. These failures showed that light energy must arrive in concentrated packets.

Key idea

The photoelectric effect is a **one photon, one electron** process. An electron cannot store the energy of two weak photons and then escape (at ordinary intensities). Hence the frequency of each photon, not the total energy of the beam, decides whether emission occurs.

4. Einstein's Photoelectric Equation

Einstein (1905) proposed that light consists of packets (quanta) of energy $h\nu$ called **photons**. A photon gives all its energy to one electron in a single collision. Part of it (φ_0) is used to escape; the rest appears as kinetic energy.

Derivation step by step

1. Energy of photon: $E = h\nu$.
2. Energy used to free the electron from the surface: $\varphi_0 = h\nu_0$.
3. Maximum kinetic energy: $K_{max} = h\nu - \varphi_0$.
4. Using $K_{max} = eV_0$: $eV_0 = h\nu - \varphi_0 \Longrightarrow V_0 = (h/e)\nu - \varphi_0/e$.

$$K_{max} = h\nu - \varphi_0 = h(\nu - \nu_0)$$ Einstein's photoelectric equation

How it explains every observation

1. K_{max} depends linearly on ν, not on intensity.
2. If $\nu < \nu_0$, $h\nu < \varphi_0$: no emission.
3. More intensity = more photons = more electrons = more current.
4. One photon, one electron, at once: no time lag.
5. V_0 vs ν is a straight line of slope h/e, intercept $-\varphi_0/e$. Millikan (1916) verified this and measured h.

Solved example (NCERT)

Q. Monochromatic light of frequency 6.0×10^{14} Hz is produced by a laser emitting 2.0×10^{-3} W. (a) Energy of a photon? (b) Photons emitted per second?

Sol. (a) $E = h\nu = 6.63 \times 10^{-34} \times 6 \times 10^{14} = \mathbf{3.98 \times 10^{-19}}$ J. (b) $N = P/E = 2 \times 10^{-3}/3.98 \times 10^{-19} = \mathbf{5.0 \times 10^{15}\ s^{-1}}$.

Solved example (NCERT)

Q. The work function of cesium is 2.14 eV. Find (a) threshold frequency, (b) wavelength of incident light if the photocurrent is stopped by 0.60 V.

Sol. (a) $\nu_0 = \varphi_0/h = 2.14 \times 1.6 \times 10^{-19}/6.63 \times 10^{-34} = \mathbf{5.16 \times 10^{14}}$ **Hz**. (b) $hc/\lambda = 2.14 + 0.60 = 2.74$ eV $\Longrightarrow \lambda = 1240/2.74 = \mathbf{454}$ **nm**.

Solved example

Q. Light of 3000 Å falls on a metal of work function 2.5 eV. Find K_{max} and V_0.

Sol. $E = 12400/3000 = 4.13$ eV; $K_{max} = \mathbf{1.63}$ **eV**; $V_0 = \mathbf{1.63}$ **V**.

Shortcut

E (in eV) = 1240/λ (in nm) = 12400/λ (in Å). Saves time in every numerical.

Paheli–Bujho

Bujho: Paheli, maine tez torch maari metal pe, phir bhi electron nahi nikle!

Paheli: Torch ki light ka frequency threshold se kam hoga. Intensity se kuch nahi hota, frequency chahiye.

Bujho: Matlab zor se chillane se nahi, sahi sur lagane se kaam banta hai!

Paheli: Exactly. Ek photon, ek electron, aur photon mein dum hona chahiye.

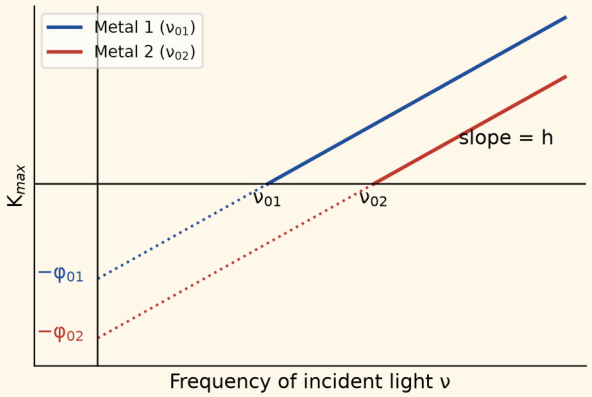


Fig. 11.A K_{max} vs ν for two metals: parallel lines of slope h; intercepts give ν_0 and $-\varphi_0$

Writing Einstein's equation as $K_{max} = h\nu - \varphi_0$ and comparing it with $y = mx + c$ shows that the graph of K_{max} against ν is a straight line of slope h and intercept $-\varphi_0$ on the energy axis. The intercept on the frequency axis is ν_0. Since h is a universal constant, the lines for all metals are parallel. For a metal with larger work function, the line is shifted to the right. Millikan (1916) verified this linear relation very carefully and obtained the value of h, confirming Einstein's theory.

Change made	*K_{max} and V_0*	*Saturation current*
Intensity doubled (same ν)	unchanged	doubled
Frequency increased (same intensity)	increase	unchanged (or slightly less, fewer photons per joule)
Metal of lower φ_0 used	increase	unchanged
Collector distance changed	unchanged	unchanged

Solved example (NCERT)

Q. The threshold frequency of a metal is 3.3×10^{14} Hz. If light of frequency 8.2×10^{14} Hz falls on it, predict the cut-off voltage.

Sol. $eV_0 = h(\nu - \nu_0) \Longrightarrow V_0 = 6.63 \times 10^{-34} \times 4.9 \times 10^{14}/1.6 \times 10^{-19} =$ **2.03 V.**

Solved example (NCERT)

Q. The work function of a metal is 4.2 eV. Will it show photoelectric emission for radiation of 330 nm?

Sol. E = 1240/330 = 3.76 eV. Since 3.76 eV < 4.2 eV, **no emission** takes place, however intense the radiation.

Solved example (NCERT)

Q. Light of frequency 7.21×10^{14} Hz falls on a metal and ejects electrons with a maximum speed of 6.0×10^5 m/s. Find the threshold frequency.

Sol. $K_{max} = \frac{1}{2}mv^2 = \frac{1}{2} \times 9.1 \times 10^{-31} \times (6 \times 10^5)^2 = 1.64 \times 10^{-19}$ J.

$h\nu_0 = h\nu - K_{max} \Longrightarrow \nu_0 = 7.21 \times 10^{14} - 1.64 \times 10^{-19}/6.63 \times 10^{-34} = 7.21 \times 10^{14} - 2.47 \times 10^{14} =$ **4.74×10^{14} Hz.**

Solved example (NCERT)

Q. Light of 488 nm from an argon laser is used in a photoelectric experiment. The stopping potential is 0.38 V. Find the work function of the emitter.

Sol. $h\nu = 1240/488 = 2.54$ eV; $\varphi_0 = h\nu - eV_0 = 2.54 - 0.38 =$ **2.16 eV.**

Solved example

Q. Light of wavelength 400 nm falls on potassium ($\varphi_0 = 2.30$ eV). Find (a) the maximum kinetic energy, (b) the maximum speed of photoelectrons, (c) the stopping potential.

Sol. (a) $h\nu = 1240/400 = 3.10$ eV; $K_{max} = 3.10 - 2.30 =$ **0.80 eV** $= 1.28 \times 10^{-19}$ J. (b) $v = \sqrt{(2K/m)} = \sqrt{(2 \times 1.28 \times 10^{-19}/9.1 \times 10^{-31})} =$ **5.3×10^5 m/s.** (c) $V_0 =$ **0.80 V.**

Solved example

Q. When light of wavelength λ falls on a metal, the stopping potential is $3V_0$. For wavelength 2λ it is V_0. Find the threshold wavelength.

Sol. $3eV_0 = hc/\lambda - \varphi$ and $eV_0 = hc/2\lambda - \varphi$. Multiply the second by 3 and subtract: $0 = hc/\lambda - 3hc/2\lambda + 2\varphi \Longrightarrow \varphi = hc/4\lambda$. Threshold $\lambda_0 = hc/\varphi =$ **4λ.**

Applications of the photoelectric effect

- → **Photocells** in automatic street lights, burglar alarms, automatic doors and counting devices: light falling on the cell completes a circuit; interrupting the beam breaks it.
- → **Light meters** in cameras measure the intensity of light, since photocurrent is proportional to intensity.
- → **Sound track of films**: variations in light passing through the sound track are converted into electrical signals and then into sound.
- → **Solar panels and light sensors** in phones work on the closely related photovoltaic effect in semiconductors.
- → **Photomultiplier tubes** detect extremely weak light, down to single photons, in scientific instruments.

Yaad karo

→ K_{max}-ν graph ka slope h, V_0-ν graph ka slope h/e.
→ Graph parallel hote hain, kyunki h sab metals ke liye same.
→ Intercept: x-axis par ν_0, y-axis par $-\varphi_0$ (ya $-\varphi_0/e$).

5. Particle Nature of Light: The Photon

1. In interaction with matter, radiation behaves as if made of photons.
2. Each photon has energy E = hν and momentum p = hν/c = h/λ.
3. All photons of a given frequency have the same energy and momentum, whatever the intensity.
4. Photons are electrically neutral and are not deflected by electric or magnetic fields.
5. In a collision (photon–electron), total energy and momentum are conserved, but the number of photons may not be (absorption or creation).
6. Rest mass of a photon is zero; it always travels at c in vacuum.

Quantity	Formula
Energy	E = hν = hc/λ
Momentum	p = h/λ = E/c
Number of photons per second	N = P/hν
Photon flux on area	N/A per second

Solved example

Q. Find the momentum of a photon of wavelength 500 nm.

Sol. $p = h/\lambda = 6.63 \times 10^{-34}/5 \times 10^{-7} = \mathbf{1.33 \times 10^{-27}}$ **kg m/s.**

NCERT Activity: photocell switch

A photocell (vacuum tube with a cesium-coated cathode) passes current only while light falls on it. Connected to a relay, it switches street lights off at dawn and on at dusk, counts people passing a beam, and opens automatic doors. Cover the cell with your hand and the relay clicks off.

Light: kabhi wave, kabhi particle. Main bhi confuse

Asli baat: Light shows wave nature in interference/diffraction and particle nature in photoelectric emission. Both are true: dual nature.

Photon: properties explained

A photon is a quantum of electromagnetic radiation. It always travels at speed c in vacuum and has zero rest mass; it cannot be at rest. Its energy and momentum depend only on its frequency. A more intense beam of the same colour simply carries more photons per second through a given area; each photon is identical. In a collision with an electron, both energy and momentum are conserved, but the number of photons need not be: a photon can be absorbed or created.

Number of photons from a source (step by step)

1. Energy of one photon: E = hc/λ.
2. Energy emitted per second = power P.
3. Number of photons per second: N = P/E = Pλ/hc.
4. Photons per unit area per second at the earth or on a surface: N/A = (intensity)/E.

Solved example (NCERT)

Q. Monochromatic light of 632.8 nm from a He-Ne laser has a power of 9.42 mW. (a) Find the energy and momentum of each photon. (b) How many photons per second reach the target? (c) How fast must a hydrogen atom move to have the same momentum as a photon?

Sol. (a) $E = hc/\lambda = 6.63 \times 10^{-34} \times 3 \times 10^8/632.8 \times 10^{-9} = \mathbf{3.14 \times 10^{-19}}$ **J**; $p = h/\lambda = \mathbf{1.05 \times 10^{-27}}$ **kg m/s.**
(b) $N = P/E = 9.42 \times 10^{-3}/3.14 \times 10^{-19} = \mathbf{3 \times 10^{16}}$ **photons/s.**
(c) $v = p/m_H = 1.05 \times 10^{-27}/1.67 \times 10^{-27} = \mathbf{0.63}$ **m/s.**

Solved example (NCERT)

Q. The energy flux of sunlight at the earth is 1.388×10^3 W/m^2. How many photons per square metre per second fall on the earth? Take the average wavelength as 550 nm.

Sol. $E = hc/\lambda = 3.61 \times 10^{-19}$ J. $N = 1.388 \times 10^3/3.61 \times 10^{-19} = \mathbf{3.8 \times 10^{21}}$ **photons m^{-2} s^{-1}.**

Solved example (NCERT)

Q. A 100 W sodium lamp radiates uniformly in all directions at 589 nm. Find (a) the energy of each photon, (b) the rate at which photons are emitted.

Sol. (a) $E = hc/\lambda = 3.38 \times 10^{-19}$ J = **2.11 eV**. (b) $N = 100/3.38 \times 10^{-19}$ = **3×10^{20} photons/s**.

Solved example

Q. Find the maximum frequency and minimum wavelength of X-rays produced by electrons accelerated through 30 kV.

Sol. The whole kinetic energy eV can become one photon: $h\nu_{max} = eV \Longrightarrow \nu_{max} = 1.6 \times 10^{-19} \times 3 \times 10^4/6.63 \times 10^{-34}$ = **7.24×10^{18} Hz**; $\lambda_{min} = c/\nu_{max}$ = **0.0414 nm**.

6. Wave Nature of Matter

Nature loves symmetry: if waves (light) can behave as particles, de Broglie (1924) proposed that moving particles should behave as waves. The wavelength associated with a particle of momentum p is:

$$\lambda = h/p = h/mv$$ de Broglie relation

Derivation step by step (electron accelerated through V)

1. Kinetic energy gained: $K = eV$.
2. Momentum: $K = p^2/2m \Longrightarrow p = \sqrt{(2mK)} = \sqrt{(2meV)}$.
3. $\lambda = h/\sqrt{(2meV)}$.
4. Put h, m_e, e: $\lambda = 1.227/\sqrt{V}$ nm.

$$\lambda = 1.227/\sqrt{V} \text{ nm}$$ for an electron accelerated through V volts

→ For a particle of kinetic energy K: $\lambda = h/\sqrt{(2mK)}$; for a gas molecule at temperature T: $\lambda = h/\sqrt{(3mkT)}$.

→ Matter waves are not electromagnetic; they exist for charged and neutral particles alike.

→ For everyday objects λ is far too small to detect (e.g. 0.12 kg ball at 20 m/s: $\lambda \approx 2.8 \times 10^{-34}$ m). For electrons it is about 0.1 nm, comparable to atomic spacing.

→ Electron microscopes use the tiny wavelength of fast electrons to see much finer detail than light microscopes.

→ Heisenberg's uncertainty principle ($\Delta x\, \Delta p \approx h/2\pi$) follows naturally from the wave picture.

Solved example (NCERT)

Q. Find the de Broglie wavelength of (a) a bullet of 0.040 kg at 1.0 km/s, (b) a ball of 0.060 kg at 1.0 m/s, (c) a dust particle of 1.0×10^{-9} kg at 2.2 m/s.

Sol. (a) $6.63 \times 10^{-34}/40$ = **1.66×10^{-35} m**. (b) **1.1×10^{-32} m**. (c) **3.0×10^{-25} m**. All too small to observe.

Solved example (NCERT)

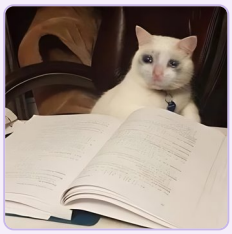
Q. An electron, an α-particle and a proton have the same kinetic energy. Which has the shortest λ?

Sol. $\lambda = h/\sqrt{(2mK)}$: the heaviest, the **α-particle**, has the shortest wavelength.

Solved example

Q. Find λ of an electron accelerated through 100 V.

Sol. $\lambda = 1.227/\sqrt{100}$ = **0.123 nm**.



Cricket ball ki bhi wavelength hoti hai? Haan, par 10^{-34} m

Asli baat: Every moving body has $\lambda = h/mv$, but for big objects it is far too small to notice.

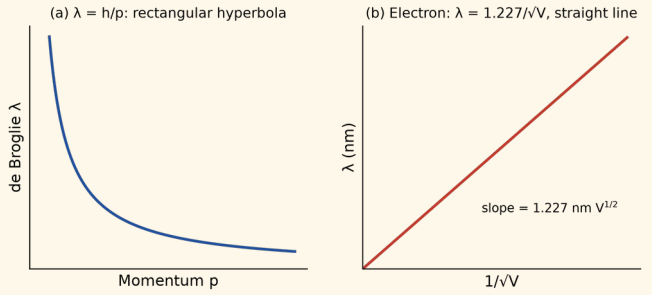


Fig. 11.B (a) λ vs p is a hyperbola; (b) for electrons, λ vs $1/\sqrt{V}$ is a straight line

Comparing wavelengths of different particles

Situation	Formula	Smallest λ for
Same velocity	$\lambda = h/mv$	heaviest particle
Same kinetic energy	$\lambda = h/\sqrt{(2mK)}$	heaviest particle
Same accelerating potential V	$\lambda = h/\sqrt{(2mqV)}$	largest mq (α-particle)
Same temperature T	$\lambda = h/\sqrt{(3mkT)}$	heaviest molecule

Comparing a photon and an electron of the same wavelength: both have the same momentum $p = h/\lambda$, but the photon energy is pc while the electron kinetic energy is $p^2/2m$, which is much smaller. This is why an electron microscope can use very short

wavelengths with modest energies and resolve far finer details than an optical microscope.

Wave nature and probability

The matter wave of a particle is not a physical vibration of a medium. The square of its amplitude at a point gives the probability of finding the particle there. A particle with a definite momentum has a perfectly defined wavelength but is spread over all space; a localised particle is a wave packet made of many wavelengths. This is the basis of Heisenberg's uncertainty principle, $\Delta x\,\Delta p \approx h/2\pi$: position and momentum cannot both be known exactly at the same time.

Solved example (NCERT)

Q. Find (a) the momentum, (b) the de Broglie wavelength of electrons accelerated through 56 V.

Sol. (a) $p = \sqrt{(2meV)} = \sqrt{(2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 56)} = \mathbf{4.04 \times 10^{-24}\ kg\ m/s}$. (b) $\lambda = h/p = \mathbf{0.164\ nm}$ (check: $1.227/\sqrt{56} = 0.164$ nm).

Solved example (NCERT)

Q. Find the momentum, speed and de Broglie wavelength of an electron of kinetic energy 120 eV.

Sol. $p = \sqrt{(2mK)} = \sqrt{(2 \times 9.1 \times 10^{-31} \times 120 \times 1.6 \times 10^{-19})} = \mathbf{5.92 \times 10^{-24}\ kg\ m/s}$; $v = p/m = \mathbf{6.5 \times 10^{6}\ m/s}$; $\lambda = h/p = \mathbf{0.112\ nm}$.

Solved example (NCERT)

Q. An electron and a photon each have a wavelength of 1.00 nm. Find (a) their momenta, (b) the energy of the photon, (c) the kinetic energy of the electron.

Sol. (a) $p = h/\lambda = \mathbf{6.63 \times 10^{-25}\ kg\ m/s}$ for both. (b) $E = pc = 1240/1$ eV $= \mathbf{1.24\ keV}$. (c) $K = p^2/2m = (6.63 \times 10^{-25})^2/(2 \times 9.1 \times 10^{-31}) = 2.4 \times 10^{-19}$ J $= \mathbf{1.51\ eV}$.

Solved example (NCERT)

Q. (a) For what kinetic energy will a neutron have a de Broglie wavelength of 1.40×10^{-10} m? (b) Find the wavelength of a thermal neutron at 300 K ($K = 3kT/2$). $m_n = 1.675 \times 10^{-27}$ kg.

Sol. (a) $K = h^2/2m\lambda^2 = (6.63 \times 10^{-34})^2/(2 \times 1.675 \times 10^{-27} \times 1.96 \times 10^{-20}) = \mathbf{6.7 \times 10^{-21}\ J \approx 0.042\ eV}$. (b) $\lambda = h/\sqrt{(3mkT)} = 6.63 \times 10^{-34}/\sqrt{(3 \times 1.675 \times 10^{-27} \times 1.38 \times 10^{-23} \times 300)} = \mathbf{0.145\ nm}$. Thermal neutrons therefore have wavelengths comparable to atomic spacings and are diffracted by crystals.

Solved example

Q. Show that the wavelength of electromagnetic radiation equals the de Broglie wavelength of its quantum (photon).

Sol. Photon momentum $p = E/c = h\nu/c$. de Broglie wavelength $= h/p = h/(h\nu/c) = c/\nu = \boldsymbol{\lambda}$, the wavelength of the radiation itself.

Yaad karo

- → Same K: $\lambda \propto 1/\sqrt{m}$; same V: $\lambda \propto 1/\sqrt{(mq)}$.
- → Electron: $\lambda = 1.227/\sqrt{V}$ nm; photon: $E(eV) = 1240/\lambda(nm)$.
- → Matter wave = probability wave, EM wave nahi.

7. Solved Numericals (Board Pattern)

N1. Threshold wavelength of a metal is 600 nm. Find φ_0.

Sol. $\varphi_0 = 1240/600 = \mathbf{2.07\ eV}$.

N2. Stopping potential is 1.5 V for light of 400 nm. Find the work function.

Sol. $E = 3.1$ eV; $\varphi_0 = 3.1 - 1.5 = \mathbf{1.6\ eV}$.

N3. From a V_0-ν graph, slope $= 4.1 \times 10^{-15}$ V s. Find h.

Sol. $h = \text{slope} \times e = 4.1 \times 10^{-15} \times 1.6 \times 10^{-19} = \mathbf{6.56 \times 10^{-34}\ J\ s}$.

N4. Find the ratio of de Broglie wavelengths of a proton and an α-particle accelerated through the same potential.

Sol. $\lambda \propto 1/\sqrt{(mq)}$: $\lambda_p/\lambda_\alpha = \sqrt{(4m \times 2e/(m \times e))} = \mathbf{2\sqrt{2}}$.

N5. The kinetic energy of an electron is doubled. How does λ change?

Sol. $\lambda \propto 1/\sqrt{K} \Longrightarrow$ becomes $\boldsymbol{\lambda/\sqrt{2}}$.

N6. A 100 W bulb emits 5% as light of 550 nm. Find photons per second.

Sol. $E = 3.6 \times 10^{-19}$ J; $N = 5/3.6 \times 10^{-19} = \mathbf{1.4 \times 10^{19}\ s^{-1}}$.

Formula sheet: Chapter 11

Quantity	Formula
Photon energy	$E = h\nu = hc/\lambda$; $E(eV) = 1240/\lambda(nm)$
Photon momentum	$p = h/\lambda$
Einstein equation	$K_{max} = h\nu - \varphi_0$
Stopping potential	$eV_0 = K_{max}$
Work function	$\varphi_0 = h\nu_0 = hc/\lambda_0$
de Broglie	$\lambda = h/p = h/mv = h/\sqrt{(2mK)}$
Electron through V	$\lambda = 1.227/\sqrt{V}$ nm
Charge q through V	$\lambda = h/\sqrt{(2mqV)}$

1 minute recap

- ✓ Photoelectric: current ∝ intensity, $K_{max} \propto (\nu - \nu_0)$, no lag.
- ✓ Wave theory fails on 3 counts.
- ✓ $K_{max} = h\nu - \varphi_0$; V_0-ν slope h/e.
- ✓ Photon: $E = h\nu$, $p = h/\lambda$, zero rest mass.
- ✓ Matter waves: $\lambda = h/p$; electron: $1.227/\sqrt{V}$ nm.

MOST IMPORTANT QUESTIONS

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM The stopping potential depends on:
(a) intensity
(b) frequency of light
(c) area of plate
(d) time of exposure
Ans: (b) frequency of light.

Q2. IM Board fav Slope of the V_0-ν graph gives:
(a) h
(b) e/h
(c) h/e
(d) φ_0
Ans: (c) h/e.

Q3. IM Photoelectric current increases with:
(a) intensity
(b) wavelength
(c) work function
(d) stopping potential
Ans: (a) intensity.

Q4. IM de Broglie wavelength of an electron accelerated through V is proportional to:
(a) V
(b) $\sqrt{V}$
(c) 1/V
(d) $1/\sqrt{V}$
Ans: (d) $1/\sqrt{V}$.

Q5. IM The rest mass of a photon is:
(a) zero
(b) 9.1×10^{-31} kg
(c) infinite
(d) $h\nu/c^2$
Ans: (a) zero.

Q6. IM If the frequency of light is doubled, K_{max}:
(a) doubles
(b) halves
(c) becomes more than double
(d) unchanged
Ans: (c) becomes more than double.

Q7. IM Board fav Define work function.
Ans: Minimum energy needed to remove an electron from a metal surface.

Q8. IM Define threshold frequency.
Ans: Minimum frequency of light that can cause photoelectric emission from a metal.

Q9. IM Does K_{max} depend on intensity?
Ans: No, only on frequency (and work function).

Q10. IM Name the experiment that verified Einstein's equation.
Ans: Millikan's experiment (1916).

Q11. IM What is the momentum of a photon of energy E?
Ans: $p = E/c$.

Q12. IM Which has a longer de Broglie wavelength at the same speed: electron or proton?
Ans: Electron (smaller mass).

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M Board fav A: Photoelectric emission is instantaneous. R: One photon transfers its energy to one electron at once.
Ans: (a)

Q14. 1M A: Increasing intensity increases stopping potential. R: More photons strike the surface.
Ans: (d)

Q15. 1M A: We do not observe wave nature of a moving car. R: Its de Broglie wavelength is extremely small.
Ans: (a)

Q16. 1M A: Photons are not deflected by magnetic fields. R: Photons are electrically neutral.
Ans: (a)

C. Short answer (2–3 marks)

Q17. 3M Board fav State three observations of the photoelectric effect that wave theory cannot explain.
Ans: K_{max} independent of intensity; threshold frequency exists; no time lag.

Q18. 2M Board fav Write Einstein's photoelectric equation and explain how it accounts for the threshold frequency.
Ans: $K_{max} = h\nu - \varphi_0$; if $h\nu < \varphi_0$, K_{max} would be negative: no emission.

Q19. 3M Board fav Draw the graph of photocurrent vs collector potential for two intensities and two frequencies.
Ans: See Figs. 11.3 and 11.4.

Q20. 2M Draw the V_0–ν graph for two metals. What do the slope and intercept give?
Ans: Parallel lines; slope h/e; intercept on ν-axis gives ν_0, on V-axis $-\varphi_0/e$.

Q21. 2M Board fav Derive $\lambda = 1.227/\sqrt{V}$ nm for an electron.
Ans: See Section 11.6.

Q22. 2M List four properties of photons.
Ans: $E = h\nu$, $p = h/\lambda$, neutral, zero rest mass, speed c.

Q23. 1M An electron and a photon have the same wavelength. Compare their momenta.
Ans: Equal ($p = h/\lambda$ for both).

Q24. 2M Light of the same frequency but double intensity falls on a metal. What changes?
Ans: Photocurrent doubles; K_{max} and V_0 unchanged.

Q25. 2M Why are alkali metals used in photocells for visible light?
Ans: Their work functions are low (≈ 2 eV), so visible photons can eject electrons.

Q26. 1M A proton and an electron have the same kinetic energy. Which has the shorter de Broglie wavelength?
Ans: Proton (larger mass): $\lambda = h/\sqrt{(2mK)}$.

D. Long answer (5 marks)

Q27. 5M Board fav (a) Describe an experiment to study the photoelectric effect. (b) Explain the effect of intensity, potential and frequency with graphs. (c) State Einstein's equation.
Ans: See Sections 11.2–11.4.

Q28. 5M (a) What are matter waves? (b) Derive the de Broglie wavelength of an electron accelerated through V. (c) Find λ for 54 V.
Ans: (c) $\lambda = 1.227/\sqrt{54} = 0.167$ nm.

E. Case-based

Q29. 4M Board fav A solar-powered calculator uses a photocell. (a) Which effect? (b) Will red light work if $\nu_{red} < \nu_0$? (c) Brighter light changes? (d) Energy of a 620 nm photon?
Ans: (a) Photoelectric. (b) No. (c) Current increases. (d) 2.0 eV.

Q30. 4M An electron microscope accelerates electrons through 10 kV. (a) λ? (b) Why better than an optical microscope? (c) λ if 40 kV? (d) Are matter waves EM waves?
Ans: (a) 0.0123 nm. (b) Much smaller λ gives better resolution. (c) Halves: 0.0061 nm. (d) No.

Q31. 1M Light of 2.5 eV gives $K_{max} = 0.5$ eV. Find φ_0.
Ans: 2.0 eV.

Q32. 1M Find the energy in eV of a photon of 248 nm.
Ans: 1240/248 = 5 eV.

Q33. 2M Board fav Threshold frequency of a metal is 5×10^{14} Hz. Light of 8×10^{14} Hz falls. Find V_0.
Ans: $eV_0 = h(\nu - \nu_0) = 6.63 \times 10^{-34} \times 3 \times 10^{14} \Longrightarrow V_0 = 1.24$ V.

Q34. 2M Find the de Broglie wavelength of a neutron ($m = 1.67 \times 10^{-27}$ kg) moving at 1.4×10^3 m/s.
Ans: $\lambda = 6.63 \times 10^{-34}/(1.67 \times 10^{-27} \times 1400) = 0.28$ nm.

Q35. 1M The work function of a metal is 4.2 eV. Will light of 400 nm cause emission?
Ans: E = 3.1 eV < 4.2 eV: no.

Q36. 1M Why does a photon of higher frequency give more energetic photoelectrons?
Ans: Photon energy $h\nu$ is larger; after paying φ_0, more is left as kinetic energy.

Q37. 1M What is the de Broglie wavelength of a photon of momentum p?
Ans: $\lambda = h/p$.

Q38. 1M Two metals A and B have threshold wavelengths 400 nm and 600 nm. Which has the larger work function?
Ans: A (smaller λ_0, larger hc/λ_0).

Q39. 1M The momentum of an electron is doubled. How does λ change?
Ans: Halves.

Q40. 2M Board fav Calculate the number of photons in 1 J of light of 600 nm.
Ans: $E = 3.3 \times 10^{-19}$ J $\Longrightarrow N = 3.0 \times 10^{18}$.

F. Extra practice (board PYQ style)

Q41. 3M Board fav Light of wavelength 2000 Å falls on aluminium ($\varphi_0 = 4.2$ eV). Find K_{max}, V_0 and the threshold wavelength.
Ans: E = 6.2 eV; $K_{max} = 2.0$ eV; $V_0 = 2.0$ V; $\lambda_0 = 12400/4.2 = 2952$ Å.

Q42. 2M Plot K_{max} vs ν. What is the slope?
Ans: Straight line starting at ν_0; slope = h.

Q43. 2M Why can't light of very long wavelength cause photoelectric emission however bright?
Ans: Each photon has energy hc/λ less than φ_0; one electron absorbs one photon only.

Q44. 1M Find the ratio of de Broglie wavelength of an electron and a proton with the same momentum.
Ans: 1 : 1.

Q45. 2M An electron and an α-particle are accelerated through the same V. Ratio λ_e/λ_α?
Ans: $\sqrt{(m_\alpha q_\alpha / m_e q_e)} = \sqrt{(2 \times 7294)} \approx 121$.

Q46. 1M Give one application each of the photoelectric effect and of matter waves.
Ans: Photocell / solar cell; electron microscope.

Yaad karo: graphs at a glance

- → I vs intensity: straight line through origin.
- → I vs V: saturates; cuts axis at $-V_0$ (same for all intensities).
- → V_0 vs ν: straight line, slope h/e, starts at ν_0.
- → K_{max} vs ν: slope h.

Board exam mein $K_{max} = h\nu + \varphi_0$ likh diya
Asli baat: It is minus: $K_{max} = h\nu - \varphi_0$. The work function is spent to escape.

Quick compare: photon vs electron

Photon	Electron
Rest mass zero	Rest mass 9.1×10^{-31} kg
Neutral	Charge −e
Speed always c	Speed less than c
$\lambda = h/p$ (EM wave)	$\lambda = h/p$ (matter wave)

Common mistakes

- → Intensity badhane se V_0 nahi badhta.
- → eV ko J mein convert karna mat bhoolo ($\times 1.6 \times 10^{-19}$).
- → $\lambda = 1.227/\sqrt{V}$ sirf electron ke liye, nm mein.
- → Saturation current intensity par depend karta hai, frequency par nahi (same intensity).

Exam Alert: last check

- → Einstein equation numerical + V_0-ν graph: har saal.
- → Wave theory failure, photon properties, de Broglie for electron.

ATOMS

Most asked: Rutherford's alpha scattering experiment and its conclusions, distance of closest approach, Bohr's postulates, radius and energy of orbits, hydrogen spectrum and spectral series, de Broglie explanation of Bohr's second postulate

What does an atom look like inside? Thomson imagined a pudding with electrons stuck in it. Rutherford fired alpha particles at gold foil and found a tiny, heavy nucleus. Bohr then explained why atoms do not collapse and why hydrogen glows in sharp colours.

1. Thomson's Model

J.J. Thomson (1898) proposed that an atom is a sphere of uniformly distributed positive charge with electrons embedded in it, like seeds in a watermelon (plum pudding model). The atom is electrically neutral overall.

→ It explained neutrality of atoms.

→ It could not explain the large-angle scattering of alpha particles or the line spectra of atoms.

J.J. Thomson (1898) imagined the atom as a sphere of uniformly distributed positive charge, about 10^{-10} m in radius, with electrons embedded in it like seeds in a watermelon (the "plum pudding" model). The total positive charge equals the total electronic charge, so the atom is neutral. In the ground state the electrons were supposed to sit at rest in stable equilibrium positions, and when disturbed they would vibrate and emit radiation.

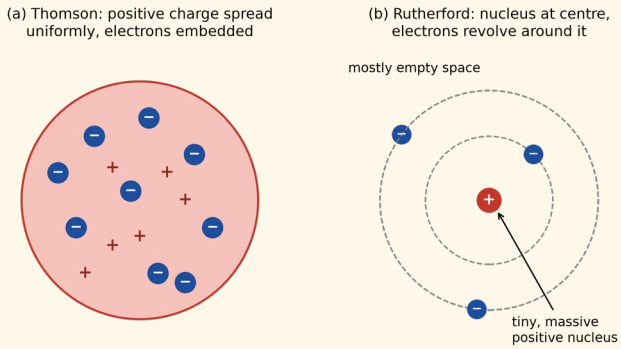


Fig. 12.A Thomson's model (positive charge spread out) versus Rutherford's nuclear model (charge and mass concentrated in a tiny nucleus)

Feature	Thomson model	Rutherford model
Positive charge	spread uniformly over the whole atom	concentrated in a tiny nucleus ($\sim 10^{-15}$ m)
Mass distribution	nearly continuous	highly non-uniform, almost all in the nucleus
Electrons	embedded, at rest in equilibrium	revolve round the nucleus; always under a net force
Size of atom	$\sim 10^{-10}$ m	$\sim 10^{-10}$ m (no different)
Large-angle alpha scattering	cannot explain	explains
Stability	stable (classically) but fails experiment	classically unstable: should collapse

Solved example (NCERT)

Q. Choose the correct alternative: (a) The size of the atom in Thomson's model is ___ the atomic size in Rutherford's model (much greater than / no different from / much less than). (b) In the ground state of ___ electrons are in stable equilibrium, while in ___ electrons always experience a net force. (c) A classical atom based on ___ is doomed to collapse. (d) An atom has a nearly continuous mass distribution in ___ but a highly non-uniform one in ___. (e) The positively charged part of the atom possesses most of the mass in ___.

Sol. (a) **no different from**; (b) **Thomson's model; Rutherford's model**; (c) **Rutherford's model**; (d) **Thomson's model; Rutherford's model**; (e) **both models.**

2. Alpha-particle Scattering and Rutherford's Model

Geiger-Marsden experiment (1911)

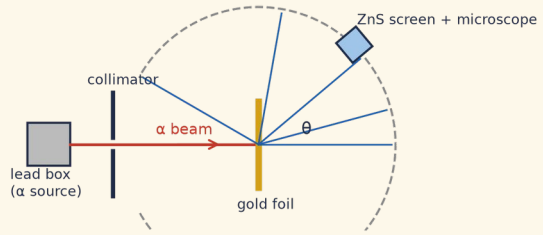


Fig. 12.1 Geiger-Marsden experiment: alpha particles from a radioactive source strike a thin gold foil; scattered particles are counted on a ZnS screen

1. A collimated beam of 5.5 MeV alpha particles from $^{214}_{83}Bi$ falls on a gold foil about 2.1 × 10^{-7} m thick.
2. Scattered alpha particles produce flashes on a rotatable zinc sulphide screen viewed through a microscope.
3. The number of particles scattered at each angle θ is counted.

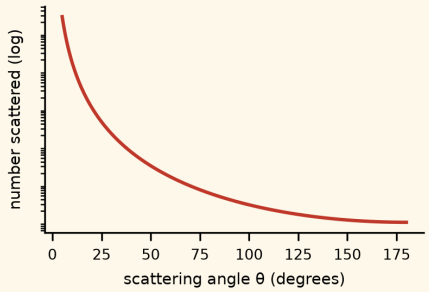


Fig. 12.2 Number of alpha particles scattered vs angle (log scale): falls steeply with θ

Observations and conclusions

Observation	Conclusion
Most alpha particles passed straight through	most of the atom is empty space
A few were deflected through small angles	positive charge is concentrated in a small region
About 1 in 8000 was deflected by more than 90°	the positive charge and almost all the mass sit in a tiny nucleus that repels alpha particles strongly

Rutherford's nuclear model

→ The atom has a tiny, dense, positively charged nucleus (radius ~10^{-15} m) containing almost all its mass.

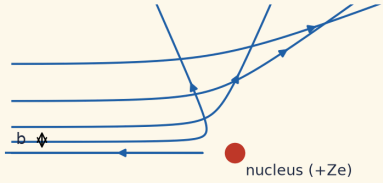
→ Electrons revolve around the nucleus in orbits, like planets around the sun; the centripetal force is the Coulomb attraction.

→ The size of the atom (~10^{-10} m) is about 10^5 times the size of the nucleus.

Atom ke andar itni khaali jagah? Phir bhi main mota dikhta hoon

Asli baat: If the nucleus were a cricket ball, the atom would be a stadium: nucleus radius ~10^{-15} m, atom ~10^{-10} m.

Alpha particle trajectory and impact parameter



smaller impact parameter b gives larger deflection

Fig. 12.3 Trajectories of alpha particles: a head-on particle (b = 0) bounces straight back

→ **Impact parameter b**: perpendicular distance of the initial velocity of the alpha particle from the centre of the nucleus.

→ Small b: large scattering angle; b = 0 (head-on): θ = 180°, the particle retraces its path.

→ Large b: hardly deflected. Since few particles have small b, large-angle scattering is rare.

Distance of closest approach (step by step)

1. A head-on alpha particle (charge 2e) with kinetic energy K approaches a nucleus of charge Ze.
2. It slows down as kinetic energy turns into electric potential energy, stopping momentarily at distance d.
3. At that point: $K = (1/4\pi\varepsilon_0)(2e)(Ze)/d$.
4. Hence:

$d = 2Ze^2/(4\pi\varepsilon_0 K)$ gives an upper limit to the size of the nucleus

Solved example (NCERT)

Q. In the Geiger-Marsden experiment, what is the distance of closest approach for a 7.7 MeV alpha particle to a gold nucleus (Z = 79)?

Sol. $d = (9 \times 10^9)(2 \times 79)(1.6 \times 10^{-19})^2/(7.7 \times 1.6 \times 10^{-13}) = 2.96 \times 10^{-14}$ m ≈ 30 fm.

Drawbacks of Rutherford's model

1. **Stability**: an orbiting electron is accelerated, so by classical electromagnetism it should radiate energy, spiral inwards and fall into the nucleus in about 10^{-8} s. Atoms are stable.

2. Line spectra: a spiralling electron would emit a continuous range of frequencies, but atoms emit sharp lines.

Exam Alert

→ Geiger-Marsden observations aur conclusions table: 3 marks.
→ Closest approach numerical aur impact parameter definition: MCQ + 2 marks.
→ Rutherford model ki do kamiyan: 2 marks pakka.

Why alpha particles and why a thin gold foil

→ Alpha particles from a radioactive source (^{214}Bi) are fast (about 1.6×10^{7} m/s) and heavy (about 7300 times the electron mass), so electrons in the foil cannot deflect them appreciably. Any large deflection must be caused by a heavy positive charge.
→ Gold is very malleable and can be beaten into a foil about 2.1×10^{-7} m thick, so each alpha particle usually suffers at most one close encounter with a nucleus.
→ Gold has a large atomic number ($Z = 79$), so the repulsion from its nucleus is strong and large deflections are easy to observe.
→ The experiment is done in vacuum so that the alpha particles are not stopped or scattered by air molecules.

What the numbers tell us

About 99.9% of alpha particles passed straight through or were deflected by less than 1°. Only about 1 in 8000 was deflected by more than 90°, and a few bounced almost straight back. Rutherford remarked that it was as surprising as firing a 15-inch shell at a piece of tissue paper and having it come back. In Thomson's model the positive charge is spread out, so the electric field inside the atom is weak and no single atom could turn an alpha particle back. Only a very small, very dense positive nucleus could produce such a strong repulsion.

Size of nucleus from the experiment

The distance of closest approach for 7.7 MeV alpha particles on gold is about 3×10^{-14} m, which puts an upper limit on the size of the nucleus. Nuclear radii are about 10^{-15} to 10^{-14} m, roughly 10^{4} to 10^{5} times smaller than the atom. If an atom were the size of a cricket stadium, the nucleus would be a small marble at the centre.

Solved example (NCERT)

Q. Suppose the alpha scattering experiment is repeated using a thin sheet of solid hydrogen instead of gold foil. What results would you expect? (Hydrogen is a solid below 14 K.)

Sol. The hydrogen nucleus is a single proton, about four times lighter than an alpha particle. A heavy alpha particle colliding with a lighter target cannot be turned back; it continues forward with only small deflections, just as a heavy ball pushes a lighter one ahead. Hence **no large-angle scattering** would be seen, and the experiment could not reveal the small size of the nucleus.

3. Atomic Spectra

A gas heated or excited by an electric discharge emits light of only certain wavelengths: an **emission line spectrum**. White light through a cool gas shows dark lines at the same wavelengths: an **absorption spectrum**. Each element has its own unique set of lines, like a fingerprint.

Spectral series of hydrogen

$$1/\lambda = R(1/n_f^2 - 1/n_i^2)$$ $R = 1.097 \times 10^{7}\ m^{-1}$ (Rydberg constant)

Series	n_f	n_i	Region
Lyman	1	2, 3, 4, ...	ultraviolet
Balmer	2	3, 4, 5, ...	visible
Paschen	3	4, 5, 6, ...	infrared
Brackett	4	5, 6, 7, ...	infrared
Pfund	5	6, 7, 8, ...	infrared

→ Balmer lines: H_α 656.3 nm (red), H_β 486.1 nm (blue-green), H_γ 434.1 nm (violet).
→ Series limit ($n_i = \infty$): shortest wavelength of a series. Balmer limit = 364.6 nm.
→ The first line of each series ($n_i = n_f + 1$) has the longest wavelength.

NCERT Activity: see hydrogen lines

Look at a hydrogen discharge tube (or a street lamp of sodium or mercury) through a diffraction grating or a CD held at an angle. Instead of a continuous rainbow you see a few separate bright coloured lines. Compare with a filament bulb, which gives a continuous spectrum.

Emission and absorption spectra

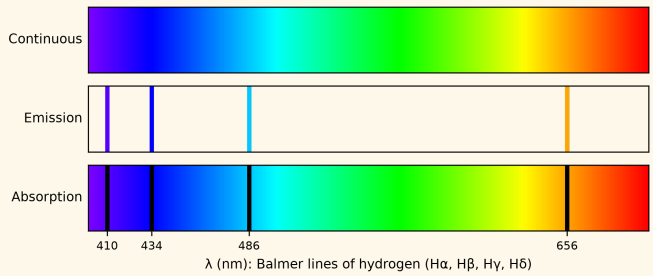


Fig. 12.B Continuous spectrum, emission line spectrum and absorption line spectrum of hydrogen (Balmer lines)

A heated solid gives a **continuous spectrum** containing all wavelengths. A gas or vapour excited by an electric discharge gives an **emission line spectrum**: bright lines at definite wavelengths on a dark background. When white light passes through a cool gas, the gas absorbs exactly those wavelengths it would emit, giving dark lines on a continuous background: an **absorption line spectrum**. The dark lines in the solar spectrum (Fraunhofer lines) are produced in this way by gases in the sun's outer atmosphere.

Each element has its own characteristic set of lines, like a fingerprint. This is how helium was first discovered in the sun, and how astronomers find the composition of stars. The regularity of the hydrogen lines, summarised by the Rydberg formula, was a key clue that led Bohr to his model.

Series	n_f	n_i	Region	Shortest λ (series limit)
Lyman	1	2, 3, …	ultraviolet	1/R = 91.2 nm
Balmer	2	3, 4, …	visible	4/R = 364.6 nm
Paschen	3	4, 5, …	infrared	9/R = 820.4 nm
Brackett	4	5, 6, …	infrared	16/R = 1459 nm
Pfund	5	6, 7, …	infrared	25/R = 2279 nm

Solved example (NCERT)

Q. What is the shortest wavelength present in the Paschen series of hydrogen?

Sol. Shortest λ for $n_i = \infty$: $1/\lambda = R(1/3^2 - 0) \Longrightarrow \lambda = 9/R = 9/1.097 \times 10^7 =$ **820.4 nm.**

4. Bohr Model of the Hydrogen Atom

Bohr's postulates (1913)

1. **Stationary orbits**: an electron revolves in certain stable orbits without radiating energy.
2. **Quantisation of angular momentum**: allowed orbits are those for which $L = mvr = nh/2\pi$, n = 1, 2, 3, …
3. **Frequency condition**: an electron emits (or absorbs) a photon only when it jumps between orbits: $h\nu = E_i - E_f$.

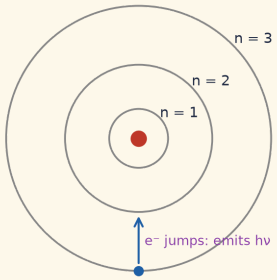


Fig. 12.4 Bohr orbits: a jump from a higher to a lower orbit emits a photon

Derivation step by step: radius of n-th orbit

1. Coulomb force provides centripetal force: $mv^2/r = (1/4\pi\varepsilon_0)e^2/r^2$.
2. Quantisation: $mvr = nh/2\pi \Longrightarrow v = nh/(2\pi mr)$.
3. Substitute v: $m(nh/2\pi mr)^2/r = e^2/(4\pi\varepsilon_0 r^2)$.
4. Solve for r:

$$r_n = n^2h^2\varepsilon_0/(\pi me^2) = n^2a_0 \quad a_0 = 0.529\ \text{Å (Bohr radius)}$$

$$v_n = e^2/(2\varepsilon_0 nh) = (2.18 \times 10^6/n)\ \text{m/s} \quad v \propto 1/n$$

Derivation step by step: energy of n-th orbit

1. Kinetic energy: $K = \frac{1}{2}mv^2 = e^2/(8\pi\varepsilon_0 r)$.
2. Potential energy: $U = -e^2/(4\pi\varepsilon_0 r) = -2K$.
3. Total: $E = K + U = -e^2/(8\pi\varepsilon_0 r)$.
4. Put r_n:

$$E_n = -me^4/(8\varepsilon_0^2 n^2h^2) = -13.6/n^2\ \text{eV}$$

→ Negative energy means the electron is bound. E = 0 at $n = \infty$ (free electron).

→ K = −E, U = 2E. For n = 1: K = 13.6 eV, U = −27.2 eV.

→ **Ionisation energy** of hydrogen (ground state) = 13.6 eV; **excitation energy** to n = 2 = 10.2 eV.

→ Ground state n = 1; higher levels are excited states. An electron stays in an excited state for about 10^{-8} s.

Quantity	Dependence on n
Radius r	$\propto n^2$
Speed v	$\propto 1/n$
Energy E	$\propto -1/n^2$
Angular momentum L	$\propto n$
Time period T	$\propto n^3$
Frequency of revolution	$\propto 1/n^3$

Energy negative kaise ho sakti hai? Bound electron ka swag

Asli baat: Negative total energy means the electron is bound to the nucleus; 13.6 eV must be supplied to free it from n = 1.

Why Bohr needed new postulates

In Rutherford's model a revolving electron is accelerated, and classical electromagnetic theory says an accelerated charge must radiate energy. The electron would lose energy continuously, spiral inward and fall into the nucleus in about 10^{-8} s, emitting a continuous spectrum of ever-increasing frequency. Real atoms are stable and emit sharp lines. Bohr kept the nuclear model but added quantum ideas: only certain orbits are allowed, the electron does not radiate in them, and radiation is emitted only during a jump between orbits.

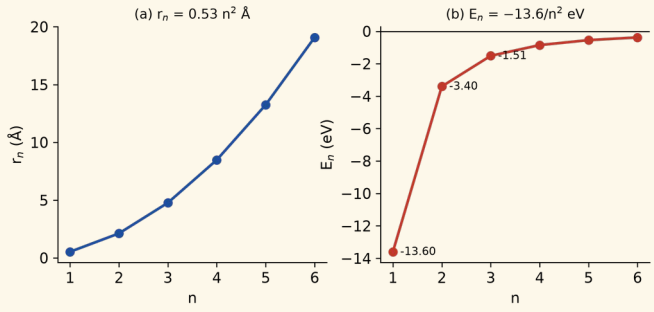


Fig. 12.C Radius grows as n^2; energy rises towards zero as $-1/n^2$, so levels crowd together at large n

Quantity	Dependence on n	Dependence on Z	Ground state of H
Radius r_n	$\propto n^2$	$\propto 1/Z$	0.53 Å
Speed v_n	$\propto 1/n$	$\propto Z$	2.19×10^6 m/s (c/137)
Total energy E_n	$\propto -1/n^2$	$\propto Z^2$	−13.6 eV
Time period T_n	$\propto n^3$	$\propto 1/Z^2$	1.52×10^{-16} s
Angular momentum	$= nh/2\pi$	independent	$h/2\pi$

Energy terms explained

→ **Kinetic energy** $K = ke^2/2r$ is positive.

→ **Potential energy** $U = -ke^2/r$ is negative and twice the kinetic energy in magnitude.

→ **Total energy** $E = K + U = -ke^2/2r = -K$. A negative total energy means the electron is bound; 13.6 eV must be supplied to free it from the ground state (ionisation energy).

→ **Excitation energy** is the energy needed to raise the electron from the ground state to a higher level: 10.2 eV for n = 2, 12.09 eV for n = 3.

→ The zero of potential energy is chosen at infinite separation. Changing this choice changes U and E, but not K or energy differences.

Solved example (NCERT)

Q. The ground state energy of hydrogen is −13.6 eV. Find the kinetic and potential energies of the electron in this state.

Sol. K = −E = **13.6 eV**; U = 2E = **−27.2 eV**.

Solved example (NCERT)

Q. Using the Bohr model, find (a) the speed of the electron in the n = 1, 2, 3 levels of hydrogen, (b) the orbital period in each.

Sol. (a) $v_n = 2.19 \times 10^6/n$ m/s: $v_1 = \mathbf{2.19 \times 10^6}$, $v_2 = \mathbf{1.09 \times 10^6}$, $v_3 = \mathbf{7.3 \times 10^5}$ **m/s.**

(b) $T_1 = 2\pi r_1/v_1 = 2\pi \times 0.53 \times 10^{-10}/2.19 \times 10^6$ $= \mathbf{1.52 \times 10^{-16}}$ **s.** $T \propto n^3$: $T_2 = \mathbf{1.22 \times 10^{-15}}$ **s**, $T_3 = \mathbf{4.1 \times 10^{-15}}$ **s.**

Solved example (NCERT)

Q. The radius of the innermost orbit of hydrogen is 5.3×10^{-11} m. Find the radii of the n = 2 and n = 3 orbits.

Sol. $r_n = n^2 r_1$: $r_2 = 4 \times 5.3 \times 10^{-11} = \mathbf{2.12 \times 10^{-10}}$ **m**; $r_3 = 9 \times 5.3 \times 10^{-11} = \mathbf{4.77 \times 10^{-10}}$ **m**.

Yaad karo: n ka khel

→ $r \propto n^2$, $v \propto 1/n$, $E \propto -1/n^2$, $T \propto n^3$.

→ K = −E, U = 2E: ground state mein 13.6, −27.2, −13.6 eV.

→ Z wale ions: $r \propto 1/Z$, $E \propto Z^2$.

5. Line Spectra Explained by Bohr

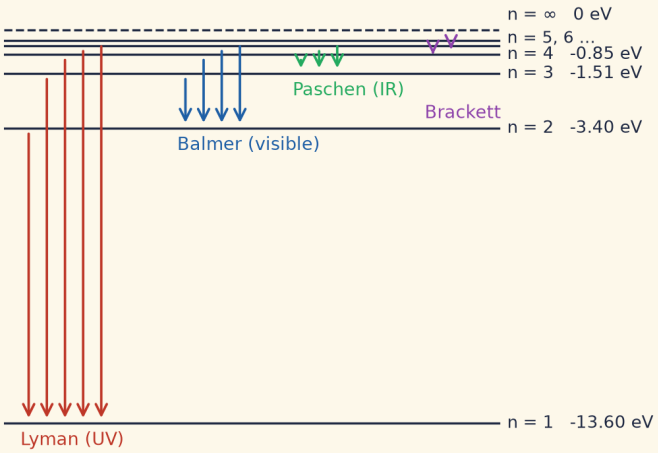


Fig. 12.5 Energy level diagram of hydrogen with the spectral series

Derivation step by step (Rydberg formula from Bohr)

1. Electron jumps from n_i to n_f ($n_i > n_f$).
2. $h\nu = E_i - E_f = (me^4/8\varepsilon_0^2 h^2)(1/n_f^2 - 1/n_i^2)$.
3. Using $\nu = c/\lambda$: $1/\lambda = (me^4/8\varepsilon_0^2 h^3 c)(1/n_f^2 - 1/n_i^2)$.
4. The constant equals $R = 1.097 \times 10^7\ m^{-1}$, exactly matching experiment.

→ Number of spectral lines possible when an electron falls from level n to the ground state: $n(n-1)/2$.

→ In absorption at room temperature, atoms are mostly in n = 1, so only Lyman lines appear in absorption.

Solved example (NCERT)

Q. Find the wavelength of the first line of the Balmer series.

Sol. $1/\lambda = R(1/4 - 1/9) = 1.097 \times 10^7 \times 5/36 \Longrightarrow \lambda =$ **656.3 nm** (red H_α).

Solved example (NCERT)

Q. The total energy of an electron in the first excited state of hydrogen is −3.4 eV. Find (a) its kinetic energy, (b) potential energy. Which would change if the zero of potential energy were changed?

Sol. (a) K = −E = **3.4 eV**. (b) U = 2E = **−6.8 eV**. Only U and E depend on the choice of zero; K does not.

Solved example

Q. An electron jumps from n = 4 to n = 2 in hydrogen. Find the photon energy and wavelength.

Sol. $\Delta E = 13.6(1/4 - 1/16) =$ **2.55 eV**; $\lambda = 1240/2.55 =$ **486 nm** (H_β).

Solved example

Q. Find the radius and speed of the electron in the third orbit of hydrogen.

Sol. $r_3 = 9 \times 0.529 =$ **4.76 Å**; $v_3 = 2.18 \times 10^6/3 =$ **7.27×10^5 m/s**.

Paheli-Bujho

Bujho: Paheli, electron gol gol ghoomta hai toh energy kyun nahi khota?

Paheli: Bohr ne kaha: kuch special orbits mein woh bina radiate kiye ghoom sakta hai. Energy tabhi nikalti hai jab woh ek orbit se doosre mein jump kare.

Bujho: Matlab seedhi pe chadhna, beech mein khade nahi ho sakte!

Paheli: Bilkul. Energy levels seedhi ke steps hain.

Hydrogen-like ions

For an ion with one electron and nuclear charge Ze (He^+, Li^{2+}, Be^{3+}), replace e^2 by Ze^2 in Bohr's derivation:

Quantity	Hydrogen-like ion (Z)
Radius	$r_n = 0.529\ n^2/Z$ Å
Speed	$v_n = 2.18 \times 10^6\ Z/n$ m/s
Energy	$E_n = -13.6\ Z^2/n^2$ eV
Wave number	$1/\lambda = RZ^2(1/n_f^2 - 1/n_i^2)$

Solved example

Q. Find the ionisation energy of Li^{2+} and the radius of its first orbit.

Sol. $E = 13.6 \times 9 = \mathbf{122.4\ eV}$; $r = 0.529/3 = \mathbf{0.176\ Å}$.

Solved example

Q. Which transition in He^+ gives the same wavelength as the $2 \to 1$ transition in hydrogen?

Sol. $Z^2(1/n_f^2 - 1/n_i^2) = 3/4$ with $Z = 2 \implies n_f = 2$, $n_i = 4$: $\mathbf{4 \to 2}$.

Solved example (NCERT)

Q. A 10 kg satellite circles the earth once every 2 h in an orbit of radius 8000 km. Assuming Bohr's angular momentum postulate applies, find the quantum number of its orbit.

Sol. $v = 2\pi r/T = 6981$ m/s; $mvr = nh/2\pi$ $\implies n = 2\pi mvr/h = 2\pi \times 10 \times 6981 \times 8 \times 10^6/6.63 \times 10^{-34} \approx \mathbf{5.3 \times 10^{45}}$. Quantisation is unnoticeable for large bodies.

Emission spectrum	*Absorption spectrum*
Bright lines on a dark background	Dark lines on a continuous bright background
Produced by hot or excited gas	Produced when white light passes through cool gas
Electrons fall to lower levels	Electrons rise to higher levels
Contains all series	Mostly Lyman series (atoms in ground state)

NCERT Activity: model of alpha scattering

Place a small heavy ball (a marble) at the centre of a table hidden under a board. Roll many small balls randomly from one side towards the hidden marble. Most roll straight through; only a few bounce back. Counting the bounces lets you estimate how small the hidden target is, just as Rutherford estimated the nucleus.

Absorption, emission and the number of lines

An atom in the ground state can absorb a photon only if its energy exactly equals the difference between the ground level and some higher level. The excited atom stays there for about 10^{-8} s and then returns, either directly or by a series of smaller jumps, emitting a photon at each jump. From level n, the total number of different lines possible is $n(n-1)/2$. A single atom emits at most $n - 1$ photons on the way down, but in a gas with many atoms all possible transitions occur, so all lines are seen together.

An electron of kinetic energy E can excite an atom in a collision by giving up only part of its energy, so electron beams excite any level up to E. A photon, however, is absorbed only if its energy matches a level difference exactly (or exceeds 13.6 eV, in which case the atom is ionised).

Solved example (NCERT)

Q. Two energy levels are separated by 2.3 eV. What is the frequency of the radiation emitted when an atom makes a transition between them?

Sol. $\nu = E/h = 2.3 \times 1.6 \times 10^{-19}/6.63 \times 10^{-34} = \mathbf{5.6 \times 10^{14}\ Hz}$.

Solved example (NCERT)

Q. A hydrogen atom in the ground state absorbs a photon that raises it to $n = 4$. Find the wavelength and frequency of the photon.

Sol. $\Delta E = 13.6(1 - 1/16) = 12.75$ eV $= 2.04 \times 10^{-18}$ J. $\lambda = 1240/12.75 = \mathbf{97.3\ nm}$; $\nu = \Delta E/h = \mathbf{3.1 \times 10^{15}\ Hz}$.

Solved example (NCERT)

Q. A 12.5 eV electron beam bombards gaseous hydrogen at room temperature. Which wavelengths will be emitted?

Sol. Energy needed to reach $n = 3$: $13.6 - 1.51 = 12.09$ eV (possible); $n = 4$ needs 12.75 eV (not possible). So atoms are excited up to $n = 3$. Lines emitted: $3 \to 1$: $\lambda = 1240/12.09 = \mathbf{102.6\ nm}$ and $2 \to 1$: $\lambda = 1240/10.2 = \mathbf{121.6\ nm}$ (Lyman, ultraviolet); $3 \to 2$: $\lambda = 1240/1.89 = \mathbf{656\ nm}$ (Balmer, red).

Solved example

Q. A muon is a particle with the same charge as the electron but 207 times its mass. In muonic hydrogen a muon orbits a proton. Find the Bohr radius and ground state energy.

Sol. $r \propto 1/m$ and $E \propto m$: $r = 0.53 \times 10^{-10}/207 = \mathbf{2.56 \times 10^{-13}\ m}$; $E = -13.6 \times 207$ eV $= \mathbf{-2.81\ keV}$.

Paheli-Bujho

Bujho: Paheli, hydrogen ke ek atom mein toh ek hi electron hai, phir itni saari lines ek saath kaise?

Paheli: Ek atom ek time pe ek hi jump karta hai. Par gas mein crore atoms hain, sab alag-alag jump kar rahe hain, isliye saari lines ek saath dikhti hain.

Bujho: Matlab ek student ek hi galti karta hai, poori class milke saari galtiyan kar deti hai!

6. de Broglie's Explanation of Bohr's Second Postulate

Fig. 12.6 Standing wave on a Bohr orbit: a whole number of wavelengths fits the circumference

1. de Broglie: an electron in orbit is a wave of wavelength $\lambda = h/mv$.
2. For a stable orbit, the wave must close on itself (standing wave): circumference = whole number of wavelengths, $2\pi r = n\lambda$.
3. Put $\lambda = h/mv$: $2\pi r = nh/mv \Longrightarrow \mathbf{mvr = nh/2\pi}$, which is exactly Bohr's quantisation condition.

Limitations of the Bohr model

- → Works only for hydrogen-like (one-electron) atoms: H, He^{+}, Li^{2+}.
- → Cannot explain relative intensities of spectral lines or fine structure.
- → Cannot explain the spectra of multi-electron atoms or chemical bonding.

Bohr ka quantisation condition rattna vs de Broglie se derive karna

Asli baat: $2\pi r = n\lambda$ with $\lambda = h/mv$ directly gives $mvr = nh/2\pi$: one-line derivation, frequently asked.

In 1913 Bohr's quantisation of angular momentum was a bold assumption without explanation. In 1924 de Broglie's matter waves gave the reason. If the electron is a wave travelling round the orbit, a steady (standing) wave is possible only if the wave joins smoothly after one revolution, that is, the circumference contains a whole number of wavelengths: $2\pi r_n = n\lambda$. With $\lambda = h/mv$ this gives $mvr_n = nh/2\pi$, which is exactly Bohr's second postulate. Any other orbit would make the wave interfere destructively with itself and die out.

Limitations of the Bohr model

- → It applies only to hydrogen-like (one-electron) atoms; it cannot explain helium or heavier atoms, where electrons interact with each other.
- → It cannot explain the relative intensities of the spectral lines.
- → It treats the electron as moving in a definite orbit, which contradicts the uncertainty principle; quantum mechanics replaces orbits with probability clouds.

Solved example (NCERT)

Q. The earth revolves round the sun in an orbit of radius 1.5×10^{11} m with a speed of 3×10^{4} m/s. Mass of earth = 6×10^{24} kg. If Bohr's rule applied, what would be its quantum number?

Sol. $mvr = nh/2\pi \Longrightarrow n = 2\pi mvr/h = 2\pi \times 6 \times 10^{24} \times 3 \times 10^{4} \times 1.5 \times 10^{11}/6.63 \times 10^{-34} = \mathbf{2.6 \times 10^{74}}$. Such a huge n means the allowed levels are so close that the motion looks continuous: quantisation is unobservable for large bodies.

7. Solved Numericals (Board Pattern)

N1. Find the shortest wavelength in the Lyman series.

Sol. $1/\lambda = R(1 - 0) \Longrightarrow \lambda = 1/R = \mathbf{91.2\ nm}$.

N2. Find the ratio of the longest wavelengths of Lyman and Balmer series.

Sol. Lyman: $1/\lambda = 3R/4$; Balmer: $5R/36 \Longrightarrow \lambda_L/\lambda_B = (5/36)/(3/4) = \mathbf{5/27}$.

N3. How many spectral lines are emitted when electrons fall from n = 5?

Sol. $5 \times 4/2 = \mathbf{10}$.

N4. Energy needed to excite a hydrogen atom from n = 1 to n = 3?

Sol. $13.6(1 - 1/9) = \mathbf{12.09\ eV}$.

N5. Find the ratio of radii of the 2nd and 3rd orbits.
Sol. $r \propto n^2 \Longrightarrow 4:9$.

N6. A 12.5 eV electron beam bombards hydrogen gas at room temperature. Which series lines are emitted?
Sol. It can raise n = 1 up to n = 3 (12.09 eV) but not n = 4 (12.75 eV). Lines: 3 → 1, 2 → 1 (Lyman) and 3 → 2 (Balmer).

Formula sheet: Chapter 12

Quantity	Formula
Closest approach	$d = 2Ze^2/(4\pi\varepsilon_0 K)$
Quantisation	$mvr = nh/2\pi$
Radius	$r_n = 0.529\ n^2$ Å
Speed	$v_n = 2.18 \times 10^6/n$ m/s
Energy	$E_n = -13.6/n^2$ eV
K and U	$K = -E,\ U = 2E$
Photon	$h\nu = E_i - E_f$
Rydberg	$1/\lambda = R(1/n_f^2 - 1/n_i^2)$
Number of lines	$n(n-1)/2$
de Broglie	$2\pi r = n\lambda$

1 minute recap

- ✓ Alpha scattering: tiny dense positive nucleus; atom mostly empty.
- ✓ Closest approach $d = 2Ze^2/4\pi\varepsilon_0 K$.
- ✓ Bohr: stationary orbits, $mvr = nh/2\pi$, $h\nu = \Delta E$.
- ✓ $r \propto n^2$, $v \propto 1/n$, $E = -13.6/n^2$ eV.
- ✓ Series: Lyman UV, Balmer visible, rest IR.

Most Important Questions

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM In Rutherford's experiment, most alpha particles pass undeviated because:
(a) nucleus is negative
(b) most of the atom is empty
(c) alpha particles are light
(d) gold is soft
Ans: (b) most of the atom is empty.

Q2. IM Board fav The radius of the n-th Bohr orbit is proportional to:
(a) n
(b) 1/n
(c) n^2
(d) $1/n^2$
Ans: (c) n^2.

Q3. IM The Balmer series lies in the:
(a) visible region
(b) UV region
(c) IR region
(d) X-ray region
Ans: (a) visible region.

Q4. IM Ionisation energy of hydrogen in the ground state is:
(a) 3.4 eV
(b) 10.2 eV
(c) 1.51 eV
(d) 13.6 eV
Ans: (d) 13.6 eV.

Q5. IM For a head-on collision the impact parameter is:
(a) zero
(b) maximum
(c) equal to atomic radius
(d) infinite
Ans: (a) zero.

Q6. IM The ratio of kinetic energy to total energy of an electron in a Bohr orbit is:
(a) 1
(b) −1
(c) 2
(d) −2
Ans: (b) −1.

Q7. IM Board fav Define impact parameter.
Ans: Perpendicular distance of the initial velocity of the alpha particle from the centre of the nucleus.

Q8. IM State Bohr's quantisation condition.
Ans: $mvr = nh/2\pi$.

Q9. IM Which series of hydrogen lies in the UV region?
Ans: Lyman.

Q10. 1M What is the Bohr radius?
Ans: 0.529 Å (radius of the first orbit of hydrogen).

Q11. 1M Why is the total energy of an electron in an atom negative?
Ans: It is bound to the nucleus; work must be done to free it.

Q12. 1M What is the speed of an electron in the first orbit of hydrogen?
Ans: 2.18×10^6 m/s.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M Board fav **A:** Rutherford's model fails to explain the stability of atoms. **R:** An accelerated charge radiates energy.
Ans: (a)

Q14. 1M **A:** Hydrogen spectrum has many lines though it has one electron. **R:** Different atoms in a sample undergo different transitions.
Ans: (a)

Q15. 1M **A:** Energy of an electron increases as n increases. **R:** $E_n = -13.6/n^2$ eV becomes less negative with n.
Ans: (a)

Q16. 1M **A:** The Bohr model explains spectra of all atoms. **R:** Bohr used quantisation of angular momentum.
Ans: (d)

C. Short answer (2–3 marks)

Q17. 3M Board fav Give the observations and conclusions of the Geiger–Marsden experiment.
Ans: See the table in Section 12.2.

Q18. 2M Derive the distance of closest approach of an alpha particle.
Ans: $K = (1/4\pi\varepsilon_0)(2Ze^2)/d \Longrightarrow d = 2Ze^2/(4\pi\varepsilon_0 K)$.

Q19. 3M Board fav State Bohr's postulates.
Ans: Stationary orbits; $mvr = nh/2\pi$; $h\nu = E_i - E_f$.

Q20. 3M Board fav Derive the expression for the radius of the n-th orbit.
Ans: See Section 12.4.

Q21. 3M Show that the total energy is $-13.6/n^2$ eV.
Ans: See Section 12.4.

Q22. 2M Board fav Use de Broglie's hypothesis to explain Bohr's second postulate.
Ans: $2\pi r = n\lambda$, $\lambda = h/mv \Longrightarrow mvr = nh/2\pi$.

Q23. 3M Draw the energy level diagram of hydrogen and show Lyman and Balmer transitions.
Ans: See Fig. 12.5.

Q24. 2M State two limitations of Rutherford's model.
Ans: Cannot explain stability; predicts continuous spectrum.

Q25. 2M Give two limitations of the Bohr model.
Ans: Only one-electron atoms; cannot explain intensities or fine structure.

Q26. 2M Find the wavelength of the series limit of the Balmer series.
Ans: $1/\lambda = R/4 \Longrightarrow \lambda = 364.6$ nm.

D. Long answer (5 marks)

Q27. 5M Board fav (a) State Bohr's postulates. (b) Derive expressions for radius and energy of the n-th orbit. (c) Find the ionisation energy of hydrogen.
Ans: (c) 13.6 eV.

Q28. 5M (a) Describe the alpha-scattering experiment with a diagram. (b) State its conclusions. (c) Why was gold used?
Ans: (c) Gold can be beaten into very thin foils and its heavy nucleus gives strong scattering.

E. Case-based

Q29. 4M Board fav Street lamps glow in specific colours. (a) Line or continuous spectrum? (b) Why discrete lines? (c) Wavelength for $3 \to 2$ in hydrogen? (d) Lines from n = 4 to ground?
Ans: (a) Line. (b) Quantised energy levels. (c) 656 nm. (d) 6.

Q30. 4M An alpha particle of 5 MeV approaches a gold nucleus head-on. (a) Closest approach? (b) What does it tell about nuclear size? (c) What happens to its KE at closest approach? (d) Path after?
Ans: (a) $d = 2 \times 79 \times 1.44$ MeV fm/5 MeV ≈ 45.5 fm. (b) Nucleus smaller than this. (c) Zero. (d) Retraces path.

Q31. 1M The energy of an electron in a hydrogen orbit is −1.51 eV. Which orbit?
Ans: $n^2 = 13.6/1.51 = 9 \Longrightarrow n = 3$.

Q32. 1M Find the wavelength of the photon emitted in the $2 \rightarrow 1$ transition.
Ans: 10.2 eV $\Longrightarrow \lambda =$ **121.6 nm**.

Q33. 2M Find the angular momentum of an electron in the 2nd orbit.
Ans: $L = 2h/2\pi = h/\pi = \mathbf{2.11 \times 10^{-34}}$ **J s**.

Q34. 1M The radius of the first orbit is 0.53 Å. Find the radius of the 4th orbit.
Ans: **8.48 Å**.

Q35. 1M Why is the hydrogen atom stable in the ground state?
Ans: There is no lower level to jump to; it cannot radiate.

Q36. 1M Which transition in hydrogen gives the photon of maximum energy?
Ans: $n = \infty \rightarrow n = 1$ (Lyman series limit).

Q37. 2M Board fav Find the frequency of revolution in the first orbit ($v = 2.18 \times 10^6$ m/s, $r = 0.53$ Å).
Ans: $f = v/2\pi r = \mathbf{6.6 \times 10^{15}}$ **Hz**.

Q38. 1M What is the ratio of speeds of the electron in the 1st and 3rd orbits?
Ans: **3 : 1**.

Q39. 1M A hydrogen atom absorbs 10.2 eV. To which level does it go?
Ans: $n = 2$.

Q40. 1M In which region does the Paschen series lie?
Ans: Infrared.

F. Extra practice (board PYQ style)

Q41. 2M Board fav The ground state energy of hydrogen is −13.6 eV. Find the energy of the photon emitted when the electron falls from the 3rd to the 2nd orbit and its wavelength.
Ans: 1.89 eV; $\lambda =$ **656 nm**.

Q42. 1M Using Bohr's model, find the ratio of the energies of the electron in the 1st and 2nd orbits.
Ans: $E \propto 1/n^2 \Longrightarrow$ **4 : 1** (E_1 more negative).

Q43. 1M The ionisation energy of He^+ is 54.4 eV. What is its ground state energy?
Ans: **−54.4 eV** ($Z^2 \times 13.6$).

Q44. 2M Why are large-angle alpha scatterings rare?
Ans: Only alpha particles with very small impact parameter (very close to the tiny nucleus) are deflected strongly.

Q45. 2M An electron in hydrogen makes a transition $4 \rightarrow 1$. Find the wavelength of the emitted photon.
Ans: $\Delta E = 12.75$ eV $\Longrightarrow \lambda =$ **97.3 nm**.

Q46. 1M Give one difference between emission and absorption spectra.
Ans: Emission: bright lines on dark background; absorption: dark lines on continuous spectrum.

Yaad karo: Bohr numbers

- $n = 1$: $r = 0.53$ Å, $v = 2.18 \times 10^6$ m/s, $E = -13.6$ eV.
- $n = 2$: $E = -3.4$ eV; $n = 3$: −1.51 eV; $n = 4$: −0.85 eV.
- Excitation $1 \rightarrow 2$: 10.2 eV; $1 \rightarrow 3$: 12.09 eV.

Jab paper mein seedha "13.6 eV" wala question aaye
Asli baat: Ionisation energy of hydrogen in the ground state is 13.6 eV: the most-used number of this chapter.

Lyman, Balmer, Paschen, Brackett, Pfund: series ka order bhool gaye?

Asli baat: Lyman (n_f = 1, UV), Balmer (2, visible), Paschen (3), Brackett (4), Pfund (5) (infrared).

Quick compare: Thomson vs Rutherford vs Bohr

Thomson	Rutherford	Bohr
positive sphere, electrons embedded	tiny nucleus, electrons orbit	fixed stationary orbits
no nucleus	unstable (radiation)	stable orbits
no spectrum explanation	continuous spectrum predicted	explains H line spectrum

Common mistakes

→ Total energy negative hai; K positive aur U = 2E.

→ Rydberg mein n_f lower level hai.

→ Lines ka count n(n – 1)/2 sirf ground state tak girne par.

→ Closest approach mein alpha ka charge 2e lo, e nahi.

Exam Alert: last check

→ Bohr radius + energy derivation (5 marks), alpha scattering diagram.

→ Series table aur energy level diagram, de Broglie explanation.

NUCLEI

Most asked: composition and size of nucleus, nuclear density, mass defect and binding energy, binding energy per nucleon curve, properties of nuclear force, nuclear fission and fusion with energy calculations

The nucleus is 10^5 times smaller than the atom but holds 99.9% of its mass. What holds protons together against their huge electric repulsion, and how does a tiny loss of mass power the sun and nuclear reactors? That is the story of this chapter.

1. Atomic Masses and Composition of Nucleus

Atomic mass unit

$$1\,u = (1/12) \times \text{mass of one } {}^{12}C \text{ atom} = 1.660539 \times 10^{-27}\ kg$$

Particle	Mass (u)	Mass (kg)	Charge
Proton	1.007276	1.6726×10^{-27}	+e
Neutron	1.008665	1.6749×10^{-27}	0
Electron	0.000549	9.109×10^{-31}	−e
Hydrogen atom	1.007825	1.6735×10^{-27}	0

Composition

- A nucleus contains Z protons and N neutrons, together called **nucleons**. Mass number A = Z + N.
- Notation: ${}^{A}_{Z}X$, e.g. ${}^{197}_{79}Au$ has 79 protons and 118 neutrons.
- The neutron was discovered by Chadwick (1932) by bombarding beryllium with alpha particles; the neutral radiation knocked protons out of paraffin.
- A free neutron is unstable (half-life about 1000 s), but stable inside a nucleus.

Term	Same	Different	Example
Isotopes	Z	N, A	${}^{1}H$, ${}^{2}H$, ${}^{3}H$
Isobars	A	Z, N	${}^{3}H$, ${}^{3}He$
Isotones	N	Z, A	${}^{198}Hg$, ${}^{197}Au$

Yaad karo

- Isotopes: same Protons (P in isotoPes).
- Isobars: same A (A in isobArs).
- Isotones: same Neutrons (N in isotoNes).

Atomic mass unit explained in detail

Atoms are far too light to be weighed in kilograms conveniently: a hydrogen atom has a mass of only about 1.67×10^{-27} kg. So atomic masses are expressed in a special unit. One **atomic mass unit (u)** is defined as one-twelfth of the mass of one neutral carbon-12 atom. Since 12 g of carbon-12 contains Avogadro number of atoms, $1\ u = (12 \times 10^{-3}\ kg)/(12 \times 6.022 \times 10^{23}) = 1.6605 \times 10^{-27}$ kg.

Atomic masses are measured very precisely with a **mass spectrometer**. These measurements show that a chemical element is usually a mixture of **isotopes**: atoms with the same atomic number Z but different mass numbers A. The atomic mass listed in the periodic table is the abundance-weighted average of its isotopes. That is why the atomic mass of chlorine (35.47 u) is not a whole number: it is a mixture of 75.4% ${}^{35}Cl$ (34.98 u) and 24.6% ${}^{37}Cl$ (36.98 u).

Solved example (NCERT exercise)

Q. Two stable isotopes of lithium, ${}^{6}Li$ and ${}^{7}Li$, have abundances of 7.5% and 92.5% and masses 6.01512 u and 7.01600 u respectively. Find the atomic mass of lithium.

Sol. $m = (7.5 \times 6.01512 + 92.5 \times 7.01600)/100 = (45.1134 + 648.98)/100 = \mathbf{6.941\ u}$.

Solved example (NCERT exercise)

Q. Boron has two stable isotopes, ${}^{10}B$ (10.01294 u) and ${}^{11}B$ (11.00931 u). Its atomic mass is 10.811 u. Find the abundances.

Sol. Let the abundance of ${}^{10}B$ be x%. Then $10.811 = [x(10.01294) + (100 - x)(11.00931)]/100$.

$1081.1 = 1100.931 - 0.99637x \Longrightarrow x = 19.831/0.99637 \approx \mathbf{19.9\%}$. Abundance of ${}^{11}B$ = **80.1%**.

Discovery of the neutron (Chadwick, 1932)

Before 1932 it was a puzzle why the mass of a nucleus was about twice the mass of its protons for most elements. Bothe and Becker, and later Irène Curie and Frédéric Joliot, found that when beryllium is bombarded with α-particles from polonium, a very penetrating, **neutral radiation** is emitted. When this radiation struck paraffin wax (rich in hydrogen), it knocked out fast protons.

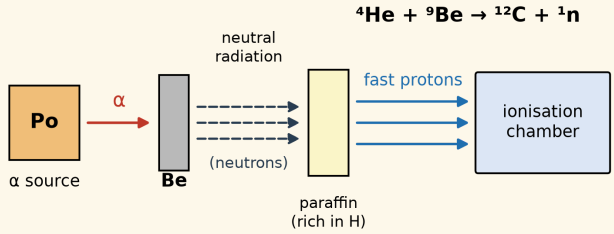


Fig. 13.1(a) Chadwick experiment: the neutral radiation from beryllium knocks protons out of paraffin

1. The radiation was first thought to be γ-rays. But a γ-photon could not knock out protons with such large energy without violating energy and momentum conservation.
2. Chadwick assumed the radiation consists of **neutral particles with mass nearly equal to that of a proton**. In a head-on collision with an equal-mass proton, such a particle can transfer almost all its kinetic energy, exactly as observed.
3. By measuring recoil speeds of hydrogen and nitrogen nuclei, he found the mass of the particle to be almost equal to the proton mass. He named it the **neutron** and received the Nobel Prize in 1935.

$$^4_2He + {}^9_4Be \rightarrow {}^{12}_6C + {}^1_0n$$

Property	Proton	Neutron
Charge	$+1.6 \times 10^{-19}$ C	0
Mass	1.007276 u	1.008665 u
Free state	stable	decays (mean life about 1000 s)
Inside nucleus	stable	stable
Discovered by	Rutherford (1919)	Chadwick (1932)

Yaad karo

Neutron proton se thoda **bhaari** hai (1.008665 u vs 1.007276 u). Free neutron unstable hai, par nucleus ke andar stable. Neutron ka charge zero hai, isliye detect karna mushkil tha.

2. Size of the Nucleus

Alpha and electron scattering experiments show that the volume of a nucleus is proportional to its mass number A:

$$R = R_0 A^{1/3} \quad R_0 = 1.2 \times 10^{-15}\ m = 1.2\ fm$$

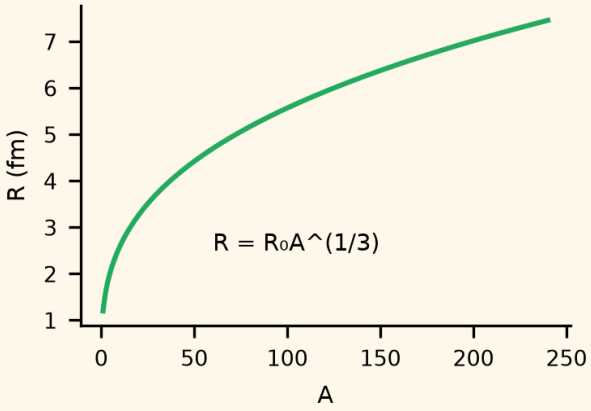


Fig. 13.1 Nuclear radius grows as the cube root of A

Derivation step by step: nuclear density

1. Mass of nucleus ≈ A × m (m = 1.66×10^{-27} kg per nucleon).
2. Volume = $(4/3)\pi R^3 = (4/3)\pi R_0^3 A$.
3. Density $\rho = Am / [(4/3)\pi R_0^3 A] = 3m/(4\pi R_0^3)$.
4. A cancels: density is the **same for all nuclei**, $\rho \approx 2.3 \times 10^{17}$ kg/m^3.

→ This is about 10^{14} times the density of water; a matchbox of nuclear matter would weigh billions of tonnes.

→ Neutron stars are made of matter at roughly this density.

Solved example (NCERT)

Q. Given the mass of iron nucleus as 55.85 u and A = 56, find the nuclear density.

Sol. $m = 55.85 \times 1.66 \times 10^{-27} = 9.27 \times 10^{-26}$ kg; $R = 1.2 \times 10^{-15} \times 56^{1/3} = 4.59$ fm; $\rho = m/(4/3\ \pi R^3) = 2.29 \times 10^{17}$ kg/m^3.

Solved example

Q. Compare the radii of ^{27}Al and ^{125}Te.

Sol. $R \propto A^{1/3} \Longrightarrow R_{Al}/R_{Te} = (27/125)^{1/3} = 3:5$.

Nucleus chhota sa, par density 10^{17} kg/m^3

Asli baat: Nuclear density $\approx 2.3 \times 10^{17}$ kg/m^3, the same for every nucleus because $R^3 \propto A$.

How do we know the size of a nucleus?

Rutherford estimated nuclear size from the distance of closest approach of α-particles. When faster α-particles were used, the scattering began to deviate from the Coulomb law at small distances, showing that the α-particle had begun to "touch" the nucleus. Later, scattering of high-energy electrons by nuclei gave precise radii. All these experiments show that the volume of a nucleus is proportional to its mass number A, which leads to $R = R_0 A^{1/3}$ with $R_0 = 1.2 \times 10^{-15}$ m.

Why is density independent of A? Since $V \propto R^3 \propto A$, each nucleon occupies the same volume whatever the size of the nucleus. The nucleus behaves like a drop of incompressible liquid: a bigger drop has more molecules but the same density. This similarity is the basis of the **liquid drop model** of the nucleus.

Object	*Typical density (kg m^{-3})*
Water	10^3
Iron	7.9×10^3
Nuclear matter	2.3×10^{17}
Neutron star	~10^{17} to 10^{18}

Nuclear matter is about 10^{14} times denser than water. A neutron star is essentially one giant nucleus held together by gravity, with density comparable to nuclear matter: a teaspoon of it would weigh hundreds of millions of tonnes.

Solved example

Q. Find the ratio of the radii of the nuclei ^{27}Al and ^{125}Te. Also find the ratio of their densities.

Sol. $R_1/R_2 = (27/125)^{1/3} = 3/5 = \mathbf{0.6}$. Density is independent of A, so the ratio of densities is **1 : 1**.

Solved example

Q. Show that nuclear density is about 2.3×10^{17} kg m^{-3}.

Sol. $\rho = Am/(4/3\ \pi R_0^3 A) = m/(4/3\ \pi R_0^3)$, with $m \approx 1.66 \times 10^{-27}$ kg and $R_0 = 1.2 \times 10^{-15}$ m: $\rho = 1.66 \times 10^{-27}/(7.24 \times 10^{-45}) \approx \mathbf{2.3 \times 10^{17}\ kg\ m^{-3}}$.

3. Mass-Energy and Nuclear Binding Energy

Mass-energy equivalence

$$E = mc^2 \quad 1\ u \equiv 931.5\ MeV$$

Einstein showed mass is a form of energy. In nuclear reactions the total mass changes measurably, and the change appears as energy. 1 kg of mass is equivalent to 9×10^{13} J.

Mass defect

The mass of a nucleus is always **less** than the total mass of its separate protons and neutrons. The difference is the mass defect:

$$\Delta M = [Zm_p + (A - Z)m_n] - M \quad M = \text{mass of nucleus}$$

Binding energy

The energy equivalent of the mass defect is the **binding energy** E_b: the energy released when the nucleus forms from separate nucleons, or the energy needed to break it completely apart.

$$E_b = \Delta M\, c^2 = \Delta M\ (\text{in u}) \times 931.5\ MeV$$

Solved example (NCERT)

Q. Find the binding energy of $^{16}_{8}$O (mass of atom 15.99493 u, m_H = 1.00783 u, m_n = 1.00867 u).

Sol. $\Delta M = 8(1.00783) + 8(1.00867) - 15.99493 = 0.137707$ u. $E_b = 0.137707 \times 931.5 = \mathbf{127.7\ MeV}$; per nucleon = **7.98 MeV**.

Solved example (NCERT)

Q. Find the energy equivalent of 1 g of substance.

Sol. $E = 10^{-3} \times (3 \times 10^8)^2 = \mathbf{9 \times 10^{13}\ J}$, about the yearly electricity of a big city for a day.

Solved example

Q. Find the binding energy of a deuteron (m = 2.014102 u for ^{2}H atom).

Sol. $\Delta M = 1.007825 + 1.008665 - 2.014102 = 0.002388$ u $\Longrightarrow E_b = \mathbf{2.22\ MeV}$; per nucleon 1.11 MeV.

Exam Alert

→ Mass defect aur binding energy numerical: 3 marks, har saal. Atomic masses use karo (electron masses cancel).

→ 1 u = 931.5 MeV yaad rakho, conversion mein galti mat karna.

Derivation: 1 u = 931.5 MeV, step by step

1. Mass of 1 u: $m = 1.6605 \times 10^{-27}$ kg.
2. Energy equivalent: $E = mc^2 = 1.6605 \times 10^{-27} \times (2.9979 \times 10^8)^2 = 1.4924 \times 10^{-10}$ J.
3. In eV: $E = 1.4924 \times 10^{-10} / 1.602 \times 10^{-19} = 9.315 \times 10^8$ eV.
4. So **1 u ≡ 931.5 MeV**. Multiply a mass in u by 931.5 to get its energy equivalent in MeV.

Why can we use atomic masses instead of nuclear masses?

Tables give the masses of **neutral atoms**, not bare nuclei. If we write $\Delta M = Z\,m_H + N\,m_n - M_{atom}$, where m_H is the mass of a hydrogen atom (1.007825 u), the Z electron masses in $Z\,m_H$ cancel the Z electron masses in M_{atom}. The binding energy of the electrons (a few eV to keV) is negligible compared to MeV, so this gives the nuclear mass defect correctly.

Physical meaning of binding energy

Binding energy can be read in two equivalent ways: (i) the energy **released** when Z protons and N neutrons come together to form the nucleus, or (ii) the minimum energy that must be **supplied** to break the nucleus completely into its free nucleons. It is the nuclear analogue of the ionisation energy of an atom, but a million times larger.

Process	Typical energy
Chemical bond (burning coal)	a few eV per molecule
Ionising hydrogen atom	13.6 eV
Binding energy of ^{4}He	28.3 MeV
One fission of U-235	about 200 MeV

Solved example (NCERT exercise)

Q. Obtain the binding energy of a nitrogen nucleus $^{14}_{7}$N. Given $m(^{14}N) = 14.00307$ u, $m_H = 1.00783$ u, $m_n = 1.00867$ u.

Sol. $\Delta M = 7(1.00783) + 7(1.00867) - 14.00307 = 7.05481 + 7.06069 - 14.00307 = 0.11243$ u.

$E_b = 0.11243 \times 931.5 =$ **104.7 MeV**.

Solved example (NCERT exercise)

Q. Find the binding energy of $^{209}_{83}$Bi (atomic mass 208.980388 u) and compare its B/A with that of ^{56}Fe (8.79 MeV).

Sol. $\Delta M = 83(1.007825) + 126(1.008665) - 208.980388 = 1.760877$ u; $E_b = 1.760877 \times 931.5 =$ **1640.3 MeV**.

B/A = 1640.3/209 = **7.85 MeV**, less than 8.79 MeV. So ^{56}Fe is more tightly bound, even though Bi has the larger total binding energy.

Solved example (NCERT exercise)

Q. A coin of mass 3.0 g is made entirely of $^{63}_{29}$Cu atoms (62.92960 u). Calculate the energy required to separate all its neutrons and protons.

Sol. ΔM per atom $= 29(1.007825) + 34(1.008665) - 62.92960 = 0.591935$ u $\Longrightarrow$ 551.4 MeV per nucleus.

Number of atoms $= (3.0/62.9296) \times 6.022 \times 10^{23} = 2.871 \times 10^{22}$.

Total $= 2.871 \times 10^{22} \times 551.4 =$ **1.58×10^{25} MeV ≈ 2.5×10^{12} J**.

Q-value of a nuclear reaction

For a reaction $a + X \rightarrow Y + b$, the **Q-value** is the energy released: $Q = [m_a + m_X - m_Y - m_b] \times 931.5$ MeV. If $Q > 0$, the reaction is **exothermic** (mass is converted into kinetic energy). If $Q < 0$, it is **endothermic** and can occur only if at least that much energy is supplied. Equivalently, Q = (total binding energy of products) − (total binding energy of reactants).

Solved example

Q. Find the Q-values and say whether each is exothermic or endothermic: (i) $^1H + {}^3H \rightarrow {}^2H + {}^2H$, (ii) $^{12}C + {}^{12}C \rightarrow {}^{20}Ne + {}^4He$. Masses: ^{1}H 1.007825, ^{2}H 2.014102, ^{3}H 3.016049, ^{4}He 4.002603, ^{20}Ne 19.992439 u.

Sol. (i) $\Delta m = 1.007825 + 3.016049 - 2(2.014102) = -0.004330$ u; Q = **−4.03 MeV**, endothermic.

(ii) $\Delta m = 24.000000 - 19.992439 - 4.002603 = 0.004958$ u; Q = **+4.62 MeV**, exothermic.

4. Binding Energy per Nucleon Curve

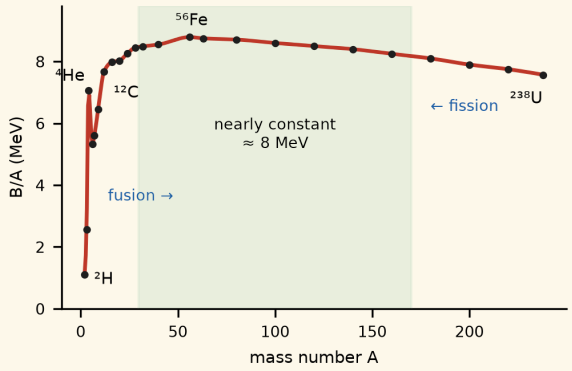


Fig. 13.2 Binding energy per nucleon vs mass number

Features of the curve

1. B/A is small for light nuclei (^{2}H: 1.1 MeV) and rises rapidly.
2. It is nearly constant, about 8 MeV, for 30 < A < 170: the nuclear force is short-ranged and **saturates** (each nucleon binds only with its near neighbours).
3. Maximum ≈ 8.75 MeV at A ≈ 56 (iron): the most stable nuclei.
4. For heavy nuclei (A > 170) B/A falls to about 7.6 MeV for uranium, because the Coulomb repulsion of many protons grows.
5. Peaks at ^{4}He, ^{12}C, ^{16}O show these nuclei are especially stable.

Conclusions

→ **Fission**: a heavy nucleus (A ≈ 240, B/A ≈ 7.6 MeV) splitting into two middle nuclei (A ≈ 120, B/A ≈ 8.5 MeV) increases total binding energy: energy is released.

→ **Fusion**: two very light nuclei joining into a heavier one also increases B/A: energy is released.

Solved example

Q. A nucleus of A = 240 (B/A = 7.6 MeV) splits into two of A = 120 (B/A = 8.5 MeV). Estimate the energy released.

Sol. Q = 240 × (8.5 − 7.6) = **216 MeV**.

Iron-56: sabse stable nucleus, baaki sab use dekhke jalte hain

Asli baat: B/A is maximum (≈ 8.75 MeV) near A = 56; both fission and fusion move nuclei towards this peak.

Why the curve has this shape: explained

1. **Nearly constant middle part (A = 30 to 170):** if every nucleon attracted every other nucleon, the total binding energy would grow as A(A − 1)/2 and B/A would grow with A. It stays almost constant (about 8 MeV), which proves that the nuclear force is **short-ranged and saturated**: a nucleon feels only its nearest neighbours.
2. **Low B/A for light nuclei:** in a small nucleus a large fraction of nucleons lie on the surface and have fewer neighbours, so they are less tightly bound.
3. **Gradual fall for heavy nuclei:** the Coulomb repulsion acts between every pair of protons (it is long-ranged) and grows roughly as Z^2. In a heavy nucleus it reduces the binding, so B/A falls to about 7.6 MeV for uranium.
4. **Peaks at ^{4}He, ^{12}C, ^{16}O:** these nuclei are especially stable, which is why helium nuclei come out as α-particles.

Nucleus	B/A (MeV)
^{2}H	1.11
^{4}He	7.07
^{12}C	7.68
^{16}O	7.98
^{56}Fe	8.79 (maximum)
^{238}U	7.57

Energy from the curve: moving towards the peak (A ≈ 56) from either side releases energy. A heavy nucleus (A ≈ 240, B/A ≈ 7.6 MeV) splitting into two medium nuclei (A ≈ 120, B/A ≈ 8.5 MeV) gains about 0.9 MeV per nucleon, i.e. about 240 × 0.9 ≈ 200 MeV: this is fission. Two very light nuclei joining to form a heavier one also increases B/A: this is fusion.

Solved example (NCERT exercise)

Q. Suppose $^{56}_{26}$Fe could split into two equal fragments $^{28}_{13}$Al. Is this energetically possible? Masses: ^{56}Fe = 55.93494 u, ^{28}Al = 27.98191 u.

Sol. Q = [55.93494 − 2(27.98191)] × 931.5 = (−0.02888) × 931.5 = **−26.9 MeV**. Q is negative, so the fission is **not possible** spontaneously. This agrees with the curve: Fe sits at the peak, and any change reduces B/A.

Yaad karo

Curve ka **peak Fe-56** pe hai. Peak ki taraf jaana = energy milti hai. Bhaari nucleus tode (fission) ya halke jode (fusion), dono mein B/A badhta hai, isliye energy release hoti hai.

5. Nuclear Force

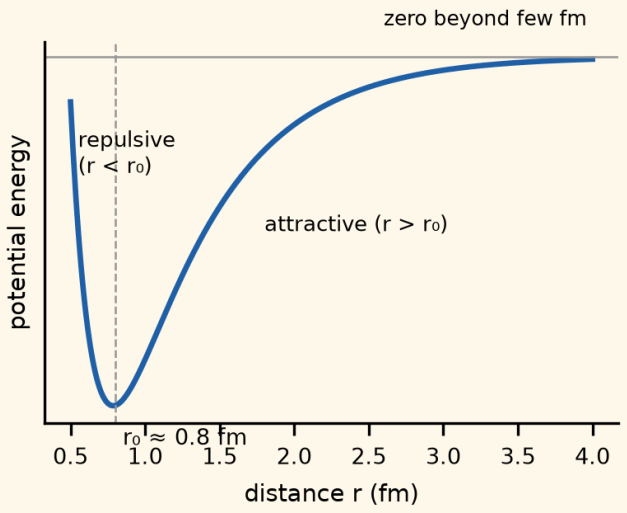


Fig. 13.3 Potential energy of a pair of nucleons vs separation

1. It is the **strongest** force in nature, much stronger than the Coulomb force (it holds protons together).
2. It is **short-ranged**: strongly attractive up to a few fermi and negligible beyond; repulsive below about 0.8 fm, which keeps nucleons from collapsing.
3. It is **charge independent**: n-n, p-p and n-p forces are about the same.
4. It **saturates**: a nucleon interacts only with its nearest neighbours.
5. It is not a central force and depends on spin.

Force	Relative strength	Range
Strong nuclear	1	~ 10^{-15} m
Electromagnetic	10^{-2}	infinite
Weak nuclear	10^{-13}	~ 10^{-18} m
Gravitational	10^{-39}	infinite

Paheli-Bujho

Bujho: Paheli, nucleus mein saare protons positive hain, toh woh ek doosre ko door kyun nahi bhagate?

Paheli: Kyunki wahan nuclear force hai, electric force se 100 guna zyada strong, par sirf 2-3 fm tak kaam karti hai.

Bujho: Matlab protons ka bhi "close friends only" wala group hai!

Paheli: Haha, bilkul. Isliye bahut bade nucleus thode unstable hote hain.

Nuclear force compared with Coulomb force

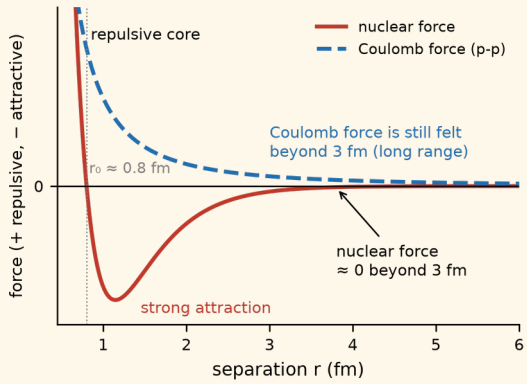


Fig. 13.3(a) Nuclear force is far stronger but acts only up to a few fm; the Coulomb force is weaker but long-ranged

Between two protons inside a nucleus both forces act. At a separation of about 1 fm, the attractive nuclear force is much larger than the Coulomb repulsion, so the protons stay bound. Beyond about 3 fm the nuclear force practically vanishes, while the Coulomb repulsion continues (it falls only as $1/r^2$). This explains why very large nuclei become unstable: every proton repels all other protons, but attracts only its nearest neighbours.

Explaining the potential energy curve, step by step

1. At large separations ($r > 3$ fm) the potential energy is nearly zero: the nucleons do not feel each other.
2. As r decreases, the potential energy becomes negative and decreases: the force is **attractive**.
3. The potential energy is minimum at $r_0 \approx 0.8$ fm. Here the force is zero: this is the equilibrium separation.

4. For $r < r_0$, the potential energy rises sharply: the force becomes strongly **repulsive**, which prevents the nucleus from collapsing.

Why heavy nuclei need more neutrons

Light stable nuclei have $N \approx Z$ (for example ^{12}C, ^{16}O). Heavy stable nuclei have $N > Z$ (for ^{238}U, $N/Z = 146/92 \approx 1.6$). Neutrons add nuclear attraction without adding any Coulomb repulsion, so extra neutrons are needed to hold many protons together.

Solved example (NCERT exercise)

Q. Calculate the height of the potential barrier for a head-on collision of two deuterons. (Treat the deuterons as spheres of radius 2.0 fm; the barrier height is the Coulomb potential energy when they just touch.)

Sol. Distance between centres $r = 2R = 4.0 \times 10^{-15}$ m.

$U = (1/4\pi\varepsilon_0)\, e^2/r = 9 \times 10^9 \times (1.6 \times 10^{-19})^2/(4.0 \times 10^{-15}) = 5.76 \times 10^{-14}$ J = **360 keV**.

Each deuteron must have at least 180 keV of kinetic energy, which is why fusion needs temperatures of the order of 10^9 K in theory (about 10^7 K in practice, helped by the fastest particles of the Maxwell distribution and quantum tunnelling).

6. Nuclear Energy

Nuclear fission

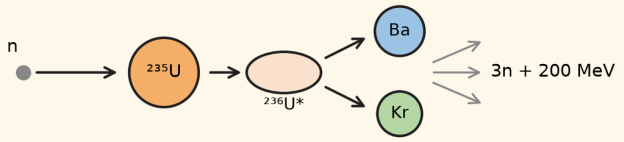


Fig. 13.4 Fission of uranium-235 by a slow neutron

$$^1_0n + {}^{235}_{92}U \rightarrow {}^{236}_{92}U^* \rightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3\,{}^1_0n + \text{energy}$$

→ Discovered by Hahn and Strassmann (1938).

→ About **200 MeV** is released per fission, mostly as kinetic energy of the fragments.

→ Fission fragments are radioactive; other pairs are possible (e.g. Sb + Nb).

Chain reaction and nuclear reactor

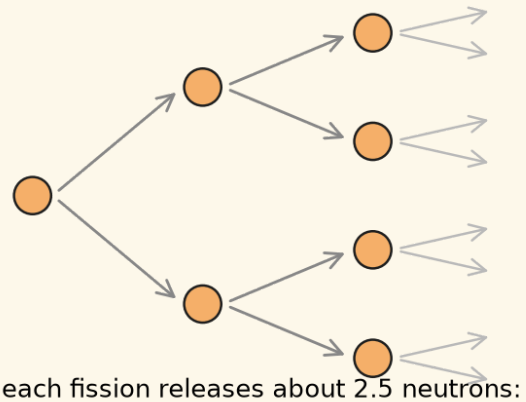


Fig. 13.5 A chain reaction: neutrons from one fission cause more fissions

→ Each fission releases 2–3 neutrons that can split more U-235 nuclei: a **chain reaction**.

→ Uncontrolled chain reaction: explosion (atom bomb). Controlled (one neutron per fission causes the next): steady power in a **nuclear reactor**.

Part of reactor	Function	Material
Fuel	undergoes fission	enriched U-235, Pu-239
Moderator	slows fast neutrons (slow neutrons fission U-235 better)	heavy water, graphite
Control rods	absorb extra neutrons, control rate	cadmium, boron
Coolant	carries heat to make steam	water, liquid sodium
Shielding	stops radiation	thick concrete

Nuclear fusion

When very light nuclei are forced together they fuse and release energy. They must overcome Coulomb repulsion, which needs temperatures of about 10^7–10^8 K: hence **thermonuclear fusion**.

$$^2_1H + {}^3_1H \rightarrow {}^4_2He + n + 17.59\ \text{MeV}$$

→ In the sun, four protons fuse into one helium nucleus (proton–proton cycle), releasing 26.7 MeV.

→ Fusion gives more energy per unit mass than fission, and its products are not radioactive, but controlled fusion on earth is still experimental (e.g. ITER).

Fission	Fusion
Heavy nucleus splits	Light nuclei join
~200 MeV per event	~17.6 MeV per event (more per unit mass)
Chain reaction possible at room temperature	Needs 10^7 K or more
Radioactive waste	Little radioactive waste
Used in reactors, bombs	Powers stars, hydrogen bomb

NCERT Activity: model of a chain reaction

Stand dominoes (or matchsticks) in a branching pattern where each one, when it falls, knocks down two more. Push the first: the number falling grows rapidly. Remove some dominoes from the branches (like control rods absorbing neutrons) so each falling domino knocks down only one: the reaction proceeds steadily instead of exploding.

Solved example (NCERT)

Q. How long can a 100 W bulb be kept glowing by fusion of 2.0 kg of deuterium? ($^2H + {}^2H \rightarrow {}^3He + n +$ 3.27 MeV)

Sol. Nuclei in 2 kg = 6.02×10^{26}; reactions = 3.01×10^{26}; energy = $3.01 \times 10^{26} \times 3.27 \times 1.6 \times 10^{-13} = 1.57 \times 10^{14}$ J; time = E/100 = 1.57×10^{12} s ≈ **4.9×10^4 years.**

Solved example

Q. A reactor produces 1000 MW using fission (200 MeV each). How many U-235 nuclei split per second?

Sol. $N = 10^9/(200 \times 1.6 \times 10^{-13}) =$ **3.1×10^{19} s^{-1}.**

Sooraj ki roshni ka secret? 4 protons ka group hug (fusion)

Asli baat: In the sun, hydrogen fuses into helium (4 protons → 1 He + 26.7 MeV), turning about 4 million tonnes of mass into energy every second.

Where do 200 MeV per fission go?

Form of energy	Approx. share (MeV)
Kinetic energy of fission fragments	about 168
Kinetic energy of prompt neutrons	about 5
Prompt γ-rays	about 7
Later β and γ decay of fragments, neutrinos	about 20
Total	about 200

Most of the energy appears as kinetic energy of the two heavy fragments. They are stopped in the fuel, and their kinetic energy turns into heat. This heat is what a reactor converts into electricity.

Multiplication factor and critical size

The **multiplication factor K** is the ratio of the number of fission neutrons in one generation to that in the preceding generation.

- → **K = 1 (critical)**: the reaction proceeds at a steady rate. This is the condition for a working reactor.
- → **K > 1 (supercritical)**: the number of fissions grows rapidly; this happens in an atom bomb.
- → **K < 1 (subcritical)**: the reaction dies out.

Neutrons are produced throughout the volume of the fuel but escape through its surface. For a small piece, surface-to-volume ratio is large, many neutrons leak out and K < 1. The minimum size (or mass) of fuel for which K = 1 is called the **critical size (critical mass)**.

Why a moderator works

Fission neutrons are fast (about 2 MeV), but U-235 is split far more readily by slow, thermal neutrons (about 0.0235 eV). In an elastic head-on collision, a neutron loses the most energy when it hits a nucleus of nearly equal mass, just as a moving billiard ball stops after a head-on hit on an identical ball. So light nuclei such as hydrogen (in water), deuterium (in heavy water) and carbon (graphite) are good moderators.

Solved example

Q. A neutron makes a head-on elastic collision with a stationary deuteron (mass 2m). What fraction of its kinetic energy is lost?

Sol. Final velocity of the neutron $v' = [(m - 2m)/(m + 2m)]v = -v/3$. Fraction of KE retained $= (1/3)^2 = 1/9$. Fraction lost = **8/9 ≈ 89%**.

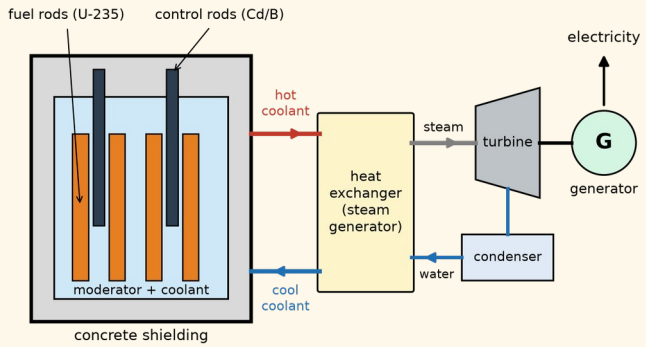


Fig. 13.5(a) Schematic of a nuclear power reactor

1. Fission in the **fuel rods** heats the core.
2. The **moderator** slows the neutrons; the **control rods** (cadmium or boron) are moved in or out to keep K = 1.
3. The **coolant** carries the heat to a **heat exchanger**, where water is turned into steam.
4. The steam drives a **turbine** coupled to a **generator**, producing electricity, and is then condensed and returned.
5. Thick concrete **shielding** protects people from neutrons and γ-rays. Spent fuel is highly radioactive and must be stored safely.

India operates reactors at Tarapur, Rawatbhata, Kalpakkam, Narora and Kudankulam, among others. Its three-stage programme aims to use its large reserves of thorium in the long run.

Solved example (NCERT exercise)

Q. The fission properties of ^{239}Pu are similar to those of ^{235}U, with about 180 MeV released per fission. How much energy is released if all the atoms in 1 kg of pure ^{239}Pu undergo fission?

Sol. $N = (1000/239) \times 6.022 \times 10^{23} = 2.52 \times 10^{24}$. $E = 2.52 \times 10^{24} \times 180 =$ **4.5×10^{26} MeV** (about 7.3×10^{13} J).

Solved example (NCERT exercise)

Q. A 1000 MW fission reactor consumes half its fuel in 5.00 years. How much ^{235}U did it contain initially? Assume it operates 80% of the time, all energy comes from ^{235}U fission (200 MeV each) and the nuclide is used up only by fission.

Sol. Energy in 5 y $= 10^9 \times 0.8 \times 5 \times 3.154 \times 10^7 = 1.26 \times 10^{17}$ J.

Fissions $= 1.26 \times 10^{17}/(200 \times 1.6 \times 10^{-13}) = 3.94 \times 10^{27}$.

Mass used $= 3.94 \times 10^{27} \times 235/(6.022 \times 10^{23})$ g = 1538 kg. This is half the fuel, so initial mass = **3076 kg**.

Solved example (NCERT exercise)

Q. Suppose a country plans 2×10^5 MW of electric power, 10% from nuclear plants with 25% efficiency and 200 MeV per fission of ^{235}U. How much uranium is needed per year?

Sol. Electric power $= 2 \times 10^{10}$ W; thermal power $= 2 \times 10^{10}/0.25 = 8 \times 10^{10}$ W.

Energy per year $= 8 \times 10^{10} \times 3.154 \times 10^7 = 2.52 \times 10^{18}$ J; fissions $= 2.52 \times 10^{18}/3.2 \times 10^{-11} = 7.88 \times 10^{28}$.

Mass $= 7.88 \times 10^{28} \times 235/(6.022 \times 10^{23})$ g ≈ **3.08×10^4 kg**.

Solved example (NCERT exercise)

Q. In the fission of ^{238}U by a fast neutron, no neutrons are emitted and the final stable products are ^{140}Ce and ^{99}Ru. Find Q. Masses: ^{238}U 238.05079, ^{140}Ce 139.90543, ^{99}Ru 98.90594, n 1.00867 u.

Sol. $\Delta m = 238.05079 + 1.00867 - 139.90543 - 98.90594 = 0.24809$ u. $Q = 0.24809 \times 931.5 =$ **231.1 MeV**.

Fusion in the sun: the proton-proton cycle

$$^1H + {}^1H \rightarrow {}^2H + e^+ + \nu + 0.42 \text{ MeV}$$

$$e^+ + e^- \rightarrow \gamma + \gamma + 1.02 \text{ MeV}$$

$$^2H + {}^1H \rightarrow {}^3He + \gamma + 5.49 \text{ MeV}$$

$$^3He + {}^3He \rightarrow {}^4He + {}^1H + {}^1H + 12.86 \text{ MeV}$$

For the last step, the first three must occur twice. The net result is **$4\,{}^1H + 2e^- \rightarrow {}^4He + 2\nu + 6\gamma + 26.7$ MeV**. The sun has been burning hydrogen this way for about 5×10^9 years. When the hydrogen in the core is

used up, the core will contract, heat up and begin to fuse helium into carbon.

Controlled fusion: the challenge

To fuse on earth, the fuel (deuterium and tritium) must be heated to over 10^8 K. At this temperature matter is a **plasma** of nuclei and electrons and no material wall can hold it. In a **tokamak**, strong magnetic fields confine the plasma in a doughnut shape (magnetic confinement). The international project ITER, in which India is a partner, aims to show that fusion power is practical. Fusion fuel is plentiful (deuterium from sea water) and produces little long-lived radioactive waste.

Solved example (NCERT exercise)

Q. Compare the energy released by (a) fusion of 1.0 kg of hydrogen in the sun (26 MeV per 4 protons) and (b) fission of 1.0 kg of ^{235}U (200 MeV per fission).

Sol. (a) Number of H atoms = 6.022×10^{26}; $E_1 = (6.022 \times 10^{26}/4) \times 26 = 3.91 \times 10^{27}$ MeV.

(b) Number of U atoms = $6.022 \times 10^{26}/235 = 2.56 \times 10^{24}$; $E_2 = 2.56 \times 10^{24} \times 200 = 5.13 \times 10^{26}$ MeV.

$E_1/E_2 \approx$ **7.6**: fusion gives about 7.6 times more energy per kg.

Yaad karo

K = 1 critical (reactor), K > 1 supercritical (bomb), K < 1 reaction band. Moderator neutron ko **slow** karta hai, control rod neutron ko **absorb** karti hai. Sun mein 4 protons se 1 He + 26.7 MeV.

7. Solved Numericals (Board Pattern)

N1. Find the radius of ^{64}Cu.

Sol. $R = 1.2 \times 4 =$ **4.8 fm**.

N2. Mass defect of a nucleus is 0.03 u. Binding energy?

Sol. $0.03 \times 931.5 =$ **27.9 MeV**.

N3. ^{7}Li: atomic mass 7.016004 u. Find B/A (m_H = 1.007825 u, m_n = 1.008665 u).

Sol. $\Delta M = 3(1.007825) + 4(1.008665) - 7.016004 = 0.042131$ u; $E_b = 39.24$ MeV; B/A = **5.61 MeV**.

N4. Calculate the Q-value: $^2H + {}^2H \rightarrow {}^4He$ (2.014102 u, 4.002603 u).

Sol. $\Delta m = 2(2.014102) - 4.002603 = 0.025601$ u $\Longrightarrow$ Q = **23.8 MeV**.

N5. Ratio of nuclear densities of ^{4}He and ^{238}U?

Sol. **1 : 1** (density independent of A).

N6. Energy released by fission of 1 g of U-235 (200 MeV per fission)?

Sol. $N = 6.02 \times 10^{23}/235 = 2.56 \times 10^{21}$; $E = 2.56 \times 10^{21} \times 200 \times 1.6 \times 10^{-13} =$ **8.2×10^{10} J**.

Formula sheet: Chapter 13

Quantity	Formula
Atomic mass unit	$1\ u = 1.66 \times 10^{-27}$ kg = 931.5 MeV
Radius	$R = 1.2\ A^{1/3}$ fm
Density	$\rho = 3m/4\pi R_0^3 \approx 2.3 \times 10^{17}$ kg/m^3
Mass-energy	$E = mc^2$
Mass defect	$\Delta M = Zm_p + Nm_n - M$
Binding energy	$E_b = \Delta M \times 931.5$ MeV
Q-value	$Q = (m_{reactants} - m_{products})\ c^2$

1 minute recap

- ✓ A = Z + N; $R \propto A^{1/3}$; density same for all nuclei.
- ✓ Mass defect × 931.5 MeV = binding energy.
- ✓ B/A ≈ 8 MeV, peak at Fe-56; fission and fusion both release energy.
- ✓ Nuclear force: strongest, short-range, charge independent, saturates.
- ✓ Fission ~200 MeV, chain reaction, reactor parts; fusion powers the sun.

MOST IMPORTANT QUESTIONS

Objective, short, Long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM The radius of a nucleus is proportional to:

(a) A
(b) A^2
(c) $A^{1/3}$
(d) $A^{2/3}$

Ans: (c) $A^{1/3}$.

Q2. IM Board fav Nuclear density is:

(a) proportional to A
(b) independent of A
(c) proportional to $A^{1/3}$
(d) inversely proportional to A

Ans: (b) independent of A.

Q3. 1M The binding energy per nucleon is maximum for:
(a) H-2
(b) He-4
(c) U-238
(d) Fe-56
Ans: (d) Fe-56.

Q4. 1M Moderators in nuclear reactors are used to:
(a) slow down neutrons
(b) absorb neutrons
(c) cool the reactor
(d) shield radiation
Ans: (a) slow down neutrons.

Q5. 1M 1 u is equivalent to:
(a) 9.315 MeV
(b) 93.15 MeV
(c) 931.5 MeV
(d) 1 MeV
Ans: (c) 931.5 MeV.

Q6. 1M Isobars have the same:
(a) atomic number
(b) mass number
(c) neutron number
(d) chemical properties
Ans: (b) mass number.

Q7. 1M Board fav Define mass defect.
Ans: Difference between the total mass of separate nucleons and the mass of the nucleus.

Q8. 1M Why is the density of nuclear matter the same for all nuclei?
Ans: Mass $\propto$ A and volume $\propto R^3 \propto$ A, so A cancels.

Q9. 1M Name the material used as control rods.
Ans: Cadmium (or boron).

Q10. 1M What is the source of energy of the sun?
Ans: Nuclear fusion of hydrogen into helium.

Q11. 1M State one property of nuclear force that differs from Coulomb force.
Ans: Short range (or charge independence).

Q12. 1M How many neutrons are there in $^{235}_{92}U$?
Ans: 143.

B. Assertion and Reason

Options: (a) Both true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M Board fav **A:** Energy is released in both fission and fusion. **R:** In both, binding energy per nucleon of products is greater than that of reactants.
Ans: (a)

Q14. 1M **A:** Nuclear force is charge independent. **R:** n-n, p-p and n-p forces are nearly equal.
Ans: (a)

Q15. 1M **A:** Fusion needs very high temperature. **R:** Nuclei must overcome Coulomb repulsion.
Ans: (a)

Q16. 1M **A:** The mass of a nucleus is greater than the sum of masses of its nucleons. **R:** Energy is needed to bind them.
Ans: (d)

C. Short answer (2-3 marks)

Q17. 2M Board fav Show that nuclear density is independent of mass number.
Ans: See Section 13.2.

Q18. 2M Define binding energy. How is it related to mass defect?
Ans: Energy to separate all nucleons; $E_b = \Delta Mc^2$.

Q19. 3M Board fav Draw the B/A vs A curve and write three conclusions.
Ans: See Fig. 13.2 and Section 13.4.

Q20. 2M Board fav Give four properties of nuclear force.
Ans: Strongest, short range, charge independent, saturating.

Q21. 3M Draw the potential energy curve of a pair of nucleons and explain it.
Ans: See Fig. 13.3: repulsive below 0.8 fm, attractive up to a few fm.

Q22. 3M Distinguish between fission and fusion (three points).
Ans: See the comparison table.

Q23. 2M What is a chain reaction? How is it controlled in a reactor?
Ans: Neutrons from one fission cause more; control rods absorb extra neutrons.

Q24. 2M Why are heavy nuclei unstable?
Ans: Coulomb repulsion among many protons reduces B/A.

Q25. 2M Why is heavy water used as a moderator?
Ans: Its deuterium nuclei have mass close to neutrons, so collisions slow neutrons efficiently, and it absorbs few neutrons.

Q26. 1M What are isotones? Give an example.
Ans: Same neutron number: ^{198}Hg and ^{197}Au (N = 118).

D. Long answer (5 marks)

Q27. 5M Board fav (a) Define mass defect and binding energy. (b) Draw the B/A curve and explain why energy is released in fission and fusion. (c) Calculate the binding energy of ^{4}He (4.002603 u).
Ans: (c) $\Delta M = 2(1.007825) + 2(1.008665) - 4.002603 = 0.030377$ u $\Longrightarrow$ **28.3 MeV**.

Q28. 5M (a) Explain nuclear fission with an example. (b) Describe the functions of the main parts of a nuclear reactor.
Ans: See Section 13.6.

E. Case-based

Q29. 4M Board fav India's Kudankulam plant uses U-235 fission. (a) Energy per fission? (b) Role of moderator? (c) What happens if control rods are pulled out? (d) Fission per second for 1 MW?
Ans: (a) 200 MeV. (b) Slows neutrons. (c) Rate increases, may run away. (d) 3.1×10^{16} s^{-1}.

Q30. 4M The sun converts hydrogen into helium. (a) Name the process. (b) Why high temperature? (c) Energy per He from 4 protons? (d) Mass lost per second for 3.8×10^{26} W?
Ans: (a) Fusion. (b) Overcome Coulomb repulsion. (c) 26.7 MeV. (d) $P/c^2 \approx 4.2 \times 10^9$ kg/s.

Q31. 1M Find the energy equivalent of 0.002 u.
Ans: **1.86 MeV**.

Q32. 1M A nucleus of A = 8 has R = 3 fm (assume). Find R for A = 64.
Ans: A ratio 8 $\Longrightarrow$ R doubles: **6 fm**.

Q33. 2M Board fav Binding energies of ^{2}H and ^{4}He are 2.22 MeV and 28.3 MeV. Find the energy released when two deuterons fuse to form He.
Ans: $28.3 - 2 \times 2.22 =$ **23.9 MeV**.

Q34. 1M Why is fusion called a thermonuclear reaction?
Ans: It needs extremely high temperatures (~10^7 K).

Q35. 1M Why do slow neutrons cause fission of U-235 more easily than fast ones?
Ans: They spend more time near the nucleus and are captured more readily.

Q36. 1M Which has more B/A, U-238 or Kr-89?
Ans: Kr-89 (middle nucleus).

Q37. 2M Find the mass defect if B/A of ^{40}Ca is 8.55 MeV.
Ans: $E_b = 342$ MeV $\Longrightarrow \Delta M = 342/931.5 =$ **0.367 u**.

Q38. 1M Name the scientist who discovered the neutron.
Ans: James Chadwick (1932).

Q39. 1M What is the order of magnitude of nuclear radius?
Ans: 10^{-15} m.

Q40. 1M Which force is responsible for the stability of a nucleus?
Ans: Strong nuclear force.

F. Extra practice (board PYQ style)

Q41. 3M Board fav Calculate the binding energy per nucleon of ^{56}Fe (atomic mass 55.934939 u).
Ans: $\Delta M = 26(1.007825) + 30(1.008665) - 55.934939 = 0.528461$ u; $E_b = 492.3$ MeV; B/A = **8.79 MeV**.

Q42. 1M A nucleus splits into two fragments of equal mass number. Compare their radii.
Ans: $R \propto A^{1/3}$, equal A $\Longrightarrow$ equal R.

Q43. 1M Why must the neutrons in a reactor be slowed down?
Ans: U-235 fission probability is far greater for slow (thermal) neutrons.

Q44. 2M Why is the binding energy per nucleon low for light nuclei?

Ans: Each nucleon has few neighbours, so surface nucleons are less bound.

931.5 MeV yaad hai? Tab hi numerical banega

Asli baat: Multiply mass defect in u by 931.5 to get binding energy in MeV.

Quick compare: binding energy vs B/A

Binding energy	Binding energy per nucleon
Total energy to split nucleus	Average per nucleon
Increases with A	Max at A ≈ 56
Tells size of energy	Tells stability

Common mistakes

→ Mass defect mein nucleus ka mass minus karo, atom ka use karo toh hydrogen ka atomic mass lo.

→ Density A par depend nahi karti.

→ Moderator neutron slow karta hai, absorb control rod karti hai.

→ Fusion mein per event energy kam, par per unit mass zyada.

Exam Alert: last check

→ B/A curve diagram + conclusions, binding energy numerical.

→ Nuclear force properties, fission vs fusion, density derivation.

Semiconductor Electronics: Materials, Devices and Simple Circuits

Most asked: energy band diagrams of conductors, semiconductors and insulators, intrinsic and extrinsic semiconductors, formation of p-n junction (depletion layer, barrier potential), forward and reverse bias, I-V characteristics, half-wave and full-wave rectifiers with waveforms

Every phone, laptop and charger runs on semiconductors. Tiny amounts of impurity turn an almost insulating crystal of silicon into a material whose current we can control. Joining p-type and n-type silicon gives the p-n junction diode, which lets current flow one way only and converts AC into DC.

1. Classification of Solids by Conductivity

On the basis of electrical conductivity σ (or resistivity $\rho = 1/\sigma$), solids are classified into three groups:

Type	ρ (Ω m)	σ (S m^{-1})	Examples
Metals	10^{-2} to 10^{-8}	10^{2} to 10^{8}	Cu, Ag, Al
Semiconductors	10^{-5} to 10^{6}	10^{5} to 10^{-6}	Si, Ge
Insulators	10^{11} to 10^{19}	10^{-11} to 10^{-19}	glass, rubber, diamond

Types of semiconductor materials

→ **Elemental**: Si and Ge (group 14).

→ **Compound**: inorganic, e.g. CdS, GaAs, CdSe, InP; organic, e.g. anthracene, doped phthalocyanines; organic polymers, e.g. polypyrrole, polyaniline.

Most present-day devices use Si and Ge, but devices based on compound and organic semiconductors (for example OLEDs) are growing fast.

2. Energy Bands in Solids

How bands form, step by step

1. In an isolated atom, electrons occupy sharp, discrete energy levels.
2. In a crystal, atoms are about 2-3 Å apart, so the outer electrons of neighbouring atoms interact strongly.
3. By the Pauli exclusion principle, no two electrons can have the same energy state, so each level splits into a very large number of closely spaced levels.
4. These levels are so close that they form a nearly continuous **energy band**.

→ **Valence band (VB)**: band of energies occupied by the valence electrons; it is completely filled at 0 K in semiconductors and insulators.

→ **Conduction band (CB)**: the next higher band; electrons in it are free to move and conduct.

→ **Energy gap E_g**: the forbidden region between the top of VB (E_V) and the bottom of CB (E_C); no electron can have energy in it.

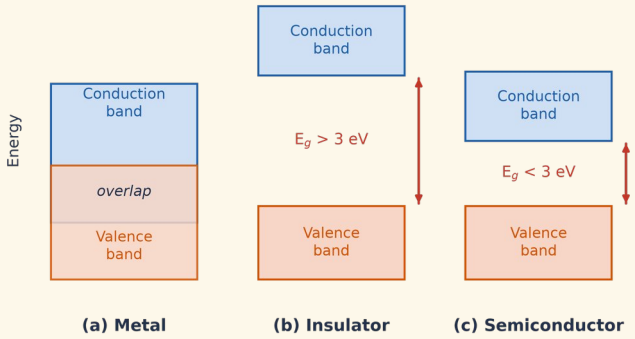


Fig. 14.1 Energy band diagrams: (a) metal, bands overlap; (b) insulator, large gap; (c) semiconductor, small gap

Explaining the three classes

→ **Metals**: either the CB is partly filled or the VB and CB overlap ($E_g = 0$). Many electrons are free, so conductivity is high.

→ **Insulators**: $E_g > 3$ eV (diamond about 5.4 eV). Thermal energy cannot lift electrons across, so the CB is empty and no conduction occurs.

→ **Semiconductors**: $E_g < 3$ eV (Si 1.1 eV, Ge 0.71 eV). At room temperature some electrons gain enough thermal energy to jump into the CB, so a small conductivity exists that increases with temperature.

Electron ka wave wala nakhra hi bands banata hai

Asli baat: Electrons behave as waves; in a periodic crystal this wave nature splits atomic levels into allowed bands separated by forbidden gaps.

Yaad karo

Gap zero: metal. Gap chhota (under 3 eV): semiconductor. Gap bada (over 3 eV): insulator. Temperature badhao toh semiconductor ka resistance **ghatta** hai, metal ka **badhta** hai.

NCERT Example 14.1. C, Si and Ge have the same lattice structure. Why is C an insulator while Si and Ge are semiconductors?

Sol. The four bonding electrons of C, Si and Ge lie in the second, third and fourth orbit respectively. The energy needed to free an electron is least for Ge, more for Si and highest for C. So the number of free electrons is negligible in C (E_g = 5.4 eV) but significant in Si and Ge.

Why temperature affects metals and semiconductors oppositely

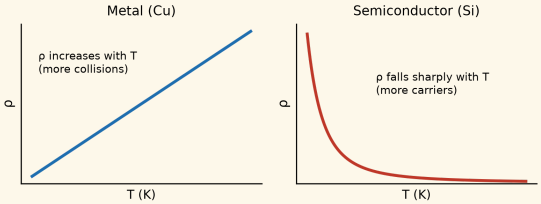


Fig. 14.2(a) Variation of resistivity with temperature: metal and semiconductor

In a metal the number of free electrons is fixed and very large. On heating, the ions vibrate more, electrons collide more often, the relaxation time falls and resistivity **increases**. In a semiconductor the number of free carriers is small and depends strongly on temperature: more thermal energy lifts more electrons across the gap, so the carrier density rises much faster than the collisions increase. Hence its resistivity **decreases** on heating (negative temperature coefficient).

Solved example

Q. The band gap of Si is 1.1 eV. Find the maximum wavelength of light that can create an electron-hole pair in it.

Sol. $\lambda_{max} = hc/E_g$ = 1240 eV nm/1.1 eV ≈ **1127 nm** (infrared). Light of shorter wavelength can also do it.

3. Intrinsic Semiconductors

A pure semiconductor (Si or Ge) is called **intrinsic**. Each atom has four valence electrons and forms four covalent bonds with its neighbours in a diamond-like lattice.

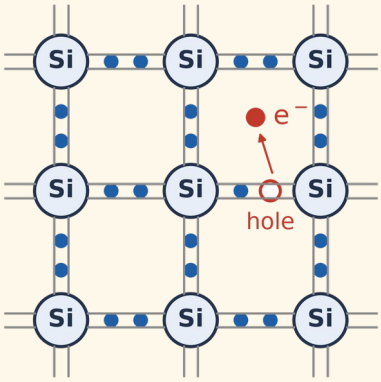


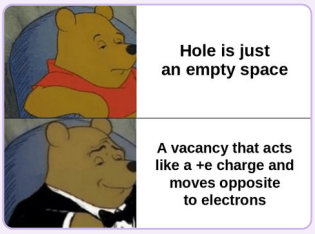
Fig. 14.2 Thermal energy breaks a covalent bond: a free electron and a hole are created as a pair

Electrons and holes, step by step

1. At 0 K all bonds are intact: no free electron, so the crystal is an insulator.
2. At room temperature thermal energy breaks a few bonds. The freed electron moves into the CB.
3. The vacancy left behind, with an effective charge +e, is called a **hole**.
4. A neighbouring bound electron can jump into this vacancy, so the hole appears to move. Holes move in the direction opposite to electrons, like positive charges.

$$n_e = n_h = n_i \text{ (intrinsic carrier concentration)}$$

Total current is the sum of electron current and hole current: $I = I_e + I_h$. Along with generation, **recombination** of electrons and holes also occurs; in equilibrium the rate of generation equals the rate of recombination.



Hole ko khaali jagah samjha?
Thoda classy socho
Asli baat: A hole is not a particle but the vacancy of a bond electron; it behaves like a positive charge +e and drifts opposite to electrons.

Generation, recombination and equilibrium

Electron-hole pairs are created continuously by thermal energy (generation) and also disappear when a free electron falls back into a hole

(recombination). At a fixed temperature the two rates become equal and the carrier concentration settles at n_i. Since n_i rises steeply with temperature, a pure semiconductor conducts far better when warm than when cold.

Feature	Electron	Hole
Charge	$-e$	$+e$
Moves in	conduction band	valence band
Direction under field	opposite to E	along E
Mobility	higher	lower

4. Extrinsic Semiconductors (Doping)

The conductivity of an intrinsic semiconductor is too low for use. Adding a tiny, controlled amount of suitable impurity (about 1 in 10^6 to 10^8 atoms) is called **doping**, and the impurity is the **dopant**. The dopant atom must be of nearly the same size as Si or Ge so that it fits into the lattice without distorting it.

(a) n-type semiconductor

1. Si is doped with a **pentavalent** impurity (valency 5): As, Sb, P.
2. Four of its five electrons form covalent bonds with neighbouring Si atoms.
3. The fifth electron is loosely bound (ionisation energy about 0.01 eV for Ge, 0.05 eV for Si), so even at room temperature it becomes free.
4. The impurity atom donates an electron, so it is called a **donor**; it becomes a fixed positive ion.
5. Electrons become **majority carriers** and holes **minority carriers**: $n_e >> n_h$.

(b) p-type semiconductor

1. Si is doped with a **trivalent** impurity (valency 3): Al, B, In.
2. Its three electrons form three bonds; the fourth bond lacks an electron, leaving a hole.
3. A neighbouring electron fills this hole, so the impurity atom accepts an electron and becomes a fixed negative ion. It is called an **acceptor**.
4. Holes are majority carriers and electrons minority carriers: $n_h >> n_e$.

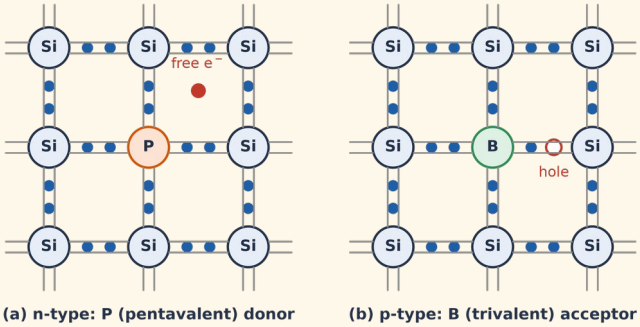


Fig. 14.3 (a) Pentavalent donor gives a free electron; (b) trivalent acceptor creates a hole

Energy band picture

In n-type, a donor energy level E_D lies just below the CB; its electrons need very little energy to enter the CB. In p-type, an acceptor level E_A lies just above the VB; electrons from the VB easily jump into it, leaving holes in the VB.

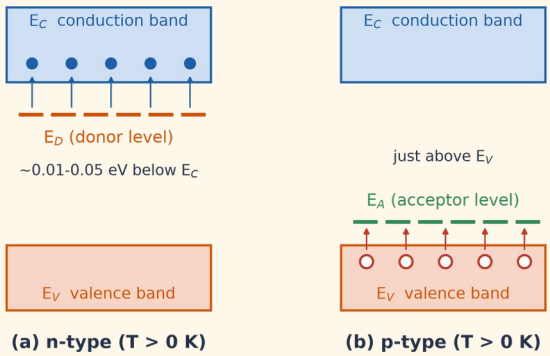


Fig. 14.4 Energy bands of (a) n-type with donor level E_D and (b) p-type with acceptor level E_A

Mass action law

Doping one type of carrier reduces the other, because the large number of majority carriers increases the rate of recombination.
In thermal equilibrium:

$$n_e n_h = n_i^2$$

Exam Alert

→ n-type aur p-type dono electrically **neutral** hote hain. "n-type negative hai" likha toh mark gaya.

→ Mass action law $n_e n_h = n_i^2$ wala numerical (NCERT Ex. 14.2) har saal ka favourite.

NCERT Example 14.2. A pure Si crystal has 5×10^{28} atoms m^{-3}. It is doped with 1 ppm concentration of pentavalent As. Find the number of electrons and holes. Given $n_i = 1.5 \times 10^{16}$ m^{-3}.

Sol. Step 1: $N_D = 10^{-6} \times 5 \times 10^{28} = 5 \times 10^{22}$ m^{-3}. Each As atom gives one electron, and thermally generated electrons (n_i) are negligible, so $n_e \approx$ **5×10^{22} m^{-3}**.
Step 2: $n_h = n_i^2/n_e = (2.25 \times 10^{32})/(5 \times 10^{22}) =$ **4.5×10^{9} m^{-3}**.

Example 3. Is an n-type semiconductor negatively charged?

Sol. No. Every free electron given by a donor leaves behind a fixed positive donor ion. Total positive charge equals total negative charge, so the crystal is neutral.

Why a doped semiconductor stays neutral

Each donor atom brings one extra electron and also one extra positive charge in its nucleus. When the electron becomes free, the donor becomes a fixed positive ion, but the total charge of the crystal is still zero. Similarly, each acceptor ion is negative but there is one hole for it. So **both n-type and p-type materials are electrically neutral**; "n" and "p" only describe the majority carriers.

Effect of temperature on an extrinsic semiconductor

At room temperature almost all donors (or acceptors) are ionised, so the majority carrier density is nearly equal to the doping density and stays constant over a wide temperature range. At very high temperatures, thermally generated pairs exceed the dopant carriers and the material behaves as intrinsic. This is why devices have a maximum operating temperature, and why Si (larger gap) works at higher temperatures than Ge.

Point	n-type	p-type
Dopant	pentavalent (P, As, Sb)	trivalent (B, Al, In)
Dopant called	donor	acceptor
Majority carriers	electrons	holes
Extra level	donor level just below E_C	acceptor level just above E_V
Relation	$n_e \gg n_h$	$n_h \gg n_e$

5. p-n Junction

A p-n junction is formed when one part of a single crystal is made p-type and the other n-type (for example by diffusing a trivalent impurity into an n-type wafer). Simply pressing two separate slabs together does not work: the surface roughness (about 1 μm) is much larger than the interatomic spacing (about 3 Å), so it is not a continuous crystal and acts as a discontinuity for carriers.

Formation step by step

1. **Diffusion**: holes are more on the p-side and electrons more on the n-side, so holes diffuse from p to n and electrons from n to p. This gives a **diffusion current** from p to n.
2. When an electron leaves the n-side, it leaves behind an uncovered positive donor ion. When a hole leaves the p-side, it leaves an uncovered negative acceptor ion.
3. So a layer of positive charge forms on the n-side and a layer of negative charge on the p-side near the junction. This region, emptied of free carriers, is the **depletion region** (about 0.1 to 1 μm thick).
4. The space charge creates an electric field from n to p. This field pushes minority carriers across: electrons from p to n, holes from n to p. This is the **drift current**, opposite to diffusion current.
5. Initially diffusion current is large. As the space charge grows, the field grows, drift increases, until **drift current = diffusion current**. In equilibrium there is no net current.

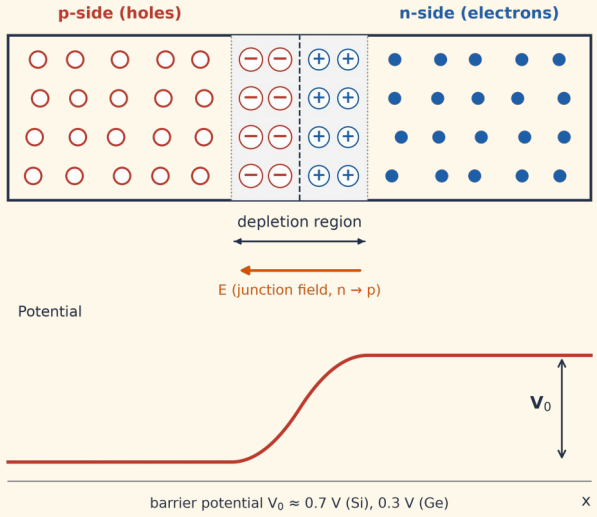


Fig. 14.5 p-n junction in equilibrium: depletion region with fixed ions, junction field and the barrier potential

Barrier potential

The potential difference across the depletion region, n-side positive with respect to p-side, opposes further diffusion of majority carriers. It is called the **barrier potential V_0**: about **0.7 V for Si** and **0.3 V for Ge**. It depends on the dopant concentration and temperature.

NCERT Example 14.3. Can we take one slab of p-type and one of n-type semiconductor and physically join them to get a p-n junction?

Sol. No. Any slab, however flat, has roughness much larger than the interatomic spacing (2-3 Å). Continuous contact at the atomic level is impossible, and the junction behaves as a discontinuity for the flowing carriers.

Paheli-Bujho

Bujho: Paheli, p aur n slab ko Fevicol se chipka dein toh diode ban jayega?

Paheli: Nahi Bujho! Surface ka khurdurapan 1 μm hai, atoms ka gap sirf 3 Å. Ek hi crystal mein doping karni padti hai.

Bujho: Accha, toh junction pe current kyun nahi bahta jab battery na ho?

Paheli: Diffusion aur drift barabar ho jaate hain. Net current zero, barrier V_0 sab rok deta hai.

Diffusion current vs drift current

Point	Diffusion current	Drift current
Cause	difference in concentration	electric field of depletion region
Carriers	majority	minority
Direction	p to n	n to p
In forward bias	large (mA)	small
In reverse bias	almost zero	small, constant (μA)

How the barrier changes with bias

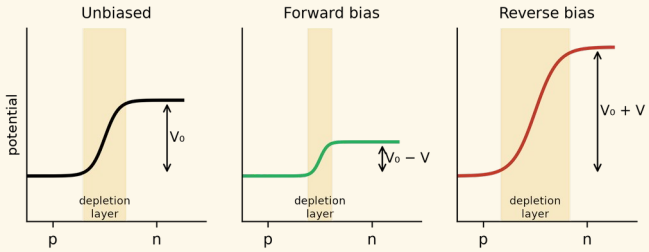


Fig. 14.5(a) Barrier height and depletion width: unbiased V_0, forward $V_0 - V$ (narrow), reverse $V_0 + V$ (wide)

The width of the depletion region depends on doping: a heavily doped junction has a **thin** depletion layer (charge is available close to the junction), while a lightly doped junction has a wide one. A thin layer gives a very strong electric field even for a small voltage.

Yaad karo

Diffusion = majority, p se n. Drift = minority, n se p. Bina battery ke dono barabar, net current zero. Forward mein barrier **ghata**, reverse mein **badha**.

6. Semiconductor Diode

A p-n junction with metallic contacts at its ends is a **semiconductor diode**, a two-terminal device. Its symbol is an arrow (p-side, anode) and a bar (n-side, cathode); the arrow shows the direction of conventional current in forward bias.

(a) Forward bias

1. The p-side is connected to the positive terminal and the n-side to the negative terminal of the battery.
2. The applied voltage V opposes the barrier, so the effective barrier height becomes **$(V_0 - V)$** and the depletion layer becomes narrower.
3. More majority carriers can now cross; electrons injected into p and holes into n become minority carriers there (**minority carrier injection**).
4. Below the threshold (cut-in or knee) voltage, about 0.7 V for Si and 0.2 V for Ge, the current is very small. Beyond it, the current rises steeply (exponentially) to a few mA even for a small rise in voltage.

(b) Reverse bias

1. The p-side is connected to the negative terminal and the n-side to the positive terminal.

2. The applied voltage adds to the barrier: barrier height becomes $(V_0 + V)$ and the depletion layer widens.
3. Diffusion of majority carriers almost stops. Only minority carriers drift across, giving a small current of a few μA.
4. This current is nearly independent of the applied voltage (up to a critical value) because it is limited by the number of minority carriers, not by the field. It is called the **reverse saturation current**.
5. At a critical voltage, the **breakdown voltage V_{br}**, the reverse current rises suddenly. An ordinary diode is damaged if the current is not limited by an external resistance.

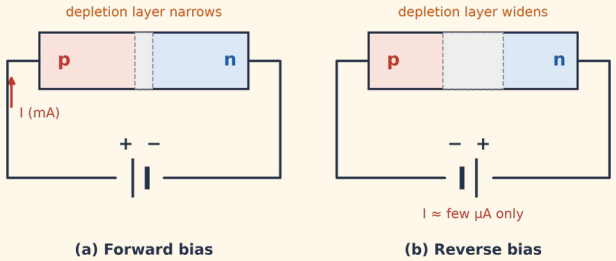


Fig. 14.6 (a) Forward bias: narrow depletion layer, large current; (b) reverse bias: wide depletion layer, tiny current

V-I characteristics

The circuit uses a battery, rheostat (potentiometer), milliammeter and voltmeter for forward bias; for reverse bias, the battery is reversed and a microammeter is used. The voltage is varied and the current is noted.

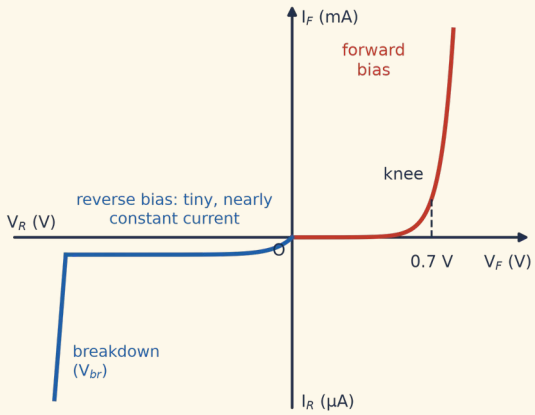


Fig. 14.7 V-I characteristics of a Si diode. Note the different scales: mA in forward bias, μA in reverse bias

→ Forward current is negligible up to the knee voltage, then rises sharply. The diode behaves like a small resistance.

→ Reverse current is tiny and nearly constant; the diode behaves like a very large resistance (nearly open switch).

→ So the diode is a **non-ohmic**, unidirectional device: it conducts easily in one direction only.

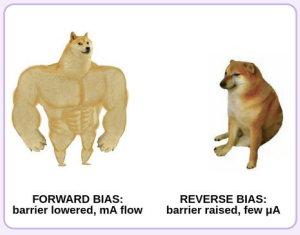
Dynamic resistance

$$r_d = \Delta V/\Delta I$$ (slope of V-I curve, not constant)

NCERT Example 14.4. For a Si diode, the forward current changes from 10 mA to 20 mA when the voltage changes from 0.7 V to 0.8 V. In reverse bias, the current is −1 μA at V = −10 V. Find (a) forward dynamic resistance at I between 10 and 20 mA, (b) reverse resistance at V = −10 V.

Sol. (a) $r_{fb} = \Delta V/\Delta I$ = 0.1 V/10 mA = **10 Ω**.

(b) r_{rb} = 10 V/1 μA = **1.0×10^7 Ω**. Reverse resistance is about a million times larger.



Same diode, bas battery ulti kar di

Asli baat: In forward bias the barrier falls to $V_0 - V$ and current rises to mA; in reverse bias it rises to $V_0 + V$ and only a few μA of minority current flows.

NCERT Activity: LED glows only in forward bias

Connect a 3 V battery, a 220 Ω resistor and an LED (light emitting diode, a special p-n junction) in series. With the longer leg (anode, p-side) towards the positive terminal, the LED glows. Reverse the LED: it does not glow, even though the battery and resistor are the same.

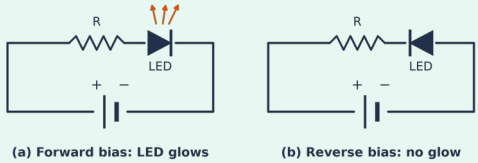


Fig. 14.8 Diode conducts only in forward bias

Conclusion: a p-n junction allows current in one direction only. The resistor limits the forward current and protects the LED.

Diode circuit example 1. A Si diode (knee 0.7 V) is connected in forward bias with a 5 V battery and a 430 Ω resistor. Find the current.

Sol. I = (5 − 0.7)/430 = 4.3/430 = **10 mA**.

Diode circuit example 2. An ideal diode and a 100 Ω resistor are connected in series with a 2 V battery: (a) p-side to +, (b) p-side to −. Find the current in each case.

Sol. (a) Ideal diode has zero forward resistance: I = 2/100 = **20 mA**. (b) Reverse biased ideal diode is open: I = **0**.

NCERT Activity: testing a diode with a multimeter

Set a digital multimeter to diode mode. Touch the red (+) probe to the anode and the black (COM) probe to the cathode: it reads about 0.6–0.7 V for Si (the forward voltage). Reverse the probes: it shows OL (open, very high resistance). A good diode shows this one-way behaviour; a damaged diode reads 0 both ways (shorted) or OL both ways (open).

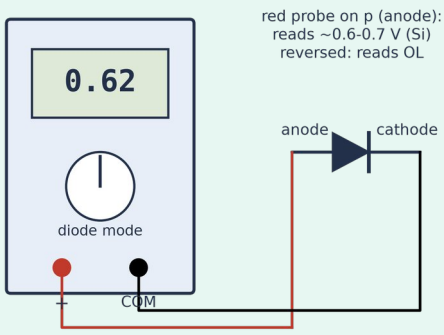


Fig. 14.9 Multimeter check: forward reading, reverse OL

Ideal diode vs practical diode

Point	Ideal diode	Practical Si diode
Forward resistance	zero	small (a few Ω to tens of Ω)
Knee voltage	zero	about 0.7 V
Reverse resistance	infinite	very large (MΩ)
Reverse current	zero	a few μA
Behaves like	perfect one-way switch	nearly one-way switch

Breakdown in reverse bias

If the reverse voltage is raised beyond the breakdown voltage, the reverse current increases sharply while the voltage stays almost constant. In an ordinary diode this large current overheats and destroys the junction, so the reverse voltage in a circuit must always remain below the breakdown voltage (also called the peak inverse voltage rating).

Solved example

Q. For a diode, the forward voltage rises from 0.70 V to 0.75 V while the current rises from 5 mA to 15 mA. Find the dynamic resistance.

Sol. $r_d = \Delta V/\Delta I = 0.05\ V/10\ mA = \mathbf{5\ \Omega}$.

7. Application: Diode as a Rectifier

Since a diode conducts only in forward bias, it can convert AC (alternating) voltage into DC (unidirectional) voltage. This process is **rectification**, and the circuit is a **rectifier**. A transformer first steps the mains voltage up or down as required.

(a) Half-wave rectifier, step by step

1. The AC input is applied to the primary of the transformer; the secondary (A and B) is connected to a diode D and load resistance R_L in series.
2. **Positive half cycle**: A is positive with respect to B. The diode is forward biased and conducts; current flows through R_L and an output voltage appears.
3. **Negative half cycle**: A is negative. The diode is reverse biased; no current flows (the reverse current is negligible), so the output is zero.
4. So output appears only in alternate half cycles. It is unidirectional but pulsating (not steady) DC. Output frequency = input frequency (50 Hz for 50 Hz mains).

The reverse breakdown voltage of the diode must be sufficiently higher than the peak AC voltage at the secondary, so that the diode is protected in the negative half cycle.

Where are rectifiers used?

Almost every electronic device that runs from the mains needs DC: mobile phone chargers, laptop adapters, LED bulbs, TV sets and battery chargers all contain a transformer (or a switching circuit), a rectifier and a filter. In a phone charger, the 230 V AC from the socket is reduced, rectified and smoothed to give a steady 5 V DC.

Yaad karo

Rectifier = AC ko DC banana. Half-wave: 1 diode, sirf aadha cycle, output f. Full-wave: 2 diode + centre-tap, dono half cycle, output 2f. Diode ka breakdown voltage peak AC se zyada hona chahiye.

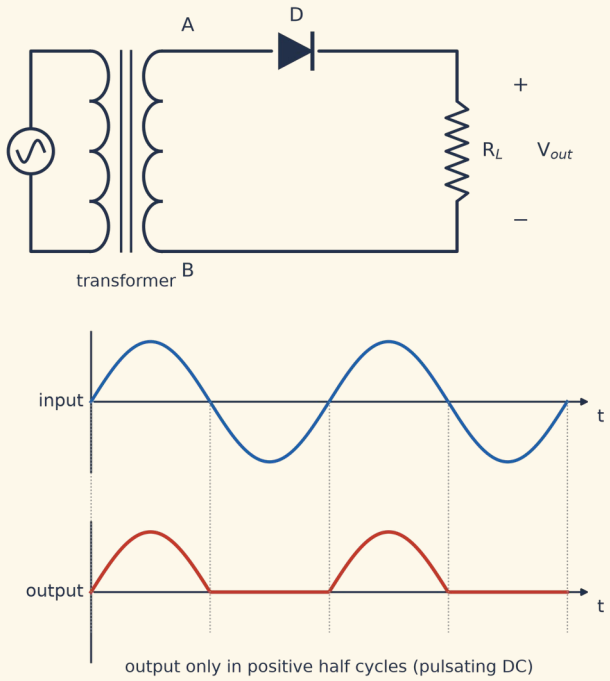


Fig. 14.10 Half-wave rectifier circuit with input and output waveforms

Note: current CBSE syllabus

In the rationalised NCERT (2023 onwards), this chapter covers energy bands, intrinsic and extrinsic semiconductors, the p-n junction, the diode and its use as a rectifier. Zener diode, optoelectronic devices (photodiode, LED, solar cell) theory, transistors and logic gates have been removed from the syllabus. The LED above is used only as an activity.

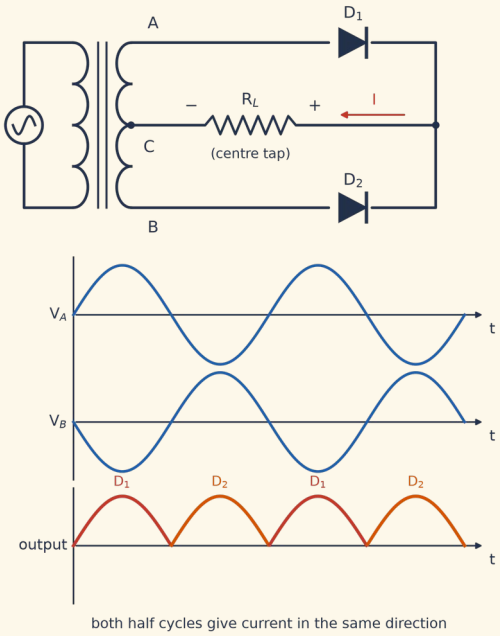


Fig. 14.11 Full-wave rectifier: D_1 conducts in positive halves, D_2 in negative halves; output in both halves

(b) Full-wave rectifier (centre-tap), step by step

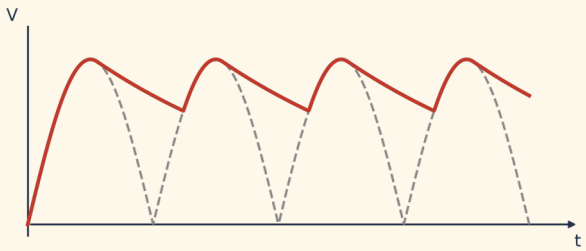
1. A centre-tapped transformer is used. The centre tap C is the reference; the voltages at A and B with respect to C are equal and out of phase by 180°.
2. The p-sides of diodes D_1 and D_2 are connected to A and B; their n-sides are joined together, and the output is taken between this common point and C through R_L.
3. **Positive half**: A is positive with respect to C, B is negative. D_1 is forward biased and conducts; D_2 is reverse biased.
4. **Negative half**: B becomes positive, A negative. Now D_2 conducts and D_1 does not.
5. In both halves, current flows through R_L in the **same direction**, so we get output in both half cycles. Output frequency = 2 × input frequency (100 Hz for 50 Hz mains).

Point	Half-wave	Full-wave
Diodes used	one	two (centre-tap)
Output in	alternate half cycles	both half cycles
Output frequency	f (50 Hz)	2f (100 Hz)
Efficiency	lower (max 40.6%)	higher (max 81.2%)
Transformer	ordinary	centre-tapped
Ripple	more	less

(c) Filter circuit

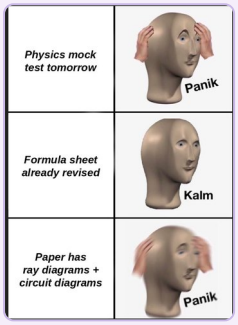
1. The rectified output still contains AC components (ripple). A **capacitor** connected across R_L smooths it.
2. While the voltage rises, the capacitor charges to the peak value.
3. When the voltage falls, the diode stops conducting and the capacitor discharges slowly through R_L, keeping the output nearly steady.

4. A large capacitance C (and large R_L) gives a large time constant CR_L, so the discharge is slow and the ripple is small. The output then is close to the peak voltage.



red: output with capacitor (small ripple); dashed: without

Fig. 14.12 Effect of a capacitor filter on full-wave output



Paper khola toh rectifier ka circuit aur
waveform dono maange
Asli baat: Draw the circuit with the transformer, diode(s) and R_L, then input and output waveforms aligned in time: that is a 3-5 mark sure question.

Why the full-wave output has double frequency

In a half-wave rectifier one output pulse appears per input cycle, so the output repeats every T (frequency f). In a full-wave rectifier one pulse appears in each half cycle, so the output repeats every T/2: its fundamental frequency is **2f**. For 50 Hz mains the ripple frequency is 100 Hz, which is easier to filter.

Solved example (NCERT exercise)

Q. In half-wave rectification, what is the output frequency if the input frequency is 50 Hz? What is the output frequency of a full-wave rectifier for the same input?

Sol. Half-wave: **50 Hz**. Full-wave: **100 Hz**.

Solved example

Q. In a centre-tap full-wave rectifier, the peak voltage across each half of the secondary is 12 V. Using ideal diodes, find the peak output voltage and the maximum reverse voltage across a non-conducting diode.

Sol. Peak output = **12 V**. When D_1 conducts, the non-conducting D_2 has the voltage of the whole secondary across it: $2 \times 12 = \mathbf{24\ V}$. Its breakdown voltage must exceed this.

8. Solved Numericals (Board Pattern)

N1. The energy gap of Si is 1.1 eV. Find the maximum wavelength of light that can create an electron-hole pair.

Sol. $\lambda = hc/E_g = 1240\ \text{eV nm}/1.1\ \text{eV} \approx \mathbf{1127\ nm}$.

N2. In an intrinsic semiconductor $n_i = 1.5 \times 10^{16}\ m^{-3}$. After doping, $n_h = 4.5 \times 10^{22}\ m^{-3}$. Find n_e and the type.

Sol. $n_e = n_i^2/n_h = 2.25 \times 10^{32}/4.5 \times 10^{22} = \mathbf{5 \times 10^{9}\ m^{-3}}$; $n_h >> n_e$, so **p-type**.

N3. A Ge crystal has 4.4×10^{28} atoms m^{-3} and is doped with 1 donor per 10^6 Ge atoms. Find the electron concentration (ignore n_i).

Sol. $n_e \approx N_D = 4.4 \times 10^{28}/10^6 = \mathbf{4.4 \times 10^{22}\ m^{-3}}$.

N4. The input frequency of a rectifier is 50 Hz. Find the output ripple frequency for (a) half-wave (b) full-wave.

Sol. (a) **50 Hz**; (b) **100 Hz**.

N5. A diode current changes from 5 mA to 15 mA when voltage changes from 0.65 V to 0.70 V. Find r_d.

Sol. $r_d = 0.05/0.010 = \mathbf{5\ \Omega}$.

N6. In a half-wave rectifier, the peak secondary voltage is 12 V. Ignoring the diode drop, what is the peak output, and what minimum reverse breakdown rating should the diode have?

Sol. Peak output = **12 V**; the diode must withstand at least **12 V** reverse (in practice choose a higher rating).

Formula sheet: Chapter 14

Quantity	Formula / value
Intrinsic	$n_e = n_h = n_i$
Mass action law	$n_e n_h = n_i^2$
n-type (full ionisation)	$n_e \approx N_D$
p-type	$n_h \approx N_A$
Total current	$I = I_e + I_h$
Energy gap	Si 1.1 eV, Ge 0.71 eV, C 5.4 eV
Barrier potential	Si ≈ 0.7 V, Ge ≈ 0.3 V
Forward / reverse barrier	$V_0 - V$ / $V_0 + V$
Dynamic resistance	$r_d = \Delta V/\Delta I$
Threshold wavelength	$\lambda = hc/E_g$
Rectifier output frequency	HWR: f, FWR: 2f

1 minute recap

- ✓ Band gap decides the class: 0 metal, under 3 eV semiconductor, over 3 eV insulator.
- ✓ Intrinsic: $n_e = n_h$; holes act as +e carriers.
- ✓ Pentavalent dopant gives n-type (donor), trivalent gives p-type (acceptor); both neutral.
- ✓ Diffusion and drift balance to form the depletion layer and barrier V_0.
- ✓ Forward bias: barrier falls, mA current; reverse bias: barrier rises, μA current.
- ✓ Half-wave: one diode, output f; full-wave: two diodes, output 2f; capacitor filter smooths ripple.

Most Important Questions

Objective, short, long and case-based, with answers

A. Objective and very short answer (1 mark)

Q1. IM The energy gap of an insulator is of the order of:
(a) 0 eV
(b) 0.7 eV
(c) 1.1 eV
(d) more than 3 eV
Ans: (d) more than 3 eV.

Q2. IM Board fav To obtain an n-type semiconductor, Si is doped with:
(a) boron
(b) phosphorus
(c) aluminium
(d) indium
Ans: (b) phosphorus.

Q3. IM In a p-type semiconductor, the majority carriers are:
(a) holes
(b) electrons
(c) positive ions
(d) negative ions
Ans: (a) holes.

Q4. IM When the temperature of a semiconductor increases, its resistance:
(a) increases
(b) decreases
(c) remains same
(d) first rises then falls
Ans: (b) decreases.

Q5. IM Board fav In forward bias, the width of the depletion layer:
(a) increases
(b) is unchanged
(c) decreases
(d) becomes infinite
Ans: (c) decreases.

Q6. IM The barrier potential of a Si p-n junction is about:
(a) 0.1 V
(b) 0.3 V
(c) 0.7 V
(d) 1.1 V
Ans: (c) 0.7 V.

Q7. IM Board fav The output frequency of a full-wave rectifier for 50 Hz input is:
(a) 50 Hz
(b) 100 Hz
(c) 25 Hz
(d) 0 Hz
Ans: (b) 100 Hz.

Q8. 1M The reverse current in a p-n junction is due to:
(a) majority carriers
(b) minority carriers
(c) donor ions
(d) acceptor ions
Ans: (b) minority carriers.

Q9. 1M Is a p-type semiconductor positively charged?
Ans: No, it is electrically neutral.

Q10. 1M Board fav Name the two processes that occur during the formation of a p-n junction.
Ans: Diffusion and drift.

Q11. 1M What is the direction of the electric field in the depletion region?
Ans: From the n-side to the p-side.

Q12. 1M Why is a semiconductor an insulator at 0 K?
Ans: All covalent bonds are intact; the conduction band is empty.

B. Assertion and Reason

(a) Both A and R true, R explains A. (b) Both true, R does not explain A. (c) A true, R false. (d) A false, R true.

Q13. 1M **A:** An n-type semiconductor is electrically neutral. **R:** Each free electron is balanced by a fixed positive donor ion.
Ans: (a)

Q14. 1M Board fav **A:** The reverse current of a diode is nearly independent of applied voltage. **R:** It is limited by the number of minority carriers.
Ans: (a)

Q15. 1M **A:** A p-n junction can be made by pressing a p-type and an n-type slab together. **R:** Surface roughness is much less than interatomic spacing.
Ans: (d) A is false; roughness (~1 μm) is much larger than spacing (~3 Å).

Q16. 1M **A:** Resistivity of a semiconductor decreases with temperature. **R:** More electron-hole pairs are generated at higher temperature.
Ans: (a)

C. Short answer (2–3 marks)

Q17. 3M Board fav Distinguish between conductors, semiconductors and insulators on the basis of energy bands. Draw diagrams.
Ans: Metal: bands overlap or CB partly filled. Semiconductor: $E_g < 3$ eV, few electrons cross at room temperature. Insulator: $E_g > 3$ eV, CB empty. (Fig. 14.1)

Q18. 3M Distinguish between intrinsic and extrinsic semiconductors (any three points).
Ans: Intrinsic: pure, $n_e = n_h$, low conductivity, depends only on temperature. Extrinsic: doped, $n_e \neq n_h$, high conductivity, controlled by doping.

Q19. 2M Explain the formation of an n-type semiconductor with a labelled diagram. Why is it neutral?
Ans: Pentavalent atom (P, As) uses four electrons in bonds; the fifth is nearly free. Donor becomes a fixed positive ion, so total charge is zero.

Q20. 2M Board fav Draw the energy band diagrams of n-type and p-type semiconductors at $T > 0$ K.
Ans: n-type: donor level E_D just below E_C; p-type: acceptor level E_A just above E_V. (Fig. 14.4)

Q21. 2M Board fav Explain the terms depletion region and barrier potential.
Ans: Depletion region: region near the junction with no free carriers, only fixed ions. Barrier potential: p.d. across it (0.7 V Si) that opposes diffusion of majority carriers.

Q22. 2M How does the width of the depletion layer change in forward and reverse bias? Explain.
Ans: Forward: barrier $V_0 - V$, carriers pushed towards junction, layer narrows. Reverse: barrier $V_0 + V$, carriers pulled away, layer widens.

Q23. 2M Board fav Draw the V-I characteristics of a p-n junction diode and mark the knee and breakdown voltages.
Ans: See Fig. 14.7: forward current rises sharply after 0.7 V; reverse current tiny until V_{br}.

Q24. 2M In a Si sample, $n_i = 1.5 \times 10^{16}\ m^{-3}$ and it is doped with 10^{22} acceptors m^{-3}. Find n_e.
Ans: $n_h \approx 10^{22}$; $n_e = 2.25 \times 10^{32}/10^{22} = 2.25 \times 10^{10}\ m^{-3}$.

Q25. 2M Why is the reverse current of a diode in μA but forward current in mA?
Ans: Forward current is due to majority carriers (large number); reverse current is due to minority carriers (very few).

Q26. 2M What is the role of a capacitor in a rectifier output? Why should CR_L be large?
Ans: It charges to the peak and discharges slowly through R_L, smoothing ripple; a large time constant keeps the discharge slow.

D. Long answer (5 marks)

Q27. 5M Board fav (a) Explain the formation of the depletion region and barrier potential in a p-n junction with diagrams. (b) Draw the circuit to study forward and reverse bias characteristics and draw the V-I graph.
Ans: (a) Diffusion of majority carriers uncovers fixed ions, creating space charge and a field n → p; drift balances diffusion at equilibrium. (b) Fig. 14.6 and 14.7.

Q28. 5M Board fav (a) With a labelled circuit, explain the working of a full-wave rectifier. Draw input and output waveforms. (b) Why is the output frequency twice the input?
Ans: (a) Centre-tap transformer, D_1 conducts in positive half, D_2 in negative half, same direction through R_L. (b) One output pulse per half cycle, i.e. two per input cycle.

Q29. 5M (a) Explain a half-wave rectifier with circuit and waveforms. (b) A HWR is fed by 50 Hz AC. Find the output frequency. (c) What happens if the diode is reversed?
Ans: (b) 50 Hz. (c) Output appears only in negative half cycles (polarity reversed).

E. Case-based

Q30. 4M Board fav A student dopes pure Si ($n_i = 1.5 \times 10^{16}\ m^{-3}$) with indium at $4.5 \times 10^{22}\ m^{-3}$. (a) Type of semiconductor? (b) Majority carriers? (c) n_e? (d) Is the sample charged?
Ans: (a) p-type. (b) Holes. (c) $n_e = 2.25 \times 10^{32}/4.5 \times 10^{22} = 5 \times 10^{9}\ m^{-3}$. (d) No, neutral.

Q31. 4M A phone charger converts 230 V, 50 Hz AC into low DC using a transformer, four or two diodes and a capacitor. (a) Which device lowers the voltage? (b) What does a diode do here? (c) Output ripple frequency for full-wave? (d) Role of capacitor?
Ans: (a) Step-down transformer. (b) Rectifies (allows one-way current). (c) 100 Hz. (d) Filters ripple to give smooth DC.

Extra numericals and practice

Q32. 1M Find the cut-off wavelength for creating an electron-hole pair in Ge ($E_g = 0.71$ eV).
Ans: $\lambda = 1240/0.71 \approx$ **1746 nm.**

Q33. 2M Board fav A diode with knee 0.7 V and $r_d = 20\ \Omega$ is connected in forward bias to 3 V through 210 Ω. Find I.
Ans: $I = (3 - 0.7)/(210 + 20) = 2.3/230 =$ **10 mA.**

Q34. 1M Two ideal diodes in series, one forward and one reverse biased, with a 6 V battery and 100 Ω. Find the current.
Ans: Reverse biased ideal diode is open: **I = 0.**

Q35. 1M Two ideal diodes in parallel (both forward biased) with 4 V and a 200 Ω series resistor. Find the battery current.
Ans: Ideal diodes have zero resistance: $I = 4/200 =$ **20 mA.**

Q36. 2M The number of holes in a Ge sample is $10^{20}\ m^{-3}$ and $n_i = 2.5 \times 10^{19}\ m^{-3}$. Find n_e.
Ans: $n_e = 6.25 \times 10^{38}/10^{20} =$ **$6.25 \times 10^{18}\ m^{-3}$.**

Q37. 1M A reverse-biased diode carries 2 μA at 8 V. Find its reverse resistance.
Ans: $R = 8/(2 \times 10^{-6}) =$ **4 MΩ.**

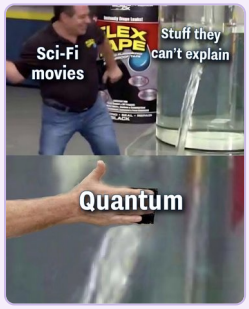
Q38. 1M Why is Si preferred over Ge in most devices?
Ans: Larger E_g (1.1 eV): much smaller reverse current and works at higher temperature.

Q39. 1M What happens to the barrier potential when a p-n junction is forward biased with 0.4 V (Si, V_0 = 0.7 V)?
Ans: Effective barrier = 0.7 − 0.4 = **0.3 V.**

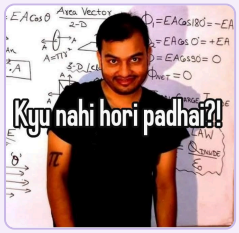
Q40. 1M Why is the doping concentration kept very small (1 in 10^6 to 10^8)?
Ans: To raise conductivity enormously without disturbing the crystal structure.

Q41. 1M Name the impurity atoms used for (a) n-type and (b) p-type Ge.
Ans: (a) As, Sb, P. (b) B, Al, In.

Q42. 1M Draw the circuit symbol of a p-n junction diode and mark the p and n sides.
Ans: Triangle (arrow) = p-side (anode), bar = n-side (cathode).



Semiconductor ka har sawaal: band theory ka flex tape lagao
Asli baat: The quantum idea of energy bands and gaps explains why Si conducts a little, why heating helps and why doping changes everything.



Reverse bias mein current kyu nahi bah raha?!
Asli baat: In reverse bias the barrier rises to V_0 + V; majority carriers cannot cross, only a few μA of minority current flows.

Common mistakes

- → Doped semiconductor ko charged mat likho: n-type aur p-type dono neutral.
- → Forward bias mein barrier V_0 − V hota hai, V_0 + V nahi.
- → Reverse current majority nahi, minority carriers ki wajah se hai.
- → Full-wave output frequency 2f hai, f nahi.
- → I-V graph mein forward axis mA aur reverse axis μA likhna mat bhoolo.

More practice

Q43. 2M Board fav Two Si diodes (knee 0.7 V each) are in series, forward biased, with a 5 V battery and 180 Ω. Find the current.
Ans: I = (5 − 1.4)/180 = 3.6/180 = **20 mA.**

Q44. 1M Why does an ordinary diode get damaged at breakdown?
Ans: Reverse current rises suddenly; large power I^2R heats and destroys the junction unless limited by a resistor.

Q45. 2M Board fav A semiconductor has equal electron and hole concentrations of 6×10^8 m^{-3}. On doping, n_e becomes 9×10^{12} m^{-3}. Find n_h and the type.
Ans: $n_h = 36 \times 10^{16}/9 \times 10^{12} = 4 \times 10^4$ m^{-3}; n-type.

Q46. 1M In which bias does the depletion region act like the dielectric of a capacitor?
Ans: Reverse bias: wide depletion layer with charges on both sides.

Q47. 1M Which diode conducts during the negative half cycle in a centre-tap full-wave rectifier?
Ans: The diode connected to end B (D_2), which is then positive with respect to C.

Quick compare: effect of temperature and doping

Property	Metal	Semiconductor	Insulator
Energy gap	0 (overlap)	under 3 eV	over 3 eV
Free carriers at 300 K	very many	few	almost none
Resistance on heating	increases	decreases	very high, falls slightly
Effect of doping	none useful	conductivity rises a lot	not used
Temperature coefficient	positive	negative	negative

Exam Alert: last check

- → Depletion layer formation aur barrier potential: 3 marks pakka.
- → Full-wave rectifier circuit + waveforms: 5 marks ka favourite. Half-wave bhi tayyar rakho.
- → Band diagrams (metal, semiconductor, insulator) aur mass action law numerical.

Formula Sheet: Ch 1–7 at One Glance

Most asked: every formula of Electrostatics, Current, Magnetism, EMI and AC

1. Electric Charges and Fields

Formula	Symbols / meaning	Unit / note
$q = \pm ne$	n = number of electrons, e = 1.6×10^{-19} C	Quantisation of charge
$F = kq_1q_2/r^2$	$k = 1/4\pi\varepsilon_0 = 9 \times 10^9$ N m^2 C^{-2}	Coulomb's law, N
$F_m = F_0/K$	$K = \varepsilon/\varepsilon_0$ = dielectric constant	Force reduces in medium
$\vec{E} = \vec{F}/q_0$	q_0 = small positive test charge	N C^{-1} = V m^{-1}

Formula	Symbols / meaning	Unit / note
$E = kq/r^2$	Field of point charge at distance r	Radially outward for +q
$p = q \times 2a$	2a = separation, direction −q to +q	Dipole moment, C m
$E_{axial} = 2kp/r^3$	On axis, $r \gg a$; along $\vec{p}$	Exact: $2kpr/(r^2-a^2)^2$
$E_{eq} = kp/r^3$	On equatorial line, $r \gg a$; opposite to $\vec{p}$	$E_{axial} = 2E_{eq}$

Formula	Symbols / meaning	Unit / note
$\tau = pE \sin\theta$, $\vec{\tau} = \vec{p} \times \vec{E}$	θ = angle between $\vec{p}$ and $\vec{E}$	Net force in uniform E = 0
$\Phi_E = \vec{E}\cdot\vec{A} = EA\cos\theta$	$\vec{A}$ = area vector (along normal)	N m^2 C^{-1}
$\Phi = q_{enc}/\varepsilon_0$	Gauss's law, closed surface	Outside charges give zero net flux
$E = \lambda/2\pi\varepsilon_0 r$	λ = linear charge density (C m^{-1})	Infinite line charge

Formula	Symbols / meaning	Unit / note
$E = \sigma/2\varepsilon_0$	σ = surface charge density (C m^{-2})	Infinite plane sheet, independent of r
$E = \sigma/\varepsilon_0$	Between two oppositely charged sheets	Zero outside the pair
$E = kq/r^2$ ($r \geq R$); 0 ($r < R$)	Uniformly charged thin spherical shell	E on surface = kq/R^2

Yaad karo

→ Axial = 2 × equatorial, dono $1/r^3$; point charge $1/r^2$, line charge 1/r, sheet r-free.

→ Gauss: flux sirf andar ke charge se, field sabke charge se.

→ Uniform E mein dipole: force zero, torque pE sin θ.

2. Electrostatic Potential and Capacitance

Formula	Symbols / meaning	Unit / note
$V = W/q_0$	Work per unit charge from infinity	Volt = J C^{-1}
$V = kq/r$	Potential of point charge	Scalar, add algebraically
$V = kp\cos\theta/r^2$	Dipole potential; 0 on equatorial line	Falls as $1/r^2$
$E = -dV/dr$	Field points to decreasing V	E ⊥ equipotential surface

Formula	Symbols / meaning	Unit / note
$W = q(V_B - V_A)$	Work by external agent A → B	Zero on equipotential surface
$U = kq_1q_2/r$	PE of two charges	For three: sum over pairs
$U = -pE\cos\theta = -\vec{p}\cdot\vec{E}$	Dipole in uniform field	Min at θ = 0 (stable), max at 180°
$V = kq/R$ ($r \leq R$); kq/r ($r > R$)	Charged conducting sphere/shell	V constant inside conductor

Formula	Symbols / meaning	Unit / note
$C = Q/V$	Capacitance	Farad = C V^{-1}
$C = \varepsilon_0 A/d$	Parallel plate, air	With dielectric: KC_0
$C = \varepsilon_0 A/(d - t + t/K)$	Slab of thickness t, constant K	Metal slab: $\varepsilon_0 A/(d-t)$
$C = 4\pi\varepsilon_0 R$	Isolated sphere	Earth ≈ 711 μF

Formula	Symbols / meaning	Unit / note
$1/C_s = 1/C_1 + 1/C_2 + \ldots$	Series: same Q	$C_s <$ smallest C
$C_p = C_1 + C_2 + \ldots$	Parallel: same V	$C_p >$ largest C
$U = Q^2/2C = \frac{1}{2}CV^2 = \frac{1}{2}QV$	Energy stored	Joule
$u = \frac{1}{2}\varepsilon_0 E^2$	Energy density	J m^{-3}

Formula	Symbols / meaning	Unit / note
$E = E_0/K$, $P = \chi_e \varepsilon_0 E$	Field in dielectric; polarisation	$K = 1 + \chi_e$
$V_c = (C_1V_1 + C_2V_2)/(C_1 + C_2)$	Common potential on sharing	Energy loss = $C_1C_2(V_1-V_2)^2/2(C_1+C_2)$

Yaad karo

→ Capacitor ulta resistor: series mein reciprocal add, parallel mein seedha add.

→ Battery juda: V fixed; battery hata: Q fixed. Pehle yahi dekho.

→ Conductor ke andar E = 0, V = constant (surface jitna).

3. Current Electricity

Formula	Symbols / meaning	Unit / note
$I = dQ/dt$	Rate of flow of charge	Ampere (A)
$I = neAv_d$	n = free electrons per m^3, v_d = drift speed	$v_d \sim 10^{-4}$ m s^{-1}
$v_d = eE\tau/m$	τ = relaxation time	Drift velocity
$\mu = v_d/E = e\tau/m$	Mobility	$m^2\ V^{-1}\ s^{-1}$

Formula	Symbols / meaning	Unit / note
$V = IR$	Ohm's law	Ohm (Ω)
$R = \rho l/A$	ρ = resistivity	Stretch n times: $R \to n^2R$
$\rho = m/ne^2\tau$, $\sigma = 1/\rho$	Resistivity, conductivity	Ω m, S m^{-1}
$\rho_T = \rho_0[1 + \alpha(T - T_0)]$	α = temperature coefficient	$\alpha > 0$ metals, $\alpha < 0$ semiconductors

Formula	Symbols / meaning	Unit / note
$R_s = R_1 + R_2$; $1/R_p = 1/R_1 + 1/R_2$	Series / parallel	Series same I, parallel same V
$P = VI = I^2R = V^2/R$	Power dissipated	Watt; $H = I^2Rt$
$V = \varepsilon - Ir$	Terminal voltage on discharge	Charging: $V = \varepsilon + Ir$
$I = \varepsilon/(R + r)$	Current from a cell	Short circuit: $I = \varepsilon/r$

Formula	Symbols / meaning	Unit / note
$\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$, $r_{eq} = r_1 + r_2$	Cells in series	n cells: $I = n\varepsilon/(R + nr)$
$\varepsilon_{eq} = (\varepsilon_1r_2 + \varepsilon_2r_1)/(r_1 + r_2)$	Cells in parallel	$r_{eq} = r_1r_2/(r_1 + r_2)$
$\Sigma I = 0$; $\Sigma IR = \Sigma\varepsilon$	Kirchhoff's junction and loop rules	Charge and energy conservation
$P/Q = R/S$	Balanced Wheatstone bridge	Galvanometer current zero

Formula	Symbols / meaning	Unit / note
$R/S = l/(100 - l)$	Metre bridge	l in cm from end of R

Yaad karo

→ Drift speed chhoti, par current turant: field poore wire mein light speed se set hota hai.

→ Max power transfer: R = r. Stretch wire: $R \propto l^2$.

→ Bulb series mein: kam watt wala zyada chamakta hai.

4. Moving Charges and Magnetism

Formula	Symbols / meaning	Unit / note
$\vec{F} = q(\vec{v} \times \vec{B})$	Magnetic Lorentz force, $F = qvB\sin\theta$	No work done; speed constant
$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$	Total Lorentz force	Velocity selector: $v = E/B$
$r = mv/qB$	Radius of circular path ($v \perp B$)	$r = \sqrt{(2mK)}/qB$
$T = 2\pi m/qB$, $\nu = qB/2\pi m$	Period, cyclotron frequency	Independent of speed

Formula	Symbols / meaning	Unit / note
$dB = (\mu_0/4\pi)\, I\, dl \sin\theta/r^2$	Biot-Savart law	$\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}
$B = \mu_0 NI/2R$	Centre of circular coil	Tesla (T)
$B = \mu_0 NIR^2/2(R^2 + x^2)^{3/2}$	On axis of loop at distance x	$x \gg R$: $\mu_0 2m/4\pi x^3$
$B = \mu_0 I/2\pi r$	Infinite straight wire (Ampere)	Right-hand thumb rule

Formula	Symbols / meaning	Unit / note
$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$	Ampere's circuital law	Needs symmetry
$B = \mu_0 nI$	Inside long solenoid, n = N/l	Ends: $\mu_0 nI/2$
$B = \mu_0 NI/2\pi r$	Inside toroid	Zero outside
$F = BIl \sin\theta$, $\vec{F} = I(\vec{l} \times \vec{B})$	Force on current-carrying wire	Fleming's left-hand rule

Formula	Symbols / meaning	Unit / note
$F/l = \mu_0 I_1 I_2/2\pi d$	Two parallel wires	Like currents attract; defines ampere
$m = NIA$, $\tau = NIAB \sin\theta$	Current loop as magnetic dipole	m in A m^2
$NIAB = k\varphi$, $I = (k/NAB)\varphi$	Moving coil galvanometer	Radial field, k = torsion const
$I_s = NAB/k$, $V_s = NAB/kR$	Current and voltage sensitivity	$V_s = I_s/R$

Formula	Symbols / meaning	Unit / note
$S = I_g G/(I - I_g)$	Shunt for ammeter (parallel)	Low resistance
$R = V/I_g - G$	Series resistance for voltmeter	High resistance
$\mu_l = evr/2 = (e/2m)L$	Orbital magnetic moment of electron	$\mu_B = 9.27 \times 10^{-24}$ A m^2

Yaad karo

→ Magnetic force kaam nahi karta: KE same, sirf direction badalti hai.

→ Wire: 1/r; loop centre: 1/R; solenoid: r-free, n pe depend.

→ Ammeter = G + chhota shunt parallel; voltmeter = G + bada R series.

5. Magnetism and Matter

Formula	Symbols / meaning	Unit / note
$m = q_m \times 2l$	Bar magnet dipole moment	S to N, A m^2
$B_{axial} = \mu_0 2m/4\pi r^3$	Field on axis of bar magnet	Like electric dipole
$B_{eq} = -\mu_0 m/4\pi r^3$	Field on equatorial line	Opposite to $\vec{m}$
$\vec{\tau} = \vec{m} \times \vec{B}$	Torque in uniform B	$\tau = mB \sin\theta$

Formula	Symbols / meaning	Unit / note
$U = -\vec{m}\cdot\vec{B} = -mB\cos\theta$	Potential energy	Stable at $\theta = 0$
$T = 2\pi\sqrt{(I/mB)}$	Oscillation of magnet in B	I = moment of inertia
$\oint \vec{B}\cdot d\vec{A} = 0$	Gauss's law for magnetism	No isolated monopole
$M = m_{net}/V$	Magnetisation	A m^{-1}

Formula	Symbols / meaning	Unit / note
$H = B_0/\mu_0 = nI$	Magnetic intensity	A m^{-1}
$B = \mu_0(H + M)$	Total field in material	Tesla
$M = \chi H$	χ = susceptibility	Dimensionless
$\mu = \mu_0(1 + \chi)$, $\mu_r = 1 + \chi$	Permeability, relative permeability	$B = \mu H$

Formula	Symbols / meaning	Unit / note
$\chi = C/T$ (para); $\chi = C/(T - T_c)$ (ferro, $T > T_c$)	Curie law, Curie-Weiss law	T_c = Curie temperature

Property	Dia	Para	Ferro
χ	small, negative	small, positive	large, positive
μ_r	slightly < 1	slightly > 1	$\gg 1$
Example	Bi, Cu, water, N_2	Al, Na, O_2	Fe, Co, Ni

Yaad karo

→ DPF: Dia ka χ negative, Para chhota positive, Ferro bahut bada.

→ Superconductor: perfect diamagnet, $\chi = -1$ (Meissner effect).

→ Permanent magnet: high retentivity + high coercivity (steel); electromagnet: soft iron, low coercivity.

6. Electromagnetic Induction

Formula	Symbols / meaning	Unit / note
$\Phi_B = \vec{B}\cdot\vec{A} = BA\cos\theta$	Magnetic flux	Weber (Wb) = T m^2
$\varepsilon = -N\, d\Phi_B/dt$	Faraday's law with Lenz's sign	Volt
$I = \varepsilon/R$, $q = N\Delta\Phi/R$	Induced current, induced charge	q independent of time
$\varepsilon = Blv$	Motional emf (rod $\perp$ B)	Direction: Fleming's right hand

Formula	Symbols / meaning	Unit / note
$F = B^2l^2v/R$	Force to move rod at constant v	$P = B^2l^2v^2/R$
$\varepsilon = \frac{1}{2}B\omega l^2$	Rod rotating about one end	Disc: same with l = R
$N_2\Phi_2 = MI_1$, $\varepsilon_2 = -M\, dI_1/dt$	Mutual inductance	Henry (H)
$M = \mu_0 n_1 n_2 \pi r_1^2 l$	Two long coaxial solenoids ($r_1 < r_2$)	$M_{12} = M_{21}$

Formula	Symbols / meaning	Unit / note
$N\Phi = LI$, $\varepsilon = -L\, dI/dt$	Self inductance (back emf)	Electrical inertia
$L = \mu_0 n^2 Al = \mu_0 N^2 A/l$	Long solenoid	With core: $\mu_r\mu_0 n^2 Al$
$U = \frac{1}{2}LI^2$, $u = B^2/2\mu_0$	Energy in inductor, energy density	J, J m^{-3}
$\varepsilon = NBA\omega \sin\omega t$, $\varepsilon_0 = NBA\omega$	AC generator	Max when plane $\parallel$ B

Yaad karo

→ Lenz = energy conservation: induced current hamesha change ka virodh karta hai.

→ Magnet coil ki taraf: paas wala face same pole banta hai (repel).

→ Eddy current uses: magnetic braking, induction furnace, damping, speedometer; loss rokne ke liye laminated core.

7. Alternating Current

Formula	Symbols / meaning	Unit / note
$v = v_m \sin\omega t$, $\omega = 2\pi\nu$	Instantaneous AC voltage	India: ν = 50 Hz
$I = i_m/\sqrt{2}$, $V = v_m/\sqrt{2}$	rms values	220 V rms $\Rightarrow$ peak 311 V
$i_{av} = 2i_m/\pi = 0.637\, i_m$	Mean over half cycle	Full cycle mean = 0

Formula	Symbols / meaning	Unit / note
$X_L = \omega L$	Inductive reactance	V leads I by $\pi/2$
$X_C = 1/\omega C$	Capacitive reactance	I leads V by $\pi/2$
$Z = \sqrt{(R^2 + (X_L - X_C)^2)}$	Impedance of series LCR	Ohm

Formula	Symbols / meaning	Unit / note
$\tan\varphi = (X_L - X_C)/R$	Phase angle of V over I	$X_L > X_C$: inductive
$V = \sqrt{(V_R^2 + (V_L - V_C)^2)}$	Phasor sum of rms voltages	Not $V_R + V_L + V_C$
$\omega_0 = 1/\sqrt{(LC)}$, $\nu_0 = 1/2\pi\sqrt{(LC)}$	Resonant frequency	Z = R, I = V/R (max)

Formula	Symbols / meaning	Unit / note
$Q = \omega_0 L/R = (1/R)\sqrt{(L/C)}$	Quality factor, $Q = \omega_0/2\Delta\omega$	Sharpness of resonance
$P = VI\cos\varphi$, $\cos\varphi = R/Z$	Average power, power factor	Pure L or C: P = 0
$I\sin\varphi$	Wattless current component	Consumes no power

Formula	Symbols / meaning	Unit / note
$\omega = 1/\sqrt{(LC)}$, $q = q_0\cos\omega t$	LC oscillations	Energy swaps between C and L
$V_s/V_p = N_s/N_p = I_p/I_s$	Ideal transformer	Step-up: $N_s > N_p$
$\eta = P_{out}/P_{in}$	Efficiency	Losses: copper, eddy, hysteresis, flux leakage

Yaad karo

→ CIVIL: C mein I leads V; L mein V leads I (E ke baad I).

→ Resonance: $X_L = X_C$, Z = R, cos φ = 1, I max.

→ Transformer sirf AC pe; V badhe toh I ghate, power same.

Exam Alert: last check

→ Formula likhte hi symbol ka matlab ek line mein likho: step marks milte hain.

→ rms aur peak mix mat karna; $\omega = 2\pi\nu$.

→ Units yaad: Wb, T, H, F, Ω, S; answer hamesha SI unit ke saath.

→ Resonance, transformer, galvanometer conversion: inke numericals har saal aate hain.

Formula Sheet Ch 8–14, Units and Constants

Most asked: formulas of EM Waves, Optics, Modern Physics, Semiconductors + SI units + constants

1. Ch 8: Electromagnetic Waves

Formula	Symbols / meaning	Unit / note
$i_d = \varepsilon_0 d\Phi_E/dt$	displacement current; Φ_E = electric flux	A; exists between capacitor plates
$\oint B\cdot dl = \mu_0(i_c + \varepsilon_0 d\Phi_E/dt)$	Ampere-Maxwell law	conduction + displacement current
$c = 1/\sqrt{(\mu_0\varepsilon_0)}$	speed of EM waves in vacuum	3×10^8 m s^{-1}
$v = 1/\sqrt{(\mu\varepsilon)} = c/n$	speed in a medium; $n = \sqrt{(\mu_r\varepsilon_r)}$	$v < c$ in matter
$E_0/B_0 = E/B = c$	amplitude ratio of fields	$E \perp B \perp$ direction of travel
$E_x = E_0 \sin(kz - \omega t)$	wave along +z; $k = 2\pi/\lambda$, $\omega = 2\pi\nu$	$c = \omega/k = \nu\lambda$
$u = \frac{1}{2}\varepsilon_0E^2 + B^2/2\mu_0$	energy density; both parts equal	J m^{-3}
$I = u_{av}c = \frac{1}{2}c\varepsilon_0E_0^2$	intensity	W m^{-2}
$P_{rad} = I/c$	radiation pressure (absorbing surface)	N m^{-2}; $2I/c$ if reflecting

Yaad karo

→ Order (λ badhta hai): Gamma, X, UV, Visible, IR, Micro, Radio: "Good Xmas Under Very Indian Mango Roof".

→ EM wave transverse hai, medium nahi chahiye; energy E aur B mein barabar bati hoti hai.

2. Ch 9: Ray Optics and Optical Instruments

Formula	Symbols / meaning	Unit / note
$f = R/2$	focal length of spherical mirror	R = radius of curvature
$1/v + 1/u = 1/f$	mirror formula (Cartesian signs)	concave f −ve, convex f +ve
$m = h'/h = -v/u$	magnification by mirror	m −ve: real, inverted
$n_{21} = \sin i / \sin r = v_1/v_2$	Snell's law	$n = c/v$
n = real depth / apparent depth	normal shift $d = t(1 - 1/n)$	object appears raised
$\sin i_c = 1/n$	critical angle (denser to rarer)	glass: $i_c \approx 42°$
$n_2/v - n_1/u = (n_2 - n_1)/R$	refraction at spherical surface	light from n_1 into n_2
$1/f = (n_{21} - 1)(1/R_1 - 1/R_2)$	lens maker's formula	f changes with medium
$1/v - 1/u = 1/f$	thin lens formula	convex f +ve, concave f −ve
$m = v/u$	magnification by lens	m +ve: virtual, erect
$P = 1/f$ (m)	power of lens	dioptre (D)
$1/F = 1/f_1 + 1/f_2$; $P = P_1 + P_2$	lenses in contact	$m = m_1m_2$
$A = r_1 + r_2$; $i + e = A + \delta$	prism relations	δ = deviation
$n = \sin[(A + D_m)/2] / \sin(A/2)$	at minimum deviation, $r = A/2$, $i = e$	D_m = minimum deviation
$\delta = (n - 1)A$	thin prism (small A)	independent of i
$m = 1 + D/f$ (near point); $m = D/f$ (infinity)	simple microscope	D = 25 cm
$m = (L/f_o)(D/f_e)$	compound microscope, normal adjustment	L = tube length
$m = -f_o/f_e$; $L = f_o + f_e$	astronomical telescope, normal adjustment	f_o large, f_e small

Yaad karo

→ Sign convention: distance pole/optical centre se; light ki direction mein +ve. Real object ka u hamesha −ve.

→ Lens ko water mein daalo: n_{21} ghatta hai, f badhta hai, power ghatti hai.

→ Microscope: dono f chhote; telescope: f_o bada, f_e chhota.

3. Ch 10: Wave Optics

Formula	Symbols / meaning	Unit / note
$\Delta x = n\lambda$ (n = 0, 1, 2 ...)	constructive interference (bright)	$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$
$\Delta x = (2n - 1)\lambda/2$	destructive interference (dark)	$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$
$I = I_1 + I_2 + 2\sqrt{(I_1 I_2)}\cos\varphi$	resultant intensity	$I_1 = I_2 = I_0$: $I = 4I_0\cos^2(\varphi/2)$
$\Delta x = yd/D$	path difference in YDSE at height y	d = slit separation

$y_n = n\lambda D/d$	position of n^{th} bright fringe	dark: $(2n - 1)\lambda D/2d$
$\beta = \lambda D/d$	fringe width (bright = dark)	in medium: $\beta' = \beta/n$
$\theta = \lambda/d$	angular fringe width	independent of D
$a \sin\theta = n\lambda$ (n = ±1, ±2 ...)	minima of single slit diffraction	central max at $\theta = 0$

$2\lambda D/a$; angular $2\lambda/a$	width of central maximum	twice a secondary max
$z_F = a^2/\lambda$	Fresnel distance	ray optics valid for $z < z_F$
$I = I_0\cos^2\theta$	Malus' law	unpolarised through polaroid: $I_0/2$
$n = \tan i_B$	Brewster's law	$i_B + r = 90^\circ$

$\lambda' = \lambda/n$; ν unchanged	wavelength in a medium	speed $v = c/n$

Yaad karo

→ YDSE ko paani mein rakho: λ aur β dono n guna kam; fringes paas-paas.

→ White light se: central fringe white, baaki coloured.

→ Polaroid crossed (90°) to I = 0.

4. Ch 11: Dual Nature of Radiation and Matter

Formula	Symbols / meaning	Unit / note
$E = h\nu = hc/\lambda$	energy of a photon	hc = 1240 eV nm
$p = h/\lambda = E/c$	momentum of a photon	rest mass of photon = 0
$N = P/h\nu = P\lambda/hc$	photons emitted per second	P = power of source
$\varphi_0 = h\nu_0 = hc/\lambda_0$	work function; ν_0 threshold frequency	eV; no emission if $\nu < \nu_0$

$K_{max} = h\nu - \varphi_0$	Einstein's photoelectric equation	$K_{max} = \frac{1}{2}mv_{max}^2$
$eV_0 = K_{max} = h(\nu - \nu_0)$	stopping potential V_0	V_0 independent of intensity
$V_0 = (h/e)\nu - \varphi_0/e$	V_0 vs ν graph: straight line	slope = h/e, intercept ν_0
$\lambda = h/p = h/mv$	de Broglie wavelength	m s^{-1} for v

$\lambda = h/\sqrt{(2mK)}$	in terms of kinetic energy	K in joule
$\lambda = h/\sqrt{(2meV)} = 1.227/\sqrt{V}$ nm	electron accelerated through V volt	V = 100 V: λ = 0.123 nm

Yaad karo

→ Intensity badhao: current badhta hai, K_{max} aur V_0 same.

→ Frequency badhao: K_{max} aur V_0 badhte hain, current same (saturation).

→ Same K par $\lambda \propto 1/\sqrt{m}$: electron ka λ proton se zyada.

5. Ch 12: Atoms

Formula	Symbols / meaning	Unit / note
$r_0 = 2kZe^2/K$	distance of closest approach (α-particle)	$k = 1/4\pi\varepsilon_0$, K = initial KE
$N(\theta) \propto 1/\sin^4(\theta/2)$	Rutherford scattering	most α pass undeflected
$mvr = nh/2\pi$	Bohr's quantisation of angular momentum	n = 1, 2, 3 ...
$2\pi r_n = n\lambda$	de Broglie explanation of Bohr orbit	standing wave in orbit

$r_n = n^2h^2\varepsilon_0/\pi me^2Z = 0.529\ n^2/Z$ Å	radius of n^{th} orbit	$r \propto n^2$
$v_n = e^2Z/2\varepsilon_0 nh = 2.18 \times 10^6\ Z/n$ m s^{-1}	speed of electron	$v \propto 1/n$
$E_n = -me^4Z^2/8\varepsilon_0^2n^2h^2 = -13.6\ Z^2/n^2$ eV	total energy	$E \propto -1/n^2$
$K = -E$, $U = 2E$	kinetic and potential energy	K = 13.6 eV for H ground state

$h\nu = E_i - E_f$	frequency of emitted photon	absorption: reverse
$1/\lambda = RZ^2(1/n_f^2 - 1/n_i^2)$	Rydberg formula	$R = 1.097 \times 10^7\ m^{-1}$
$N = n(n-1)/2$	number of spectral lines from level n	n = 4: 6 lines

Yaad karo

→ H atom: ionisation energy 13.6 eV, ionisation potential 13.6 V; first excitation 10.2 eV.

→ Balmer H_α $(3 \to 2)$ = 656.3 nm; Lyman shortest λ = 91.2 nm.

→ Series order: "Lazy Boys Play Billiards Properly".

6. Ch 13: Nuclei

Formula	Symbols / meaning	Unit / note
$R = R_0 A^{1/3}$	nuclear radius; R_0 = 1.2 fm	1 fm = 10^{-15} m
$\rho = 3m/4\pi R_0^3 \approx 2.3 \times 10^{17}\ kg\ m^{-3}$	nuclear density	same for all nuclei
$E = mc^2$	mass-energy equivalence	1 u = 931.5 MeV/c^2
$\Delta m = [Zm_p + (A-Z)m_n] - M$	mass defect	in u or kg

$E_b = \Delta m\, c^2 = \Delta m(u) \times 931.5$ MeV	binding energy	energy to split nucleus
$E_{bn} = E_b/A$	binding energy per nucleon	max ≈ 8.75 MeV at A = 56 (Fe)
$Q = (m_{reactants} - m_{products})c^2$	Q-value of nuclear reaction	Q > 0: exothermic
$^{235}U + n \to {}^{141}Ba + {}^{92}Kr + 3n$	typical fission	≈ 200 MeV per fission

$4\,{}^1H \to {}^4He + 2e^+ + 2\nu + 26.7$ MeV	fusion in the sun (p-p cycle)	needs ~10^7 K
$K = N_{n+1}/N_n$	multiplication factor	K = 1 critical (reactor)

Yaad karo

→ $R \propto A^{1/3}$, isliye volume $\propto$ A aur density constant.

→ Light nuclei: fusion; heavy nuclei: fission; dono mein E_{bn} badhta hai, energy release.

→ Nuclear force: strongest, short range (2-3 fm), charge independent.

7. Ch 14: Semiconductor Electronics

Formula	Symbols / meaning	Unit / note
$n_e = n_h = n_i$	intrinsic semiconductor	n_i rises with T
$n_e n_h = n_i^2$	mass action law (equilibrium)	holds for doped too
$n_e \approx N_D \gg n_h$	n-type (pentavalent: P, As, Sb)	electrons majority
$n_h \approx N_A \gg n_e$	p-type (trivalent: B, Al, In)	holes majority

E_g: Si 1.1 eV, Ge 0.7 eV, C 5.4 eV	band gap	insulator E_g > 3 eV
V_0: Si ≈ 0.7 V, Ge ≈ 0.3 V	barrier potential (knee voltage)	forward bias reduces it
$r_d = \Delta V/\Delta I$	dynamic resistance of diode	from slope of I-V curve
$\nu_{out} = \nu_{in}$	half-wave rectifier ripple frequency	50 Hz for mains

$\nu_{out} = 2\nu_{in}$	full-wave rectifier ripple frequency	100 Hz for mains
$\lambda = hc/E_g = 1240/E_g(eV)$ nm	LED colour / photodiode cut-off	$h\nu \geq E_g$ to detect

Yaad karo

→ Forward bias: p ko +ve; depletion layer patli, current mA mein.

→ Reverse bias: depletion layer moti, current μA mein (minority carriers).

→ Doping ke baad bhi crystal electrically neutral rehta hai.

8. Physical Quantities: SI Units and Dimensions

Quantity	SI unit	Dimensions
Force	newton (N)	$[MLT^{-2}]$
Work, energy	joule (J)	$[ML^2T^{-2}]$
Power	watt (W)	$[ML^2T^{-3}]$
Pressure, stress	pascal (Pa)	$[ML^{-1}T^{-2}]$

Momentum	kg m s^{-1}	$[MLT^{-1}]$
Angular momentum, h	J s	$[ML^2T^{-1}]$
Frequency	hertz (Hz)	$[T^{-1}]$
Electric charge	coulomb (C)	[AT]

Electric current	ampere (A)	[A]
Electric field	N C^{-1} = V m^{-1}	$[MLT^{-3}A^{-1}]$
Potential, emf	volt (V)	$[ML^2T^{-3}A^{-1}]$
Electric dipole moment	C m	[LTA]

Capacitance	farad (F)	$[M^{-1}L^{-2}T^{4}A^{2}]$
Permittivity ε_0	$C^2\ N^{-1}\ m^{-2}$ = $F\ m^{-1}$	$[M^{-1}L^{-3}T^{4}A^{2}]$
Resistance, reactance, impedance	ohm (Ω)	$[ML^{2}T^{-3}A^{-2}]$
Resistivity	Ω m	$[ML^{3}T^{-3}A^{-2}]$

Magnetic field B	tesla (T)	$[MT^{-2}A^{-1}]$
Magnetic flux	weber (Wb)	$[ML^{2}T^{-2}A^{-1}]$
Magnetic moment	$A\ m^2$ = $J\ T^{-1}$	$[L^{2}A]$
Magnetic intensity H, magnetisation M	$A\ m^{-1}$	$[L^{-1}A]$

Permeability μ_0	$T\ m\ A^{-1}$ = $H\ m^{-1}$	$[MLT^{-2}A^{-2}]$
Inductance L, M	henry (H)	$[ML^{2}T^{-2}A^{-2}]$
Susceptibility, refractive index, Q-factor	none	dimensionless
Power of a lens	dioptre (D) = m^{-1}	$[L^{-1}]$

Intensity of wave	$W\ m^{-2}$	$[MT^{-3}]$
Wave number, Rydberg constant	m^{-1}	$[L^{-1}]$
Decay constant λ	s^{-1}	$[T^{-1}]$
Activity	becquerel (Bq)	$[T^{-1}]$; 1 Ci = 3.7×10^{10} Bq

9. Physical Constants at a Glance

Constant	Symbol	Value
Speed of light in vacuum	c	$2.998 \times 10^{8}\ m\ s^{-1} \approx 3 \times 10^{8}$
Elementary charge	e	1.602×10^{-19} C
Electron mass	m_e	9.109×10^{-31} kg = 0.511 MeV/c^2
Proton mass	m_p	1.6726×10^{-27} kg = 1.007276 u

Specific charge of electron	e/m_e	$1.759 \times 10^{11}\ C\ kg^{-1}$
Planck constant	h	6.626×10^{-34} J s = 4.136×10^{-15} eV s
Product hc	hc	1.986×10^{-25} J m ≈ 1240 eV nm
Permittivity of free space	ε_0	$8.854 \times 10^{-12}\ C^2\ N^{-1}\ m^{-2}$

Coulomb constant	$1/4\pi\varepsilon_0$	$8.988 \times 10^{9}\ N\ m^2\ C^{-2} \approx 9 \times 10^{9}$
Permeability of free space	μ_0	$4\pi \times 10^{-7}\ T\ m\ A^{-1}$
Boltzmann constant	k_B	$1.381 \times 10^{-23}\ J\ K^{-1}$
Avogadro number	N_A	$6.022 \times 10^{23}\ mol^{-1}$

Universal gas constant	R	$8.314\ J\ mol^{-1}\ K^{-1}$
Faraday constant	$F = N_A e$	$9.649 \times 10^{4}\ C\ mol^{-1}$
Gravitational constant	G	$6.674 \times 10^{-11}\ N\ m^2\ kg^{-2}$
Atomic mass unit	1 u	1.6605×10^{-27} kg

Energy equivalent of 1 u	1 u × c^2	931.5 MeV
Rydberg constant	R	$1.097 \times 10^{7}\ m^{-1}$
Bohr radius	a_0	0.529×10^{-10} m = 0.529 Å
Wien constant	b	2.898×10^{-3} m K

Stefan constant	σ	$5.67 \times 10^{-8}\ W\ m^{-2}\ K^{-4}$
Acceleration due to gravity	g	$9.8\ m\ s^{-2}$

Useful conversions

- 1 Å = 10^{-10} m; 1 nm = 10^{-9} m; 1 fm = 10^{-15} m.
- 1 MeV = 1.602×10^{-13} J; 1 kWh = 3.6×10^{6} J.
- 1 T = 10^{4} gauss; 1 μF = 10^{-6} F; 1 pF = 10^{-12} F.
- 1 Ci = 3.7×10^{10} Bq; 1 year ≈ 3.15×10^{7} s.

Quick checks with dimensions

- $1/\sqrt{(\mu_0\varepsilon_0)}$ has dimensions of speed $[LT^{-1}]$.
- L/R and RC both have dimensions of time [T].
- E/B has dimensions of speed; $\varepsilon_0 E^2$ and B^2/μ_0 have dimensions of energy density.
- h/e (slope of the V_0–ν graph) is measured in V s.

Exam Alert: last check

- Units hamesha final answer ke saath likho, warna mark katega.
- e, h, c, m_e, ε_0 ki values exam se pehle ek baar zor se bolo.
- eV ko J mein badalna: × 1.6×10^{-19}.
- $1/4\pi\varepsilon_0 = 9 \times 10^{9}$ aur $\mu_0/4\pi = 10^{-7}$: ye do values sabse zyada kaam aati hain.
- Answer ko significant figures mein likho, data jitne digits ka ho.

Derivations, Differences, Graphs and Most Expected Questions

Most asked: must-do derivations list, key differences, graphs, and most expected board questions

1. Must-do Derivations: Chapter-wise

Derivation	Marks	Key steps in one line
Ch1 Field on axial line of dipole	3	$E_+ - E_-$ at distance r; for $r \gg a$, $E = 2kp/r^3$ along p
Field on equatorial line of dipole	3	Resolve both fields; sine parts cancel; $E = kp/r^3$ opposite to p
Field of infinite line charge (Gauss)	3	Cylindrical surface; flux $E \cdot 2\pi rl = \lambda l/\varepsilon_0$; $E = \lambda/2\pi\varepsilon_0 r$
Ch2 Potential due to point charge	2	$V = -\int E \cdot dr$ from ∞ to r; $V = kq/r$
Potential due to dipole	3	$V = kq/r_1 - kq/r_2$; $r \gg a$ gives $V = kp\cos\theta/r^2$
Capacitance of parallel plate capacitor	2	$E = \sigma/\varepsilon_0$, $V = Ed$; $C = \varepsilon_0 A/d$
Energy stored in capacitor	2	$dW = (q/C)dq$; $U = Q^2/2C = \frac{1}{2}CV^2$; $u = \frac{1}{2}\varepsilon_0 E^2$
Ch3 Drift velocity and $I = neAv_d$	3	$a = -eE/m$, $v_d = -eE\tau/m$; charge crossing area in Δt: $neAv_d\Delta t$
Ch4 Biot-Savart: field on axis of loop	3	dB from element; cos parts cancel; $B = \mu_0 IR^2/2(R^2 + x^2)^{3/2}$
Ampere's law: field of long straight wire	2	Circle of radius r; $B \cdot 2\pi r = \mu_0 I$; $B = \mu_0 I/2\pi r$
Field inside long solenoid	3	Rectangular loop; $Bl = \mu_0 nlI$; $B = \mu_0 nI$
Force between parallel conductors	3	$B_1 = \mu_0 I_1/2\pi d$; $F/l = \mu_0 I_1 I_2/2\pi d$; like currents attract
Torque on current loop in B	3	Forces on arms; $\tau = NIAB\sin\theta = m \times B$
Ch5 Dipole oscillating in B	2	$\tau = -mB\sin\theta \approx -mB\theta$; SHM; $T = 2\pi\sqrt{(I/mB)}$
Ch6 Motional emf (rod on rails)	3	Flux Blx; $\varepsilon = -d\Phi/dt = Blv$; also from Lorentz force qvB
Mutual inductance of coaxial solenoids	3	Field of inner: $\mu_0 n_2 I_2$; $M = \mu_0 n_1 n_2 \pi r_1^2 l$
AC generator emf	3	$\Phi = NBA\cos\omega t$; $\varepsilon = NBA\omega\sin\omega t$; $\varepsilon_0 = NBA\omega$
Ch7 rms value of AC	2	Mean of $\sin^2 = \frac{1}{2}$; $I_{rms} = i_m/\sqrt{2}$
Series LCR: impedance and phase	5	Phasors: $V^2 = V_R^2 + (V_L - V_C)^2$; $Z = \sqrt{(R^2 + (X_L - X_C)^2)}$, $\tan\varphi = (X_L - X_C)/R$
Resonance and Q factor	3	$X_L = X_C$; $\omega_0 = 1/\sqrt{LC}$; $Q = \omega_0 L/R$
Transformer ratio	3	$\varepsilon_s/\varepsilon_p = N_s/N_p$; ideal: $I_p/I_s = N_s/N_p$
Ch8 Displacement current	2	Between capacitor plates; $i_d = \varepsilon_0 d\Phi_E/dt$
Ch9 Mirror formula (concave)	3	Similar triangles; $1/v + 1/u = 1/f$ with sign convention
Lens maker's formula	3	Two refracting surfaces in sequence; $1/f = (n-1)(1/R_1 - 1/R_2)$

Derivation	Marks	Key steps in one line
Magnifying power: microscope, telescope	3	$m = (L/f_o)(D/f_e)$; telescope $m = f_o/f_e$ (normal adjustment)
Ch10 Snell's law by Huygens	3	Wavefront AB to CE in time τ; $\sin i/\sin r = v_1/v_2$
YDSE fringe width	5	Path difference xd/D; bright: $n\lambda$, dark: $(n + ½)\lambda$; $\beta = \lambda D/d$
Ch11 Einstein's photoelectric equation	3	$h\nu = \varphi_0 + K_{max}$; $eV_0 = h(\nu - \nu_0)$

Derivation	Marks	Key steps in one line
de Broglie wavelength of electron	2	$\lambda = h/p = h/\sqrt{(2meV)} = 1.227/\sqrt{V}$ nm
Ch12 Bohr radius and energy	5	$mv^2/r = ke^2/r^2$, $mvr = nh/2\pi$; $r_n \propto n^2$, $E_n = -13.6/n^2$ eV
Ch13 Nuclear density is constant	2	$R = R_0A^{1/3}$; $\rho = Am_u/(4/3)\pi R_0^3A$, A cancels
Ch14 Full wave rectifier working	3	Diodes conduct in alternate halves; same direction through load; output frequency 2ν

Yaad karo: derivation strategy

→ Pehle labelled diagram, phir formula: diagram ke 1 mark alag milte hain.

→ Har step mein assumption likho: $r \gg a$, small angle, uniform field.

→ Last line box karo aur SI unit likho.

2. Key Differences: Board Favourites

Diamagnetic vs paramagnetic vs ferromagnetic

Property	Dia	Para	Ferro
χ	small, negative	small, positive	large, positive
μ_r	slightly < 1	slightly > 1	$\gg 1$ (10^3–10^5)
In non-uniform B	moves to weaker field	moves to stronger field	strongly to stronger field
Temperature	independent	$\chi \propto 1/T$ (Curie)	para above T_C
Examples	Bi, Cu, water, N_2	Al, Na, O_2	Fe, Co, Ni

Interference vs diffraction

Interference	Diffraction
Superposition of waves from two coherent sources	Superposition of wavelets from different parts of the same wavefront
Fringes of equal width	Central maximum twice as wide as others
All bright fringes of equal intensity	Intensity falls rapidly for higher orders

n-type vs p-type semiconductor

n-type	p-type
Doped with pentavalent atom (P, As, Sb)	Doped with trivalent atom (B, Al, In)
Donor impurity; majority carriers electrons	Acceptor impurity; majority carriers holes
$n_e \gg n_h$; donor level just below conduction band	$n_h \gg n_e$; acceptor level just above valence band

Conductor vs semiconductor vs insulator

Property	Conductor	Semiconductor	Insulator
Band gap E_g	zero (bands overlap)	small (< 3 eV)	large (> 3 eV)
Resistivity (Ω m)	10^{-8}–10^{-6}	10^{-5}–10^{6}	10^{11}–10^{19}
Effect of rising T on R	increases	decreases	decreases (slightly)
Examples	Cu, Al, Ag	Si, Ge	diamond, glass

AC vs DC

AC	DC
Magnitude and direction change periodically	Constant direction
Stepped up or down by transformer	Cannot use a transformer
Lower transmission loss at high voltage	High loss over long distance

Self vs mutual inductance

Self inductance L	Mutual inductance M
Emf induced in the same coil by its own changing current	Emf induced in one coil by changing current in a nearby coil
$\varepsilon = -L\,dI/dt$; $L = \mu_0n^2Al$	$\varepsilon_2 = -M\,dI_1/dt$; $M = \mu_0n_1n_2Al$
Depends on one coil's geometry and core	Also depends on separation and orientation of coils

Real vs virtual image

Real image	Virtual image
Rays actually meet	Rays appear to diverge from the point
Can be taken on a screen	Cannot be taken on a screen
Always inverted (single mirror or lens)	Always erect

Compound microscope vs astronomical telescope

Microscope	Telescope
Views tiny nearby objects	Views distant objects
$f_o < f_e$; both small	$f_o \gg f_e$; large aperture objective
$m = (L/f_o)(D/f_e)$	$m = f_o/f_e$

Nuclear fission vs fusion

Fission	Fusion
Heavy nucleus splits into two medium nuclei	Light nuclei combine into a heavier nucleus
Triggered by slow neutrons; room temperature	Needs about 10^7 K (thermonuclear)
About 200 MeV per event; 0.9 MeV/nucleon	Less per event but more per nucleon (about 6.7 MeV/nucleon for D-T)

Emf vs terminal voltage

Emf ε	Terminal voltage V
Potential difference across terminals when no current is drawn	Potential difference when current I flows
Independent of external resistance	$V = \varepsilon - Ir$ (discharging), $\varepsilon + Ir$ (charging)
Cause of current	Effect; always $\leq \varepsilon$ while discharging

3. Important Graphs at One Glance

Graph	Shape	What it shows
E vs r (charged shell)	zero inside, then $1/r^2$ fall	No field inside a conducting shell
V vs I (ohmic conductor)	straight line through origin	Slope = R; Ohm's law holds
ρ vs T (semiconductor)	falls exponentially	More carriers at higher T
B vs x (axis of loop)	peak at centre, falls on both sides	$B_{max} = \mu_0 I/2R$ at x = 0

Graph	Shape	What it shows
B vs H (hysteresis)	closed loop	Retentivity, coercivity, energy loss = area
ε vs t (AC generator)	sine curve	$\varepsilon = NBA\omega \sin \omega t$
I vs ω (series LCR)	peak at ω_0	Resonance; sharper for small R (high Q)
δ vs i (prism)	U-shaped, minimum D_m	Minimum deviation at i = e

Graph	Shape	What it shows
Intensity vs position (YDSE)	equal maxima, equal spacing	Fringe width $\beta = \lambda D/d$
Photocurrent vs anode potential	saturates; cuts axis at $-V_0$	V_0 independent of intensity, depends on ν
V_0 vs ν	straight line, slope h/e	Intercept on ν-axis = ν_0
B/A vs A	rises fast, flat near 8.75 MeV at A ≈ 56, falls slowly	Fission and fusion both release energy

Graph	Shape	What it shows
I-V of p-n diode	knee at 0.7 V (Si) forward; tiny reverse current	Rectifying action

4. Most Expected Questions 2026-27

Ch	Question	Marks
1	Derive the electric field at a point on the equatorial line of a dipole.	3
1	State Gauss's law; find the field of an infinite plane sheet of charge.	3
2	Find capacitance of a parallel plate capacitor partly filled with a dielectric slab.	3
2	Derive energy stored in a capacitor; find energy density.	3

Ch	Question	Marks
3	Derive $I = neAv_d$ and the expression for resistivity.	3
3	State Kirchhoff's rules; derive Wheatstone bridge balance.	3
3	Graph of V vs I of a cell; find emf and internal resistance.	2
4	Derive B on the axis of a circular loop using Biot-Savart law.	3

Ch	Question	Marks
4	Force between two parallel conductors; define ampere.	3
4	Moving coil galvanometer: principle, radial field, conversion to ammeter.	5
5	Compare dia, para and ferro magnetic materials (3 points).	3
5	Draw hysteresis loop; define retentivity and coercivity.	2

Ch	Question	Marks
6	Derive motional emf; find force, power and heat for a sliding rod.	3
6	Mutual inductance of two long coaxial solenoids.	3
6	State Lenz's law; show it obeys energy conservation.	2
7	Series LCR: phasor diagram, impedance, phase angle, resonance.	5

Ch	Question	Marks
7	Transformer: principle, ratio, four energy losses.	3
7	Wattless current; why a choke is preferred over a resistor.	2
8	Displacement current; how is it different from conduction current?	2
8	Arrange EM waves by wavelength; one use each.	3

Ch	Question	Marks
9	Derive the lens maker's formula.	3
9	Compound microscope: ray diagram and magnifying power.	5
9	Prism: derive n in terms of A and D_m.	3
10	YDSE: derive fringe width; effect of changes on pattern.	5

Ch	Question	Marks
10	Prove Snell's law using Huygens' principle.	3
10	Single slit diffraction: width of central maximum; Brewster's law.	3
11	Einstein's equation; explain the three photoelectric laws.	3
11	Graphs: photocurrent vs potential for different intensities and frequencies.	3

Ch	Question	Marks
12	Bohr's postulates; derive radius and energy of the n^{th} orbit.	5
12	Spectral series of hydrogen; find the shortest wavelength of Lyman series.	3
12	Alpha scattering experiment: observations and conclusions.	3
13	Draw the B/A curve; explain fission and fusion from it.	3

Ch	Question	Marks
13	Mass defect and binding energy numerical (in MeV).	3
14	Full wave rectifier: circuit, working, waveforms.	3
14	p-n junction: formation of depletion layer and barrier potential.	3
14	I-V characteristic of a diode in forward and reverse bias.	2

Exam Alert: last night

→ 5 mark wale pakke karo: LCR, YDSE, Bohr model, microscope, galvanometer.

→ Numericals mein unit aur significant figures check karo.

→ Har chapter ka kam se kam ek derivation aur ek graph tayyar rakho.

ALL THE BEST FOR YOUR BOARD EXAM!

Most asked: revise derivations, draw every ray diagram once, solve one timed paper, sleep well

Yeh notebook yahin khatam hoti hai, par tumhari **tayyari** ab final round mein hai.

1. Paheli aur Bujho ki taraf se

Paheli-Bujho

Bujho: Paheli, poori Physics khatam! Ab kya?

Paheli: Ab derivations likh ke revise karo aur ek timed sample paper.

Bujho: Aur numerical mein atak gaye toh?

Paheli: Formula likho, values SI unit mein daalo, step by step. Step marks milte hain!

Bujho: All the best, sabko!

2. Exam Day Checklist

- → Admit card, two blue pens, pencil, eraser, scale: raat ko hi bag mein.
- → Reach the centre 30 minutes early.
- → Reading time: choice questions tick karo, long answers ka order plan karo.
- → Har numerical mein formula, substitution, unit ke saath answer.
- → Last 10 minutes: units, signs and arrow directions check karo.

3. 10 Lines for the Exam Hall Corridor

1. $E = kq/r^2$, $V = kq/r$; inside a conductor $E = 0$.
2. $C = \varepsilon_0 A/d$; with dielectric $C = K\varepsilon_0 A/d$; $U = Q^2/2C$.
3. Drift velocity $v_d = I/neA$; $\rho = m/ne^2\tau$.
4. B at centre of loop $= \mu_0 I/2R$; solenoid $B = \mu_0 nI$.
5. Faraday: $\varepsilon = -d\Phi/dt$; motional emf $= Blv$.
6. Resonance $\omega_0 = 1/\sqrt{(LC)}$; power $P = V_{rms} I_{rms} \cos\varphi$.
7. Mirror/lens: $1/v \pm 1/u = 1/f$, sign convention carefully.
8. YDSE fringe width $\beta = \lambda D/d$.
9. Photoelectric: $K_{max} = h\nu - \varphi_0$; $\lambda = h/p$.
10. Bohr: $r \propto n^2$, $E_n = -13.6/n^2$ eV.

4. Must-Draw Diagrams: Tick Karo

Diagram checklist

Ch	Diagrams
1-2	field lines, dipole field, equipotentials, capacitor with dielectric
3	Wheatstone bridge, metre bridge, potentiometer
4-5	moving coil galvanometer, solenoid field, hysteresis loop
6-7	AC generator, transformer, LCR phasor diagram
8	EM wave (E and B perpendicular)
9	ray diagrams, microscope, telescope, prism
10	Huygens wavefronts, YDSE, single slit pattern
11-13	photocell, I-V graphs, energy levels, B/A curve
14	p-n junction, half and full wave rectifier, diode I-V

Nitesh Bhaiya ki baat

Tumne poori Class 12 Physics ek notebook mein padh li, yeh khud mein badi baat hai. Ab bas revise karo, likh ke practice karo, aur khud pe bharosa rakho. Board exam tumhara hai!

Nitesh Bhaiya | Eduspark

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